

# QUANTUM COMPUTING SOFTWARE MARKET

GLOBAL FORECAST TO 2026

BY COMPONENT (SOFTWARE, SERVICES), DEPLOYMENT MODE (CLOUD, ON-PREMISES),  
ORGANIZATION SIZE, TECHNOLOGY, APPLICATION (OPTIMIZATION, SIMULATION),  
VERTICAL (BFSI, GOVERNMENT), AND REGION

MarketsandMarkets™ is the world's largest revenue impact company, serving over 7500 customers. 80% of top 2000 companies globally rely on us for identifying new high growth and niche revenue opportunities. In the face of constant technology innovation and market disruption, we help organizations plan and operationalize their future revenue mix decisions by identifying over 30,000 high growth opportunities ranging from \$1B to \$500B across 90+ industry trends and markets. Organizations choose MarketsandMarkets™ to stay ahead of the curve and accelerate their revenue decisions and implementations by 6 – 12 months, giving them a unique, first-mover advantage. Our revenue impact methodology provides quantified and actionable insights on converged, granular and connected market ecosystems that result from disruptive technologies and high-growth markets. We provide an extended lens on not only what will impact our client's revenue but also what will impact their clients' revenues, continually uncovering latent opportunities.

We work across all major B2B industries with C-level executives in functions such as Strategy, Marketing, Sales, R&D, Product, and M&A. MarketsandMarkets™ brings exclusive high-growth markets intelligence generated by over 850 SMEs and analysts along with its proprietary Revenue Impact platform (Knowledge Store).

**Copyright © 2021 MarketsandMarkets™**

All Rights Reserved. This document contains highly confidential information and is the sole property of MarketsandMarkets™. No part of it shall be circulated, copied, quoted, or otherwise reproduced without the prior written approval of MarketsandMarkets™.

## TABLE OF CONTENTS

---

|          |                                                                                |           |
|----------|--------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>INTRODUCTION .....</b>                                                      | <b>26</b> |
| 1.1      | INTRODUCTION TO COVID-19.....                                                  | 26        |
| 1.2      | COVID-19 HEALTH ASSESSMENT .....                                               | 26        |
| 1.3      | COVID-19 ECONOMIC ASSESSMENT .....                                             | 28        |
| 1.3.1    | COVID-19 ECONOMIC IMPACT—SCENARIO ASSESSMENT.....                              | 29        |
| 1.4      | OBJECTIVES OF STUDY.....                                                       | 31        |
| 1.5      | MARKET DEFINITION.....                                                         | 31        |
| 1.5.1    | INCLUSIONS AND EXCLUSIONS .....                                                | 32        |
| 1.6      | MARKET SCOPE.....                                                              | 33        |
| 1.6.1    | MARKET SEGMENTATION.....                                                       | 33        |
| 1.6.2    | YEARS CONSIDERED FOR STUDY .....                                               | 34        |
| 1.7      | CURRENCY.....                                                                  | 34        |
| 1.8      | STAKEHOLDERS .....                                                             | 34        |
| <b>2</b> | <b>RESEARCH METHODOLOGY .....</b>                                              | <b>36</b> |
| 2.1      | RESEARCH DATA .....                                                            | 36        |
| 2.1.1    | SECONDARY DATA.....                                                            | 37        |
| 2.1.2    | PRIMARY DATA.....                                                              | 37        |
| 2.1.2.1  | Breakup of primary profiles.....                                               | 38        |
| 2.1.2.2  | Key industry insights.....                                                     | 38        |
| 2.2      | MARKET BREAKUP AND DATA TRIANGULATION .....                                    | 39        |
| 2.3      | MARKET SIZE ESTIMATION.....                                                    | 40        |
| 2.4      | MARKET FORECAST .....                                                          | 42        |
| 2.5      | ASSUMPTIONS FOR STUDY .....                                                    | 43        |
| 2.6      | LIMITATIONS OF STUDY.....                                                      | 44        |
| <b>3</b> | <b>EXECUTIVE SUMMARY.....</b>                                                  | <b>45</b> |
| <b>4</b> | <b>PREMIUM INSIGHTS .....</b>                                                  | <b>50</b> |
| 4.1      | ATTRACTIVE OPPORTUNITIES IN QUANTUM COMPUTING SOFTWARE MARKET .....            | 50        |
| 4.2      | ASIA PACIFIC QUANTUM COMPUTING SOFTWARE MARKET, BY COMPONENT AND COUNTRY ..... | 51        |
| 4.3      | QUANTUM COMPUTING SOFTWARE MARKET: MAJOR COUNTRIES.....                        | 51        |

|          |                                                                                                             |           |
|----------|-------------------------------------------------------------------------------------------------------------|-----------|
| <b>5</b> | <b>MARKET OVERVIEW.....</b>                                                                                 | <b>52</b> |
| 5.1      | INTRODUCTION .....                                                                                          | 52        |
| 5.2      | MARKET DYNAMICS.....                                                                                        | 52        |
| 5.2.1    | DRIVERS .....                                                                                               | 53        |
| 5.2.1.1  | Growing adoption of quantum computing software in banking, financial services, and insurance vertical ..... | 53        |
| 5.2.1.2  | Government support for development and deployment of quantum computing technology .....                     | 53        |
| 5.2.1.3  | Increasing number of strategic alliances for research and development .....                                 | 54        |
| 5.2.2    | RESTRAINTS.....                                                                                             | 54        |
| 5.2.2.1  | Technical and implementation challenges.....                                                                | 54        |
| 5.2.3    | OPPORTUNITIES.....                                                                                          | 54        |
| 5.2.3.1  | Growing adoption of quantum computing technology in healthcare vertical.....                                | 54        |
| 5.2.3.2  | Growing need for high-level computing performance .....                                                     | 55        |
| 5.2.4    | CHALLENGES .....                                                                                            | 55        |
| 5.2.4.1  | Lack of highly skilled professionals .....                                                                  | 55        |
| 5.3      | IMPACT OF COVID-19 ON QUANTUM COMPUTING SOFTWARE MARKET .....                                               | 55        |
| 5.3.1    | PRE-COVID-19 .....                                                                                          | 55        |
| 5.3.2    | POST-COVID-19 .....                                                                                         | 55        |
| 5.4      | CASE STUDY ANALYSIS .....                                                                                   | 56        |
| 5.4.1    | USE OF QUANTUM COMPUTING IN ADVANCED MACHINE LEARNING.....                                                  | 56        |
| 5.4.2    | EXPLORING QUANTUM COMPUTING IN FINANCIAL SERVICES .....                                                     | 56        |
| 5.4.3    | USE OF QUANTUM COMPUTING TO IMPROVE ANALYSIS OF MEDICAL IMAGES .....                                        | 56        |
| 5.4.4    | DEVELOPMENT OF NEW CHEMICALS WITH QUANTUM COMPUTING.....                                                    | 57        |
| 5.4.5    | TRADING OPTIMIZATION WITH QUANTUM COMPUTING .....                                                           | 57        |
| 5.5      | VALUE CHAIN ANALYSIS .....                                                                                  | 57        |
| 5.6      | ECOSYSTEM .....                                                                                             | 58        |
| 5.7      | PORTER'S FIVE FORCES ANALYSIS .....                                                                         | 59        |
| 5.7.1    | THREAT OF NEW ENTRANTS .....                                                                                | 60        |
| 5.7.2    | THREAT OF SUBSTITUTES.....                                                                                  | 60        |
| 5.7.3    | BARGAINING POWER OF SUPPLIERS .....                                                                         | 60        |
| 5.7.4    | BARGAINING POWER OF BUYERS .....                                                                            | 60        |
| 5.7.5    | INTENSITY OF COMPETITIVE RIVALRY .....                                                                      | 61        |
| 5.8      | PATENT ANALYSIS .....                                                                                       | 61        |
| 5.9      | PRICING MODEL ANALYSIS.....                                                                                 | 63        |
| 5.10     | TRADE ANALYSIS .....                                                                                        | 63        |
| 5.11     | REGULATORY LANDSCAPE .....                                                                                  | 64        |
| 5.11.1   | GENERAL DATA PROTECTION REGULATION.....                                                                     | 64        |
| 5.11.2   | HEALTH INSURANCE PORTABILITY AND ACCOUNTABILITY ACT OF 1996 .....                                           | 65        |
| 5.11.3   | GOVERNANCE, RISK, AND COMPLIANCE.....                                                                       | 65        |
| 5.11.4   | P1913 -SOFTWARE-DEFINED QUANTUM COMMUNICATION .....                                                         | 65        |

|          |                                                                                      |           |
|----------|--------------------------------------------------------------------------------------|-----------|
| 5.11.5   | P7130 - STANDARD FOR QUANTUM TECHNOLOGIES DEFINITIONS .....                          | 65        |
| 5.11.6   | P7131 - STANDARD FOR QUANTUM COMPUTING PERFORMANCE METRICS<br>AND BENCHMARKING ..... | 66        |
| <b>6</b> | <b>QUANTUM COMPUTING SOFTWARE MARKET, BY COMPONENT.....</b>                          | <b>67</b> |
| 6.1      | INTRODUCTION .....                                                                   | 68        |
| 6.1.1    | COMPONENT: COVID-19 IMPACT.....                                                      | 68        |
| 6.2      | SOFTWARE.....                                                                        | 70        |
| 6.2.1    | SOFTWARE: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS.....                             | 70        |
| 6.3      | SERVICES .....                                                                       | 71        |
| 6.3.1    | QUANTUM COMPUTING AS A SERVICE (QAAS).....                                           | 73        |
| 6.3.2    | QUANTUM AS A SERVICE: QUANTUM COMPUTING SOFTWARE MARKET<br>DRIVERS .....             | 73        |
| 6.3.3    | CONSULTING SERVICES .....                                                            | 74        |
| 6.3.4    | CONSULTING SERVICES: QUANTUM COMPUTING SOFTWARE MARKET<br>DRIVERS .....              | 74        |
| <b>7</b> | <b>QUANTUM COMPUTING SOFTWARE MARKET, BY ORGANIZATION SIZE .....</b>                 | <b>76</b> |
| 7.1      | INTRODUCTION .....                                                                   | 77        |
| 7.1.1    | ORGANIZATION SIZE: COVID-19 IMPACT .....                                             | 77        |
| 7.2      | LARGE ENTERPRISES.....                                                               | 79        |
| 7.2.1    | LARGE ENTERPRISES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                   | 79        |
| 7.3      | SMALL AND MEDIUM ENTERPRISES .....                                                   | 80        |
| 7.3.1    | SMES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                                | 80        |
| <b>8</b> | <b>QUANTUM COMPUTING SOFTWARE MARKET, BY DEPLOYMENT MODE.....</b>                    | <b>82</b> |
| 8.1      | INTRODUCTION .....                                                                   | 83        |
| 8.1.1    | DEPLOYMENT MODE: COVID-19 IMPACT.....                                                | 83        |
| 8.2      | ON-PREMISES.....                                                                     | 84        |
| 8.2.1    | ON-PREMISES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                         | 85        |
| 8.3      | CLOUD .....                                                                          | 85        |
| 8.3.1    | CLOUD: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS.....                                | 85        |
| <b>9</b> | <b>QUANTUM COMPUTING SOFTWARE MARKET, BY APPLICATION .....</b>                       | <b>87</b> |
| 9.1      | INTRODUCTION .....                                                                   | 88        |
| 9.1.1    | APPLICATION: COVID-19 IMPACT.....                                                    | 88        |
| 9.2      | OPTIMIZATION .....                                                                   | 89        |
| 9.2.1    | OPTIMIZATION: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                        | 90        |
| 9.3      | MACHINE LEARNING .....                                                               | 91        |
| 9.3.1    | MACHINE LEARNING: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                    | 91        |
| 9.4      | SIMULATION.....                                                                      | 92        |
| 9.4.1    | SIMULATION: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                          | 92        |
| 9.5      | OTHERS .....                                                                         | 93        |

|           |                                                                             |            |
|-----------|-----------------------------------------------------------------------------|------------|
| <b>10</b> | <b>QUANTUM COMPUTING SOFTWARE MARKET, BY TECHNOLOGY .....</b>               | <b>95</b>  |
| 10.1      | QUANTUM ARTIFICIAL INTELLIGENCE .....                                       | 95         |
| 10.2      | QUANTUM MACHINE LEARNING .....                                              | 95         |
| 10.3      | QUANTUM COMPUTING AND INTERNET OF THINGS .....                              | 95         |
| 10.4      | QUANTUM COMMUNICATION TECHNOLOGY .....                                      | 96         |
| 10.5      | QUANTUM COMPUTING ACADEMIA .....                                            | 96         |
| 10.6      | CYBERSECURITY TRANSFORMATION .....                                          | 96         |
| <b>11</b> | <b>QUANTUM COMPUTING SOFTWARE MARKET, BY VERTICAL .....</b>                 | <b>97</b>  |
| 11.1      | INTRODUCTION .....                                                          | 98         |
| 11.1.1    | VERTICAL: COVID-19 IMPACT .....                                             | 98         |
| 11.2      | AEROSPACE & DEFENSE .....                                                   | 99         |
| 11.2.1    | AEROSPACE & DEFENSE: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....        | 100        |
| 11.3      | BANKING, FINANCIAL SERVICES, AND INSURANCE .....                            | 101        |
| 11.3.1    | BFSI: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                       | 101        |
| 11.4      | HEALTHCARE & LIFE SCIENCES .....                                            | 102        |
| 11.4.1    | HEALTHCARE & LIFE SCIENCES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS ..... | 103        |
| 11.5      | ENERGY & UTILITIES .....                                                    | 103        |
| 11.5.1    | ENERGY & UTILITIES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....         | 104        |
| 11.6      | CHEMICAL .....                                                              | 105        |
| 11.6.1    | CHEMICAL: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                   | 105        |
| 11.7      | TRANSPORTATION & LOGISTICS .....                                            | 106        |
| 11.7.1    | TRANSPORTATION & LOGISTICS: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS ..... | 106        |
| 11.8      | GOVERNMENT .....                                                            | 107        |
| 11.8.1    | GOVERNMENT: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                 | 108        |
| 11.9      | EDUCATION .....                                                             | 108        |
| 11.9.1    | EDUCATION: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....                  | 109        |
| <b>12</b> | <b>QUANTUM COMPUTING SOFTWARE MARKET, BY REGION .....</b>                   | <b>110</b> |
| 12.1      | INTRODUCTION .....                                                          | 111        |
| 12.2      | NORTH AMERICA .....                                                         | 112        |
| 12.2.1    | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....              | 112        |
| 12.2.2    | NORTH AMERICA: COVID-19 IMPACT .....                                        | 113        |
| 12.2.3    | NORTH AMERICA: REGULATORY IMPLICATIONS .....                                | 113        |
| 12.2.3.1  | Health Insurance Portability and Accountability Act of 1996 .....           | 113        |
| 12.2.3.2  | California Consumer Privacy Act .....                                       | 113        |
| 12.2.3.3  | Gramm–Leach–Bliley Act .....                                                | 113        |
| 12.2.3.4  | Health Information Technology for Economic and Clinical Health Act .....    | 114        |
| 12.2.3.5  | Sarbanes Oxley Act .....                                                    | 114        |

|        |                                                                    |                                                           |     |
|--------|--------------------------------------------------------------------|-----------------------------------------------------------|-----|
|        | 12.2.3.6                                                           | United States Securities and Exchange Commission.....     | 114 |
|        | 12.2.3.7                                                           | International Organization for Standardization 27001..... | 114 |
|        | 12.2.3.8                                                           | California Consumer Privacy Act.....                      | 115 |
|        | 12.2.3.9                                                           | Federal Information Security Management Act .....         | 115 |
|        | 12.2.3.10                                                          | Federal Information Processing Standards.....             | 115 |
| 12.2.4 | US .....                                                           |                                                           | 121 |
| 12.2.5 | CANADA.....                                                        |                                                           | 125 |
| 12.3   | EUROPE .....                                                       |                                                           | 129 |
| 12.3.1 | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS.....             |                                                           | 130 |
| 12.3.2 | EUROPE: COVID-19 IMPACT .....                                      |                                                           | 130 |
| 12.3.3 | EUROPE: REGULATORY IMPLICATIONS.....                               |                                                           | 130 |
|        | 12.3.3.1                                                           | General Data Protection Regulation .....                  | 130 |
|        | 12.3.3.2                                                           | European Committee for Standardization.....               | 131 |
|        | 12.3.3.3                                                           | European Technical Standards Institute .....              | 131 |
| 12.3.4 | GERMANY .....                                                      |                                                           | 135 |
| 12.3.5 | UK .....                                                           |                                                           | 139 |
| 12.3.6 | FRANCE .....                                                       |                                                           | 144 |
| 12.3.7 | REST OF EUROPE .....                                               |                                                           | 144 |
| 12.4   | ASIA PACIFIC.....                                                  |                                                           | 145 |
| 12.4.1 | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS .....      |                                                           | 145 |
| 12.4.2 | ASIA PACIFIC: COVID-19 IMPACT.....                                 |                                                           | 145 |
| 12.4.3 | ASIA PACIFIC: REGULATORY IMPLICATIONS .....                        |                                                           | 145 |
|        | 12.4.3.1                                                           | Privacy Commissioner for Personal Data .....              | 145 |
|        | 12.4.3.2                                                           | Act on the Protection of Personal Information.....        | 146 |
|        | 12.4.3.3                                                           | Critical Information Infrastructure .....                 | 146 |
|        | 12.4.3.4                                                           | International Organization for Standardization 27001..... | 146 |
|        | 12.4.3.5                                                           | Personal Data Protection Act .....                        | 146 |
| 12.4.4 | CHINA.....                                                         |                                                           | 152 |
| 12.4.5 | JAPAN.....                                                         |                                                           | 156 |
| 12.4.6 | SOUTH KOREA.....                                                   |                                                           | 156 |
| 12.4.7 | REST OF ASIA PACIFIC .....                                         |                                                           | 157 |
| 12.5   | REST OF THE WORLD .....                                            |                                                           | 157 |
| 12.5.1 | REST OF THE WORLD: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS ..... |                                                           | 157 |
| 12.5.2 | REST OF THE WORLD: COVID-19 IMPACT.....                            |                                                           | 157 |
| 12.5.3 | LATIN AMERICA.....                                                 |                                                           | 162 |
| 12.5.4 | MIDDLE EAST AND AFRICA .....                                       |                                                           | 162 |

|            |                                                |            |
|------------|------------------------------------------------|------------|
| <b>13</b>  | <b>COMPETITIVE LANDSCAPE .....</b>             | <b>163</b> |
| 13.1       | OVERVIEW .....                                 | 163        |
| 13.2       | MARKET EVALUATION FRAMEWORK.....               | 163        |
| 13.3       | MARKET RANKING .....                           | 164        |
| 13.4       | MARKET SHARE ANALYSIS.....                     | 164        |
| 13.5       | KEY MARKET DEVELOPMENTS.....                   | 166        |
| 13.5.1     | PRODUCT LAUNCHES AND ENHANCEMENTS.....         | 167        |
| 13.5.2     | DEALS.....                                     | 168        |
| 13.6       | COMPANY EVALUATION MATRIX.....                 | 171        |
| 13.6.1     | STAR.....                                      | 174        |
| 13.6.2     | EMERGING LEADER.....                           | 174        |
| 13.6.3     | PERVASIVE.....                                 | 174        |
| 13.6.4     | PARTICIPANT.....                               | 174        |
| 13.7       | START-UP/SME EVALUATION MATRIX, 2021 .....     | 175        |
| 13.7.1     | PROGRESSIVE COMPANY .....                      | 175        |
| 13.7.2     | RESPONSIVE COMPANY .....                       | 176        |
| 13.7.3     | DYNAMIC COMPANY.....                           | 176        |
| 13.7.4     | STARTING BLOCK.....                            | 176        |
| <b>14</b>  | <b>COMPANY PROFILES.....</b>                   | <b>177</b> |
| 14.1       | INTRODUCTION .....                             | 177        |
| 14.2       | MAJOR PLAYERS .....                            | 177        |
| 14.2.1     | IBM .....                                      | 177        |
| 14.2.1.1   | Business overview .....                        | 177        |
| 14.2.1.2   | Recent developments.....                       | 179        |
| 14.2.1.3   | IBM: COVID-19-related developments .....       | 180        |
| 14.2.1.4   | MnM view.....                                  | 180        |
| 14.2.1.4.1 | Key strengths/Right to win .....               | 180        |
| 14.2.1.4.2 | Strategic choices made.....                    | 181        |
| 14.2.1.4.3 | Weaknesses and competitive threats.....        | 181        |
| 14.2.2     | MICROSOFT .....                                | 182        |
| 14.2.2.1   | Business overview .....                        | 182        |
| 14.2.2.2   | Recent developments.....                       | 184        |
| 14.2.2.3   | Microsoft: COVID-19-related developments ..... | 184        |
| 14.2.2.4   | MnM view.....                                  | 185        |
| 14.2.2.4.1 | Key strengths/Right to win .....               | 185        |
| 14.2.2.4.2 | Strategic choices made.....                    | 185        |
| 14.2.2.4.3 | Weaknesses and competitive threats.....        | 185        |

|            |                                                     |     |
|------------|-----------------------------------------------------|-----|
| 14.2.3     | AWS.....                                            | 186 |
| 14.2.3.1   | Business overview .....                             | 186 |
| 14.2.3.2   | Recent developments .....                           | 187 |
| 14.2.3.3   | AWS: COVID-19-related developments .....            | 187 |
| 14.2.3.4   | MnM view.....                                       | 187 |
| 14.2.3.4.1 | Key strengths/Right to win .....                    | 187 |
| 14.2.3.4.2 | Strategic choices made.....                         | 188 |
| 14.2.3.4.3 | Weaknesses and competitive threats.....             | 188 |
| 14.2.4     | D-WAVE SYSTEMS .....                                | 189 |
| 14.2.4.1   | Business overview .....                             | 189 |
| 14.2.4.2   | Recent developments.....                            | 190 |
| 14.2.4.3   | D-Wave Systems: COVID-19-related developments ..... | 192 |
| 14.2.4.4   | MnM view.....                                       | 192 |
| 14.2.4.4.1 | Key strengths/Right to win .....                    | 192 |
| 14.2.4.4.2 | Strategic choices made.....                         | 193 |
| 14.2.4.4.3 | Weaknesses and competitive threats.....             | 193 |
| 14.2.5     | RIGETTI COMPUTING .....                             | 194 |
| 14.2.5.1   | Business overview .....                             | 194 |
| 14.2.5.2   | Recent developments.....                            | 195 |
| 14.2.5.3   | MnM view.....                                       | 196 |
| 14.2.5.3.1 | Key strengths/Right to win .....                    | 196 |
| 14.2.5.3.2 | Strategic choices made.....                         | 196 |
| 14.2.5.3.3 | Weaknesses and competitive threats.....             | 197 |
| 14.2.6     | GOOGLE.....                                         | 198 |
| 14.2.6.1   | Business overview .....                             | 198 |
| 14.2.6.2   | Recent developments.....                            | 200 |
| 14.2.7     | HONEYWELL.....                                      | 201 |
| 14.2.7.1   | Business overview .....                             | 201 |
| 14.2.7.2   | Recent developments.....                            | 203 |
| 14.2.8     | QC WARE .....                                       | 204 |
| 14.2.8.1   | Business overview .....                             | 204 |
| 14.2.8.2   | Recent developments.....                            | 204 |
| 14.2.9     | 1QBIT .....                                         | 206 |
| 14.2.9.1   | Business overview .....                             | 206 |
| 14.2.9.2   | Recent developments.....                            | 207 |
| 14.2.10    | CAMBRIDGE QUANTUM COMPUTING .....                   | 208 |
| 14.2.10.1  | Business overview .....                             | 208 |
| 14.2.10.2  | Recent developments.....                            | 209 |
| 14.2.11    | FUJITSU .....                                       | 211 |
| 14.2.11.1  | Business overview .....                             | 211 |
| 14.2.11.2  | Recent developments.....                            | 213 |

|           |                                                              |     |
|-----------|--------------------------------------------------------------|-----|
| 14.2.12   | HUAWEI .....                                                 | 214 |
| 14.2.12.1 | Business overview .....                                      | 214 |
| 14.2.12.2 | Recent developments .....                                    | 215 |
| 14.2.13   | ACCENTURE .....                                              | 216 |
| 14.2.13.1 | Business overview .....                                      | 216 |
| 14.2.13.2 | Recent developments .....                                    | 217 |
| 14.3      | OTHER PLAYERS .....                                          | 218 |
| 14.3.1    | RIVERLANE .....                                              | 218 |
| 14.3.2    | ZAPATA COMPUTING .....                                       | 218 |
| 14.3.3    | QUANTUM CIRCUITS.....                                        | 219 |
| 14.3.4    | QUANTICA COMPUTACAO .....                                    | 219 |
| 14.3.5    | XANADU .....                                                 | 220 |
| 14.3.6    | QUANTASTICA .....                                            | 220 |
| 14.3.7    | VERIQLOUD.....                                               | 221 |
| 14.3.8    | AVANETIX .....                                               | 221 |
| 14.3.9    | KUANO.....                                                   | 222 |
| 14.3.10   | RAHKO.....                                                   | 222 |
| 14.3.11   | KETITA LABS.....                                             | 223 |
| 14.3.12   | ALIRO QUANTUM.....                                           | 223 |
| 15        | APPENDIX.....                                                | 224 |
| 15.1      | DISCUSSION GUIDE .....                                       | 224 |
| 15.2      | KNOWLEDGE STORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL..... | 228 |
| 15.3      | AVAILABLE CUSTOMIZATIONS.....                                | 230 |
| 15.4      | RELATED REPORTS .....                                        | 230 |
| 15.5      | AUTHOR DETAILS.....                                          | 231 |

## LIST OF TABLES

|                 |                                                                                                  |    |
|-----------------|--------------------------------------------------------------------------------------------------|----|
| <b>TABLE 1</b>  | UNITED STATES DOLLAR EXCHANGE RATE, 2018–2020                                                    | 34 |
| <b>TABLE 2</b>  | PRIMARY INTERVIEWS WITH EXPERTS                                                                  | 37 |
| <b>TABLE 3</b>  | FACTOR ANALYSIS                                                                                  | 42 |
| <b>TABLE 4</b>  | ECOSYSTEM: QUANTUM COMPUTING SOFTWARE MARKET                                                     | 58 |
| <b>TABLE 5</b>  | IMPACT OF EACH FORCE ON QUANTUM COMPUTING SOFTWARE MARKET                                        | 59 |
| <b>TABLE 6</b>  | RECENT PATENTS RELATED TO QUANTUM COMPUTING SOFTWARE MARKET                                      | 61 |
| <b>TABLE 7</b>  | AVERAGE SELLING PRICE RANGE OF QUANTUM COMPUTING SYSTEMS                                         | 63 |
| <b>TABLE 8</b>  | AVERAGE SELLING PRICE RANGE OF QUANTUM COMPUTING AS A SERVICE                                    | 63 |
| <b>TABLE 9</b>  | IMPORT DATA FOR AUTOMATIC DATA-PROCESSING MACHINES, BY COUNTRY, 2015–2019 (USD MILLION)          | 63 |
| <b>TABLE 10</b> | EXPORT DATA FOR AUTOMATIC DATA-PROCESSING MACHINES, BY COUNTRY, 2015–2019 (USD MILLION)          | 64 |
| <b>TABLE 11</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)                    | 69 |
| <b>TABLE 12</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)                    | 69 |
| <b>TABLE 13</b> | SOFTWARE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)             | 70 |
| <b>TABLE 14</b> | SOFTWARE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)             | 71 |
| <b>TABLE 15</b> | SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY TYPE, 2017–2019 (USD MILLION)               | 71 |
| <b>TABLE 16</b> | SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY TYPE, 2020–2026 (USD MILLION)               | 72 |
| <b>TABLE 17</b> | SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)             | 72 |
| <b>TABLE 18</b> | SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)             | 72 |
| <b>TABLE 19</b> | QUANTUM AS A SERVICE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION) | 73 |
| <b>TABLE 20</b> | QUANTUM AS A SERVICE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION) | 74 |
| <b>TABLE 21</b> | CONSULTING SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)  | 74 |
| <b>TABLE 22</b> | CONSULTING SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)  | 75 |
| <b>TABLE 23</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)            | 78 |
| <b>TABLE 24</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)            | 78 |
| <b>TABLE 25</b> | LARGE ENTERPRISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)    | 79 |
| <b>TABLE 26</b> | LARGE ENTERPRISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)    | 80 |
| <b>TABLE 27</b> | SMES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                 | 81 |

|                 |                                                                                                        |     |
|-----------------|--------------------------------------------------------------------------------------------------------|-----|
| <b>TABLE 28</b> | SMES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                       | 81  |
| <b>TABLE 29</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)                    | 84  |
| <b>TABLE 30</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)                    | 84  |
| <b>TABLE 31</b> | ON-PREMISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                | 85  |
| <b>TABLE 32</b> | ON-PREMISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                | 85  |
| <b>TABLE 33</b> | CLOUD: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                      | 86  |
| <b>TABLE 34</b> | CLOUD: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                      | 86  |
| <b>TABLE 35</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)                        | 89  |
| <b>TABLE 36</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)                        | 89  |
| <b>TABLE 37</b> | OPTIMIZATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)               | 90  |
| <b>TABLE 38</b> | OPTIMIZATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)               | 90  |
| <b>TABLE 39</b> | MACHINE LEARNING: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)           | 91  |
| <b>TABLE 40</b> | MACHINE LEARNING: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)           | 92  |
| <b>TABLE 41</b> | SIMULATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                 | 93  |
| <b>TABLE 42</b> | SIMULATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                 | 93  |
| <b>TABLE 43</b> | OTHERS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                     | 94  |
| <b>TABLE 44</b> | OTHERS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                     | 94  |
| <b>TABLE 45</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)                           | 98  |
| <b>TABLE 46</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)                           | 99  |
| <b>TABLE 47</b> | AEROSPACE & DEFENSE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)        | 100 |
| <b>TABLE 48</b> | AEROSPACE & DEFENSE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)        | 100 |
| <b>TABLE 49</b> | BFSI: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                       | 101 |
| <b>TABLE 50</b> | BFSI: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                       | 102 |
| <b>TABLE 51</b> | HEALTHCARE & LIFE SCIENCES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION) | 103 |
| <b>TABLE 52</b> | HEALTHCARE & LIFE SCIENCES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION) | 103 |
| <b>TABLE 53</b> | ENERGY & UTILITIES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)         | 104 |

|                 |                                                                                                        |     |
|-----------------|--------------------------------------------------------------------------------------------------------|-----|
| <b>TABLE 54</b> | ENERGY & UTILITIES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)         | 104 |
| <b>TABLE 55</b> | CHEMICAL: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                   | 105 |
| <b>TABLE 56</b> | CHEMICAL: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                   | 106 |
| <b>TABLE 57</b> | TRANSPORTATION & LOGISTICS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION) | 107 |
| <b>TABLE 58</b> | TRANSPORTATION & LOGISTICS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION) | 107 |
| <b>TABLE 59</b> | GOVERNMENT: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                 | 108 |
| <b>TABLE 60</b> | GOVERNMENT: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                 | 108 |
| <b>TABLE 61</b> | EDUCATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                  | 109 |
| <b>TABLE 62</b> | EDUCATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                  | 109 |
| <b>TABLE 63</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)                             | 111 |
| <b>TABLE 64</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)                             | 112 |
| <b>TABLE 65</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)           | 116 |
| <b>TABLE 66</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)           | 117 |
| <b>TABLE 67</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)             | 117 |
| <b>TABLE 68</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)             | 117 |
| <b>TABLE 69</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)     | 117 |
| <b>TABLE 70</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)     | 118 |
| <b>TABLE 71</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)   | 118 |
| <b>TABLE 72</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)   | 118 |
| <b>TABLE 73</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)         | 119 |
| <b>TABLE 74</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)         | 119 |
| <b>TABLE 75</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)            | 119 |
| <b>TABLE 76</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)            | 120 |
| <b>TABLE 77</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)             | 120 |
| <b>TABLE 78</b> | NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)             | 120 |
| <b>TABLE 79</b> | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)                      | 121 |

|                  |                                                                                               |     |
|------------------|-----------------------------------------------------------------------------------------------|-----|
| <b>TABLE 80</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)             | 121 |
| <b>TABLE 81</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)               | 122 |
| <b>TABLE 82</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)               | 122 |
| <b>TABLE 83</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)       | 122 |
| <b>TABLE 84</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)       | 122 |
| <b>TABLE 85</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)     | 123 |
| <b>TABLE 86</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)     | 123 |
| <b>TABLE 87</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)           | 123 |
| <b>TABLE 88</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)           | 124 |
| <b>TABLE 89</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)              | 124 |
| <b>TABLE 90</b>  | US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)              | 125 |
| <b>TABLE 91</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)         | 125 |
| <b>TABLE 92</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)         | 126 |
| <b>TABLE 93</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)           | 126 |
| <b>TABLE 94</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)           | 126 |
| <b>TABLE 95</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)   | 126 |
| <b>TABLE 96</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)   | 127 |
| <b>TABLE 97</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION) | 127 |
| <b>TABLE 98</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION) | 127 |
| <b>TABLE 99</b>  | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)       | 128 |
| <b>TABLE 100</b> | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)       | 128 |
| <b>TABLE 101</b> | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)          | 128 |
| <b>TABLE 102</b> | CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)          | 129 |
| <b>TABLE 103</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)         | 131 |
| <b>TABLE 104</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)         | 131 |
| <b>TABLE 105</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)           | 131 |

|                  |                                                                                                |     |
|------------------|------------------------------------------------------------------------------------------------|-----|
| <b>TABLE 106</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)            | 132 |
| <b>TABLE 107</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)    | 132 |
| <b>TABLE 108</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)    | 132 |
| <b>TABLE 109</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)  | 132 |
| <b>TABLE 110</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)  | 133 |
| <b>TABLE 111</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)        | 133 |
| <b>TABLE 112</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)        | 133 |
| <b>TABLE 113</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)           | 134 |
| <b>TABLE 114</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)           | 134 |
| <b>TABLE 115</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)            | 135 |
| <b>TABLE 116</b> | EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)            | 135 |
| <b>TABLE 117</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)         | 136 |
| <b>TABLE 118</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)         | 136 |
| <b>TABLE 119</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)           | 136 |
| <b>TABLE 120</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)           | 137 |
| <b>TABLE 121</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)   | 137 |
| <b>TABLE 122</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)   | 137 |
| <b>TABLE 123</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION) | 137 |
| <b>TABLE 124</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION) | 138 |
| <b>TABLE 125</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)       | 138 |
| <b>TABLE 126</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)       | 138 |
| <b>TABLE 127</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)          | 139 |
| <b>TABLE 128</b> | GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)          | 139 |
| <b>TABLE 129</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)              | 140 |
| <b>TABLE 130</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)              | 140 |
| <b>TABLE 131</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)                | 141 |

|                  |                                                                                                     |     |
|------------------|-----------------------------------------------------------------------------------------------------|-----|
| <b>TABLE 132</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)                     | 141 |
| <b>TABLE 133</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)             | 141 |
| <b>TABLE 134</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)             | 141 |
| <b>TABLE 135</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)           | 142 |
| <b>TABLE 136</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)           | 142 |
| <b>TABLE 137</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)                 | 142 |
| <b>TABLE 138</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)                 | 143 |
| <b>TABLE 139</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)                    | 143 |
| <b>TABLE 140</b> | UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)                    | 144 |
| <b>TABLE 141</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)         | 148 |
| <b>TABLE 142</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)         | 148 |
| <b>TABLE 143</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)           | 148 |
| <b>TABLE 144</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)           | 148 |
| <b>TABLE 145</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)   | 149 |
| <b>TABLE 146</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)   | 149 |
| <b>TABLE 147</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION) | 149 |
| <b>TABLE 148</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION) | 149 |
| <b>TABLE 149</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)       | 150 |
| <b>TABLE 150</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)       | 150 |
| <b>TABLE 151</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)          | 150 |
| <b>TABLE 152</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)          | 151 |
| <b>TABLE 153</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)           | 151 |
| <b>TABLE 154</b> | ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)           | 151 |
| <b>TABLE 155</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)                | 152 |
| <b>TABLE 156</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)                | 152 |
| <b>TABLE 157</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)                  | 153 |

|                  |                                                                                              |     |
|------------------|----------------------------------------------------------------------------------------------|-----|
| <b>TABLE 158</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)           | 153 |
| <b>TABLE 159</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)   | 153 |
| <b>TABLE 160</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)   | 153 |
| <b>TABLE 161</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION) | 154 |
| <b>TABLE 162</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION) | 154 |
| <b>TABLE 163</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)       | 154 |
| <b>TABLE 164</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)       | 155 |
| <b>TABLE 165</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)          | 155 |
| <b>TABLE 166</b> | CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)          | 156 |
| <b>TABLE 167</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)           | 158 |
| <b>TABLE 168</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)           | 158 |
| <b>TABLE 169</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)             | 158 |
| <b>TABLE 170</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)             | 158 |
| <b>TABLE 171</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)     | 159 |
| <b>TABLE 172</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)     | 159 |
| <b>TABLE 173</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)   | 159 |
| <b>TABLE 174</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)   | 159 |
| <b>TABLE 175</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)         | 160 |
| <b>TABLE 176</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)         | 160 |
| <b>TABLE 177</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)            | 160 |
| <b>TABLE 178</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)            | 161 |
| <b>TABLE 179</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)             | 161 |
| <b>TABLE 180</b> | ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)             | 161 |
| <b>TABLE 181</b> | QUANTUM COMPUTING SOFTWARE MARKET: DEGREE OF COMPETITION                                     | 164 |
| <b>TABLE 182</b> | QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS, 2019–2021              | 167 |
| <b>TABLE 183</b> | QUANTUM COMPUTING SOFTWARE MARKET: DEALS, 2019–2021                                          | 168 |
| <b>TABLE 184</b> | COMPANY PRODUCT FOOTPRINT                                                                    | 171 |

|                  |                                                                                         |     |
|------------------|-----------------------------------------------------------------------------------------|-----|
| <b>TABLE 185</b> | COMPANY APPLICATION FOOTPRINT                                                           | 172 |
| <b>TABLE 186</b> | COMPANY REGION FOOTPRINT                                                                | 172 |
| <b>TABLE 187</b> | COMPANY COMPONENT FOOTPRINT                                                             | 173 |
| <b>TABLE 188</b> | IBM: BUSINESS OVERVIEW                                                                  | 177 |
| <b>TABLE 189</b> | IBM: PRODUCTS OFFERED                                                                   | 178 |
| <b>TABLE 190</b> | IBM: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS               | 179 |
| <b>TABLE 191</b> | IBM: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                                           | 179 |
| <b>TABLE 192</b> | MICROSOFT: BUSINESS OVERVIEW                                                            | 182 |
| <b>TABLE 193</b> | MICROSOFT: PRODUCTS OFFERED                                                             | 183 |
| <b>TABLE 194</b> | MICROSOFT: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS         | 184 |
| <b>TABLE 195</b> | MICROSOFT: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                                     | 184 |
| <b>TABLE 196</b> | AWS: BUSINESS OVERVIEW                                                                  | 186 |
| <b>TABLE 197</b> | AWS: PRODUCTS OFFERED                                                                   | 187 |
| <b>TABLE 198</b> | AWS: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS               | 187 |
| <b>TABLE 199</b> | D-WAVE SYSTEMS: BUSINESS OVERVIEW                                                       | 189 |
| <b>TABLE 200</b> | D-WAVE SYSTEMS: PRODUCTS OFFERED                                                        | 189 |
| <b>TABLE 201</b> | D-WAVE SYSTEMS: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS    | 190 |
| <b>TABLE 202</b> | D-WAVE SYSTEM: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                                 | 191 |
| <b>TABLE 203</b> | D-WAVE SYSTEMS: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS                   | 192 |
| <b>TABLE 204</b> | RIGETTI COMPUTING: BUSINESS OVERVIEW                                                    | 194 |
| <b>TABLE 205</b> | RIGETTI COMPUTING: PRODUCTS OFFERED                                                     | 194 |
| <b>TABLE 206</b> | RIGETTI COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS | 195 |
| <b>TABLE 207</b> | RIGETTI COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                             | 196 |
| <b>TABLE 208</b> | RIGETTI COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS                | 196 |
| <b>TABLE 209</b> | GOOGLE: BUSINESS OVERVIEW                                                               | 198 |
| <b>TABLE 210</b> | GOOGLE: PRODUCTS OFFERED                                                                | 199 |
| <b>TABLE 211</b> | GOOGLE: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS            | 200 |
| <b>TABLE 212</b> | GOOGLE: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS                           | 200 |
| <b>TABLE 213</b> | HONEYWELL: BUSINESS OVERVIEW                                                            | 201 |
| <b>TABLE 214</b> | HONEYWELL: PRODUCTS OFFERED                                                             | 202 |
| <b>TABLE 215</b> | HONEYWELL: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS         | 203 |
| <b>TABLE 216</b> | HONEYWELL: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                                     | 203 |
| <b>TABLE 217</b> | QC WARE: BUSINESS OVERVIEW                                                              | 204 |
| <b>TABLE 218</b> | QC WARE: PRODUCTS OFFERED                                                               | 204 |
| <b>TABLE 219</b> | QC WARE: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS           | 204 |
| <b>TABLE 220</b> | QC WARE: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                                       | 205 |
| <b>TABLE 221</b> | 1QBIT: BUSINESS OVERVIEW                                                                | 206 |

|                  |                                                                                                   |     |
|------------------|---------------------------------------------------------------------------------------------------|-----|
| <b>TABLE 222</b> | 1QBIT: PRODUCTS OFFERED                                                                           | 206 |
| <b>TABLE 223</b> | 1QBIT: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                                                   | 207 |
| <b>TABLE 224</b> | 1QBIT: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS                                      | 207 |
| <b>TABLE 225</b> | CAMBRIDGE QUANTUM COMPUTING: BUSINESS OVERVIEW                                                    | 208 |
| <b>TABLE 226</b> | CAMBRIDGE QUANTUM COMPUTING: PRODUCTS OFFERED                                                     | 208 |
| <b>TABLE 227</b> | CAMBRIDGE QUANTUM COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS | 209 |
| <b>TABLE 228</b> | CAMBRIDGE QUANTUM COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                             | 209 |
| <b>TABLE 229</b> | CAMBRIDGE QUANTUM COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS                | 210 |
| <b>TABLE 230</b> | FUJITSU: BUSINESS OVERVIEW                                                                        | 211 |
| <b>TABLE 231</b> | FUJITSU: PRODUCTS OFFERED                                                                         | 212 |
| <b>TABLE 232</b> | FUJITSU: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS                     | 213 |
| <b>TABLE 233</b> | FUJITSU: QUANTUM COMPUTING SOFTWARE MARKET: DEALS                                                 | 213 |
| <b>TABLE 234</b> | HUAWEI: BUSINESS OVERVIEW                                                                         | 214 |
| <b>TABLE 235</b> | HUAWEI: PRODUCTS OFFERED                                                                          | 215 |
| <b>TABLE 236</b> | HUAWEI: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS                      | 215 |
| <b>TABLE 237</b> | ACCENTURE: BUSINESS OVERVIEW                                                                      | 216 |
| <b>TABLE 238</b> | ACCENTURE: PRODUCTS OFFERED                                                                       | 217 |
| <b>TABLE 239</b> | ACCENTURE: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS                                  | 217 |

## LIST OF FIGURES

|                  |                                                                                                                                                     |    |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>FIGURE 1</b>  | COVID-19: GLOBAL PROPAGATION                                                                                                                        | 26 |
| <b>FIGURE 2</b>  | COVID-19 PROPAGATION: SELECT COUNTRIES                                                                                                              | 27 |
| <b>FIGURE 3</b>  | REVISED GROSS DOMESTIC PRODUCT FORECASTS FOR SELECT G20 COUNTRIES IN 2020                                                                           | 28 |
| <b>FIGURE 4</b>  | CRITERIA IMPACTING THE GLOBAL ECONOMY                                                                                                               | 29 |
| <b>FIGURE 5</b>  | SCENARIOS IN TERMS OF RECOVERY OF THE GLOBAL ECONOMY                                                                                                | 30 |
| <b>FIGURE 6</b>  | GLOBAL QUANTUM COMPUTING SOFTWARE MARKET: RESEARCH DESIGN                                                                                           | 36 |
| <b>FIGURE 7</b>  | BREAKUP OF PRIMARY INTERVIEWS: BY COMPANY, DESIGNATION, AND REGION                                                                                  | 38 |
| <b>FIGURE 8</b>  | DATA TRIANGULATION                                                                                                                                  | 39 |
| <b>FIGURE 9</b>  | RESEARCH METHODOLOGY: APPROACH                                                                                                                      | 40 |
| <b>FIGURE 10</b> | MARKET SIZE ESTIMATION METHODOLOGY—APPROACH 1 (SUPPLY SIDE): REVENUE OF SOFTWARE/SERVICES IN QUANTUM COMPUTING SOFTWARE MARKET                      | 41 |
| <b>FIGURE 11</b> | MARKET SIZE ESTIMATION METHODOLOGY—APPROACH 2 BOTTOM-UP (SUPPLY SIDE): COLLECTIVE REVENUE OF SOFTWARE/SERVICES IN QUANTUM COMPUTING SOFTWARE MARKET | 41 |
| <b>FIGURE 12</b> | QUANTUM COMPUTING SOFTWARE MARKET SIZE, 2021–2026                                                                                                   | 46 |
| <b>FIGURE 13</b> | SERVICES SEGMENT, BASED ON COMPONENT, TO LEAD QUANTUM COMPUTING SOFTWARE MARKET DURING FORECAST PERIOD                                              | 46 |
| <b>FIGURE 14</b> | QUANTUM AS A SERVICE SEGMENT, BASED ON SERVICES, TO REGISTER HIGHER CAGR DURING FORECAST PERIOD                                                     | 47 |
| <b>FIGURE 15</b> | CLOUD-BASED DEPLOYMENT MODE TO LEAD QUANTUM COMPUTING SOFTWARE MARKET DURING FORECAST PERIOD                                                        | 47 |
| <b>FIGURE 16</b> | SMES, BASED ON ORGANIZATION SIZE, TO EXHIBIT HIGHER GROWTH RATE DURING FORECAST PERIOD                                                              | 48 |
| <b>FIGURE 17</b> | OPTIMIZATION SEGMENT, BASED ON APPLICATION, TO LEAD QUANTUM COMPUTING SOFTWARE MARKET DURING FORECAST PERIOD                                        | 48 |
| <b>FIGURE 18</b> | GROWING SEGMENTS IN QUANTUM COMPUTING SOFTWARE MARKET, 2021–2026                                                                                    | 49 |
| <b>FIGURE 19</b> | QUANTUM COMPUTING SOFTWARE MARKET: REGIONAL SNAPSHOT                                                                                                | 49 |
| <b>FIGURE 20</b> | GROWING GOVERNMENT SUPPORT FOR DEVELOPMENT AND DEPLOYMENT OF QUANTUM COMPUTING TECHNOLOGY TO DRIVE MARKET GROWTH DURING FORECAST PERIOD             | 50 |
| <b>FIGURE 21</b> | SERVICES SEGMENT AND CHINA TO HOLD LARGEST SHARES OF ASIA PACIFIC MARKET IN 2021                                                                    | 51 |
| <b>FIGURE 22</b> | SOUTH KOREA TO GROW AT HIGHEST RATE DURING FORECAST PERIOD                                                                                          | 51 |
| <b>FIGURE 23</b> | QUANTUM COMPUTING SOFTWARE MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES                                                               | 52 |
| <b>FIGURE 24</b> | VALUE CHAIN ANALYSIS: QUANTUM COMPUTING SOFTWARE MARKET                                                                                             | 57 |
| <b>FIGURE 25</b> | PORTER'S FIVE FORCES ANALYSIS: QUANTUM COMPUTING SOFTWARE MARKET                                                                                    | 59 |
| <b>FIGURE 26</b> | SERVICES SEGMENT TO REGISTER HIGHER CAGR DURING FORECAST PERIOD                                                                                     | 69 |
| <b>FIGURE 27</b> | SMES SEGMENT TO EXHIBIT HIGHER CAGR DURING FORECAST PERIOD                                                                                          | 78 |
| <b>FIGURE 28</b> | CLOUD-BASED DEPLOYMENT MODE TO REGISTER HIGHER CAGR DURING FORECAST PERIOD                                                                          | 83 |
| <b>FIGURE 29</b> | MACHINE LEARNING SEGMENT TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD                                                                            | 88 |

|                  |                                                                                     |     |
|------------------|-------------------------------------------------------------------------------------|-----|
| <b>FIGURE 30</b> | HEALTHCARE & LIFE SCIENCES VERTICAL TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD | 98  |
| <b>FIGURE 31</b> | APAC TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD                                | 111 |
| <b>FIGURE 32</b> | NORTH AMERICA MARKET SNAPSHOT                                                       | 116 |
| <b>FIGURE 33</b> | ASIA PACIFIC: MARKET SNAPSHOT                                                       | 147 |
| <b>FIGURE 34</b> | QUANTUM COMPUTING SOFTWARE MARKET EVALUATION FRAMEWORK                              | 163 |
| <b>FIGURE 35</b> | MARKET RANKING IN 2020                                                              | 164 |
| <b>FIGURE 36</b> | MARKET SHARE ANALYSIS OF COMPANIES IN QUANTUM COMPUTING SOFTWARE MARKET             | 165 |
| <b>FIGURE 37</b> | KEY DEVELOPMENTS IN QUANTUM COMPUTING SOFTWARE MARKET FOR 2019–2021                 | 166 |
| <b>FIGURE 38</b> | GLOBAL QUANTUM COMPUTING SOFTWARE MARKET, COMPANY EVALUATION MATRIX                 | 175 |
| <b>FIGURE 39</b> | GLOBAL QUANTUM COMPUTING SOFTWARE: STARTUP/SME EVALUATION MATRIX, 2021              | 176 |
| <b>FIGURE 40</b> | IBM: COMPANY SNAPSHOT                                                               | 178 |
| <b>FIGURE 41</b> | MICROSOFT: COMPANY SNAPSHOT                                                         | 183 |
| <b>FIGURE 42</b> | AWS: COMPANY SNAPSHOT                                                               | 186 |
| <b>FIGURE 43</b> | GOOGLE: COMPANY SNAPSHOT                                                            | 199 |
| <b>FIGURE 44</b> | HONEYWELL: COMPANY SNAPSHOT                                                         | 202 |
| <b>FIGURE 45</b> | FUJITSU: COMPANY SNAPSHOT                                                           | 212 |
| <b>FIGURE 46</b> | HUAWEI: COMPANY SNAPSHOT                                                            | 215 |
| <b>FIGURE 47</b> | ACCENTURE: COMPANY SNAPSHOT                                                         | 217 |

## LIST OF ABBREVIATIONS

| ABBREVIATION | FULL FORM                                     |
|--------------|-----------------------------------------------|
| AGI          | Artificial General Intelligence               |
| AI           | Artificial Intelligence                       |
| ANZ          | Australia and New Zealand                     |
| APAC         | Asia Pacific                                  |
| API          | Application Programming Interface             |
| APPI         | Act on the Protection of Personal Information |
| ASP          | Average Selling Price                         |
| AWS          | Amazon Web Services                           |
| BFSI         | Banking, Financial Services and Insurance     |
| BU           | Business Units                                |
| CAGR         | Compound Average Growth Rate                  |
| CAL          | Client Access License                         |
| CAPA         | Corrective Action/Preventive Action           |
| CCPA         | California Consumer Privacy Act               |
| CEO          | Chief Executive Officer                       |
| CHPC         | Centre for High Performance Computing         |
| CII          | Critical Information Infrastructure           |
| COO          | Chief Operating Officers                      |
| CVaR         | Conditional Value at Risk                     |
| DEWA         | Dubai Electricity and Water Authority         |
| DoE          | Department of Energy                          |
| EMEA         | Europe, Middle East and Africa                |
| ETSI         | European Technical Standards Institute        |
| EU           | European Union                                |
| FET          | Future and Emerging Technologies              |
| FIPS         | Federal Information Processing Standards      |
| FISMA        | Federal Information Security Management Act   |
| FOA          | Funding Opportunity Announcements             |

|         |                                                                                   |
|---------|-----------------------------------------------------------------------------------|
| GCC     | Gulf Cooperation Council                                                          |
| GDP     | Gross Domestic Product                                                            |
| GDPR    | General Data Protection Regulation                                                |
| HIPAA   | Health Insurance Portability and Accountability Act                               |
| HITECH  | Health Information Technology for Economic and Clinical Health                    |
| IBM     | International Business Machines                                                   |
| ICT     | Information and Communication Technology                                          |
| IIoT    | Industrial IoT                                                                    |
| IJCSITS | International Journal of Computer Science and Information Technology and Security |
| IoT     | Internet of Things                                                                |
| ISMS    | Information Security Management System                                            |
| ISO     | International Organization for Standardization                                    |
| ISV     | Independent Software Vendors                                                      |
| IT      | Information Technology                                                            |
| Li-S    | Lithium Sulfur                                                                    |
| M&A     | Mergers and Acquisitions                                                          |
| M2M     | Machine-to-Machine                                                                |
| MD      | Managing Directors                                                                |
| MEA     | Middle East and Africa                                                            |
| MENA    | Middle East and North Africa                                                      |
| ML      | Machine Learning                                                                  |
| MRI     | Magnetic Resonance Imaging                                                        |
| MSIT    | Ministry of Science and ICT                                                       |
| NISQ    | Noisy Intermediate-Scale Quantum                                                  |
| NIST    | National Institute of Standards and Technology's                                  |
| NLP     | Natural Language Processing                                                       |
| NQIT    | Networked Quantum Information Technologies                                        |
| NQIT    | Networked Quantum Information Technologies                                        |
| NSA     | National Security Agency                                                          |
| OEM     | Original Equipment Manufacturers                                                  |
| PCPD    | Privacy Commissioner for Personal Data                                            |
| PDPA    | Personal Data Protection Act                                                      |

|        |                                                        |
|--------|--------------------------------------------------------|
| QaaS   | Quantum as a Service                                   |
| QAE    | Quantum Application Environment                        |
| QAI    | Quantum artificial intelligence                        |
| QC     | Quantum Computing                                      |
| QCaaS  | Quantum Computing as a Service                         |
| QCI    | Quantum Communication Infrastructure                   |
| QCK    | Quantum Collective Knowledge                           |
| QEMIST | Quantum-enabled Molecular Ab Initio Simulation Toolkit |
| QIP    | Quantum Information Processing                         |
| QKD    | Quantum Key Distribution                               |
| QML    | Quantum Machine Learning                               |
| QVM    | Quantum Virtual Machine                                |
| QVS    | Quantum Virtual Simulators                             |
| R&D    | Research and Development                               |
| RoI    | Return on Investment                                   |
| RoW    | Rest of the World                                      |
| SEC    | Securities and Exchange Commission                     |
| SI     | System Integrator                                      |
| SMS    | Short Message Service                                  |
| SP     | Service Providers                                      |
| SQL    | Structured Query Language                              |
| SSL    | Secure Sockets Layer                                   |
| TLS    | Transport Layer Security                               |
| UAE    | United Arab Emirates                                   |
| UK     | United Kingdom                                         |
| US     | United States                                          |
| US DOT | United States Department of Transportation             |
| USD    | United States Dollar                                   |
| VAR    | Value-Added Reseller                                   |
| VaR    | Value at Risk                                          |
| VP     | Vice President                                         |
| VR     | Virtual Reality                                        |

|       |                           |
|-------|---------------------------|
| WFH   | Work From Home            |
| WHO   | World Health Organization |
| Y-O-Y | Year-on-Year              |

# 1 INTRODUCTION

## 1.1 INTRODUCTION TO COVID-19

COVID-19 is an infectious disease caused by the recently discovered novel coronavirus. Largely unknown before the outbreak began in Wuhan, China, in December 2019, COVID-19 has moved from a regional crisis to a global pandemic in just a few weeks. The World Health Organization (WHO) declared COVID-19 as a pandemic on March 11, 2020.

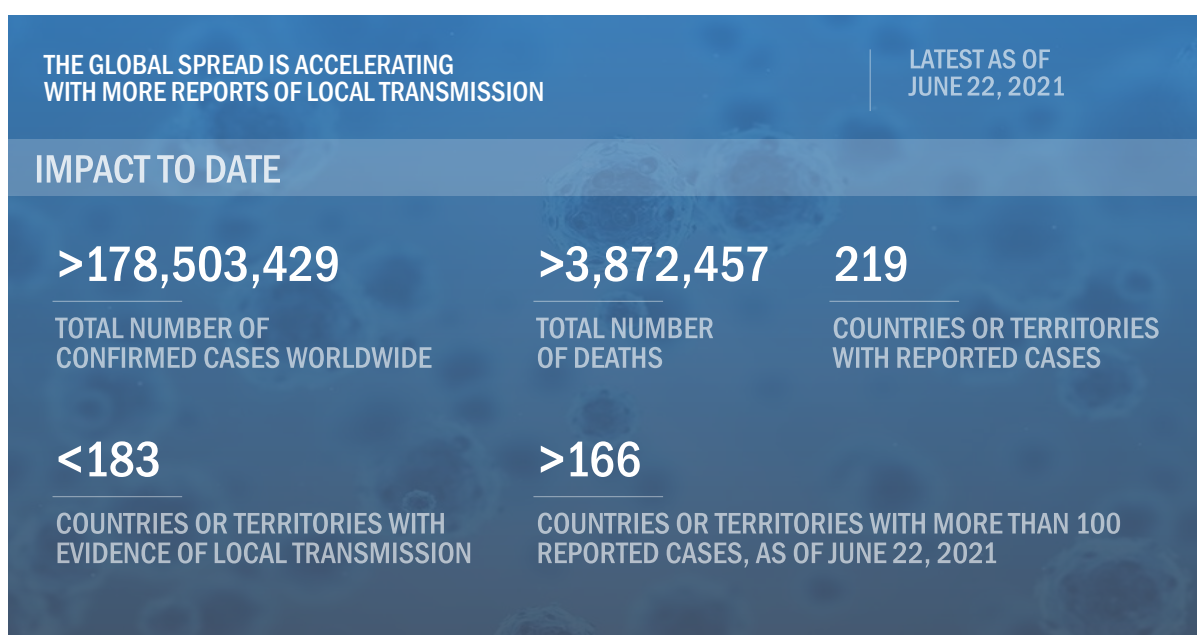
As this virus is spreading rapidly at a global level, countries across the globe are trying to prevent further contagion by adopting measures such as social distancing, contact tracing, self-quarantining, surveillance, communication, and testing. So far, China, South Korea, Japan, and Singapore have managed to flatten the curve, while the US, Italy, Spain, Germany, France, and Iran are imposing drastic measures to slow down the spread and control fatalities.

Government agencies have announced special financial aid packages toward the development of preventive and curative drugs, purchase of critical care medical devices, and fast-track approval of diagnostic tests. Various organizations across the globe, such as the Bill and Melinda Gates Foundation (BMGF) and Wellcome Trust, are also coming together to accelerate and strengthen efforts to fight the COVID-19 pandemic.

## 1.2 COVID-19 HEALTH ASSESSMENT

This is an unprecedented crisis putting an enormous strain on healthcare providers across the globe. According to the WHO, within 67 days, the first 100,000 confirmed cases were reported worldwide. It took another 11 days to reach 200,000, an additional four days to reach 300,000, and the number of confirmed cases reached one million in 12 days.

### FIGURE 1 COVID-19: GLOBAL PROPAGATION

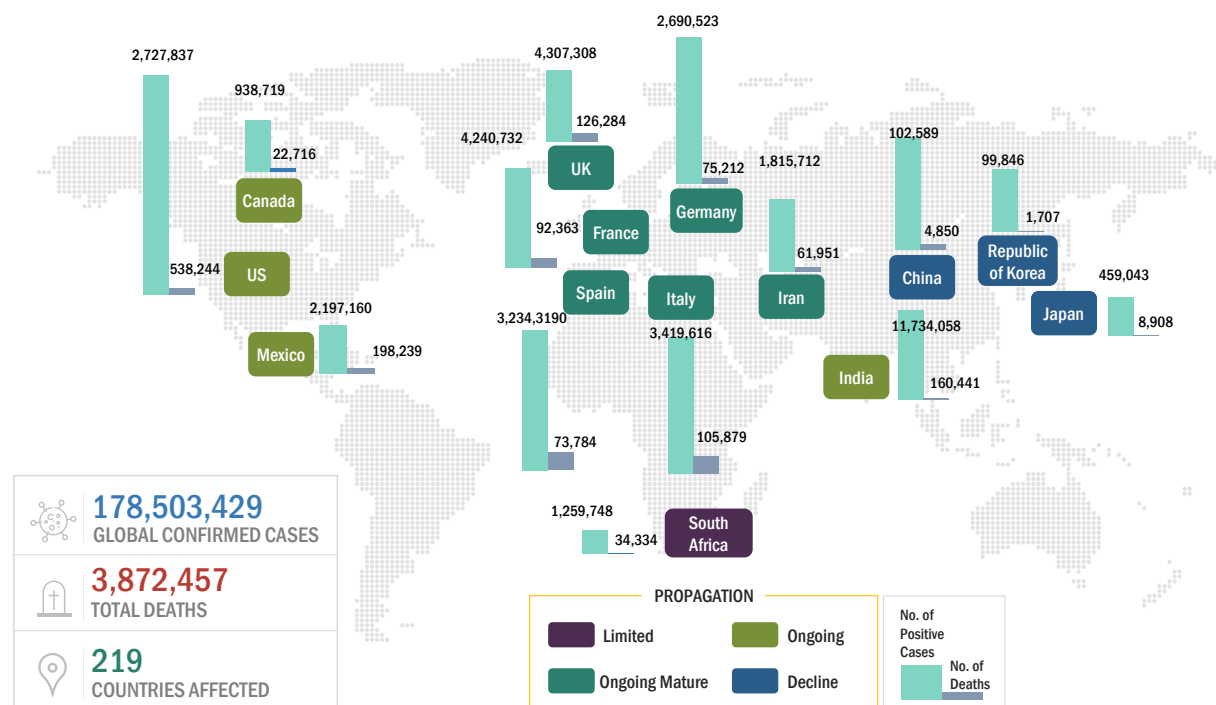


Source: WHO

As of June 22, 2021, the virus has spread to more than 219 countries, with over 178,503,429 reported cases and close to 3,872,457 deaths.

The actual number of infected people is probably higher than the official total, as some mild and asymptomatic cases are not tested and counted.

**FIGURE 2** COVID-19 PROPAGATION: SELECT COUNTRIES



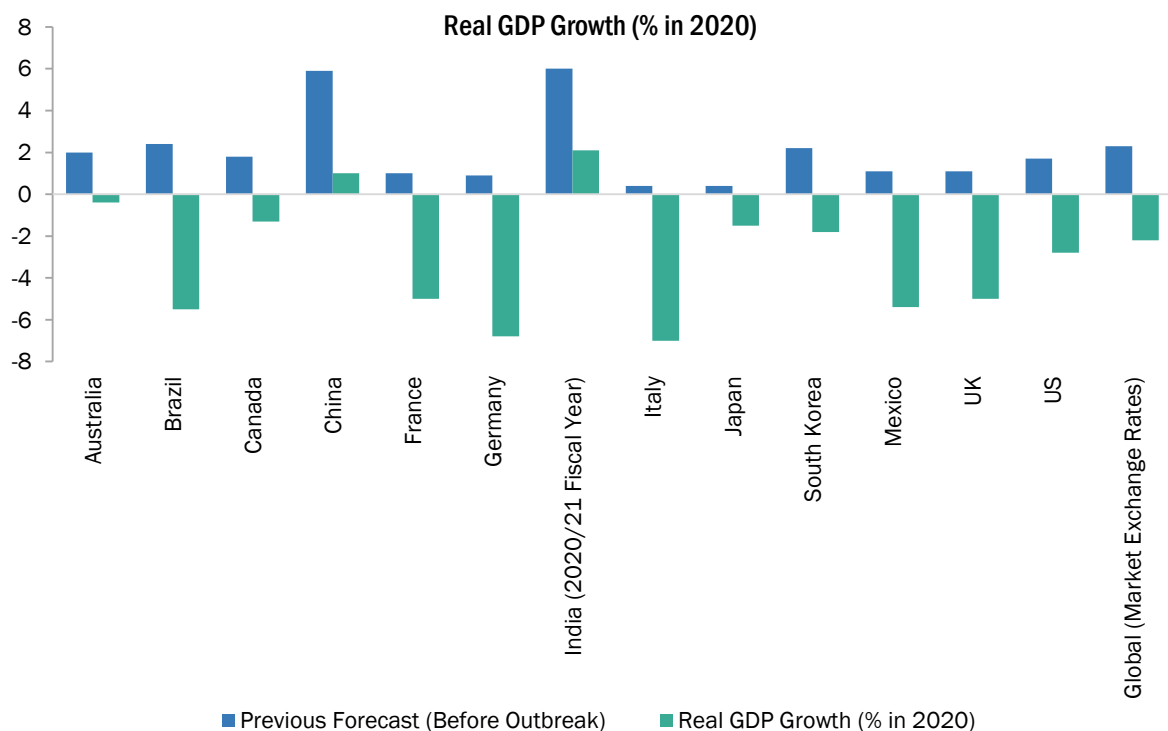
Source: WHO Coronavirus Disease 2019; data as of June 22, 2021

Disease progression patterns are similar in most countries, but flattening of the curve and disease containment are largely dependent on measures adopted by each country. China and South Korea, for example, have managed to control the spread of the disease by implementing severe countermeasures, such as restricting travel; imposing city-wide quarantines, curfews, and lockdowns; building specialty hospitals; increasing medical aid; and ramping up testing. Most countries are trying to emulate these measures for containing the spread of the disease.

### 1.3 COVID-19 ECONOMIC ASSESSMENT

COVID-19 has not only affected the healthcare sector but also impacted the global economy. It has had a significant economic impact on various financial as well as industrial sectors, such as travel & tourism, manufacturing, and aviation. It is predicted that the world economy will go into recession due to the loss of trillions of dollars. With the increasing number of countries imposing and extending lockdowns, economic activities are declining, impacting the global economy.

**FIGURE 3** REVISED GROSS DOMESTIC PRODUCT FORECASTS FOR SELECT G20 COUNTRIES IN 2020



Source: The Economist Intelligent Unit

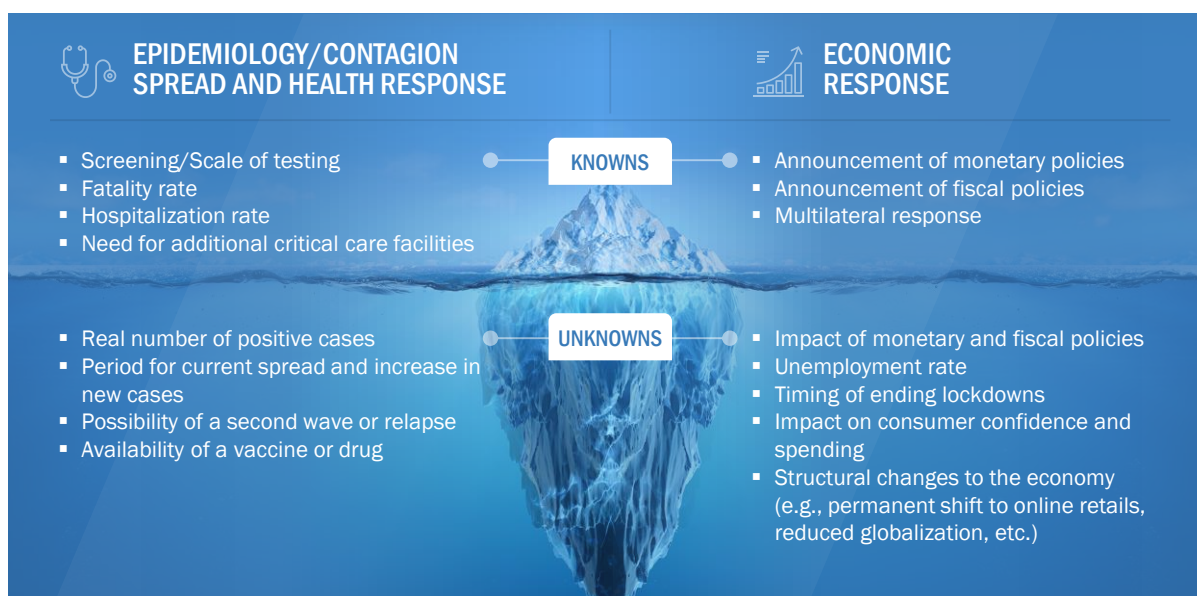
With combined efforts to restrict the spread of the pandemic and the slump in global oil prices, investments are expected to contract sharply in 2020 this year, especially in the energy sector. Export growth is also expected to reduce drastically. A recovery is expected in the second half of the year, provided there is no emergence of another wave of the epidemic, and countries quickly manage to control the spread and resume routine.

China has managed to contain the spread and is currently on track to resume economic activities. However, the Eurozone, one of the hardest-hit regions, had to undergo recession throughout 2020. Within the Eurozone, Germany, which is a manufacturing hub, is projected to recover at a much slower pace than other countries. The US economy witnessed a slump in 2020. With increased unemployment in 2021, the recovery of the economy is slow and will only be observed in the second half of 2021.

### 1.3.1 COVID-19 ECONOMIC IMPACT—SCENARIO ASSESSMENT

Recovery from this crisis largely depends on various knowns and unknowns, which most countries would have to deal with and respond to. The speed and strength of economic recovery would depend on health and economic responses as well as the agility and speed of implementing various measures.

**FIGURE 4** CRITERIA IMPACTING GLOBAL ECONOMY



In the recent past, the global economy became substantially more interconnected. The adverse consequences of various steps related to containment are evident from global supply chain disruptions, weaker demand for imported products and services, and increased unemployment. Risk aversion has increased in the financial market, with all-time low-interest rates and sharp declines in equity and commodity prices. Consumer and business confidence have also reduced significantly.

This has magnified over time, and the spillover is seen in other countries as well. The latest forecasts by multiple sources predicted a 4% decrease in the global Gross Domestic Product (GDP) in 2020 and a slow recovery, making the recession caused by this pandemic worse than the 2009 global recession. Based on these forecasts, the global economy is expected to recover by end of 2021; however, for some countries, including the US, Germany, and Italy, recovery is predicted to take longer.

In this fast-changing environment, the full impact of the pandemic on the global economy may not be entirely known. MarketsandMarkets has used three scenario-based approaches (epidemiology, health response, and economic response) to assess the economic impact and recovery period at the global level. Countries and regions are expected to have different impacts and recovery periods.

**FIGURE 5** SCENARIOS IN TERMS OF RECOVERY OF GLOBAL ECONOMY

| SCENARIO                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>OPTIMISTIC</b>                                                                                                                                                                               |  <b>REALISTIC</b>                                                                                                                                                                                   |  <b>PESSIMISTIC</b>                                                                                                                                                                    |
| EPIDEMIOLOGY/CONTAGION SPREAD AND HEALTH RESPONSE                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                           |
| <b>FATALITY RATE</b><br><b>1.5–2%</b>                                                                                                                                                                                                                                             | <b>FATALITY RATE</b><br><b>2–3.5%</b>                                                                                                                                                                                                                                                | <b>FATALITY RATE</b><br><b>MORE THAN 3.5%</b>                                                                                                                                                                                                                             |
| <b>HOSPITALIZATION RATE:</b><br><b>LESS THAN 15%</b>                                                                                                                                                                                                                              | <b>HOSPITALIZATION RATE:</b><br><b>15–20%</b>                                                                                                                                                                                                                                        | <b>HOSPITALIZATION RATE:</b><br><b>MORE THAN 20%</b>                                                                                                                                                                                                                      |
| <b>ADDITIONAL ICU BEDS AND LESS THAN</b><br><b>2.5 - 5%</b>                                                                                                                                                                                                                       | <b>ADDITIONAL ICU BEDS AND</b><br><b>5%</b>                                                                                                                                                                                                                                          | <b>ADDITIONAL ICU BEDS AND</b><br><b>15%</b>                                                                                                                                                                                                                              |
| <b>ADDITIONAL VENTILATORS</b><br><b>15%</b>                                                                                                                                                                                                                                       | <b>ADDITIONAL VENTILATORS</b><br><b>15–20%</b>                                                                                                                                                                                                                                       | <b>ADDITIONAL VENTILATORS</b><br><b>25%</b>                                                                                                                                                                                                                               |
| ECONOMIC RESPONSE                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                           |
| <b>FISCAL STIMULUS TO THE TUNE OF</b><br><b>USD 7.5–8</b><br><b>TRILLION GLOBALLY</b>                                                                                                                                                                                             | <b>FISCAL STIMULUS TO THE TUNE OF</b><br><b>USD 5–7.5</b><br><b>TRILLION GLOBALLY</b>                                                                                                                                                                                                | <b>FISCAL STIMULUS TO THE TUNE OF LESS THAN</b><br><b>USD 5</b><br><b>TRILLION GLOBALLY</b>                                                                                                                                                                               |
| <b>EMERGENCY FUNDING TO SUPPORT</b><br><b>EFFORTS TO FIGHT COVID-19 TO THE</b><br><b>TUNE OF</b><br><b>USD 100</b><br><b>BILLION GLOBALLY</b>                                                                                                                                     | <b>EMERGENCY FUNDING TO SUPPORT</b><br><b>EFFORTS TO FIGHT COVID-19 TO THE</b><br><b>TUNE OF</b><br><b>USD 75–100</b><br><b>BILLION GLOBALLY</b>                                                                                                                                     | <b>EMERGENCY FUNDING TO SUPPORT</b><br><b>EFFORTS TO FIGHT COVID-19 TO THE</b><br><b>TUNE OF LESS THAN</b><br><b>USD 75</b><br><b>BILLION GLOBALLY</b>                                                                                                                    |
| <b>MULTILATERAL AGENCIES FINANCING TO</b><br><b>THE TUNE OF</b><br><b>USD 100</b><br><b>BILLION GLOBALLY</b>                                                                                                                                                                      | <b>MULTILATERAL AGENCIES FINANCING</b><br><b>TO THE TUNE OF</b><br><b>USD 50–75</b><br><b>BILLION GLOBALLY</b>                                                                                                                                                                       | <b>MULTILATERAL AGENCIES FINANCING</b><br><b>TO THE TUNE OF LESS THAN</b><br><b>USD 50</b><br><b>BILLION GLOBALLY</b>                                                                                                                                                     |
| RECOVERY SCENARIO                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>▪ Milder recession in 2020 Q1 and Q2; stronger recovery in 2020 Q3 and after</li> <li>▪ US: Recovery by 2020 Q3</li> <li>▪ EU5: Recovery by 2020 Q3</li> <li>▪ China: Recovery by 2020 Q2</li> <li>▪ India: Recovery by 2020 Q3</li> </ul> | <ul style="list-style-type: none"> <li>▪ Recession in 2020 Q1 and Q2; partial bounce back in 2020 Q3, then slow growth</li> <li>▪ US: Recovery by 2020 Q4</li> <li>▪ EU5: Recovery by 2020 Q4</li> <li>▪ China: Recovery by 2020 Q2</li> <li>▪ India: Recovery by 2020 Q3</li> </ul> | <ul style="list-style-type: none"> <li>▪ Deeper recession in 2020 Q1, Q2, and Q3; modest rebound in 2020 Q4</li> <li>▪ US: Recovery by 2021 Q1</li> <li>▪ EU5: Recovery by 2021 Q1</li> <li>▪ China: Recovery by 2020 Q3</li> <li>▪ India: Recovery by 2020 Q4</li> </ul> |

Source: UNCTAD, Moody's, FAS, OECD, and MarketsandMarkets Analysis

With more than 4 million confirmed cases of COVID-19, businesses are coping with lost revenues and disrupted supply chains as factory shutdowns and quarantine measures are spreading across the globe, restricting movement and business activities. The economic impact is mounting, and the top economies of the world are expected to enter a global recession.

To provide a perspective on various scenarios, we have assume several factors that would impact economic recovery from the global recession. In a realistic scenario, we assume that possible health and economic responses will help economies bounce back partially in Q3 of 2021. In Q1 and Q2 of 2021, countries will ramp up testing capacities and critical healthcare services as well as increase the availability of financial aid. This would help contain the spread of the virus. The economic recession will last until Q2 of 2021; slow recovery will be seen from Q3. This is expected to be a “U-shaped” economic recovery, with a gradual improvement in the economy. In an optimistic scenario, the economy is expected to recover much faster and will continue from Q3 of 2021. A quick return to routine will be seen after the curve of infection

flattens toward the end of 2021 due to time-bound and efficient health and economic measures. In this scenario, most economies would experience a mild recession, and the growth would accelerate in 2022. This is expected to be a “V-shaped” economic recovery, with a strong and fast improvement in the economy.

In a pessimistic scenario, we assume that all measures, both health and economic, would neither help control the spread of the virus nor lead to economic recovery. Most economies would experience unprecedented economic losses, along with social and political turmoil. In this scenario, most economies will face recession till Q3, and modest recovery will begin in Q4 of 2021. This is expected to be an “L-shaped” economic recovery, with countries taking longer to return to their pre-crisis levels.

Similar scenario-based approaches are considered in this report to assess the impact of COVID-19 and provide market forecasts.

## 1.4 OBJECTIVES OF STUDY

- To determine and forecast the global quantum computing software market based on component (software and services), organization size, deployment mode, application, technology, vertical, and region from 2020 to 2026, and analyze various macroeconomic and microeconomic factors that affect market growth
- To forecast the size of the market segments with respect to four major regions: North America, Europe, Asia Pacific (APAC), and Rest of the World (RoW)
- To provide detailed information about the major factors (drivers, restraints, opportunities, and challenges) influencing the growth of the quantum computing software market
- To analyze each submarket with respect to individual growth trends, prospects, and contributions to the quantum computing software market
- To analyze the opportunities in the market for stakeholders by identifying the high-growth segments of the quantum computing software market
- To profile the key market players; provide a comparative analysis on the basis of business overviews, regional presence, product offerings, business strategies, and key financials; and illustrate the competitive landscape of the market
- To track and analyze competitive developments, such as mergers and acquisitions, product launches and developments, partnerships and collaborations, and research and development (R&D), in the market

## 1.5 MARKET DEFINITION

Quantum computing involves phenomena such as quantum entanglement and quantum mechanics superposition that are used by quantum computers for computation. Quantum computers exhibit improved computing power as the data is provided in a representative way. Conventional computers use bits that can either be 1s or 0s, while quantum computers use qubits (quantum bits) that can be both 0s and 1s simultaneously due to superposition.

Quantum computing software comprises quantum computing tools and platforms that help in solving problems related to quantum computers. The software stack implements the computations needed to transform an arbitrarily posed problem to a form solvable on a quantum solver. Quantum computing software are of two types, i.e., quantum software for optimization problems and classical computing software enabling quantum computing. The quantum software tools and computational platforms provide solutions for several end users by helping develop and test their quantum algorithms. Quantum computers have performance issues due to random errors, owing to which error-correcting software have been built to

rectify errors. An error-correcting software or firmware is a low-level program that increases the stability of quantum computers.

### 1.5.1 INCLUSIONS AND EXCLUSIONS

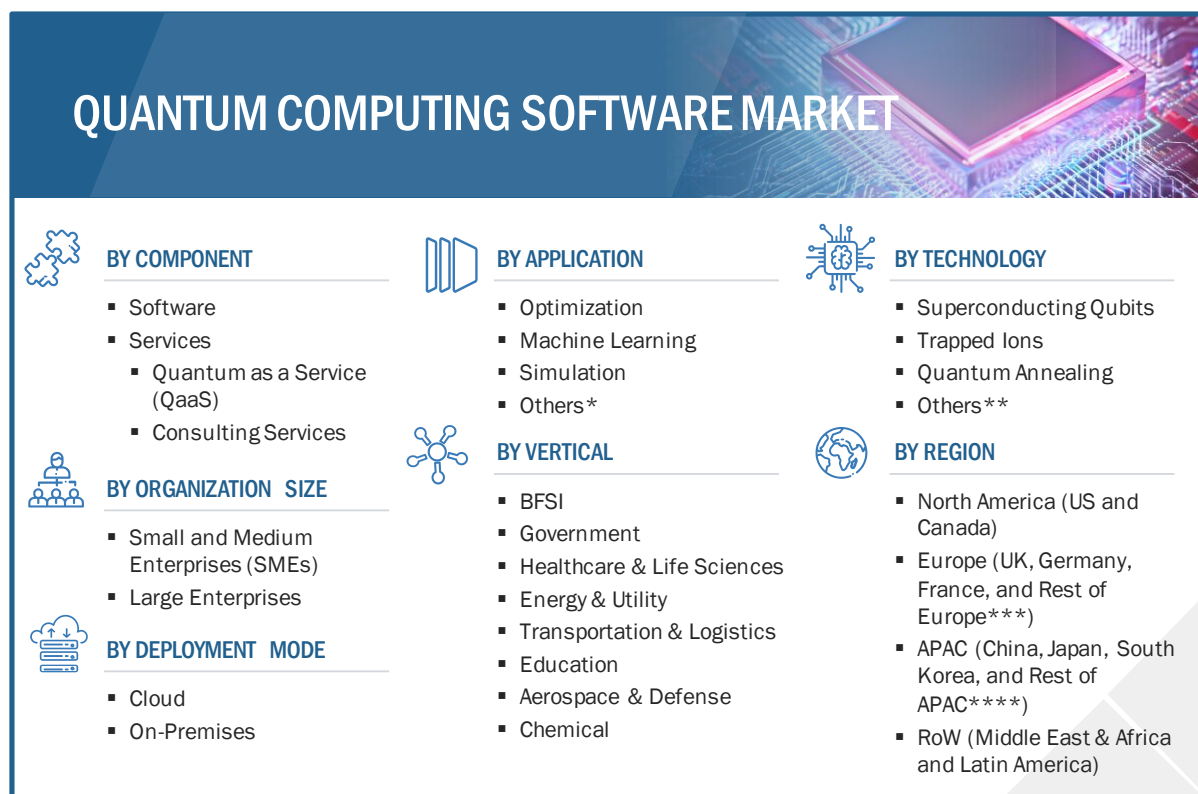
| CATEGORY                                | INCLUSION                                                                                                                                                                                                                                                                              | EXCLUSION                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantum Computing Software and Services | <ul style="list-style-type: none"> <li>The market sizes of the quantum computing software and services segments have been included in this study.</li> <li>The quantum computing software market only incorporates the software services associated with quantum computers.</li> </ul> | <ul style="list-style-type: none"> <li>Hardware components in the quantum computing software market are not considered in the scope.</li> <li>Maintenance services have been excluded as very few quantum computers are sold presently, and companies mostly provide quantum computing as a service.</li> </ul> |
| Deployment                              | <ul style="list-style-type: none"> <li>On-premise and cloud-based deployment modes have been considered in the scope of the report.</li> </ul>                                                                                                                                         | NA                                                                                                                                                                                                                                                                                                              |
| Organization Size                       | <ul style="list-style-type: none"> <li>SMEs and large enterprises have been incorporated in the scope of the report.</li> </ul>                                                                                                                                                        | NA                                                                                                                                                                                                                                                                                                              |
| Technology                              | <ul style="list-style-type: none"> <li>The market sizes of trapped ions, quantum annealing, superconducting qubits, and others (topological and photonic) technologies have been included in the study.</li> </ul>                                                                     | NA                                                                                                                                                                                                                                                                                                              |
| Application                             | <ul style="list-style-type: none"> <li>The market sizes of optimization, simulation, machine learning, and others (quantum chemistry and quantum finance) applications have been included in this study.</li> </ul>                                                                    | <ul style="list-style-type: none"> <li>The market sizes of cryptography and sensing applications have not been considered in this study.</li> </ul>                                                                                                                                                             |
| Vertical                                | <ul style="list-style-type: none"> <li>The market sizes of the aerospace &amp; defense, BFSI, healthcare &amp; life sciences, energy &amp; utility, chemical, transportation &amp; logistics, government, and education verticals have been included in this study.</li> </ul>         | NA                                                                                                                                                                                                                                                                                                              |
| Vendors                                 | <ul style="list-style-type: none"> <li>Vendors that provide quantum computing software and services are considered in the scope of the report.</li> </ul>                                                                                                                              | NA                                                                                                                                                                                                                                                                                                              |

Source: MarketsandMarkets Analysis

## 1.6 MARKET SCOPE

The study provides an in-depth analysis of the global quantum computing software market since 2020 based on the contemporary market trends and developments and its potential growth from 2021 to 2026. It also provides detailed market trends, competitive landscape, market size, forecasts, and comprehensive analysis of the quantum computing software market.

### 1.6.1 MARKET SEGMENTATION



\*Others includes Quantum chemistry and Quantum Finance

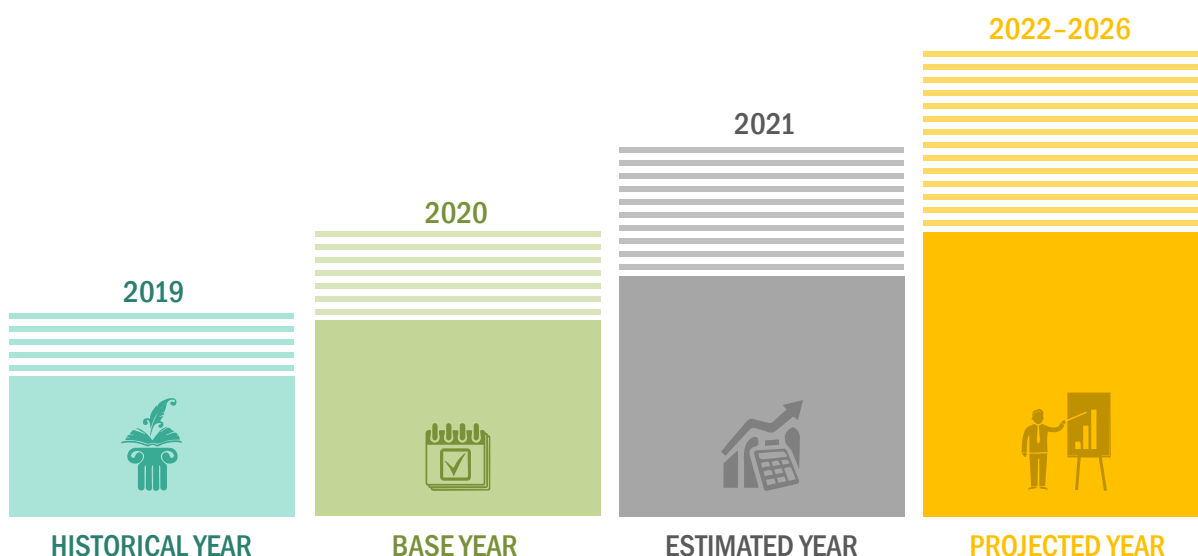
\*\*Others includes Topological technologies and Photonic technologies

\*\*\*Rest of Europe includes Netherlands, Italy, Belgium, Finland, and Russia

\*\*\*\*Rest of APAC includes India, Australia and New Zealand (ANZ), Singapore, Malaysia, and Taiwan

Source: Industry Experts and MarketsandMarkets Analysis

## 1.6.2 YEARS CONSIDERED FOR STUDY



Note: The forecast period under consideration is 2021–2026.

## 1.7 CURRENCY

The base currency used in the report is the United States Dollar (USD), with the market size indicated in USD billion.

- Revenues of companies were sourced from their annual reports.
- The average annual currency conversion rate was used for companies that reported their revenues in currencies other than USD.

**TABLE 1** UNITED STATES DOLLAR EXCHANGE RATE, 2018–2020

| Currency    | Period           | 2018   | 2019   | 2020    |
|-------------|------------------|--------|--------|---------|
| YEN to USD  | January–December | 110.42 | 109.01 | 106.725 |
| Yuan to USD | January–December | 6.90   | 6.91   | 6.62    |

Source: oanda.com

## 1.8 STAKEHOLDERS

- Quantum Computing Software Providers
- System Integrators (SIs)
- Consultancy Firms/Advisory Firms
- Quantum Computing Service Providers
- Independent Software Vendors (ISVs)
- Government Authorities
- Research Organizations and Universities
- Original Equipment Manufacturers (OEMs)
- Technology Standard Organizations, Forums, Alliances, and Associations

- Analysts and Strategic Business Planners
- Government Authorities
- Investors and Venture Capitalists
- Value-Added Resellers (VARs) and Distributors
- Private Equity Firms
- End-use Industries

## 2 RESEARCH METHODOLOGY

### 2.1 RESEARCH DATA

This research study involved the use of extensive secondary sources, directories, and databases, such as D&B Hoovers and Bloomberg BusinessWeek, to identify and collect information useful for this technical, market-oriented, and commercial study of the global quantum computing software market. The primary sources were several industry experts from core and related industries and preferred suppliers, manufacturers, distributors, service providers (SPs), technology developers, and organizations related to all the segments of this industry's value chain. In-depth interviews were conducted with various primary respondents that included key industry participants, subject-matter experts, C-level executives of key market players, and industry consultants to obtain and verify critical qualitative and quantitative information as well as assess prospects. The following illustrative figure shows the market research methodology applied in making this report on the quantum computing software market.

**FIGURE 6** GLOBAL QUANTUM COMPUTING SOFTWARE MARKET: RESEARCH DESIGN



Source: Industry Experts and MarketsandMarkets Analysis

### 2.1.1 SECONDARY DATA

In the secondary research process, various secondary sources were referred to for identifying and collecting information for the study. The secondary sources included annual reports, press releases, and investor presentations of companies, and white papers, certified publications, and articles from recognized associations and government publishing sources. Several journals were also referred to, such as the International Journal of Computer Science and Information Technology and Security (IJCSITS), ScienceDirect, ResearchGate, Academic Journals, and Scientific.Net. Secondary research was mainly used to obtain key information about industry insights, market's monetary chain, overall pool of key players, market classification, and segmentation according to industry trends to the bottom-most level, regional markets, and key developments from both market- and technology-oriented perspectives.

### 2.1.2 PRIMARY DATA

In the primary research process, various primary sources from both supply and demand sides were interviewed to obtain qualitative and quantitative information for this report. The primary sources from the supply side included industry experts, such as chief executive officers (CEOs), chief technology officers (CTOs), chief operating officers (COOs), vice presidents (VPs), managing directors (MDs), technology and innovation directors, and related key executives from different key companies and organizations operating in the quantum computing software market.

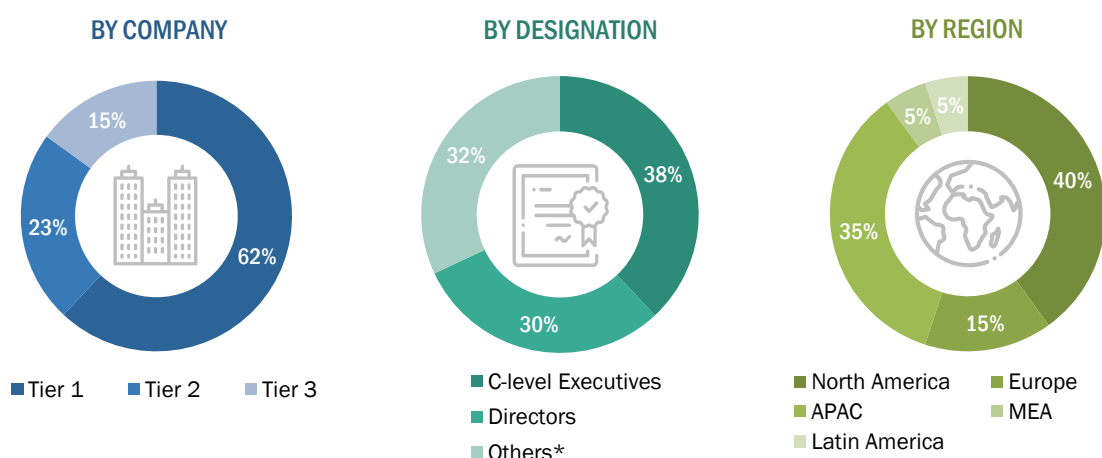
**TABLE 2** PRIMARY INTERVIEWS WITH EXPERTS

| KEY OPINION LEADERS         | HEADQUARTERS |
|-----------------------------|--------------|
| IBM                         | US           |
| Microsoft                   | US           |
| QC Ware                     | US           |
| D-Wave Systems              | Canada       |
| Cambridge Quantum Computing | US           |

*Note: The list of companies is non-exhaustive.*

### 2.1.2.1 Breakup of primary profiles

**FIGURE 7** BREAKUP OF PRIMARY INTERVIEWS: BY COMPANY, DESIGNATION, AND REGION

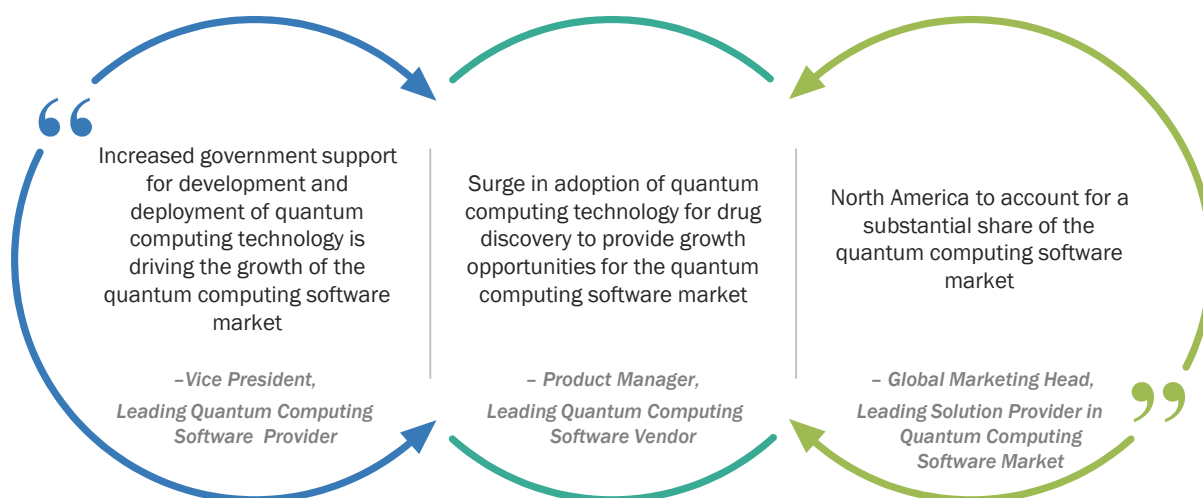


\*Others include sales managers, marketing managers, and product managers

Note: Tier 1 company—recorded overall annual revenues of more than USD 10 billion of overall revenues; Tier 2 company—USD 1–10 billion; and Tier 3 company—USD 500 million–1 billion

Source: Industry Experts and MarketsandMarkets Analysis

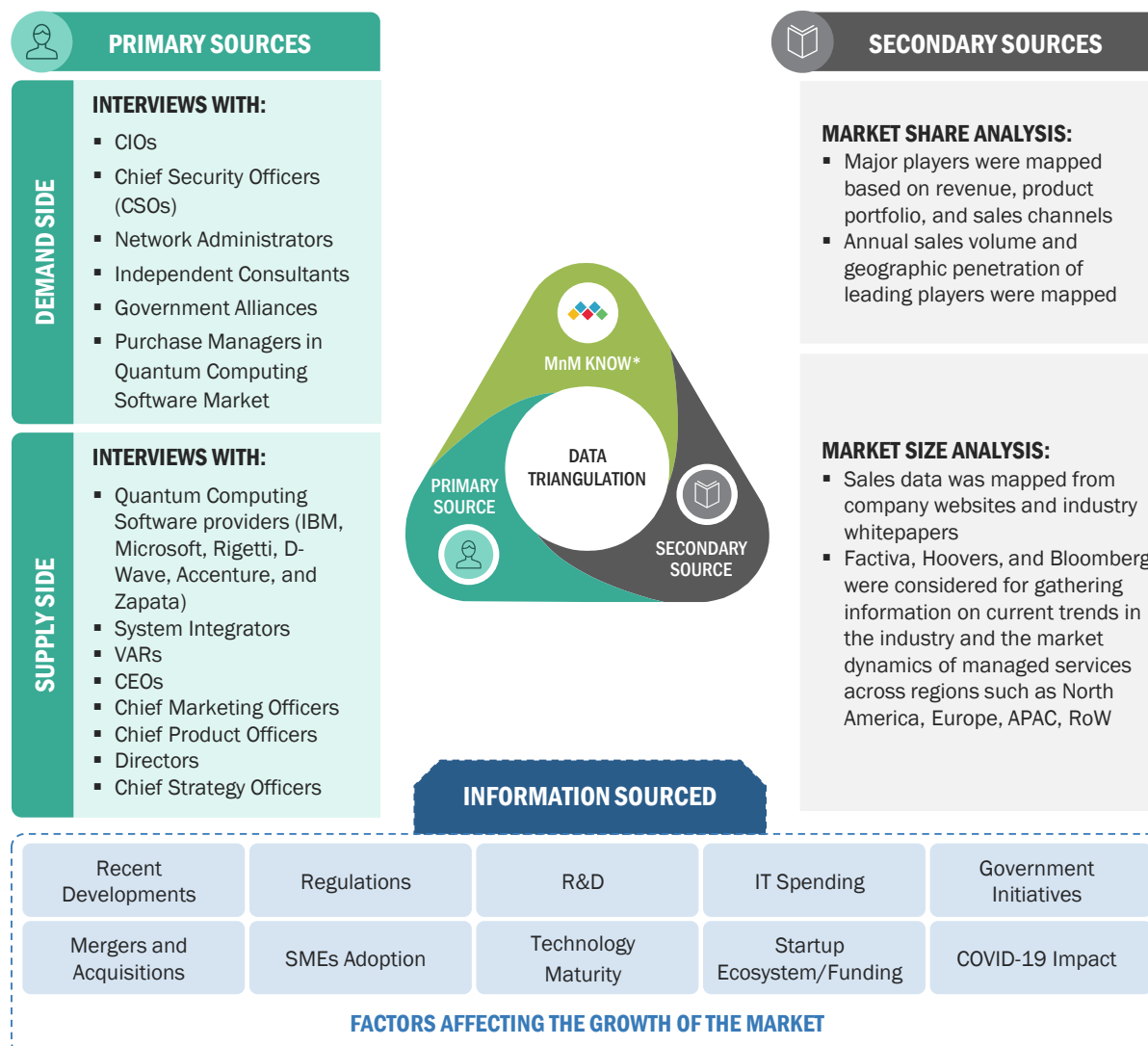
### 2.1.2.2 Key industry insights



Source: Industry Experts

## 2.2 MARKET BREAKUP AND DATA TRIANGULATION

**FIGURE 8** DATA TRIANGULATION



\*MnM KNOW stands for MarketsandMarkets' "Knowledge Asset Management" framework. In this context, it stands for existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "Knowledge Store", subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

Source: MarketsandMarkets Analysis

## 2.3 MARKET SIZE ESTIMATION

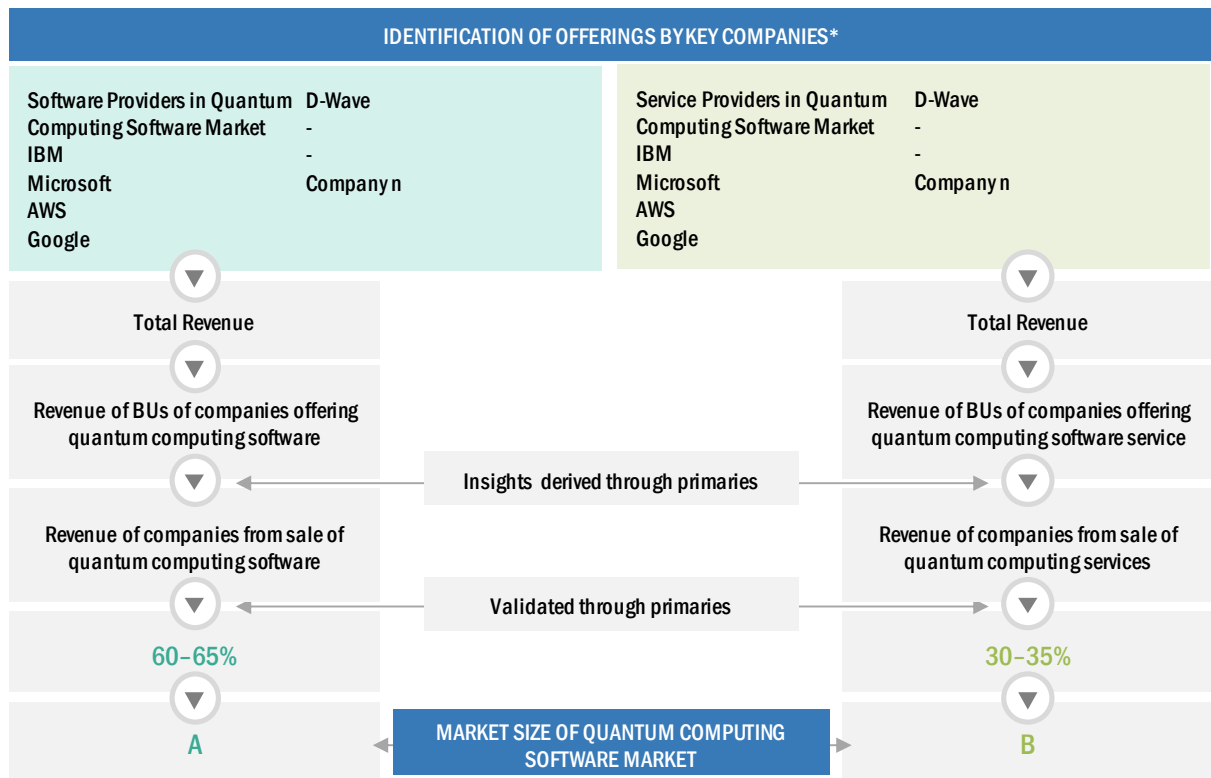
Multiple approaches were adopted for the estimation and forecasting of the quantum computing software market. The first approach involved the estimation of the market size by summation of a company's revenue generated through different software and services in the quantum computing software market.

**FIGURE 9** RESEARCH METHODOLOGY: APPROACH

| STEP 1                                                                                                                                   | STEP 2                                                                                                            | STEP 3                                                                                                             | STEP 4                                                                                                                     | STEP 5                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Identification of companies operating in the quantum computing software market, such as IBM, Microsoft, AWS, Google, D-Wave, and Rigetti | Computation of company's revenue in the quantum computing software market                                         | Identification of combined percentage share of top companies in the quantum computing software market              | Estimation of total size of the quantum computing software market using combined share of top companies and their revenues | Revalidation of total market size and market share estimations through primary respondents.                        |
| KEY SOURCE OF INFORMATION                                                                                                                |                                                                                                                   |                                                                                                                    |                                                                                                                            |                                                                                                                    |
| Secondary sources such as company financials, press releases, paid database (e.g., Factiva), and primary research                        | Secondary sources such as company financials, press releases, paid database (e.g., Factiva), and primary research | Primary research with stakeholders such as CEOs, VPs, marketing directors, and technology and innovation directors | MarketsandMarkets Calculations                                                                                             | Primary research with stakeholders such as CEOs, VPs, marketing directors, and technology and innovation directors |

Source: MarketsandMarkets Analysis

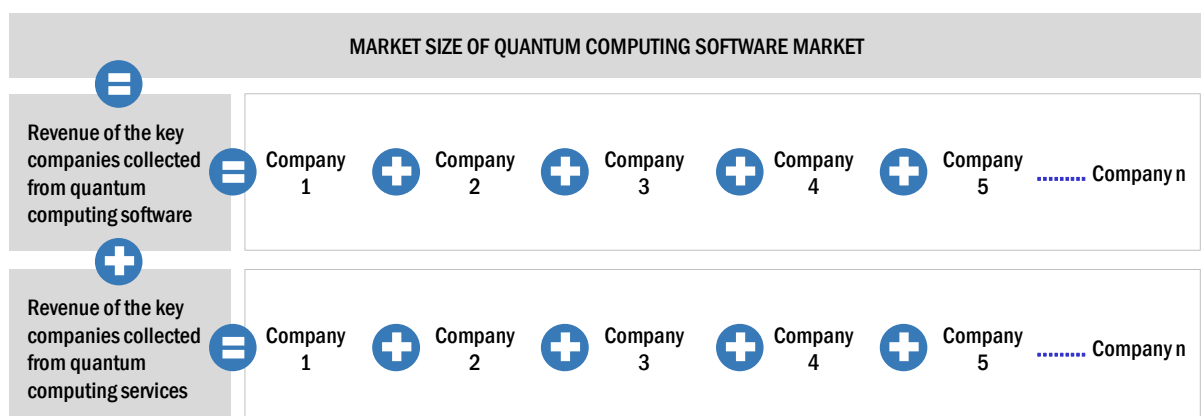
**FIGURE 10** MARKET SIZE ESTIMATION METHODOLOGY—APPROACH 1 (SUPPLY SIDE): REVENUE OF SOFTWARE/SERVICES IN QUANTUM COMPUTING SOFTWARE MARKET



Note: \*Key companies would contribute to more than 50% of the total market

Source: MarketsandMarkets Analysis

**FIGURE 11** MARKET SIZE ESTIMATION METHODOLOGY—APPROACH 2 BOTTOM-UP (SUPPLY SIDE): COLLECTIVE REVENUE OF SOFTWARE/SERVICES IN QUANTUM COMPUTING SOFTWARE MARKET



Note: Key companies contribute to more than 60% of the total market

Source: MarketsandMarkets Analysis

In this approach for market estimation, we identified the key companies offering software and services in the quantum computing software market, such as IBM, Microsoft, AWS, Google, D-Wave, Rigetti, QC Ware, 1QBit, Cambridge Quantum Computing, and Huawei, which are the major contributors to the quantum computing software market. After finalizing these companies, validation of the data was done from industry

experts through primary interviews related to the market's leading vendors. Total revenue through annual reports was estimated by US Securities and Exchange Commission (SEC) filings and paid databases. Companies' revenue pertaining to the Business Units (BUs) that offer the quantum computing services were identified through similar sources. Through primaries, revenue data generated through software and services of quantum computing software was collected. The collective revenue of key companies that offer quantum computing software comprised 50% of the market, which was again confirmed through primary interviews with industry experts. With the assumption that the rest of the market is contributed by smaller players (part of the unorganized market), the market size of organized players (60–65%) and unorganized players (30–35%) collectively was assumed to be the market size of the global quantum computing software market for FY 2020.

## 2.4 MARKET FORECAST

Factors that have a direct or indirect impact on the market were analyzed to estimate the trend of the market during the forecast period. Each factor was analyzed on a scale of 1–3, 1 being the lowest and 3 being the highest. After analyzing each factor and its impact on the market, various forecasting techniques were used to arrive at the market size.

**TABLE 3** FACTOR ANALYSIS

| FACTOR                 | INFERENCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | IMPACT   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Regulations            | Regulations have a mild impact on the quantum computing software market. Globally, regional and local regulatory authorities have enforced rules and regulations that help run quantum computing smoothly. These regulations assist ecosystem vendors in functioning within a regulatory framework.                                                                                                                                                                                                                 | Medium   |
| Government Initiatives | Government authorities have aggressively and actively promoted digital transformations in their countries. For instance, government firms are actively promoting the use of disruptive technologies, such as Artificial Intelligence (AI) and Machine Learning (ML), and to adhere to the regulations provided by different countries while executing quantum computing. Hence, government organizations have also updated rules and policies to accelerate the growth of the quantum computing software ecosystem. | Moderate |
| Recent Developments    | The growth of the quantum computing software market is mainly driven by partnerships, collaborations, and product launches and developments. Key vendors have adopted inorganic and organic growth strategies to offer core quantum computing software to different industry verticals.                                                                                                                                                                                                                             | Medium   |
| IT Spending            | The top three verticals in the quantum computing software market contribute more than 45% of the overall market share. Various stakeholders across verticals in different regions are looking forward to adopting quantum computing software to gain a competitive advantage in the global market.                                                                                                                                                                                                                  | Medium   |
| R&D Spending           | The key players operating in the quantum computing software market allocate a significant part of their revenues for R&D to innovate their software and meet the emerging demand from customers.                                                                                                                                                                                                                                                                                                                    | Medium   |
| Enterprises Spending   | Over the past few years, the spending by enterprises on quantum computing software and services has shown a growth trajectory, specifically by small and medium enterprises. Organizations are embracing new technologies to streamline business processes and drive business revenue. Digital transformation initiatives have been influencing enterprises to increase their capital spending on quantum computing software.                                                                                       | Medium   |

|                                |                                                                                                                                                                                                                                                                                                                           |        |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| COVID-19 Implications          | The recent COVID-19 pandemic is projected to have a minimal impact on the growth of the quantum computing software market. The adoption of quantum computing software has increased among the BFSI and healthcare and life sciences verticals for data protection, vaccine development, and patient monitoring worldwide. | Medium |
| Technology Maturity            | The Quantum computing software market is in the growth phase. With the increase in the adoption of disruptive technologies such as AI and ML, the quantum computing software market is projected to gain traction in the coming years and would reach near its maturity stage in the next 5–10 years.                     | Medium |
| Mergers and Acquisitions (M&A) | The quantum computing software market has experienced several mergers and acquisitions over the last few years. The market growth is mainly driven by leading and emerging vendors.                                                                                                                                       | Medium |

Source: MarketsandMarkets Analysis

## 2.5 ASSUMPTIONS FOR STUDY

The following assumptions have been taken into consideration for the quantum computing software market report.

| PARAMETER                                     | DESCRIPTION                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GLOBAL QUANTUM COMPUTING SOFTWARE MARKET SIZE | Market size estimates are conservative/realistic as opposed to optimistic/aggressive.                                                                                                                                                                                                                                     |
| EXCHANGE RATE                                 | Dollar fluctuations are not expected to be serious enough to affect forecasts to a significant extent. Fall in the value of dollar is expected to enhance revenue growth opportunities for the US multinationals, whereas the stability in the value of dollar is expected to stabilize the price of imports and exports. |
| QUANTITATIVE ASSUMPTION                       | The market size may not add up to the total, as it has been rounded off.                                                                                                                                                                                                                                                  |
| GLOBAL ECONOMIC CONDITION                     | The global economy is assumed to be stable, exhibiting a strong positive growth. This is also expected to pertain among the economies of countries.                                                                                                                                                                       |
| FINANCIAL STATISTICS                          | All the revenues and financial insights (related developments) have been extracted from the company website or annual reports, and the financials considered for the market estimation have been calculated as per the financial year, which are true to the best of MarketsandMarkets' knowledge.                        |

Source: MarketsandMarkets Analysis

## 2.6 LIMITATIONS OF STUDY

### METHODOLOGY- RELATED LIMITATIONS



- The report mainly focuses on the offerings of different software and services related to the quantum computing software market by the vendors. The market size estimation was done on the basis of approximations of the market rankings/contributions of these vendors, along with corresponding investments by various governments in different countries.
- The regional market sizing was derived based on the weightages assigned to the four regions; the qualitative insights from C-level executives, leaders, and industry experts; and the technology adoption trends in regional markets.
- All the forecast values are based on the historic data. The dynamic pricing model may change the projected market size.
- Industry players maintain confidentiality regarding certain quantitative information on some market segments. Hence, qualitative insights gathered during the study were used to arrive at the market size of such market segments.
- The market sizing for the overall market has been arrived at using a holistic and systematic approach.
- Revenue from quantum computing software and services alone has been considered while estimating the market size. No hardware components have been considered in the scope of the report.

Source: MarketsandMarkets Analysis

### 3 EXECUTIVE SUMMARY

---

Quantum computing is the manipulation of collective properties of quantum states, such as entanglement and superposition, to perform computation. Quantum computers solve computational problems, e.g., integer factorization, considerably faster than classical computers. Quantum computing harnesses the phenomena of quantum mechanics to deliver a high leap forward in computation to solve certain problems. Currently, rather than storing information using bits represented by 0s or 1s as conventional digital computers do, quantum computers use quantum bits or qubits to encode information as 0s, 1s, or both at the same time. This superposition of states along with the other quantum mechanical phenomena of entanglement and tunneling, enables quantum computers to manipulate enormous combinations of states at once.

The outbreak and widespread of the COVID-19 pandemic have led all companies to reassess their supply chain strategies, anticipate potential disruptions in their business operations, and come up with quick responses to stabilize their supply chains. Most of the companies offering quantum computing software and services across the world had witnessed less impact of the pandemic on their revenue growth in 2020, however, some witnessed a considerable impact of the COVID-19 on their sales and manufacturing activities.

Quantum software plays a critical role in exploiting the full potential of quantum computing systems. With the rapid development of advanced quantum hardware and the growing accessibility of universal quantum devices to researchers and professionals via Quantum as a Service (QaaS), it is time to focus on engineering quantum software systems to derive the benefits of quantum technology. Several applications of quantum computing need quantum software engineering methodologies and techniques to deal with specific problems. Thus, several organizations are coming forward to set up a community for quantum software engineering that focuses on devising methods, tools, and processes for developing quantum software systems efficiently.

Quantum computing is anticipated to solve today's problems faster and effectively. Recent developments in quantum computing (QC), e.g., Google achieving quantum supremacy, have increased the popularity of quantum computing. Companies such as Google, IBM, and Microsoft are taking measures to provide quantum computing software. Besides, many other companies are aiming to lead pioneering research in their areas of specialization thanks to quantum computing.

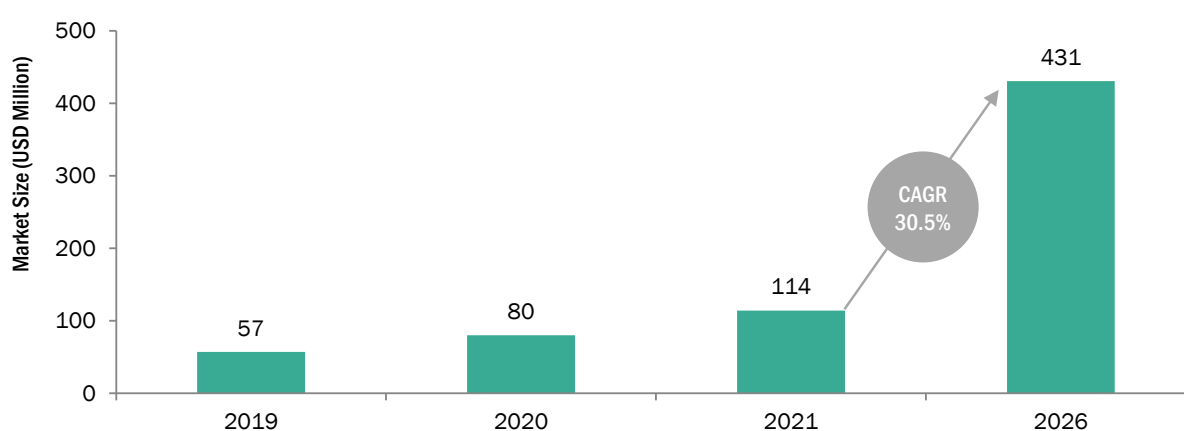
The growing adoption of quantum computing software in the BFSI vertical, support by governments for the development and deployment of the technology, and increasing number of strategic alliances for research and development are expected to boost the growth of the quantum computing software market. Likewise, the rising adoption of quantum computing technology in the healthcare industry and the growing need for high-level computing performance are expected to create lucrative opportunities for the market players. However, the technical and implementation challenges in real-time applications are expected to restrain the market growth, and the lack of highly skilled professionals is expected to create challenges for the providers of quantum computing software.

Major vendors offering quantum computing software across the globe are IBM Corporation (IBM), Microsoft Corporation (Microsoft), Amazon Web Services, Inc. (AWS), D-Wave Systems Inc (D-Wave Systems), Rigetti Computing (Rigetti), Google LLC (Google), Honeywell International Inc. (Honeywell), QC Ware (QC Ware), 1QB Information Technologies, Inc. (1Qbit), Huawei Technologies Co., Ltd. (Huawei), Accenture plc (Accenture), Cambridge Quantum Computing (Cambridge Quantum Computing), Fujitsu Limited (Fujitsu), Riverlane (Riverlane), Zapata Computing (Zapata), Quantum Circuits, Inc. (Quantum Circuits), Quantica Computacao (Quantica Computacao), XANADU Quantum Technologies (XANADU), VeriQloud (VeriQloud), Quantastica (Quantastica), AVANETIX (AVANETIX), Kuano (Kuano), Rahko (Rahko), Ketita Labs (Ketita

Labs), and Aliro Quantum (Aliro Quantum). These companies adopt various strategies, including acquisitions; product launches and enhancements; and partnerships, agreements, and collaborations, to cater to the growing demand for quantum computing software and services across the globe as well as to strengthen their position in the market. These strategies are tracked and mentioned in the report.

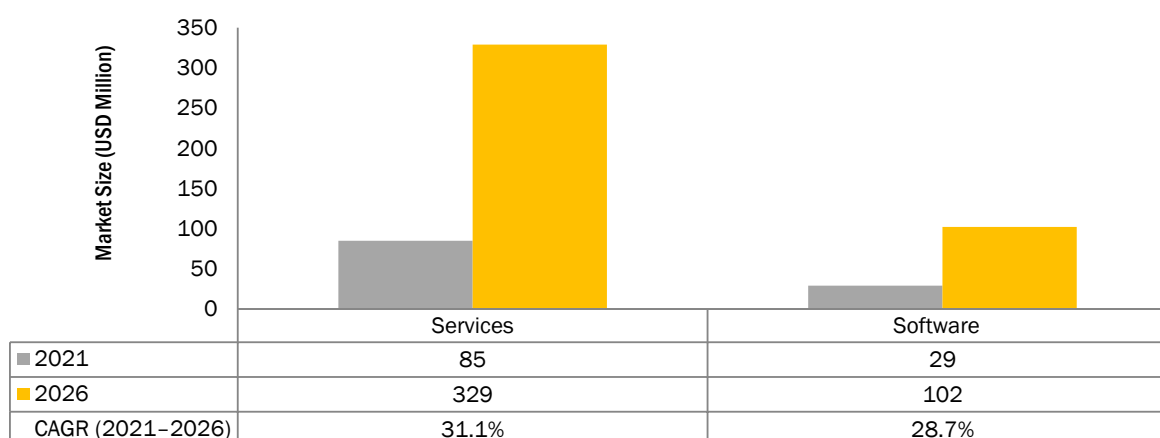
The quantum computing software market has been segmented based on component (software and services [Quantum as a Service and consulting services]), deployment mode (cloud and on-premises), organization size (SMEs and large enterprises), application (optimization, machine learning, simulation, and others), technology (superconducting qubits, trapped ions, quantum annealing, and other technologies), and vertical (aerospace & defense, BFSI, healthcare & life sciences, energy & utility, chemical, transportation & logistics, government, and education). The regional analysis of the quantum computing software market covers North America, Europe, APAC, and RoW.

**FIGURE 12** QUANTUM COMPUTING SOFTWARE MARKET SIZE, 2021–2026



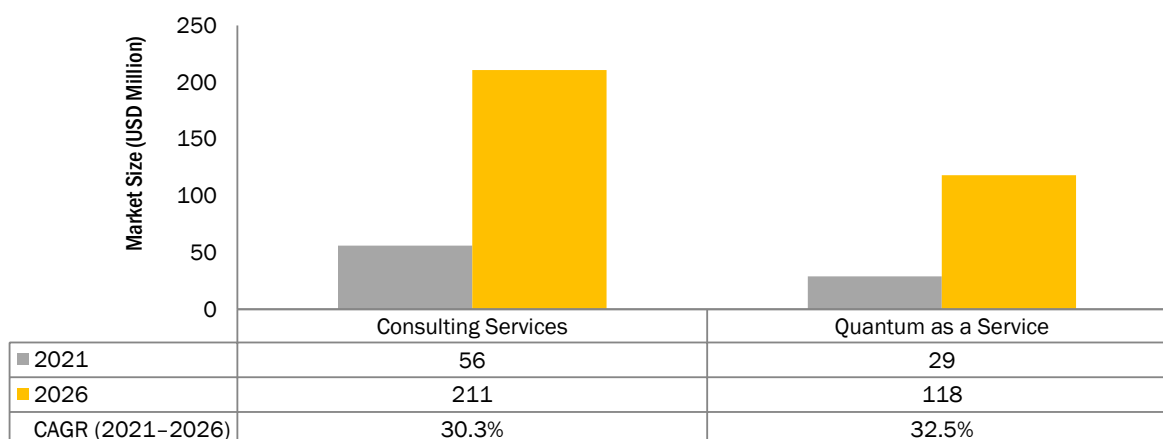
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**FIGURE 13** SERVICES SEGMENT, BASED ON COMPONENT, TO LEAD QUANTUM COMPUTING SOFTWARE MARKET DURING FORECAST PERIOD



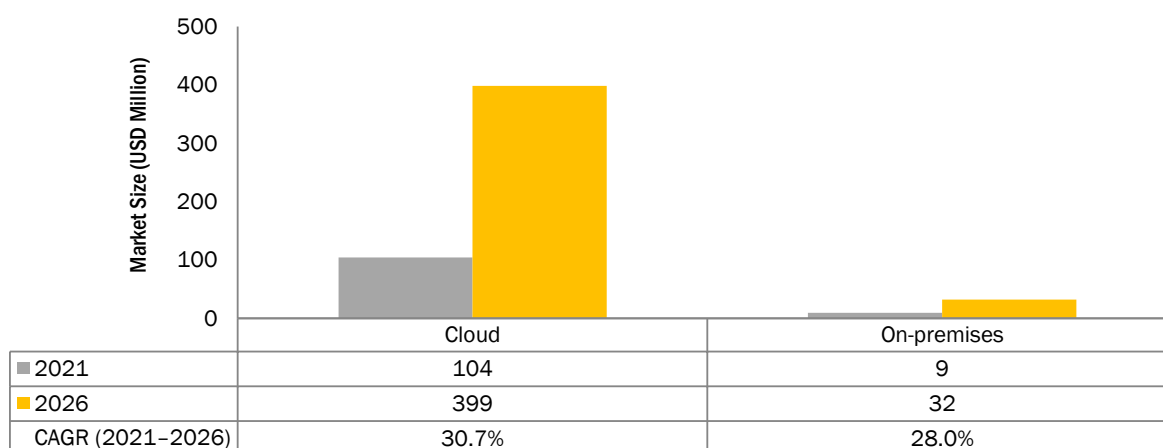
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**FIGURE 14** QUANTUM AS A SERVICE SEGMENT, BASED ON SERVICES, TO REGISTER HIGHER CAGR DURING FORECAST PERIOD



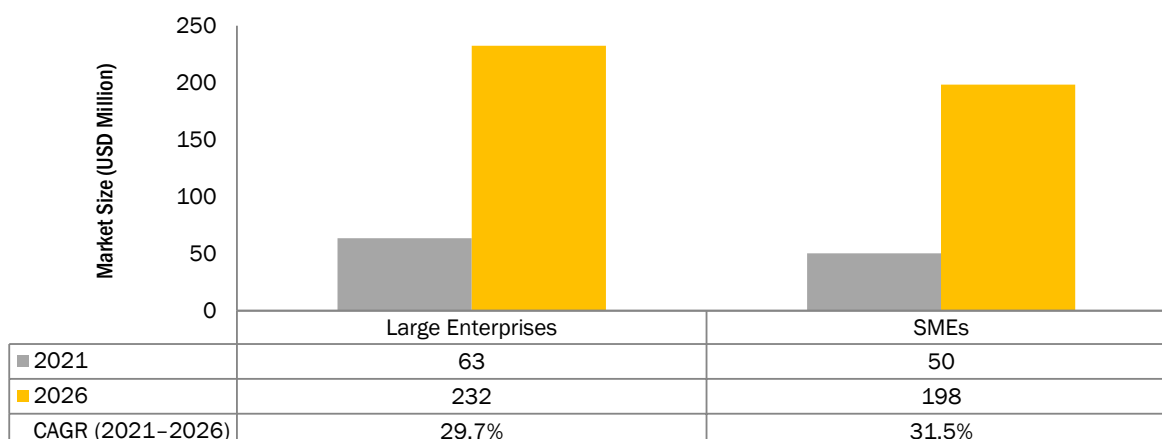
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**FIGURE 15** CLOUD-BASED DEPLOYMENT MODE TO LEAD QUANTUM COMPUTING SOFTWARE MARKET DURING FORECAST PERIOD



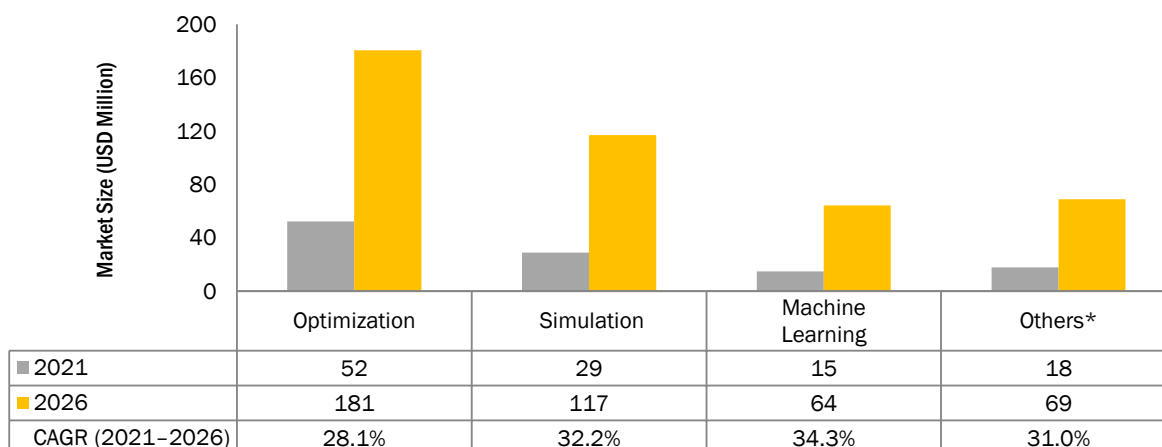
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**FIGURE 16** SMES, BASED ON ORGANIZATION SIZE, TO EXHIBIT HIGHER GROWTH RATE DURING FORECAST PERIOD



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

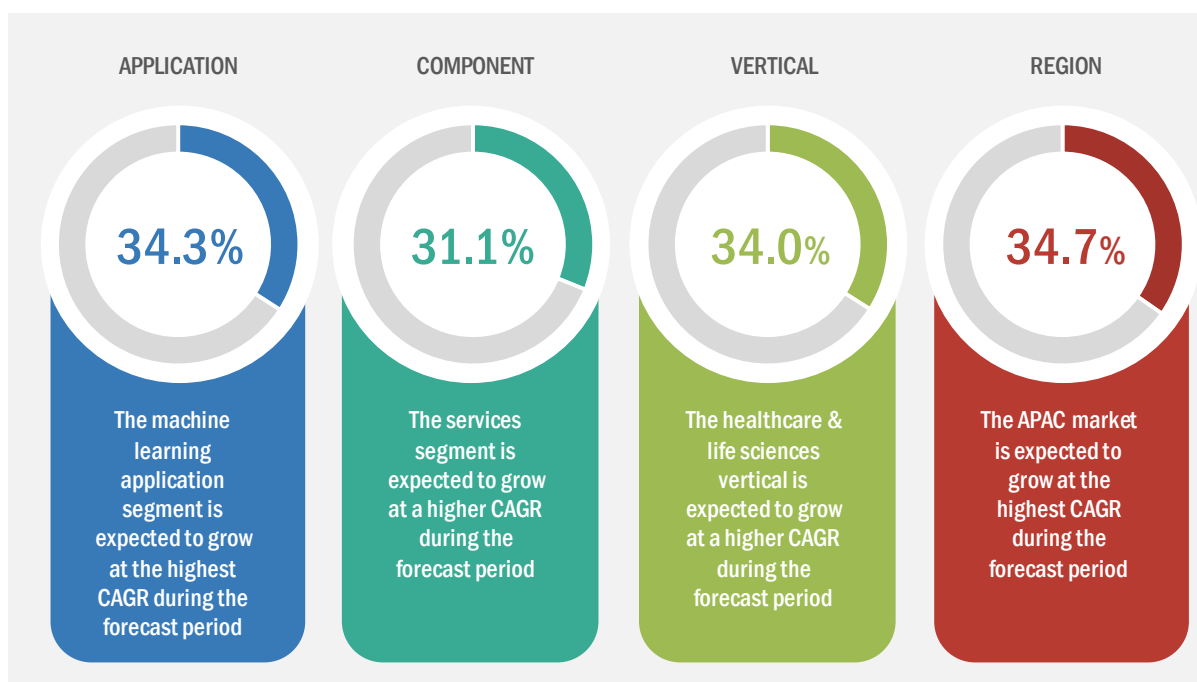
**FIGURE 17** OPTIMIZATION SEGMENT, BASED ON APPLICATION, TO LEAD QUANTUM COMPUTING SOFTWARE MARKET DURING FORECAST PERIOD



Note: \*Others includes quantum chemistry and quantum finance.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**FIGURE 18** GROWING SEGMENTS IN QUANTUM COMPUTING SOFTWARE MARKET, 2021-2026



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**FIGURE 19** QUANTUM COMPUTING SOFTWARE MARKET: REGIONAL SNAPSHOT

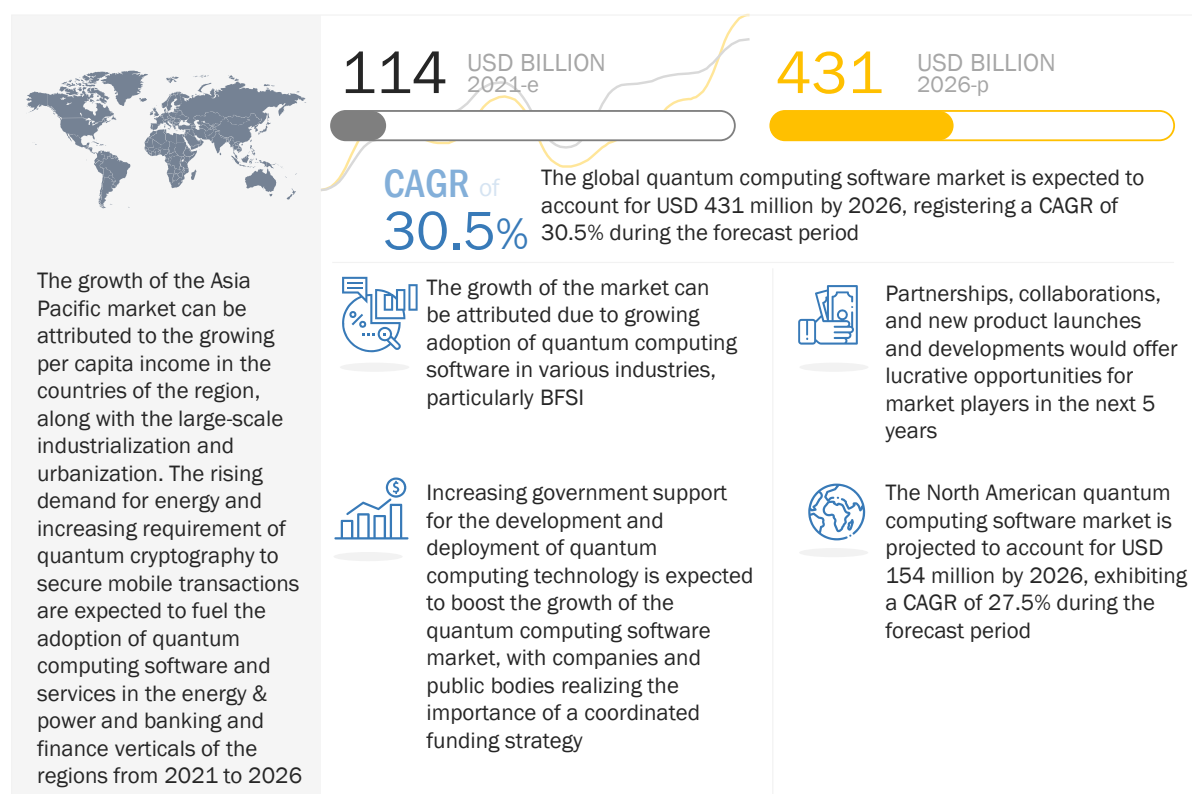
| QUANTUM COMPUTING SOFTWARE MARKET, BY REGION                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NORTH AMERICA                                                                                                                                                                                                                                                                                                                                                                                                     | EUROPE                                                                                                                                                                                                                                                                                                                                                                                                            | APAC                                                                                                                                                                                                                                                                                                                                                                                                                    | ROW                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div></div> <div><div> 40.2</div><div> 27.5</div><div> 46</div></div> | <div></div> <div><div> 24.4</div><div> 28.9</div><div> 28</div></div> | <div></div> <div><div> 33.3</div><div> 34.7</div><div> 38</div></div> | <div></div> <div><div> 2.1</div><div> 32.7</div><div> 2</div></div> |
| <div> MARKET SHARE% (2021)</div>                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                   | <div><div> CAGR% (2021-2026)</div><div> MARKET SIZE, 2021 (USD MILLION)</div></div>                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 4 PREMIUM INSIGHTS

### 4.1 ATTRACTIVE OPPORTUNITIES IN QUANTUM COMPUTING SOFTWARE MARKET

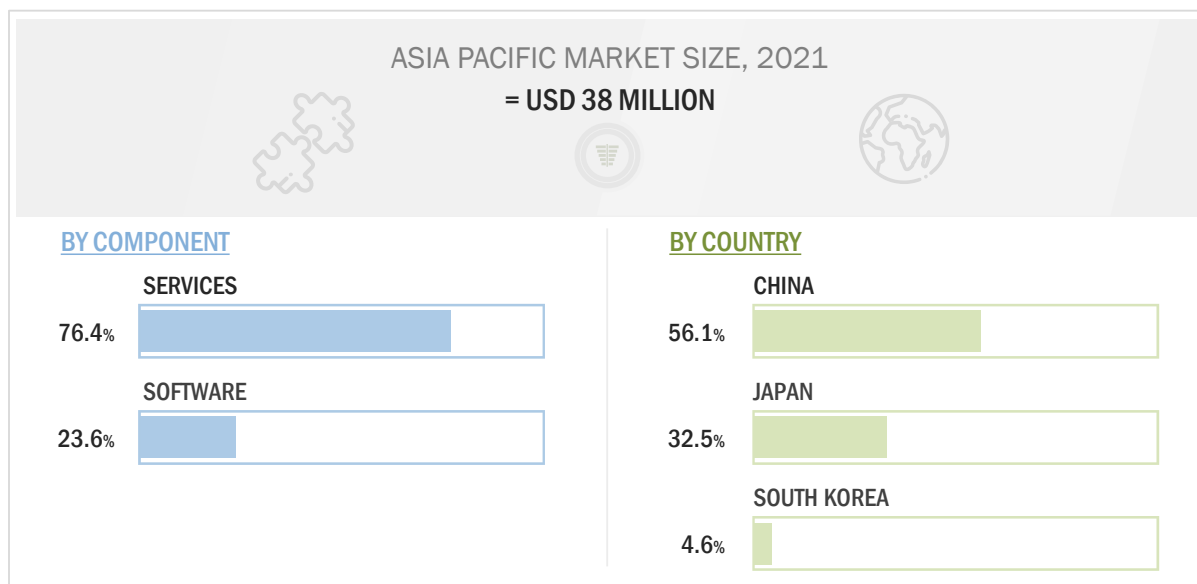
**FIGURE 20** GROWING GOVERNMENT SUPPORT FOR DEVELOPMENT AND DEPLOYMENT OF QUANTUM COMPUTING TECHNOLOGY TO DRIVE MARKET GROWTH DURING FORECAST PERIOD



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 4.2 ASIA PACIFIC QUANTUM COMPUTING SOFTWARE MARKET, BY COMPONENT AND COUNTRY

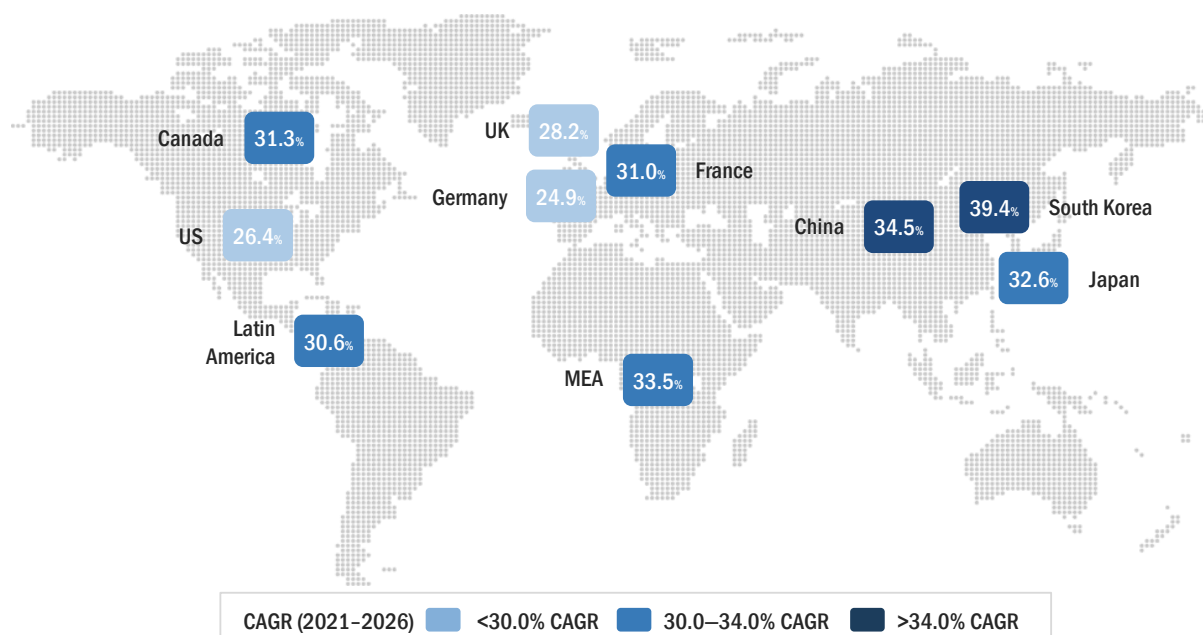
**FIGURE 21** SERVICES SEGMENT AND CHINA TO HOLD LARGEST SHARES OF ASIA PACIFIC MARKET IN 2021



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 4.3 QUANTUM COMPUTING SOFTWARE MARKET: MAJOR COUNTRIES

**FIGURE 22** SOUTH KOREA TO GROW AT HIGHEST RATE DURING FORECAST PERIOD



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 5 MARKET OVERVIEW

### 5.1 INTRODUCTION

In this section, we have described the drivers, restraints, opportunities, and challenges influencing the growth of the quantum computing software market. The market is distinct and competitive, with a large presence of major players, such as IBM, Microsoft, D-Wave Systems, and AWS, and small players, such as Riverlane, Zapata Computing, and XANADU.

Quantum computing is an emerging technology that uses the laws of quantum physics to provide significant computational speed for solving complex problems. It finds applications such as simulation, optimization, and machine learning in various industries, such as aerospace & defense, banking and finance, healthcare, pharmaceutical, and chemical.

### 5.2 MARKET DYNAMICS

**FIGURE 23** QUANTUM COMPUTING SOFTWARE MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

|                                                                                                                 |                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>DRIVERS</b></p>        | <ul style="list-style-type: none"> <li>▪ Growing adoption of quantum computing software in banking, financial services, and insurance vertical</li> <li>▪ Government support for development and deployment of quantum computing technology</li> <li>▪ Increasing number of strategic alliances for research and development</li> </ul> |
|  <p><b>RESTRAINTS</b></p>    | <ul style="list-style-type: none"> <li>▪ Technical and implementation challenges</li> </ul>                                                                                                                                                                                                                                             |
|  <p><b>OPPORTUNITIES</b></p> | <ul style="list-style-type: none"> <li>▪ Growing adoption of quantum computing technology in healthcare industry</li> <li>▪ Growing need for high level computing performance</li> </ul>                                                                                                                                                |
|  <p><b>CHALLENGES</b></p>    | <ul style="list-style-type: none"> <li>▪ Lack of highly skilled professionals</li> </ul>                                                                                                                                                                                                                                                |

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 5.2.1 DRIVERS

### 5.2.1.1 Growing adoption of quantum computing software in banking, financial services, and insurance vertical

Quantum computing is gaining traction in the banking, financial services, and insurance (BFSI) vertical, where companies are highly focusing on increasing the speed of trade activities, transactions, and data processing. One of the potential applications of quantum computing is simulation. Quantum computing helps to identify effective and efficient ways to manage financial risks. The processing time and costs of high-quality solutions can increase exponentially if conventional computers are used in financial institutions, while quantum computers can carry out operations faster at optimized costs, resulting in cost savings and the creation of new opportunities for revenue generation.

Financial institutions that can harness quantum computing software are likely to observe significant benefits. They will be able to more effectively analyze large or unstructured data sets, which could help banks and other related domains to make better decisions and improve customer service. The potential benefit of quantum computing in the financial services segment includes providing relevant and required solutions for cybersecurity to safeguard the financial data of consumers using next-generation cryptography. Moreover, the detection of fraudulent activities by recognizing the behavior patterns of consumers is increasing with the help of quantum computing technology, which leads to proactive fraud risk management. The optimization of portfolio management of assets with interdependencies and predictive analytics in customer behavior can be done by combining quantum computing with artificial intelligence (AI). In addition, a combination of quantum computing and blockchain technology is expected to lead to the development of the most hack-proof technology in this era of IoT. The combination is also expected to significantly increase the transaction speed and reduce processing costs in the BFSI vertical, thereby reducing infrastructure downtime.

JP Morgan Chase and Barclays Bank have been giving major consideration to quantum computing. They began experimenting with this technology in 2017. In December 2017, JP Morgan partnered with IBM to explore the use of quantum computing technology in the financial services segment with direct access to cloud-based IBM Q systems for performing corporate experiments. Moreover, in May 2018, Japan-based banks, MUFG and Mizuho tied up with IBM Q Hub at the Keio University to experiment with the future applications of quantum computing technology in the financial sector.

### 5.2.1.2 Government support for development and deployment of quantum computing technology

Increasing government support for the development and deployment of quantum computing technology is expected to boost the growth of the quantum computing software market, with companies and public bodies realizing the importance of a coordinated funding strategy. For instance, the National Quantum Initiative Act, introduced in December 2018, included a funding of USD 1.2 billion from the US House of Representatives for the National Quantum Initiative Program. The funding aimed to facilitate the development of applications and quantum information science, over a 10-year period, by setting the priorities and goals.

Likewise, governments of various countries worldwide are making significant investments to support their research institutes in developing quantum computing technology. China is highly investing in research and development related to quantum computing technology. Governments of the US, Australia, and European Union countries are forging ahead with quantum computing initiatives. For instance, in October 2019, the Netherlands unveiled its national plan to become a prominent player in quantum computing development. Under the program named Quantum Delta NL, the country has allocated a budget of ~USD 113 million for already existing programs and new investments in new action lines.

### 5.2.1.3 Increasing number of strategic alliances for research and development

Several companies, such as Intel, Google, and IBM, are collaborating with universities of different countries to carry out research and development related to quantum computing technology. IBM has announced a series of partnerships with corporations and academic institutions to explore the practical aspects of this technology. Similarly, the Massachusetts Institute of Technology, Princeton University, and University of Waterloo are working collaboratively on the development of quantum computer prototypes. Moreover, the retail bank, NatWest, partnered with Fujitsu on a proof-of-concept project that aims to optimize its mix of high-quality liquid assets, including bonds, cash, and government securities.

In this fast-growing marketplace, it is difficult for quantum computing solution providers to specialize in all the aspects of this ecosystem. Thus, alliances or partnerships are essential for growth in the market. In addition, to defend their current position in the quantum computing software market or to introduce new distribution channels, it is very crucial to have new dimensions in the connectivity of applications. Thus, companies can interface or open up with business partners to capitalize on new technologies in the business. With these strategic alliances, companies also focus on innovations as customers continue to demand improved solutions with high capabilities.

## 5.2.2 RESTRAINTS

### 5.2.2.1 Technical and implementation challenges

The engineering of quantum computing is still in the research stage. Besides, the implementation of quantum computing solutions in real-time applications has various technical challenges. Quantum computers are susceptible to interaction with the environment since any interaction can collapse the state function. The development of universal computers, which can be used for real-time applications, is only theoretically proved, and the exact contribution of quantum computing solutions to the existing computing methods is not yet clear. These factors have developed a gap between research labs, investment companies, and the enterprise ecosystem, which consequently is expected to hamper the growth of the global quantum computing market.

## 5.2.3 OPPORTUNITIES

### 5.2.3.1 Growing adoption of quantum computing technology in healthcare vertical

The research and development related to biopharmaceuticals, right from the drug discovery to its production, is expensive, lengthy, and risky. A new drug typically takes 10–15 years to progress from the discovery to the launching stage, and the capital related to it exceeds USD 2.0 billion. The success rate of developing new drugs is less than 10% from their entry into the clinical development stage to their launch. Biopharmaceutical companies count on a few blockbuster drugs to realize the payback of more than USD 180.0 billion that the industry spends annually on research and development activities related to new drugs.

Quantum computers are powerful tools for studying complex systems such as human physiology and the impact of drugs on biological systems and living organisms. These computers are expected to be used in a number of applications in pharmaceutical research and development, especially during the early phases of drug discovery and development. Computational tools are the key components in drug discovery and development. In a number of instances, they have significantly shortened the time consumed by companies on drug optimization. Researchers rely on high-performance computing of powerful supercomputers or massive, parallel processing systems for carrying out in silico modeling of molecular structures, mapping the interactions between a drug and its target, and developing a simulation of the metabolism, distribution, and interaction of a drug in a human body. The researchers in the biopharmaceutical industry apply quantum mechanics for energy calculations and structural optimization,

especially during molecular docking and quantitative structure-activity relationship analyses. Quantum mechanics-enabled synthetic chemistry gives researchers the tools to preclude potentially inactive compounds and support the synthesis of highly challenging compounds. Quantum-based virtual screening and optimization leverage molecular simulations. Hybrid quantum-classical approaches that can predict molecule structure are expected to be available in the market within the next 5 years. They are expected to allow highly effective structure-based drug designs of small molecules.

#### **5.2.3.2 Growing need for high-level computing performance**

The growing need for high-level computing performance is expected to create lucrative opportunities for the players operating in the quantum computing software market. Increasing data center workloads, growing preference for Software-as-a-Service (SaaS) business models, coupled with increasing complexity in processor design of classical binary computing systems, are the key factors driving the demand for quantum computing software. A classical binary computing system stores data in tiny transistors and each of these transistors hold a single “bit” of information such as a 0 or a 1, whereas quantum computing works on a phenomenon called superposition that stores/process both a 1 and a 0 simultaneously, which is significantly more powerful compared to the classical binary computing system.

### **5.2.4 CHALLENGES**

#### **5.2.4.1 Lack of highly skilled professionals**

Quantum computing is a new and promising technology. The implementation of this technology requires a pool of technically skilled professionals. Hence, finding professionals with the required skillset is a major challenge for the industry players. Governments of different countries and manufacturers of quantum computers are planning to start training sessions related to the technology to tackle this challenge. For instance, D-Wave Systems is providing training modules to its customers to make them aware of the functioning of quantum computers. However, the high cost associated with these modules is a challenge for businesses aiming to train their employees. Moreover, as these modules are complex, understanding them is a time-consuming and difficult process. As a result, manufacturers of quantum computers have to reduce their profit margins by additionally investing in different post-sales activities.

## **5.3 IMPACT OF COVID-19 ON QUANTUM COMPUTING SOFTWARE MARKET**

The outbreak and widespread of the COVID-19 pandemic have increased the adoption of quantum computing systems and services in drug discovery. Companies operating in different sectors are revisiting their financial trajectories and framing new strategies to tackle the impact of the pandemic on their revenues and businesses.

#### **5.3.1 PRE-COVID-19**

Quantum computing technology was witnessing increasing global demand during the pre-COVID-19 period, as companies were making strategic partnerships and collaborations and undertaking patent registrations to enhance their position in the market. During the pre-COVID-19 scenario, the key factor driving the growth of the quantum computing market was the rising investments by governments of different countries in the development of quantum computing technology.

#### **5.3.2 POST-COVID-19**

The growth of the quantum computing market is expected to be primarily driven by technological advancements in quantum computing technology. The demand for quantum computing systems and services post-COVID-19 is expected to increase owing to the rise in the adoption of quantum computing technology in drug discovery. However, the stability and error correction issues in quantum computing technology are expected to restrain the market growth.

## 5.4 CASE STUDY ANALYSIS

In this section, we have described the use cases across companies deploying quantum computing software of various vendors. Case studies include various quantum computing software solutions provided by different vendors and their applications.

### 5.4.1 USE OF QUANTUM COMPUTING IN ADVANCED MACHINE LEARNING

|                |                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client         | NASA, Google, and the Universities Space Research Association (USRA)                                                                                                                                                                             |
| Description    | D-Wave Two Quantum Computer was selected for new Quantum Artificial Intelligence Initiative. The purpose of this effort is to use quantum computing to advance machine learning to solve some of the most challenging computer science problems. |
| Solution       | D-Wave Two (512 qubits) was installed at the Quantum Artificial Intelligence Laboratory that is operational since the third quarter of 2013.                                                                                                     |
| Benefit Gained | This helped researchers develop highly efficient and accurate models for everything, right from speech recognition and web search to protein folding.                                                                                            |

Source: Company Website and Press Releases

### 5.4.2 EXPLORING QUANTUM COMPUTING IN FINANCIAL SERVICES

|                  |                                                                                                                                                                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client           | BBVA                                                                                                                                                                                                                                                                                                         |
| Description      | BBVA, a multinational financial services company, wanted to explore the use of quantum computing. Accenture was ideally positioned to assist.                                                                                                                                                                |
| Solution Offered | A team of researchers from Accenture worked with D-Wave and the innovation team at BBVA to explore how best to use quantum computing to gain a competitive advantage. Quantum algorithms were constructed to find opportunities for currency arbitrage, credit scoring, and optimizing trading trajectories. |
| Benefit Gained   | BBVA's quantum experimentation for financial services contributed to BBVA's position at the forefront of the exploration of quantum technologies and has helped the firm forge a path of innovation in the field of quantum computing.                                                                       |

Source: Company Website and Press Releases

### 5.4.3 USE OF QUANTUM COMPUTING TO IMPROVE ANALYSIS OF MEDICAL IMAGES

|             |                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | Medical imaging techniques, such as CT, MRI, and X-ray scans, have become crucial diagnostic tools for medical practitioners over the last century. Computer-aided detection and diagnosis of health issues through medical images have been witnessing rapid adoption globally. However, these crucial images are sometimes impacted by noise, poor resolution, and low replicability. |
| Solution    | Quantum computing has the potential to improve the quality of medical images, right from their processing stages that include edge detection and image matching to their interpretation. These improvements are expected to considerably enhance image-aided medical diagnostics.                                                                                                       |

Source: Press Releases

#### 5.4.4 DEVELOPMENT OF NEW CHEMICALS WITH QUANTUM COMPUTING

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | The chemical industry contributes ~7% (USD 5.7 trillion) to the global gross domestic product (GDP), along with providing employment to ~120 million people. The development of new chemicals is an expensive and time-consuming process. As such, classical simulations of chemistry are carried out for the laboratory testing of chemicals. However, the accuracy of calculations decreases as the complexity of molecular interactions in chemicals increases. |
| Solution    | The use of quantum computers by chemical and petroleum companies has the potential to accelerate the discovery and development of new chemicals.                                                                                                                                                                                                                                                                                                                   |

Source: Press Releases

#### 5.4.5 TRADING OPTIMIZATION WITH QUANTUM COMPUTING

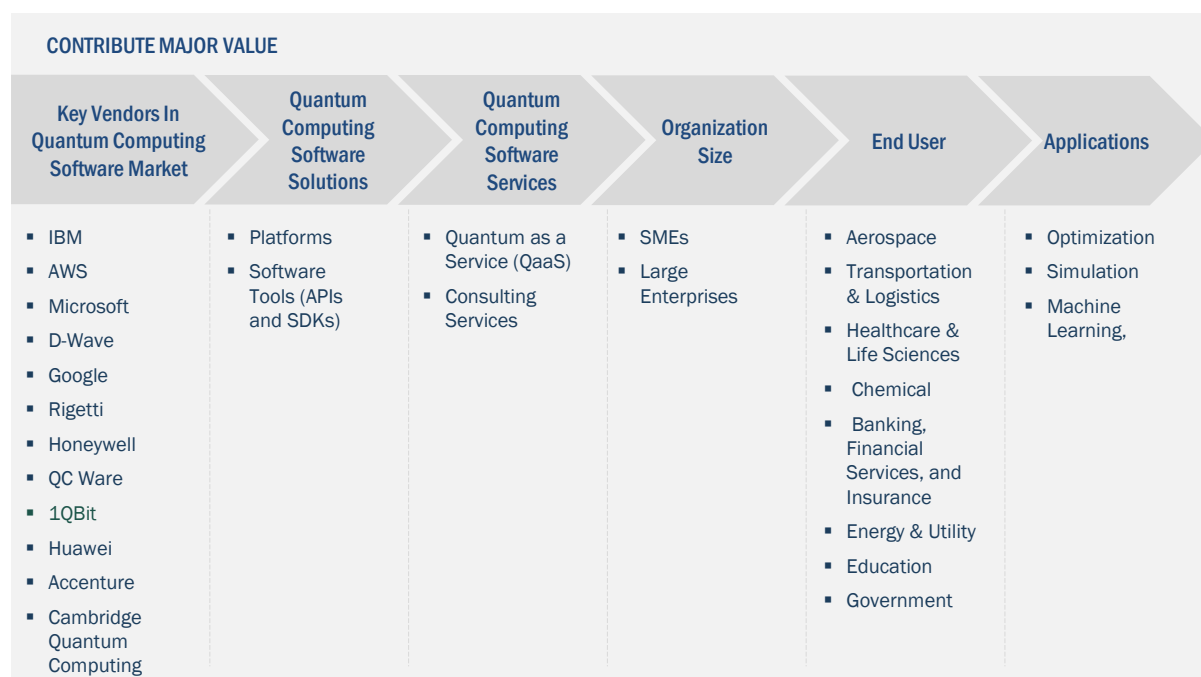
|             |                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | Investment managers struggle to incorporate real-life constraints, such as market volatility and customer life-event changes, into portfolio optimization. Currently, rebalancing investment portfolios that keep up with market movements is significantly impacted by computational limitations and transaction costs.                                                                       |
| Solution    | Quantum technology can help cut through the complexity of today's trading environments. Quantum computing's combinatorial optimization capabilities may enable investment managers to improve portfolio diversification, rebalance portfolio investments to more precisely respond to market conditions and investor goals, and more cost-effectively streamline trading settlement processes. |

Source: Press Releases

### 5.5 VALUE CHAIN ANALYSIS

The value chain model works by segregating a firm into its strategically relevant activities to understand its cost behavior and identify any source of differentiation in terms of its product costs and innovations, as well as system integration. Thus, a firm requires to achieve cost reduction in these strategically important activities to gain a competitive advantage. The value chain model of the quantum computing market has been depicted in the figure below.

**FIGURE 24** VALUE CHAIN ANALYSIS: QUANTUM COMPUTING SOFTWARE MARKET



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 5.6 ECOSYSTEM

The growing adoption of quantum computing software and services in the banking and finance industry is driving the growth of the quantum computing software market across the world. Various industries worldwide are witnessing digital disruptions. They are leveraging advanced technologies, such as cloud computing, industrial IoT (IIoT), and smart logistics, to modify their business models and enhance their operational efficiency. This has also led to the development of smart supply chains, improved manufacturing processes, and an efficient end-to-end ecosystem.

**TABLE 4** ECOSYSTEM: QUANTUM COMPUTING SOFTWARE MARKET

| COMPANY                 | ROLE IN ECOSYSTEM                                              |
|-------------------------|----------------------------------------------------------------|
| IBM                     | System Manufacturer/QCaaS Provider/Consulting Service Provider |
| D-Wave Systems          | System Manufacturer/QCaaS Provider/Consulting Service Provider |
| Rigetti Computing       | System Manufacturer/QCaaS Provider                             |
| Microsoft               | QCaaS Provider                                                 |
| Amazon                  | QCaaS Provider                                                 |
| Accenture               | Consulting Service Provider                                    |
| Google                  | QCaaS Provider                                                 |
| IonQ                    | System Manufacturer/Consulting Service Provider                |
| Honeywell International | System Manufacturer                                            |
| XANADU                  | QCaaS Provider                                                 |
| Huawei                  | QCaaS Provider                                                 |
| Zapata Computing        | QCaaS Provider/Consulting Service Provider                     |
| Riverlane               | Consulting Service Provider                                    |
| 1QBit                   | Consulting Service Provider                                    |

*Note: This list of vendors is incomprehensive. There are many more vendors that can play a role in the quantum computing software ecosystem.*

*Source: Secondary Research, Articles, Press Releases, Company Website, MarketsandMarkets Analysis*

## 5.7 PORTER'S FIVE FORCES ANALYSIS

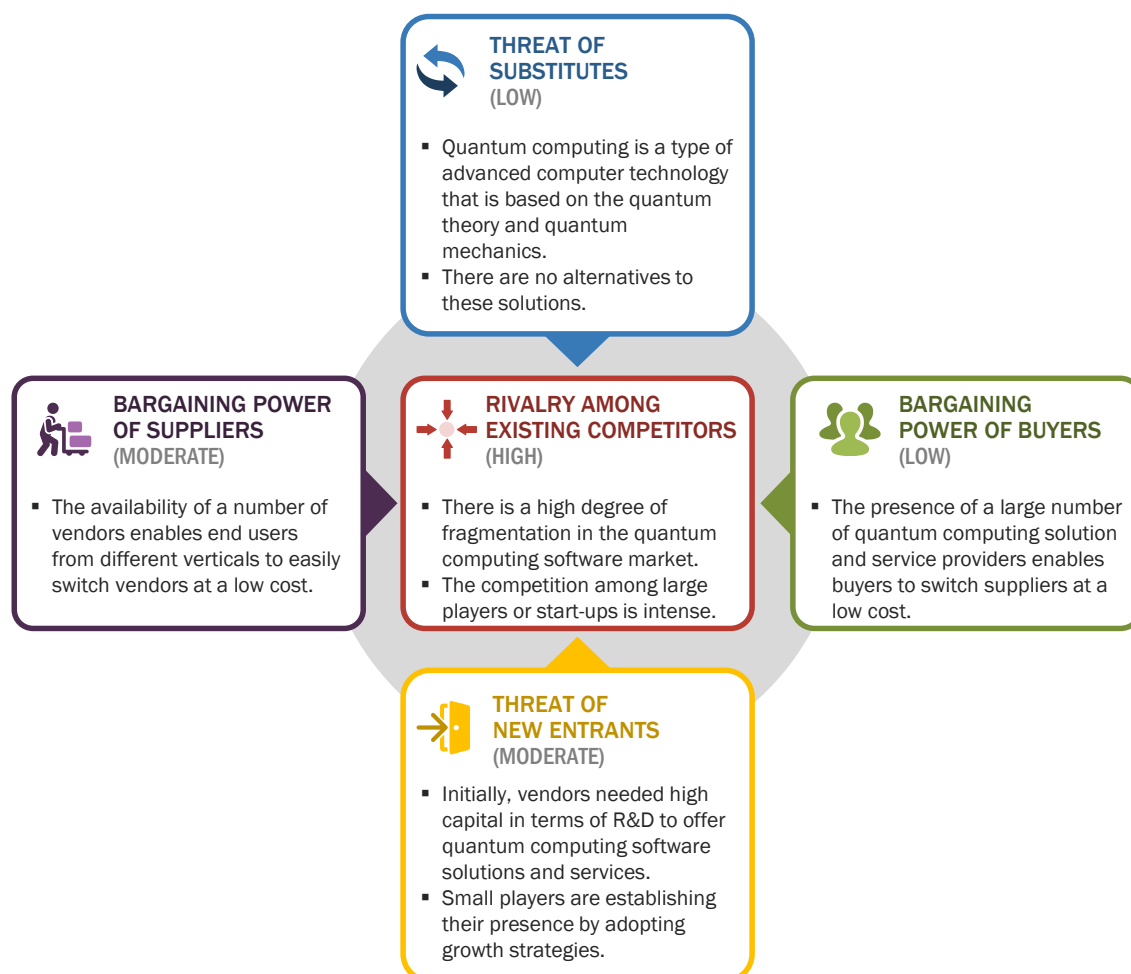
Porter's five forces analysis is a framework that determines the competitive intensity of a market. This section analyzes the quantum computing software market from five different perspectives: the competitive rivalry within industries, threat from new entrants, bargaining power of suppliers, bargaining power of buyers, and threat from substitutes. It draws insights from organizations' economics to find five forces that determine the competitive intensity and, therefore, the attractiveness of the quantum computing software market in terms of profitability.

**TABLE 5** IMPACT OF EACH FORCE ON QUANTUM COMPUTING SOFTWARE MARKET

| PORTER'S FIVE FORCES             | IMPACT   |
|----------------------------------|----------|
| Intensity of Competitive Rivalry | High     |
| Bargaining Power of Suppliers    | Moderate |
| Bargaining Power of Buyers       | Low      |
| Threat of Substitutes            | Low      |
| Threat of New Entrants           | Moderate |

Source: Industry Journals, Expert Interviews, and MarketsandMarkets Analysis

**FIGURE 25** PORTER'S FIVE FORCES ANALYSIS: QUANTUM COMPUTING SOFTWARE MARKET



Source: MarketsandMarkets Analysis

### 5.7.1 THREAT OF NEW ENTRANTS

The threat of new entrants refers to the force of new potential competitors in any market. The new entrants can capture the market share of existing companies. Currently, the threat of new entrants in the quantum computing software market is moderate attributed to the requirement of high R&D expenditure and the capital-intensive nature of the industry. There are many global and domestic players with an established base in the market. These market players have a global presence and offer a wide range of quantum computing software solutions for various applications. They have extensive R&D facilities to deliver application-specific solutions and maintain strong customer relationships. These established players have several approved and pending patents, which describe their competitive advantage in the market. Proprietary technologies and stringent government regulations are entry barriers in the market.

Various companies enter this market by adopting strategies such as acquisitions and partnerships. Due to new entrants and increasing investments in R&D to develop low-cost solutions, experts believe that the cost of solutions will decline in the near future. The growing demand for quantum computing software in emerging applications is enabling start-ups to establish their presence in the market.

### 5.7.2 THREAT OF SUBSTITUTES

The threat of substitutes refers to the possibility of alternative products replacing the existing ones in the market. Switching to substitutes will be expensive, especially for giant corporations. An increase in spending on the software and solutions would translate to an increase in pricing. Loyal customers may not approve or be able to afford the new prices, which would, in turn, empower them to opt for new substitutes. For this reason, companies often remain with their system even after they have discovered an alternative that has proven to be more efficient. A full switch would require years of research and preparation. Hence, the threat from substitutes in the quantum computing software market is expected to be low.

### 5.7.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers refers to the extent to which solution and service suppliers/providers pressurize the end users of quantum computing software and services. Quantum computing software suppliers can influence or manipulate the prices of quantum computing software solutions and services. Solutions required for the development of quantum computing software include standalone platforms or software tools (APIs and SDKs). Companies providing these solutions are abundant in the market, which enables the end users of quantum computing software solutions and services to switch suppliers/providers at a low cost. There are minimal logistics costs involved, as most of the quantum computing software solutions are moving to the cloud. Moreover, in the case of on-premise deployment, support can be provided virtually. As a result, the bargaining power of suppliers is expected to be moderate.

### 5.7.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers refers to the pressure that buyers exert on quantum computing software and service providers. Buyers of quantum computing software in emerging countries are price sensitive, which increases their bargaining power. However, buyers lose their power when there is a requirement of customized quantum computing software, i.e., a requirement-based platform to be integrated with the existing system or a specific business requirement. This makes quantum computing software providers price setters, decreasing the bargaining power of buyers. Therefore, the bargaining power of buyers in this market is expected to be low.

### 5.7.5 INTENSITY OF COMPETITIVE RIVALRY

The competitive rivalry within the industry refers to the extent to which market players exert pressure on each other. The presence of various well-established players has intensified the level of competitive rivalry in the quantum computing software market. Some of these players are IBM, Microsoft, and AWS. These vendors have a strong foothold in local and global markets due to their strong product portfolios and well-established strategic alliances. Besides, in the unorganized sector, there are several vendors who collectively hold a prominent share in the market. Thus, the intensity of competitive rivalry in the market is expected to be high.

## 5.8 PATENT ANALYSIS

Below is the list of a few patents filed related to quantum computing methods, devices, and software.

**TABLE 6** RECENT PATENTS RELATED TO QUANTUM COMPUTING SOFTWARE MARKET

| MONTH AND YEAR | TITLE                                                                                            | PATENT NO. | DESCRIPTION                                                                                                                                                                                            | COMPANY/<br>ORGANIZATION/<br>INVENTOR NAME | COUNTRY |
|----------------|--------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------|
| December 2019  | Visualizing arbitrary pulse shapes and schedules in quantum computing applications               | 10,790,912 | According to an embodiment, a system can have a processor that can execute computer-executable components stored in memory units.                                                                      | IBM                                        | US      |
| November 2019  | Quantum computing devices with Majorana Hexon qubits                                             | 10,777,605 | Various embodiments of a modular unit for a topologic qubit, as well as of scalable quantum computing architectures using such modular units, are disclosed herein.                                    | Microsoft Technology Licensing, LLC        | US      |
| August 2019    | Fabrication of interlayer dielectrics with high-quality interfaces for quantum computing devices | 10,770,638 | This method includes providing a first wafer, including a first substrate, a first insulator layer on the first substrate, and a first dielectric layer on the first insulator layer.                  | Google LLC                                 | US      |
| April 2019     | Performing a calibration process in a quantum computing system                                   | 10,643,143 | Quantum logic control parameters for the subset of quantum circuit devices of the domain are obtained based on a set of measurements that are stored in memory units.                                  | Rigetti & Co, Inc.                         | US      |
| April 2019     | Quantum computing in a three-dimensional device lattice                                          | 10,878,332 | A three-dimensional color code is applied in a three-dimensional device lattice to detect errors in the data qubits residing in the multiple layers. A two-dimensional color code is applied in a two- | Rigetti & Co., Inc.                        | US      |

|               |                                                                                                          |            |                                                                                                                                                                                                                                                                              |                                     |       |
|---------------|----------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------|
|               |                                                                                                          |            | dimensional device lattice in each respective layer to detect errors in one or more data qubits residing in the respective layer.                                                                                                                                            |                                     |       |
| April 2019    | Hardware-efficient variational quantum eigenvalue solver for quantum computing machines                  | 10,839,306 | The method includes the response to an estimated energy state not converging with an expected energy state, computing a new trial state for the VQE, and iterating to compute the estimated energy using a new trial state.                                                  | IBM                                 | US    |
| February 2019 | Monolithic signal carrier device implemented in cryogenic quantum computing applications                 | 10,681,842 | Devices, systems, methods, and computer-implemented methods to facilitate the transmission of microwave signals to one or more cryogenic stages of a cryogenic refrigerator are provided.                                                                                    | IBM                                 | US    |
| December 2018 | Phase shifter, quantum logic gate apparatus, optical quantum computing apparatus, and phase shift method | 10,860,944 | A phase shifter, a quantum logic gate apparatus, an optical quantum computing apparatus, and a phase shift method, wherein the phase shifter includes an optical resonant cavity and a quantum point, while the resonance frequency of the optical resonant cavity is omega. | Huawei Technologies Co., Ltd.       | China |
| June 2018     | Generating quantum computing circuits by distributing approximation errors in a quantum algorithm        | 10,664,761 | Methods for generating quantum computing circuits by distributing approximation errors in a quantum algorithm.                                                                                                                                                               | Microsoft Technology Licensing, LLC | US    |
| June 2018     | Methods and systems for quantum computing                                                                | 10,826,845 | The present disclosure provides methods, systems, and media for quantum computing by allowing access to quantum-ready and/or quantum-enabled computers in a distributed computing environment (for instance, the cloud).                                                     | 1QB Information Technologies Inc.   | US    |

Source: US Patents and Trade Organization

## 5.9 PRICING MODEL ANALYSIS

**TABLE 7** AVERAGE SELLING PRICE RANGE OF QUANTUM COMPUTING SYSTEMS

| Company        | Product | AVERAGE SELLING PRICE RANGE |
|----------------|---------|-----------------------------|
| D-Wave Systems | 200Q    | USD 10–15 Million           |
| D-Wave Systems | 500Q    | USD 15–20 Million           |

Source: Digikey, Mouser, Arrow, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 8** AVERAGE SELLING PRICE RANGE OF QUANTUM COMPUTING AS A SERVICE

| Duration | Charges Range   |
|----------|-----------------|
| 1 minute | Free of Cost    |
| 1 hour   | USD 2,000–5,000 |

Note: Some companies provide free access to their systems on a trial basis.

Source: Press Releases, Company Websites, and Secondary Research

## 5.10 TRADE ANALYSIS

The import and export data provided below is for the HS code 847180. It includes automatic data-processing machines (excluding processing units, input or output units, and storage units) imported and exported from 2015 to 2019.

**TABLE 9** IMPORT DATA FOR AUTOMATIC DATA-PROCESSING MACHINES, BY COUNTRY, 2015–2019 (USD MILLION)

| Country           | 2015          | 2016          | 2017          | 2018          | 2019          |
|-------------------|---------------|---------------|---------------|---------------|---------------|
| Hong Kong, China  | 3,024         | 2,775         | 3,046         | 3,505         | 3,250         |
| China             | 1,343         | 1,371         | 1,815         | 2,656         | 3,142         |
| US                | 4,119         | 3,820         | 2,955         | 3,367         | 2,831         |
| Netherlands       | 1,156         | 1,342         | 1,720         | 1,856         | 1,708         |
| Germany           | 1,042         | 1,044         | 1,199         | 1,325         | 1,429         |
| Czech Republic    | 355           | 350           | 500           | 595           | 1,004         |
| UK                | 620           | 597           | 542           | 639           | 672           |
| Mexico            | 333           | 294           | 338           | 432           | 648           |
| Japan             | 340           | 392           | 485           | 510           | 584           |
| Canada            | 446           | 449           | 445           | 537           | 490           |
| Rest of the World | 4,965         | 4,991         | 6,663         | 6,935         | 6,385         |
| <b>Total</b>      | <b>17,743</b> | <b>17,425</b> | <b>19,709</b> | <b>22,358</b> | <b>22,141</b> |

Source: ITC Trade Map

**TABLE 10** EXPORT DATA FOR AUTOMATIC DATA-PROCESSING MACHINES, BY COUNTRY, 2015–2019 (USD MILLION)

| Country           | 2015          | 2016          | 2017          | 2018          | 2019          |
|-------------------|---------------|---------------|---------------|---------------|---------------|
| China             | 4,460         | 4,294         | 5,073         | 6,165         | 5,458         |
| Hong Kong, China  | 1,833         | 2,255         | 3,426         | 4,131         | 4,826         |
| Malaysia          | 2,626         | 2,595         | 3,416         | 3,945         | 3,136         |
| US                | 2,140         | 2,215         | 2,529         | 2,621         | 2,026         |
| Netherlands       | 1,066         | 1,294         | 1,668         | 1,678         | 1,772         |
| Taipei, China     | 375           | 391           | 421           | 534           | 1,267         |
| Mexico            | 430           | 430           | 459           | 694           | 1,153         |
| Czech Republic    | 363           | 406           | 521           | 632           | 890           |
| Germany           | 667           | 587           | 619           | 678           | 687           |
| UK                | 564           | 568           | 537           | 639           | 667           |
| Rest of the World | 3,566         | 3,812         | 3,671         | 3,242         | 3,134         |
| <b>Total</b>      | <b>18,089</b> | <b>18,846</b> | <b>22,339</b> | <b>24,958</b> | <b>25,017</b> |

Source: ITC Trade Map

The US, China, the Netherlands, and Germany were the largest importers of automatic data-processing machines in 2019, followed by the UK, Mexico, and Japan. In 2019, these countries accounted for shares of ~29%, ~13%, ~8%, and ~6%, respectively, of the overall automatic data-processing machines imported globally. The capability of companies based in the US and China to manufacture semiconductor chips has increased the demand for automatic data-processing machines. Countries such as China, Malaysia, the US, and the Netherlands were the largest exporters of automatic data-processing machines in 2019. The presence of a large consumer base in these countries and the surged demand for advanced technologies are expected to increase the export of automatic data-processing machines in the coming years.

## 5.11 REGULATORY LANDSCAPE

This section consists of details related to laws and regulations implemented by government bodies to safeguard personal data and privacy. Europe and North America, from where a massive amount of data is generated, have been following the standards introduced by regulatory bodies. Some of the regulations are General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). These regulations and acts help define standards for quantum computing software, solutions, and services in organizations. Companies are also acknowledging the importance of complying with data safety regulations and are trying to adhere to them.

### 5.11.1 GENERAL DATA PROTECTION REGULATION

GDPR (Regulation [EU] 2016/679) is a regulation by which the European Parliament, the Council of the European Union (EU), and the European Commission ensure the safety of the EU citizens from privacy and data breaches. The GDPR extends the scope of the EU data protection law for all foreign organizations that process the data of EU citizens. It delivers the coordination of data protection regulations throughout the EU, thereby making it easier for non-European companies to comply with the regulations. However, if the regulations are not complied with, the company may have to pay 4% of its worldwide turnover as a penalty. Key privacy and data protection requirements of the GDPR include:

- Confidentiality of collected data to ensure data protection
- Safe transfer of data across borders
- Seeking consent of subjects for data processing
- Providing data breach notifications

The GDPR would impose consistent data security laws for all the EU members and the companies marketing goods and services to EU residents, whatever the location maybe.

### 5.11.2 HEALTH INSURANCE PORTABILITY AND ACCOUNTABILITY ACT OF 1996

The HIPAA was signed and brought into law by President Bill Clinton in 1996. The act was passed onto the US legislation to safeguard and protect medical information by maintaining data privacy. HIPAA was implemented for protecting the rights of workers regarding health insurance at the time of job change or job loss. This act also covers other healthcare issues related to the private information of patients. HIPAA governs many healthcare organizations and provides them guidance related to healthcare plans and the transmission of private information of patients within HIPAA transaction standards. The five sections included in the act are as follows:

- HIPAA Title I protects the health insurance cover of individuals and prohibits group health plans from denying coverage to any individual.
- HIPAA Title II establishes national standards for processing electronic healthcare transactions.
- HIPAA Title III includes provisions related to tax and guidelines for medical care.
- HIPAA Title IV includes provisions for individuals with pre-existing conditions and those seeking continued coverage.
- HIPAA Title V includes provisions on company-owned life insurance and treatment of those who lose their US citizenship for income tax purposes.

### 5.11.3 GOVERNANCE, RISK, AND COMPLIANCE

Governance, Risk, and Compliance (GRC) standards have been designed to bring crucial aspects of enterprises' security that synchronizes compliance with policy, audit, and incident management. Organizations are heavily dependent on a digitally enhanced infrastructure, and therefore GRC, with regard to data security, fulfills the increasing security needs. Features of GRC include robust information and data governance through a defined set of ownership and classification, secured infrastructure applications through procedure analysis, and enhanced security checks by generating a comprehensive view of hidden risks to data and digital assets.

### 5.11.4 P1913 -SOFTWARE-DEFINED QUANTUM COMMUNICATION

The P1913 standard defines the software-defined quantum communication (SDQC) protocol that enables the configuration of quantum endpoints in a communication network to dynamically create, modify, or remove quantum protocols or applications. This protocol resides at the application layer and communicates over transmission control protocol/internet protocol.

### 5.11.5 P7130 - STANDARD FOR QUANTUM TECHNOLOGIES DEFINITIONS

The P7130 standard addresses quantum technology-specific terminology and establishes definitions necessary to facilitate clarity of understanding among users to enable compatibility and interoperability.

#### 5.11.6 P7131 - STANDARD FOR QUANTUM COMPUTING PERFORMANCE METRICS AND BENCHMARKING

The P7131 standard covers quantum computing performance metrics for standardizing the performance benchmarking of quantum computing hardware and software. These metrics and performance tests include everything necessary to benchmark quantum computers (standalone and by/for comparison). They are also used to benchmark quantum computers against classical computers using a methodology that takes into account parameters such as dedicated solvers.

## 6 QUANTUM COMPUTING SOFTWARE MARKET, BY COMPONENT

---

### KEY FINDINGS

- The global quantum computing software market by component is estimated to reach USD 114 million in 2021 and USD 431 million by 2026, at a CAGR of 30.5% during the forecast period.
- Quantum computing software and services are used in optimization, simulation, and machine learning applications, thereby leading to optimum utilization costs and highly efficient operations in various industries.
- The software segment is estimated to be valued at USD 29 million in 2021 and reach USD 102 million by 2026, at a CAGR of 28.7% during the forecast period.
- The services segment is projected to register a higher CAGR of 31.1% during the forecast period.
- The growth of the services segment can be attributed to the increasing investments by start-ups in research and development related to quantum computing technology.

## 6.1 INTRODUCTION

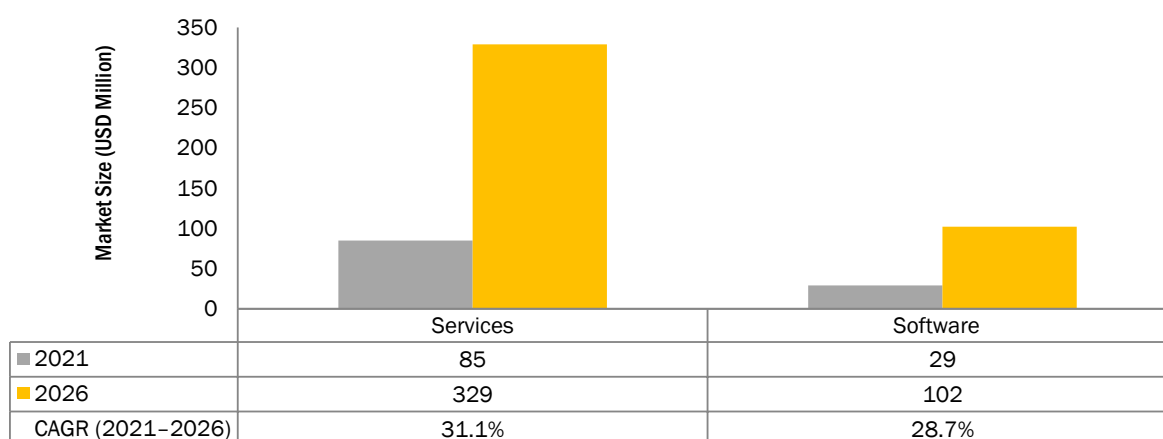
Quantum computing being an emerging technology, performs high-intensive computing tasks that are needed in different areas, such as data search from a high-volume data repository, prime factorization, number theory, cybersecurity, polynomial evaluation, interpolation, machine learning, and artificial intelligence. One of the most promising applications of quantum computers is for simulating the behavior of matter down to the molecular level. Automobile manufacturers such as Volkswagen and Daimler are using quantum computers to simulate the chemical composition of electric vehicle batteries to help find new ways to improve their performance. Pharmaceutical companies are also leveraging quantum computing technology to analyze and compare compounds, which could help develop new drugs, especially in the COVID-19 scenario.

Advancement in quantum computing technology was one of the emerging market trends in 2020. For instance, in February 2020, D-Wave launched Leap-2, which is the first quantum cloud service designed for developers and organizations to easily build and deploy real-world hybrid quantum applications with a practical impact. In March 2020, researchers at the US Army Research Laboratory developed a quantum sensor to detect communications over the entire radio frequency spectrum. In October 2020, IonQ launched its next-generation quantum computer system with 32 qubits and an expected quantum volume greater than 4 million. China developed a quantum computing system called Jiuzhang, which is reported to be 10 billion times faster than Google's Sycamore. Using Gaussian Boson Sampling (GBS), Jiuzhang had captured 76 photons and also achieved quantum supremacy. These latest advancements in quantum computing are signs of major investments in commercial and scientific reality.

Quantum computing plays a central role in enterprise computing strategies. Quantum annealers developed by Canada-based D-Wave Systems are being deployed across various industries. They are deployed on the cloud for various applications such as optimization, simulation, and machine learning. Enterprises are making efforts to add new revenue lines to their businesses by using quantum computing software and services through the cloud. This facilitates easy access of quantum computers to researchers and physicists who lack the experience of working on quantum computers owing to the requirements of investments in their deployment.

### 6.1.1 COMPONENT: COVID-19 IMPACT

- Major vendors in the quantum computing software market are adopting numerous business strategies with research institutes and universities to find a solution to fight the pandemic. For instance, in May 2021, CINECA, the Italian inter-university consortium and one of the world's leading global supercomputing centers, and D-Wave, the leader in quantum computing systems, software, and services, have collaborated to offer Italian universities, researchers, and developers access to practical quantum computing technology and resources through D-Wave's Leap (TM) quantum cloud service.
- During the COVID-19 situation, quantum computing technology has been used in vaccine development, optimization of solutions, and identifying and managing the spread of viruses.
- The growth of the quantum computing market is primarily driven by technological advancements in quantum computing technology.

**FIGURE 26 SERVICES SEGMENT TO REGISTER HIGHER CAGR DURING FORECAST PERIOD**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 11 QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017-2019 (USD MILLION)**

| Component    | 2017      | 2018      | 2019      | CAGR (2017-2019) |
|--------------|-----------|-----------|-----------|------------------|
| Software     | 8         | 11        | 15        | 33.4%            |
| Services     | 23        | 30        | 42        | 35.6%            |
| <b>Total</b> | <b>31</b> | <b>41</b> | <b>57</b> | <b>35.0%</b>     |

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 12 QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020-2026 (USD MILLION)**

| Component    | 2020      | 2021-e     | 2022-p     | 2023-p     | 2024-p     | 2025-p     | 2026-p     | CAGR (2021-2026) |
|--------------|-----------|------------|------------|------------|------------|------------|------------|------------------|
| Software     | 21        | 29         | 36         | 46         | 61         | 78         | 102        | 28.7%            |
| Services     | 59        | 85         | 107        | 140        | 188        | 248        | 329        | 31.1%            |
| <b>Total</b> | <b>80</b> | <b>114</b> | <b>142</b> | <b>186</b> | <b>249</b> | <b>326</b> | <b>431</b> | <b>30.5%</b>     |

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The services segment is expected to lead the global quantum computing software market in terms of market share and growth rate. The growth of the services segment can be attributed to the increasing investments by start-ups in the research and development related to quantum computing technology. The quantum computing software and services are used in optimization, simulation, and machine learning applications, thereby leading to optimum utilization costs and highly efficient operations of end-use industries.

## 6.2 SOFTWARE

Rapid advancements in quantum technology have helped to solve complex computational problems in a significantly lesser amount of time. On account of its high potential, experimentation and research work in quantum computing have gained momentum in recent years. The quantum computing software help to define the right way to build and connect to quantum computers. Quantum computing software provide users access to quantum computers to perform quantum computations via the cloud. Building a new quantum computer is highly expensive, thus quantum computing software help organizations to experiment with quantum computing. The software is provided by quantum computer companies for developers to test their code on real quantum computers.

Not only are tech giants dominating are the quantum software market but there are also many start-ups that provide software solutions for quantum computing. D-wave is the earliest quantum start-up and the first company to sell a commercial quantum computer. It is also one of the leading companies that work on the software development part of quantum computers.

On-premise quantum computers are systems that are deployed at the sites of clients. D-Wave Systems has launched quantum computing systems that can be deployed on premises by clients operating in digital marketing, drug and protein designing, and manufacturing, as well as the e-commerce, finance, operations and logistics, telecommunication, and transportation industries. In September 2020, D-Wave Systems announced the general availability of its next-generation quantum computing platform, which incorporates new hardware, software, and tools. Continuous investments in the development of quantum computing technology are expected to push the boundaries of this technology to address the real-world problems being faced by the customers. Moreover, companies such as IBM and Rigetti Computing have developed quantum computing software that are offered as a service to customers.

### 6.2.1 SOFTWARE: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Quantum computing is gaining popularity in most industries and research fields with its different promising applications.
- Quantum computer software with several other technologies that need immense computation power, such as machine learning and 5G, are helping to increase the computing speed, particularly in the BFSI and healthcare industries.

**TABLE 13** SOFTWARE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018         | 2019         | CAGR (2017–2019) |
|---------------|-------------|--------------|--------------|------------------|
| North America | 3.84        | 4.86         | 6.56         | 30.6%            |
| Europe        | 2.07        | 2.66         | 3.63         | 32.3%            |
| APAC          | 2.22        | 2.99         | 4.29         | 38.9%            |
| RoW           | 0.17        | 0.22         | 0.31         | 36.3%            |
| <b>Total</b>  | <b>8.31</b> | <b>10.74</b> | <b>14.79</b> | <b>33.4%</b>     |

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 14** SOFTWARE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p        | CAGR<br>(2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------------|
| North America | 8.90         | 12.31        | 14.91        | 18.92        | 24.41        | 30.89        | 39.26         | 26.1%               |
| Europe        | 4.99         | 6.95         | 8.50         | 10.90        | 14.23        | 18.24        | 23.53         | 27.6%               |
| APAC          | 6.17         | 8.94         | 11.35        | 15.13        | 20.51        | 27.32        | 36.62         | 32.6%               |
| RoW           | 0.44         | 0.64         | 0.80         | 1.06         | 1.42         | 1.88         | 2.49          | 31.4%               |
| <b>Total</b>  | <b>20.50</b> | <b>28.84</b> | <b>35.57</b> | <b>46.00</b> | <b>60.57</b> | <b>78.33</b> | <b>101.90</b> | <b>28.7%</b>        |

e: estimated; p: projected

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 6.3 SERVICES

With technological advancements, the demand for quantum computing software is expected to rise, which is projected to fuel the demand for quantum computing services. The services offered by most of the quantum computing vendors include Quantum as a Service (QaaS) and consulting services. Quantum computing services play a crucial role as clients expect seamless integration of the software, as well as support of the provided software with the enterprise systems. Services are sometimes provided by channel partners that help strengthen the geographical reach of software vendors.

Quantum computers are designed to solve complex problems outside the realm of classic computing. Classic computers use conventional bits, which can be easily managed and are stored only as a 1 or 0, while quantum computers use qubits that can be stored as a 1 or 0, or both. These qubits can be multiplied, resulting in the excellent computing potential of quantum computers. While quantum computing is still in its initial stage, it is undergoing a number of significant developments. The quantum computing service providers are catering to the requirements of industries such as healthcare, pharmaceutical, and aerospace & defense by providing quantum computers or quantum computing as a service (QCaaS). Companies that provide QCaaS include QC Ware, 1QB Information Technologies, and Cambridge Quantum Computing.

**TABLE 15** SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY TYPE, 2017–2019 (USD MILLION)

| Type                | 2017      | 2018      | 2019      | CAGR (2017–2019) |
|---------------------|-----------|-----------|-----------|------------------|
| QaaS                | 7         | 10        | 14        | 37.5%            |
| Consulting Services | 16        | 20        | 28        | 34.7%            |
| <b>Total</b>        | <b>23</b> | <b>30</b> | <b>42</b> | <b>35.6%</b>     |

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 16** SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY TYPE, 2020–2026 (USD MILLION)

| Type                | 2020      | 2021-e    | 2022-p     | 2023-p     | 2024-p     | 2025-p     | 2026-p     | CAGR<br>(2021–2026) |
|---------------------|-----------|-----------|------------|------------|------------|------------|------------|---------------------|
| QaaS                | 20        | 29        | 37         | 49         | 66         | 88         | 118        | 32.5%               |
| Consulting Services | 39        | 56        | 70         | 92         | 122        | 160        | 211        | 30.3%               |
| <b>Total</b>        | <b>59</b> | <b>85</b> | <b>107</b> | <b>140</b> | <b>188</b> | <b>248</b> | <b>329</b> | <b>31.1%</b>        |

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 17** SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 9.9         | 12.7        | 17.3        | 32.4%            |
| Europe        | 5.9         | 7.7         | 10.6        | 34.0%            |
| APAC          | 6.7         | 9.1         | 13.4        | 41.5%            |
| RoW           | 0.4         | 0.6         | 0.8         | 38.0%            |
| <b>Total</b>  | <b>22.9</b> | <b>30.1</b> | <b>42.1</b> | <b>35.6%</b>     |

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 18** SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|---------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| North America | 23.9        | 33.5        | 41.1         | 52.9         | 69.2         | 88.9         | 114.7        | 27.9%               |
| Europe        | 14.8        | 20.8        | 25.8         | 33.6         | 44.4         | 57.7         | 75.5         | 29.4%               |
| APAC          | 19.6        | 28.9        | 37.5         | 51.0         | 70.6         | 96.0         | 131.4        | 35.3%               |
| RoW           | 1.2         | 1.7         | 2.2          | 2.9          | 4.0          | 5.3          | 7.2          | 33.1%               |
| <b>Total</b>  | <b>59.4</b> | <b>85.0</b> | <b>106.6</b> | <b>140.4</b> | <b>188.2</b> | <b>248.0</b> | <b>328.8</b> | <b>31.1%</b>        |

e: estimated; p: projected

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

### 6.3.1 QUANTUM COMPUTING AS A SERVICE (QAAS)

Quantum computing technology is being deployed as a service so that researchers, physicists, and consulting companies worldwide can develop a large number of applications and explore the possibilities for providing quantum solutions in the market. For instance, IBM has several real quantum devices and simulators available for use as a service. These devices are accessed and used through Qiskit, an open-source quantum software development kit, and IBM Q Experience, which offers a virtual interface for coding a quantum computer. Canada-based D-Wave Systems launched D-Wave Leap and Leap2 cloud services that enable developers to run their open-source applications on the company's quantum computers. Consulting service providers use these quantum computers to develop algorithms and applications according to the requirements of their clients. AWS Braket aims at providing a fully managed quantum service to customers as quantum computers are not easily available in the market and require large capital for development. Thus, both AWS and Microsoft are making efforts to avail the benefits of quantum computers for their clients. They provide fully managed and easily available quantum computing solutions to clients in the form of QCaaS.

### 6.3.2 QUANTUM AS A SERVICE: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Increase in the number of companies offering QaaS owing to the growing demand for cloud-based software and services.
- Faster operation time offered by the superconducting qubits technology used in QCaaS than other technologies.

**TABLE 19** QUANTUM AS A SERVICE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019         | CAGR (2017–2019) |
|---------------|-------------|-------------|--------------|------------------|
| North America | 3.36        | 4.40        | 6.13         | 35.0%            |
| Europe        | 1.89        | 2.50        | 3.53         | 36.7%            |
| APAC          | 2.00        | 2.76        | 4.06         | 42.5%            |
| RoW           | 0.13        | 0.17        | 0.24         | 37.2%            |
| <b>Total</b>  | <b>7.38</b> | <b>9.83</b> | <b>13.96</b> | <b>37.5%</b>     |

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 20** QUANTUM AS A SERVICE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p        | CAGR<br>(2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------------|
| North America | 8.59         | 12.26        | 15.33        | 20.07        | 26.70        | 34.84        | 45.66         | 30.1%               |
| Europe        | 5.01         | 7.21         | 9.09         | 12.03        | 16.19        | 21.39        | 28.44         | 31.6%               |
| APAC          | 6.00         | 8.93         | 11.65        | 15.94        | 22.20        | 30.40        | 41.88         | 36.2%               |
| RoW           | 0.35         | 0.50         | 0.64         | 0.85         | 1.15         | 1.53         | 2.05          | 32.4%               |
| <b>Total</b>  | <b>19.95</b> | <b>28.90</b> | <b>36.70</b> | <b>48.88</b> | <b>66.24</b> | <b>88.16</b> | <b>118.02</b> | <b>32.5%</b>        |

e: estimated; p: projected

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

### 6.3.3 CONSULTING SERVICES

Consulting services are offered by consultants, scientists, and industry experts that assist clients in recognizing new business values through the implementation of quantum computing technology. They also deliver customized roadmaps to help clients in adopting quantum computing technology. IBM offers consulting services, along with quantum computers and QCaaS. 1QBit offers consulting services to customers to solve large-scale and difficult computational problems using complex algorithms and software development tools. These complex algorithms and software development tools use both classical methods and quantum computers, as well as quantum annealing hardware.

### 6.3.4 CONSULTING SERVICES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Consulting service providers offer customized roadmaps to clients to help them in the adoption of quantum computing technology. For instance, IBM Quantum Consulting brings together consultants, scientists, and industry experts to help clients realize new business value through the application of quantum computing technology and delivers customized roadmaps to help them become quantum ready.

**TABLE 21** CONSULTING SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 6.5         | 8.3         | 11.2        | 31.1%            |
| Europe        | 4.0         | 5.2         | 7.1         | 32.7%            |
| APAC          | 4.7         | 6.4         | 9.3         | 41.1%            |
| RoW           | 0.3         | 0.4         | 0.6         | 38.3%            |
| <b>Total</b>  | <b>15.5</b> | <b>20.2</b> | <b>28.2</b> | <b>34.7%</b>     |

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 22** CONSULTING SERVICES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|---------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|---------------------|
| North America | 15.3        | 21.2        | 25.8        | 32.8        | 42.5         | 54.1         | 69.1         | 26.6%               |
| Europe        | 9.8         | 13.6        | 16.7        | 21.5        | 28.2         | 36.4         | 47.1         | 28.1%               |
| APAC          | 13.6        | 20.0        | 25.9        | 35.0        | 48.4         | 65.6         | 89.5         | 34.9%               |
| RoW           | 0.8         | 1.2         | 1.6         | 2.1         | 2.8          | 3.8          | 5.1          | 33.5%               |
| <b>Total</b>  | <b>39.4</b> | <b>56.1</b> | <b>69.9</b> | <b>91.5</b> | <b>122.0</b> | <b>159.8</b> | <b>210.8</b> | <b>30.3%</b>        |

e: estimated; p: projected

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 7 QUANTUM COMPUTING SOFTWARE MARKET, BY ORGANIZATION SIZE

---

### KEY FINDINGS

- The large enterprises segment is expected to account for a larger market share during the forecast period. It is estimated to grow from USD 63 million in 2021 to USD 232 million by 2026, at a CAGR of 29.7% during the forecast period.
- The market size of the SMEs segment is expected to grow from USD 50 million in 2021 to USD 198 million by 2026, at a higher CAGR of 31.5% during the forecast period.
- The adoption of quantum computing software related to cloud-based deployment is expected to grow at a significant rate in SMEs during the forecast period due to their cost-effectiveness.
- The adoption of quantum computing software and services is expected to be the highest by large enterprises in APAC during the forecast period.

## 7.1 INTRODUCTION

Quantum computing is nearing a phase of commercialization that may change the upcoming world. Early adopters of quantum computing technologies were able to solve only certain types of problems. However, with the emergence of disruptive technologies, quantum computing software may achieve breakthroughs that will enable new business models in the quantum computing market. Large organizations are already aligning with the emerging quantum computing ecosystem to become “quantum ready.” They are exploring use cases and associated algorithms that address complex business problems. While quantum computing is still in its “theoretical phase,” there is ongoing progress in efforts to close the gap between research labs and real-world applications. Applications for quantum computing-related patents are increasing steadily every year—the highest in the US. IBM alone has filed 554 patent applications in 2020, followed by D-Wave and Google.

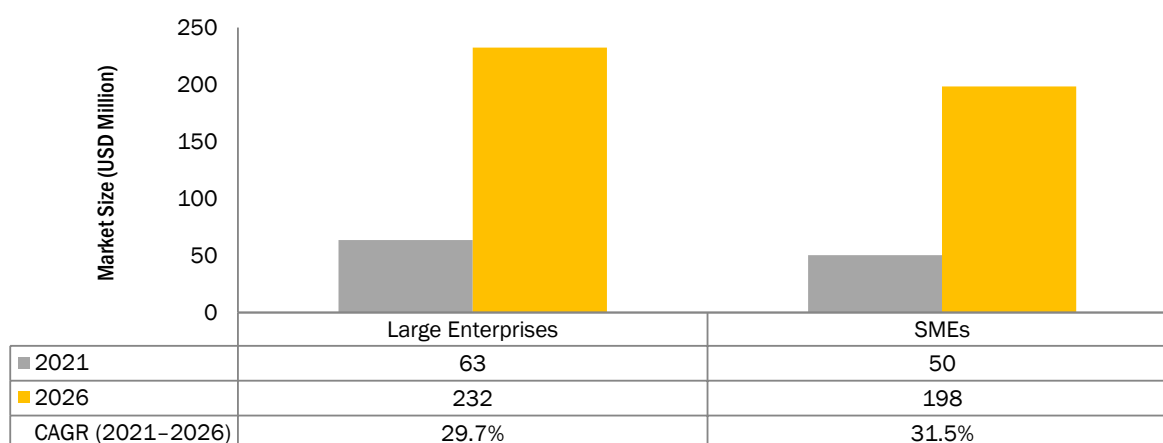
Quantum computing software and its associated services will find more opportunities in the near future in real-world applications. However, like any emerging technology, different companies are following different approaches to commercialize quantum computing. Based on organization size, the quantum computing software market has been segmented into SMEs and large enterprises. Large enterprises are the major adopters of quantum computing software and services. Although quantum computing technology is still in the nascent stage, SMEs and large enterprises are realizing and learning the importance of the technology to solve complex optimization problems. These enterprises are focusing on delivering enhanced customer experiences and gaining a leading edge in the market. In this section, we have discussed the quantum computing software market size and growth trends across organization size. The organization size is identified on the basis of the number of employees working in an organization as per global standards given below:

- SMEs: Less than 1,000 employees
- Large Enterprises: More than 1,000 employees

Both organizations are focusing on adopting quantum computing software and services as quantum computing has become a necessity for both SMEs and large enterprises. Several start-ups are partnering with researchers and technology providers to translate quantum research into capabilities suited to the business world. Technology firms developing quantum computers are partnering with businesses to identify potential use cases, develop quantum algorithms, and test solutions on actual quantum computers. The growing engagement of businesses with quantum technology will accelerate the arrival of the first commercial applications. Enterprises across industries are at the forefront in adopting quantum computing software due to digital transformation. SMEs are expected to experience a high adoption rate of quantum computing software and services, whereas the present market size of large enterprises is expected to be larger than SMEs.

### 7.1.1 ORGANIZATION SIZE: COVID-19 IMPACT

- The outbreak and widespread of the COVID-19 pandemic have increased the adoption of quantum computing software and services across various industries, especially in the drug industry. For instance, Accenture Labs worked with 1QBit to adopt their pre-developed structural molecular comparison algorithm and cloud-based API to include Biogen’s additional pharmacophore requirements. Quantum computing technology allowed Biogen’s scientists and researchers to see exactly how, where, and why molecule bonds matched, offering better insights and potential to help expedite drug discovery.
- Organizations are including quantum computing technology into their encrypted data as part of their risk mitigation strategy and plan.

**FIGURE 27** SMES SEGMENT TO EXHIBIT HIGHER CAGR DURING FORECAST PERIOD

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 23** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017-2019 (USD MILLION)

| Organization Size | 2017      | 2018      | 2019      | CAGR (2017-2019) |
|-------------------|-----------|-----------|-----------|------------------|
| SMEs              | 13        | 18        | 25        | 36.1%            |
| Large Enterprises | 18        | 23        | 32        | 34.2%            |
| <b>Total</b>      | <b>31</b> | <b>41</b> | <b>57</b> | <b>35.0%</b>     |

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 24** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020-2026 (USD MILLION)

| Organization Size | 2020      | 2021-e     | 2022-p     | 2023-p     | 2024-p     | 2025-p     | 2026-p     | CAGR (2021-2026) |
|-------------------|-----------|------------|------------|------------|------------|------------|------------|------------------|
| SMEs              | 35        | 50         | 63         | 84         | 113        | 149        | 198        | 31.5%            |
| Large Enterprises | 45        | 63         | 79         | 103        | 136        | 177        | 232        | 29.7%            |
| <b>Total</b>      | <b>80</b> | <b>114</b> | <b>142</b> | <b>186</b> | <b>249</b> | <b>326</b> | <b>431</b> | <b>30.5%</b>     |

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The large enterprises segment is expected to dominate the quantum computing software market in terms of revenue in 2021. The quantum computing technology being at the latest stage, several tech giants, namely, Google, IBM, Microsoft, D-Wave, Rigetti, and IonQ, are focusing on developing their noisy intermediate-scale quantum (NISQ) devices. Hence, the market is being dominated by large enterprises in terms of revenue. On the other hand, the quantum computing software market comprises a large number of start-ups as compared to enterprises. Thus, the SMEs segment is estimated to grow at the highest rate of 31.5% during the forecast period.

## 7.2 LARGE ENTERPRISES

Organizations with more than 1,000 employees are categorized as large enterprises. The traction of quantum computing software in large enterprises is higher as compared to SMEs. This is due to the affordability and high economies at stake that enable organizations to adopt quantum computing software. Companies leave no stone unturned to compete in the market; hence, they spend a significant amount to adopt tactical quantum computing techniques. The quantum computing software market being at the theoretical stage is witnessing significant technological transitions, such as enhancement in quantum communication technology, quantum computing academics, and the use of certain disruptive technologies. All these advancements are highly preferred in large organizations.

Several major players in the quantum computing software market provide both hardware and software. For instance, IBM designs and develops quantum computer systems and software to solve very specific, complex computational problems that are impossible to solve using classic supercomputers. The company's full-stack approach brings the best of IBM's quantum computing systems, together with extensive software tools and cloud services, to carry out and extend the technological research study.

### 7.2.1 LARGE ENTERPRISES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- The growing need for enhancing operational efficiency is driving the demand for quantum computing software in large enterprises.
- Large enterprises are developing quantum computers by adopting different business strategies to identify potential use cases, develop quantum algorithms, and test solutions on actual quantum computers. This growing engagement of businesses with quantum technology will fasten the arrival of the first commercial applications.

**TABLE 25** LARGE ENTERPRISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017         | 2018         | 2019         | CAGR (2017–2019) |
|---------------|--------------|--------------|--------------|------------------|
| North America | 7.55         | 9.59         | 12.98        | 31.1%            |
| Europe        | 4.63         | 5.95         | 8.15         | 32.7%            |
| APAC          | 5.34         | 7.22         | 10.40        | 39.6%            |
| RoW           | 0.31         | 0.42         | 0.59         | 37.0%            |
| <b>Total</b>  | <b>17.83</b> | <b>23.17</b> | <b>32.12</b> | <b>34.2%</b>     |

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 26** LARGE ENTERPRISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p        | 2024-p        | 2025-p        | 2026-p        | CAGR<br>(2021–2026) |
|---------------|--------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------------|
| North America | 17.68        | 24.55        | 29.86        | 38.06         | 49.31         | 62.69         | 80.07         | 26.7%               |
| Europe        | 11.24        | 15.72        | 19.29        | 24.83         | 32.55         | 41.90         | 54.29         | 28.1%               |
| APAC          | 15.06        | 21.95        | 28.07        | 37.65         | 51.43         | 69.02         | 93.23         | 33.5%               |
| RoW           | 0.84         | 1.21         | 1.53         | 2.03          | 2.74          | 3.64          | 4.86          | 32.2%               |
| <b>Total</b>  | <b>44.82</b> | <b>63.43</b> | <b>78.76</b> | <b>102.57</b> | <b>136.03</b> | <b>177.25</b> | <b>232.44</b> | <b>29.7%</b>        |

e: estimated; p: projected

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 7.3 SMALL AND MEDIUM ENTERPRISES

Organizations with employees ranging between 1 and 999 are stated as SMEs. Quantum computing software are essential in all organizations for competing in the market, irrespective of the size. Cost-effectiveness is a major focus area of SMEs, as they are always constrained by limited budgets, which leaves them with fewer ways to promote themselves and gain proper visibility. SMEs usually opt for cloud-based quantum computing software due to cost constraints. Besides, the cloud model minimizes IT requirements. However, SMEs are expected to have a high potential to flourish in the global quantum computing software market as it comprises a large number of start-ups.

The use of quantum computing technology is growing across different application areas, specifically for optimization, simulation, and machine learning. Initially, SMEs were facing various issues, such as tangible Return on Investment (RoI) and less capital to develop and maintain quantum computing software. Although SMEs have not yet understood the hype of quantum computing technologies, their adoption has been increasing due to their potential benefits.

### 7.3.1 SMES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- The growing need for cost-effective and comprehensive solutions is expected to drive the growth of the quantum computing software market for SMEs.
- SMEs are adopting quantum computing technologies to remain competitive in the quantum computing software market. The number of start-ups in verticals such as BFSI, healthcare, retail, e-commerce, chemical, and automotive is increasing rapidly due to favorable market conditions and government support. They are adopting different quantum computing software and services for delivering personalized customer experiences and reducing operational costs.

**TABLE 27** SMES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017         | 2018         | 2019         | CAGR (2017–2019) |
|---------------|--------------|--------------|--------------|------------------|
| North America | 6.18         | 7.95         | 10.92        | 32.9%            |
| Europe        | 3.35         | 4.37         | 6.08         | 34.7%            |
| APAC          | 3.56         | 4.92         | 7.24         | 42.6%            |
| RoW           | 0.29         | 0.39         | 0.55         | 38.1%            |
| <b>Total</b>  | <b>13.37</b> | <b>17.63</b> | <b>24.78</b> | <b>36.1%</b>     |

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 28** SMES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p        | 2025-p        | 2026-p        | CAGR (2021–2026) |
|---------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------|------------------|
| North America | 15.08        | 21.21        | 26.15        | 33.77        | 44.35         | 57.12         | 73.91         | 28.4%            |
| Europe        | 8.51         | 12.08        | 15.04        | 19.63        | 26.10         | 34.08         | 44.78         | 30.0%            |
| APAC          | 10.70        | 15.93        | 20.79        | 28.45        | 39.65         | 54.27         | 74.75         | 36.2%            |
| RoW           | 0.79         | 1.15         | 1.47         | 1.97         | 2.68          | 3.58          | 4.83          | 33.2%            |
| <b>Total</b>  | <b>35.08</b> | <b>50.37</b> | <b>63.44</b> | <b>83.83</b> | <b>112.77</b> | <b>149.05</b> | <b>198.26</b> | <b>31.5%</b>     |

e: estimated; p: projected

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 8 QUANTUM COMPUTING SOFTWARE MARKET, BY DEPLOYMENT MODE

---

### KEY FINDINGS

- The cloud deployment mode segment is expected to account for a larger market share during the forecast period. It is estimated to grow from USD 104 million in 2021 to USD 399 million by 2026, at a CAGR of 30.7% during the forecast period.
- The market size of the on-premises deployment mode is expected to grow from USD 9 million in 2021 to USD 32 million by 2026, at a higher CAGR of 28.0% during the forecast period.
- The on-premises deployment mode segment in APAC is expected to register a higher CAGR of 32.7% during the forecast period.
- Organizations are adopting on-premise quantum computers to ensure data security.
- The cloud-based approach is expected to become a significant revenue source for quantum computing software and service providers, with users paying for access to noisy intermediate-scale quantum (NISQ) systems that can solve real-world problems.

## 8.1 INTRODUCTION

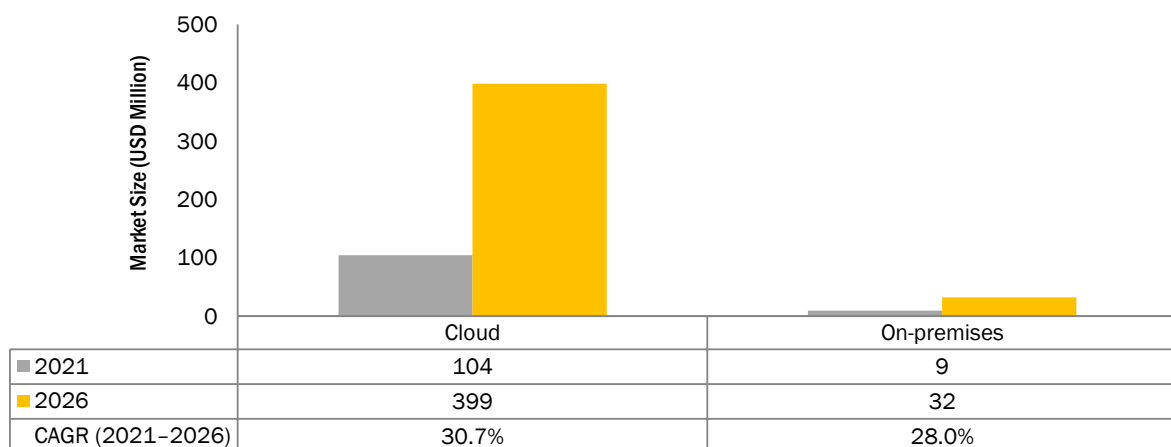
Based on deployment mode, the quantum computing software market has been segmented into on-premises and cloud-based. In the COVID-19 situation, quantum computing, along with other disruptive technologies, is being used at the service level to control the pandemic effectively. Several major players have come forward to fight against the pandemic. For instance, IBM Summit has been deployed by researchers to identify chemical compounds that can contribute to the fight against coronavirus. Researchers are using emergency computation time on machines to perform analysis at an unmatched pace. IBM Summit reportedly generates results within 1–2 days, compared to the months that the standard computing systems could have taken to yield equivalent results. Thus, quantum computers help researchers identify 77 small-molecule compounds that can undergo further investigations as laboratories and medical institutions worldwide are making efforts to develop COVID-19 vaccines.

Quantum computing technology is still in the early days of its improvement and implementation. Though quantum computing is emerging as an unrelenting concept, the commercialization of quantum computers is considerably limited. However, moving forward into the future, tech giants are trying their hands to streamline quantum advantages with the help of cloud computing services. This has opened the door to the cloud-based quantum computing software market. Developers are using quantum computing to encode problems as qubits, which compute multiple combinations of variables at once rather than exploring each possibility discretely. Quantum computing software in the cloud has the potential to enhance operations in industries in a similar way as other emerging technologies, such as AI and machine learning.

### 8.1.1 DEPLOYMENT MODE: COVID-19 IMPACT

- The outbreak and widespread of the COVID-19 pandemic have increased the adoption of quantum computing software and services across various industries.
- Rising investments by governments of different countries in the development of quantum computing technology related to cloud and on-premises modes are expected to fuel market growth. For instance, by 2030, China aims to expand its national quantum communications infrastructure, develop a general quantum computer prototype, and construct a practical quantum simulator. It is estimated that the Chinese government has invested around USD 10 billion in quantum initiatives.

**FIGURE 28** CLOUD-BASED DEPLOYMENT MODE TO REGISTER HIGHER CAGR DURING FORECAST PERIOD



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 29** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017      | 2018      | 2019      | CAGR (2017–2019) |
|-----------------|-----------|-----------|-----------|------------------|
| On-Premises     | 3         | 4         | 5         | 32.6%            |
| Cloud           | 28        | 37        | 52        | 35.3%            |
| <b>Total</b>    | <b>31</b> | <b>41</b> | <b>57</b> | <b>35.0%</b>     |

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 30** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020      | 2021-e     | 2022-p     | 2023-p     | 2024-p     | 2025-p     | 2026-p     | CAGR (2021–2026) |
|-----------------|-----------|------------|------------|------------|------------|------------|------------|------------------|
| On-Premises     | 7         | 9          | 11         | 15         | 19         | 25         | 32         | 28.0%            |
| Cloud           | 73        | 104        | 131        | 172        | 229        | 301        | 399        | 30.7%            |
| <b>Total</b>    | <b>80</b> | <b>114</b> | <b>142</b> | <b>186</b> | <b>249</b> | <b>326</b> | <b>431</b> | <b>30.5%</b>     |

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The cloud deployment mode dominates the quantum computing software market both in terms of revenue and growth rate. As highly powerful systems are being developed, the cloud-based approach is expected to become a significant revenue source for quantum computing software providers, with users paying for access to noisy intermediate-scale quantum (NISQ) systems that can solve real-world problems. The flexibility offered by quantum computing software to users is another factor favoring the adoption of cloud-based quantum computing software and services. Cloud-based quantum computing software can provide users with access to different devices and simulators from their laptops.

## 8.2 ON-PREMISES

Organizations deploy on-premise quantum computing software as they offer high data security. For organizations that work with classified data or are developing computational chemistry algorithms to discover new proprietary drugs or materials, there is the concern that their calculations or data can be leaked. The concern of these organizations is that the technicians of cloud providers can intercept their quantum programs and data while they are running on the cloud and use them for unauthorized purposes.

Another reason for the deployment of on-premise quantum computing software on quantum computers is to bypass the job queues that may occur with shared quantum computers of cloud providers. Moreover, cloud providers focus on ensuring a high utilization rate of their expensive machines, thereby resulting in an increased load on them. However, this can result in a long wait of several hours to receive the results of a run after the initial submission of a job. This can slow down any quantum software development effort, especially when programmers require to submit multiple runs to debug programs or try out different parameters to identify the one that works the best. Some quantum hardware suppliers offer reservation of time on machines so that users can solely use them during the reserved period. However, it is still not the same as having on-premise machines that can be used any time by programmers and enterprises. Though the use of quantum access through the cloud is expected to increase in the near future, there will always be some demand for on-premise machines. This demand is expected to come from government agencies that work with classified data and are ready to pay the required amount to support the deployment of on-premise secure machines.

### 8.2.1 ON-PREMISES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Increasing deployment of on-premise quantum computing software by organizations to ensure data security.

**TABLE 31** ON-PREMISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 1.10        | 1.37        | 1.83        | 29.2%            |
| Europe        | 0.80        | 1.01        | 1.36        | 30.6%            |
| APAC          | 0.80        | 1.08        | 1.54        | 38.9%            |
| RoW           | 0.07        | 0.09        | 0.13        | 35.6%            |
| <b>Total</b>  | <b>2.77</b> | <b>3.56</b> | <b>4.87</b> | <b>32.6%</b>     |

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 32** ON-PREMISES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|------------------|
| North America | 2.46        | 3.36        | 4.02         | 5.03         | 6.40         | 7.99         | 10.01        | 24.4%            |
| Europe        | 1.84        | 2.53        | 3.05         | 3.86         | 4.95         | 6.25         | 7.92         | 25.6%            |
| APAC          | 2.22        | 3.22        | 4.09         | 5.46         | 7.40         | 9.87         | 13.24        | 32.7%            |
| RoW           | 0.19        | 0.27        | 0.33         | 0.44         | 0.59         | 0.77         | 1.02         | 30.7%            |
| <b>Total</b>  | <b>6.71</b> | <b>9.38</b> | <b>11.50</b> | <b>14.78</b> | <b>19.35</b> | <b>24.88</b> | <b>32.19</b> | <b>28.0%</b>     |

e: estimated; p: projected

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 8.3 CLOUD

The enterprises that carry out data processing within their facilities have been shifting a significant portion of their IT load to specialized cloud services, such as Amazon Web Services, Microsoft Azure, and Google Cloud, over the past decade. A few enterprises are also using the concept of hybrid cloud, which uses both private computer resources belonging to enterprises and public cloud-based services.

The high costs and complexity of quantum computing software and services have led enterprises to avail the software and services of cloud providers such as Amazon, Google, and Microsoft. This leads to saving the time and efforts required to deploy technologies on premises as quantum computers are still in the development stage. These computers are generally sensitive to the local environment and require highly trained technical experts for operation.

### 8.3.1 CLOUD: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- High costs and complexity of the quantum computing system, software, and services are influencing enterprises to deploy cloud-based machines.

**TABLE 33** CLOUD: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 12.6        | 16.2        | 22.1        | 32.2%            |
| Europe        | 7.2         | 9.3         | 12.9        | 33.9%            |
| APAC          | 8.1         | 11.1        | 16.1        | 41.0%            |
| RoW           | 0.5         | 0.7         | 1.0         | 37.8%            |
| <b>Total</b>  | <b>28.4</b> | <b>37.2</b> | <b>52.0</b> | <b>35.3%</b>     |

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 34** CLOUD: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| North America | 30.3        | 42.4         | 52.0         | 66.8         | 87.3         | 111.8        | 144.0        | 27.7%            |
| Europe        | 17.9        | 25.3         | 31.3         | 40.6         | 53.7         | 69.7         | 91.1         | 29.3%            |
| APAC          | 23.5        | 34.7         | 44.8         | 60.6         | 83.7         | 113.4        | 154.7        | 34.9%            |
| RoW           | 1.4         | 2.1          | 2.7          | 3.6          | 4.8          | 6.5          | 8.7          | 32.9%            |
| <b>Total</b>  | <b>73.2</b> | <b>104.4</b> | <b>130.7</b> | <b>171.6</b> | <b>229.5</b> | <b>301.4</b> | <b>398.5</b> | <b>30.7%</b>     |

e: estimated; p: projected

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 9 QUANTUM COMPUTING SOFTWARE MARKET, BY APPLICATION

---

### KEY FINDINGS

- The application segment of the quantum computing software market is estimated to grow from USD 114 million in 2021 to USD 431 million by 2026, registering a CAGR of 30.5% from 2021 to 2026.
- The optimization segment of the quantum computing software market is expected to be valued at USD 52 million in 2021 and reach USD 181 million by 2026, exhibiting a CAGR of 28.1% from 2021 to 2026.
- Quantum computing is one of the most accurate techniques that provide a quick resolution to complex problems in every vertical.
- The machine learning segment of the quantum computing software market is estimated to be valued at USD 15 million in 2021 and reach USD 64 million by 2026, registering a CAGR of 34.3% during the forecast period.
- Companies such as D-Wave Systems and 1QB Information Technologies are developing platforms to increase the availability, usability, and accessibility of quantum computers in the simulation application in the near future.

## 9.1 INTRODUCTION

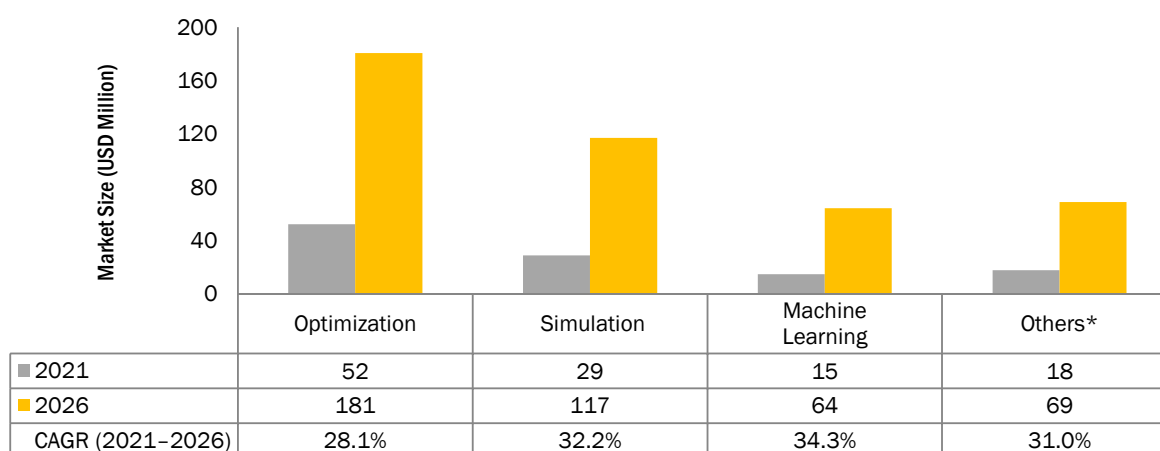
With the exponential growth in computing power, the applications of quantum computing technology are increasing rapidly in different areas. Quantum computers are ideally suited for solving complex problems, which are difficult for classical computers but are easy to factor into a quantum computer.

Quantum computing is central to the computing strategies of enterprises. Quantum annealers developed by Canada-based D-Wave Systems are being deployed across various industries, such as aerospace and automotive. Besides, these computing systems are also accessible to customers as a service for optimization, simulation, and machine learning applications. Businesses are making efforts to add new lines of revenues by offering quantum computing as a service to their clients without deploying quantum systems. Many start-ups and major technology companies, such as Microsoft, IBM, Google, D-Wave, and IonQ, are investing in QCaaS to help their clients use the power of qubits.

### 9.1.1 APPLICATION: COVID-19 IMPACT

- Quantum computing applications are exhibiting the potential of bridging the large computational capability gap.
- Quantum phase estimation is a game-changer in the simulation of systems with major interaction between electrons. The impact of the COVID-19 pandemic can be seen across all industries, economies, and societies. However, there has been an increase in the number of use cases of quantum computing software and services in vaccine development.

**FIGURE 29** MACHINE LEARNING SEGMENT TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD



\*"Others" includes quantum chemistry and quantum finance applications.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 35** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017      | 2018      | 2019      | CAGR (2017–2019) |
|------------------|-----------|-----------|-----------|------------------|
| Optimization     | 15        | 20        | 27        | 33.0%            |
| Machine Learning | 4         | 5         | 7         | 39.0%            |
| Simulation       | 8         | 10        | 14        | 36.6%            |
| Others*          | 5         | 6         | 9         | 36.2%            |
| <b>Total</b>     | <b>31</b> | <b>41</b> | <b>57</b> | <b>35.0%</b>     |

Note: "Others" includes quantum chemistry and quantum finance applications.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 36** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020      | 2021-e     | 2022-p     | 2023-p     | 2024-p     | 2025-p     | 2026-p     | CAGR (2021–2026) |
|------------------|-----------|------------|------------|------------|------------|------------|------------|------------------|
| Optimization     | 37        | 52         | 64         | 83         | 108        | 140        | 181        | 28.1%            |
| Machine Learning | 10        | 15         | 19         | 25         | 35         | 47         | 64         | 34.3%            |
| Simulation       | 20        | 29         | 37         | 49         | 66         | 87         | 117        | 32.2%            |
| Others*          | 12        | 18         | 22         | 29         | 39         | 52         | 69         | 31.0%            |
| <b>Total</b>     | <b>80</b> | <b>114</b> | <b>142</b> | <b>186</b> | <b>249</b> | <b>326</b> | <b>431</b> | <b>30.5%</b>     |

e: estimated; p: projected

Note: "Others" includes quantum chemistry and quantum finance applications.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The optimization segment is expected to lead the global quantum computing software market in terms of market share. Optimization problems exist across all industries and business functions. Some of these problems take too long to be solved optimally with traditional computers, where the usage of quantum computing technology is expected to be an optimum solution. Several optimization problems require a global minimal point solution. By using quantum annealing, the optimization problems can be solved earlier as compared to supercomputers.

## 9.2 OPTIMIZATION

Optimization is a way to resolve problems that are otherwise hard to solve owing to the availability of a large number of possible solutions. Quantum computers solve problems with the best combination of solutions under certain constraints. In the industrial sector, these problems can occur in various processes such as system designing, mission planning, and airline scheduling, as well as financial analysis and web searches. Quantum annealing is an optimization technique that is expected to surpass classical computers in certain optimization problems. Quantum annealing can be implemented on specialized quantum annealers that are far easier to build than a universal quantum computer. Currently, such computers are available, however, their supremacy over classical computers is yet to be decisively proven.

Optimization problems frequently occur in real-world settings. For instance, the logistics industry requires to identify the most efficient routes between locations. Networked Quantum Information Technologies Hub (NQIT) is expanding to incorporate optimization solutions for resolving problems faced by the practical applications of quantum computing technology. Industries such as banking & finance, aerospace &

defense, and transportation use QCaaS provided by leading players such as D-Wave Systems, IBM, and Rigetti Computing to solve various optimization problems. With the help of quantum annealer-based QCaaS of D-Wave Systems, Volkswagen has tapped the power of quantum computing to develop a traffic management system that can process transportation information in an improved manner, thereby enhancing the performance of its fleet services, such as taxis and public buses. Companies such as JP Morgan, Airbus, and Daimler are testing quantum computing systems for optimizing various business processes.

### 9.2.1 OPTIMIZATION: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- The growing adoption of quantum computing technology in the optimization application to resolve real-world problems is expected to fuel the growth of the quantum computing software market.
- Quantum annealing being one of the advanced optimization techniques is used in optimization application.

**TABLE 37** OPTIMIZATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 6.6         | 8.3         | 11.2        | 30.6%            |
| Europe        | 3.7         | 4.8         | 6.5         | 31.3%            |
| APAC          | 4.6         | 6.2         | 8.7         | 37.4%            |
| RoW           | 0.3         | 0.4         | 0.6         | 35.6%            |
| <b>Total</b>  | <b>15.3</b> | <b>19.7</b> | <b>27.0</b> | <b>33.0%</b>     |

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 38** OPTIMIZATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|------------------|
| North America | 15.2        | 21.1        | 25.5        | 32.3        | 41.6         | 52.6         | 66.5         | 25.9%            |
| Europe        | 8.8         | 12.2        | 14.8        | 18.8        | 24.4         | 31.0         | 39.6         | 26.6%            |
| APAC          | 12.4        | 17.8        | 22.4        | 29.6        | 39.7         | 52.5         | 69.7         | 31.4%            |
| RoW           | 0.9         | 1.2         | 1.5         | 2.0         | 2.7          | 3.6          | 4.7          | 30.9%            |
| <b>Total</b>  | <b>37.4</b> | <b>52.3</b> | <b>64.2</b> | <b>82.7</b> | <b>108.4</b> | <b>139.6</b> | <b>180.6</b> | <b>28.1%</b>     |

e: estimated; p: projected

Note: The market size is mentioned in single decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 9.3 MACHINE LEARNING

Machine learning relies on samples obtained from a complex probability distribution. Sampling is a technique of selecting units from a large set of samples to get accurate results from the respective dataset. This technique is used in the banking and finance, energy & power, and chemical industries. In the case of complex models, it becomes very difficult to sort a large number of variables systematically. Thus, users can opt for quantum computing technology to resolve this complexity with accuracy and speed. For instance, in the Monte Carlo method, sampling is used to estimate numerical quantities from a large dataset. A quantum algorithm can be used to increase the execution speed of quantum computations. It can enhance the performance of the Monte Carlo method and can reduce the computational time.

The potential use of quantum computing and machine learning is expected to lead to advancements in clinical research and the transformation of medical practices. Machine learning applications are capable of eliminating most diagnostic errors related to image reading. 1QB Information Technologies collaborated with D-Wave Systems to develop a reinforcement learning application using the 2000Q system of D-Wave Systems. The experimental results showed that machine learning is a promising method for harnessing the power of quantum sampling in reinforcement learning tasks. Machine learning is frequently subjected to a cost-quality or time-quality trade-off. Certain machine learning principles yield provably accurate results but are computationally not feasible. Even in modern computer clusters, users have to wait long to obtain approximate but computationally feasible solutions.

### 9.3.1 MACHINE LEARNING: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Quantum computers significantly accelerate the machine learning processes.
- Quantum computers with better algorithms could transform machine learning across industries as diverse as pharma and automotive.
- Quantum computing with the help of machine learning can help in developing various techniques to combat cybersecurity threats.

**TABLE 39** MACHINE LEARNING: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 1.64        | 2.15        | 3.00        | 35.3%            |
| Europe        | 0.80        | 1.07        | 1.52        | 38.1%            |
| APAC          | 1.07        | 1.50        | 2.24        | 44.9%            |
| RoW           | 0.08        | 0.12        | 0.17        | 41.0%            |
| <b>Total</b>  | <b>3.59</b> | <b>4.83</b> | <b>6.93</b> | <b>39.0%</b>     |

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 40** MACHINE LEARNING: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| North America | 4.22         | 6.05         | 7.59         | 9.99         | 13.36        | 17.52        | 23.10        | 30.7%               |
| Europe        | 2.19         | 3.18         | 4.07         | 5.45         | 7.43         | 9.96         | 13.43        | 33.4%               |
| APAC          | 3.37         | 5.09         | 6.76         | 9.41         | 13.34        | 18.57        | 26.04        | 38.6%               |
| RoW           | 0.25         | 0.36         | 0.47         | 0.65         | 0.90         | 1.23         | 1.70         | 36.0%               |
| <b>Total</b>  | <b>10.02</b> | <b>14.69</b> | <b>18.89</b> | <b>25.49</b> | <b>35.03</b> | <b>47.29</b> | <b>64.26</b> | <b>34.3%</b>        |

e: estimated; p: projected

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 9.4 SIMULATION

Simulation can help scientists to gain an improved understanding of molecule and sub-molecule level interactions that can lead to breakthroughs in the fields of chemistry, biology, healthcare, and nanotechnology. Molecular biology and healthcare fields use processes similar to chemical research. They can replace laboratory experiments with quantum computing simulations. Releasing a drug is a challenging process that takes a number of years and costs ~USD 2.7 billion. Thus, quantum computers can be used to carry out drug discovery, development, and testing by enabling investigations into the effects of diseases on the human body and simulating these effects on the molecular level.

Simulation is used in the healthcare, banking and finance, and defense industries. Companies such as D-Wave Systems and 1QB Information Technologies are developing platforms that enhance the availability, usability, and accessibility of quantum computers in simulation applications in the near future. Such developments are expected to drive the growth of the quantum computing market for simulation applications.

The existing method of performing a risk analysis of investment portfolios in the banking and finance industry is the Monte Carlo simulation. This classical method is applied when computational expectation values or risk measures of functions depend on random parameters. One such use case involves computing the risks associated with investment portfolios, which contain a number of assets. The values of these assets are modeled using a large number of random variables. Monte Carlo simulations are employed to compute frequently-used risk measures such as the value at risk (VaR) or the conditional value at risk (CVaR), also called expected shortfall.

### 9.4.1 SIMULATION: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Quantum simulation can help scientists better understand molecule and sub-molecule level interactions, which can lead to breakthroughs in various fields, such as chemistry, biology, healthcare, and nanotechnology.
- Simulation act as a tool to anticipate possible errors and take action. Quantum computing methods can be used to simulate complex systems.

**TABLE 41** SIMULATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018         | 2019         | CAGR (2017–2019) |
|---------------|-------------|--------------|--------------|------------------|
| North America | 3.35        | 4.33         | 5.95         | 33.3%            |
| Europe        | 2.15        | 2.82         | 3.93         | 35.0%            |
| APAC          | 1.96        | 2.72         | 4.04         | 43.6%            |
| RoW           | 0.12        | 0.16         | 0.23         | 39.3%            |
| <b>Total</b>  | <b>7.58</b> | <b>10.03</b> | <b>14.15</b> | <b>36.6%</b>     |

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 42** SIMULATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p        | CAGR (2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|------------------|
| North America | 8.25         | 11.64        | 14.40        | 18.66        | 24.58        | 31.77        | 41.27         | 28.8%            |
| Europe        | 5.51         | 7.84         | 9.79         | 12.82        | 17.10        | 22.40        | 29.52         | 30.4%            |
| APAC          | 6.01         | 9.01         | 11.85        | 16.36        | 22.98        | 31.72        | 44.08         | 37.4%            |
| RoW           | 0.34         | 0.50         | 0.64         | 0.86         | 1.19         | 1.60         | 2.18          | 34.4%            |
| <b>Total</b>  | <b>20.11</b> | <b>28.99</b> | <b>36.68</b> | <b>48.70</b> | <b>65.85</b> | <b>87.50</b> | <b>117.04</b> | <b>32.2%</b>     |

e: estimated; p: projected

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 9.5 OTHERS

The scope of the others segment of the quantum computing market includes quantum chemistry and quantum finance applications. It is believed that the number of quantum states, even in the tiniest of a molecule, is extremely vast, and as such, it is difficult for conventional computing memory systems to process them. The ability of quantum computers to focus on the existence of both 1 and 0 simultaneously is expected to provide significant power to machines to successfully map the molecules. This, in turn, is expected to be helpful in pharmaceutical research. Quantum computing technology can help to enhance the nitrogen-fixation process for manufacturing ammonia-based fertilizers, make developments in room-temperature superconductors, in removing excess carbon dioxide from the atmosphere, and develop solid-state batteries.

Furthermore, carrying out financial data analysis, developing customized reports on financial data and business information, offering financial brokerage services using deep learning, and carrying out AI-assisted fund management and research activities on stock pricing and trends are some of the other possible benefits of quantum computing. Companies such as IBM, Microsoft, and Cambridge Quantum Computing are developing quantum computing systems for use in finance applications.

**TABLE 43** OTHERS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 2.15        | 2.73        | 3.70        | 31.2%            |
| Europe        | 1.28        | 1.66        | 2.31        | 34.6%            |
| APAC          | 1.24        | 1.76        | 2.64        | 45.6%            |
| RoW           | 0.07        | 0.09        | 0.13        | 39.1%            |
| <b>Total</b>  | <b>4.74</b> | <b>6.25</b> | <b>8.78</b> | <b>36.2%</b>     |

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 44** OTHERS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| North America | 5.04         | 7.01         | 8.53         | 10.88        | 14.11        | 17.95        | 23.07        | 26.9%            |
| Europe        | 3.23         | 4.57         | 5.67         | 7.37         | 9.75         | 12.64        | 16.48        | 29.2%            |
| APAC          | 3.96         | 5.96         | 7.83         | 10.76        | 15.02        | 20.53        | 28.15        | 36.4%            |
| RoW           | 0.18         | 0.27         | 0.34         | 0.46         | 0.62         | 0.83         | 1.10         | 32.7%            |
| <b>Total</b>  | <b>12.42</b> | <b>17.81</b> | <b>22.38</b> | <b>29.48</b> | <b>39.50</b> | <b>51.95</b> | <b>68.81</b> | <b>31.0%</b>     |

e: estimated; p: projected

Note: The market size is mentioned in double decimal to demarcate the values.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 10 QUANTUM COMPUTING SOFTWARE MARKET, BY TECHNOLOGY

---

### 10.1 QUANTUM ARTIFICIAL INTELLIGENCE

Quantum artificial intelligence (QAI) is an interdisciplinary field that focuses on building quantum algorithms for improving computational tasks within artificial intelligence and its subfields, such as machine learning, deep learning, and NLP. Quantum computing and AI are both transformational technologies, and artificial intelligence is likely to require quantum computing to achieve significant progress. Although AI develops functional applications with classical computers, it is limited by the computational capabilities of classical computers. Quantum computing can provide a computation boost to artificial intelligence, enabling it to tackle more complex problems and handle Artificial General Intelligence (AGI).

Quantum computing can be used for the rapid training of machine learning models and to create optimized algorithms. An optimized and stable AI provided by quantum computing can complete years of analysis in a short time and lead to advances in technology. Neuromorphic cognitive models, adaptive machine learning, or reasoning under uncertainty are some of the fundamental challenges of today's AI. Quantum AI is one of the most likely solutions for next-generation AI. Recently, Google announced the launch of TensorFlow Quantum (TFQ), an open-source library for quantum machine learning, in collaboration with the University of Waterloo, X, and Volkswagen. The aim of TFQ is to provide the necessary tools to control and model natural or artificial quantum systems. TFQ is an example of a suite of tools that combines quantum modeling and ML techniques.

### 10.2 QUANTUM MACHINE LEARNING

Quantum ML is the integration of quantum algorithms within ML programs. The most common use of the term refers to ML algorithms for the analysis of classical data executed on a quantum computer, i.e., quantum-enhanced ML. While ML algorithms are used to compute immense quantities of data, quantum ML utilizes qubits and quantum operations or specialized quantum systems to improve computational speed and data storage in a program. This includes hybrid methods that involve both classical and quantum processing, where computationally difficult subroutines are outsourced to a quantum device.

### 10.3 QUANTUM COMPUTING AND INTERNET OF THINGS

The Internet of Things (IoT) is actively shaping both the industrial and consumer worlds, and by 2023, consumers, companies, and governments will install 40 billion IoT devices globally. Smart tech finds its way to every business and consumer domains—from retail to healthcare, from finances to logistics. With the lack of privacy standards and end-to-end security solutions being an ongoing concern for conventional IoT deployment, there is a need to secure all the four components of the IoT Model, i.e., Sensors, Networks (Communications), Analytics (Cloud), and Applications. With its capabilities, quantum computing can help address the challenges and issues that hamper the growth of IoT.

IoT is expanding as people are highly depending on their digital devices day by day. Furthermore, the WFH (Work from Home) concept resulted from COVID-19 lockdowns has accelerated the deployment of many IoT devices and shortened the learning curves of using such devices. When IoT converges with quantum computing under “Quantum IoT” or QIoT, which will push other technologies to use quantum computing and add “Quantum” or “Q” to their products and service labels, the adoption of quantum hardware and software applications is expected to increase in addition to quantum services such as QSaaS and QPaaS as parts of Quantum Cloud and QAI.

## 10.4 QUANTUM COMMUNICATION TECHNOLOGY

With 5G networks becoming operational, the amount of personal data moving globally is expected to increase substantially, as the use of 5G networks will cause a significant increase in the number of connected devices and sensors. All this implies that the requirement of data security against breaches and thefts is expected to increase. This will lead to demand for quantum-safe solutions that are expected to play a significant role in ensuring data protection.

Quantum key distribution (QKD) is a component of quantum cryptography that uses principles of quantum physics instead of mathematical algorithms to generate and distribute encryption keys used to safeguard the transmission of data over unprotected networks. QKD enables secure data transmission as signals traveling over fiber-optic cables are attenuated. Moreover, encrypted messages can be sent to a distance of 100 kilometers only, after which they need to be detected and retransmitted.

## 10.5 QUANTUM COMPUTING ACADEMIA

The top institutes, including the University of Chicago and the Massachusetts Institute of Technology (MIT), are at the forefront of quantum computing research. The University of Maryland spin-out firm IonQ is vying to lead in the age of quantum computing. The University of Maryland is a hub for the fundamental research in hardware, software, and materials that is necessary to make robust, large-scale quantum computing possible. The research work being carried out by academia is contributing to research activities related to quantum computing technology. Quantum computing empowers researchers and applied scientists across academia and enterprises to tackle challenges long viewed as intractable.

## 10.6 CYBERSECURITY TRANSFORMATION

Quantum computing poses a serious threat to the cybersecurity systems on which virtually all companies rely. Presently, most of the online account passwords, as well as secure transactions and communications, are protected through encryption algorithms such as Rivest–Shamir–Adleman (RSA) or Secure Sockets Layer (SSL)/Transport Layer Security (TLS). These systems make it easy for businesses to create data that can be shared by authorized users while also being protected from outsiders. Breaking through these encryptions requires massive computational power. It is virtually impossible for the computers being used presently to solve the mathematical problems behind well-architected encryptions quickly for practical use. Data thefts occur because of the poor implementation of cybersecurity protocols. Since quantum computers can perform multiple calculations simultaneously, they have the potential to break any classical encryption system.

## 11 QUANTUM COMPUTING SOFTWARE MARKET, BY VERTICAL

---

### KEY FINDINGS

- The healthcare & life sciences vertical is expected to grow from USD 8.63 million in 2021 to USD 37.35 million by 2026, at the highest CAGR of 34.0% during the forecast period.
- The BFSI vertical is estimated to dominate the quantum computing software market during the forecast period, reaching a market size of USD 146.95 by 2026, at a CAGR of 30.9% during the forecast period.
- In the BFSI vertical, quantum computing software and services are used to understand the impact of risks and solve the most complex problems faced during the delivery of financial services. Thus, the banking and finance segment of the quantum computing market is expected to witness significant growth during the forecast period.
- The government vertical is expected to register the second-highest CAGR of 32.2% during the forecast period.
- Several companies are expected to focus on the adoption of QCaaS in the post-COVID-19 period, which consequently is expected to contribute to the growth of the quantum computing software market.

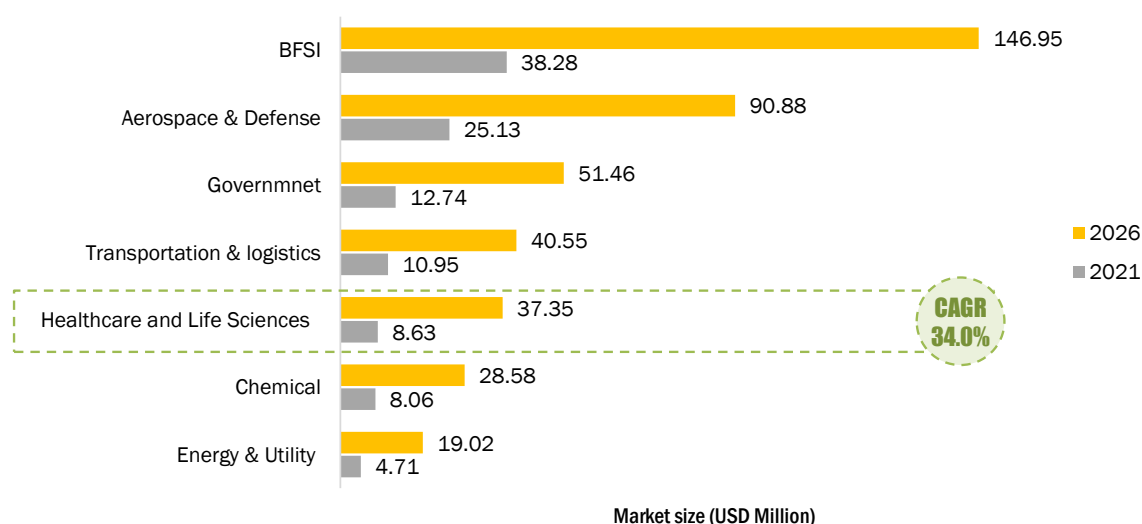
## 11.1 INTRODUCTION

In this section, we have segmented the quantum computing software market based on verticals. The verticals include aerospace & defense, BFSI, healthcare & life sciences, energy & utility, chemical, transportation & logistics, government, and education. The market size and CAGR of each vertical across regions have been discussed in this section.

### 11.1.1 VERTICAL: COVID-19 IMPACT

- Looking at the impact that COVID-19 is having on society, economy, and healthcare, we can envision future use cases for the role of quantum computing in vaccine development, optimization solutions, and identifying and managing the spread of viruses.
- D-Wave's partners and customers are providing expertise to help researchers use quantum tools to study the virus and find ways to stop it.

**FIGURE 30 HEALTHCARE & LIFE SCIENCES VERTICAL TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD**



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 45 QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017-2019 (USD MILLION)**

| Vertical                   | 2017      | 2018      | 2019      | CAGR (2017-2019) |
|----------------------------|-----------|-----------|-----------|------------------|
| Aerospace & Defense        | 7.14      | 9.26      | 12.80     | 33.9%            |
| BFSI                       | 10.44     | 13.67     | 19.08     | 35.2%            |
| Healthcare & Life Sciences | 2.11      | 2.85      | 4.09      | 39.1%            |
| Energy & Utility           | 1.22      | 1.61      | 2.29      | 37.1%            |
| Chemical                   | 2.35      | 3.02      | 4.15      | 33.1%            |
| Transportation & Logistics | 3.10      | 4.02      | 5.56      | 33.9%            |
| Government                 | 3.17      | 4.19      | 6.20      | 40.0%            |
| Education                  | 1.68      | 2.18      | 2.73      | 27.4%            |
| <b>Total</b>               | <b>31</b> | <b>41</b> | <b>57</b> | <b>35.0%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 46** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020      | 2021-e     | 2022-p     | 2023-p     | 2024-p     | 2025-p     | 2026-p     | CAGR<br>(2021–2026) |
|----------------------------|-----------|------------|------------|------------|------------|------------|------------|---------------------|
| Aerospace & Defense        | 17.81     | 25.13      | 31.12      | 40.41      | 53.45      | 69.47      | 90.88      | 29.3%               |
| BFSI                       | 26.84     | 38.28      | 47.95      | 63.02      | 84.36      | 110.97     | 146.95     | 30.9%               |
| Healthcare & Life Sciences | 5.90      | 8.63       | 11.07      | 15.06      | 20.57      | 27.57      | 37.35      | 34.0%               |
| Energy & Utility           | 3.26      | 4.71       | 5.96       | 7.92       | 10.72      | 14.24      | 19.02      | 32.2%               |
| Chemical                   | 5.75      | 8.06       | 9.93       | 12.97      | 17.05      | 22.01      | 28.58      | 28.8%               |
| Transportation & Logistics | 7.74      | 10.95      | 13.60      | 17.83      | 23.66      | 30.87      | 40.55      | 29.9%               |
| Government                 | 8.83      | 12.74      | 16.13      | 21.41      | 28.95      | 38.46      | 51.46      | 32.2%               |
| Education                  | 3.77      | 5.29       | 6.46       | 7.76       | 10.04      | 12.71      | 15.90      | 24.6%               |
| <b>Total</b>               | <b>80</b> | <b>114</b> | <b>142</b> | <b>186</b> | <b>249</b> | <b>326</b> | <b>431</b> | <b>30.5%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 11.2 AEROSPACE & DEFENSE

The defense system of any country primarily involves its army, navy, and air force. In certain countries, it also includes marine and coast guard forces. The military budgets proposed by the governments of various countries are mainly dependent on the requirements of their respective armed forces. According to the US Army Contracting Command, in 2019, the Government of the US approved an additional amount of USD 82.0 billion in its military budget that was approved in 2017. Such increased investments in the defense industry of the key countries are expected to drive the adoption of quantum computing software in this industry. Over the past 10 years, the governments of countries such as the US, France, and Canada have increased their spending on military forces to strengthen their armed forces. In 2018, the US Senate voted for the allocation of USD 716.0 billion for use by the military of the country in 2019. The 2019 military budget, approved by an 85-to-10 vote, provided a hike of USD 82.0 billion over the budget allocated in 2017 for the US armed forces. Quantum computing technology is expected to play a significant role in the aerospace & defense industry; hence, the US Armed Forces, Navy, and Air Force are working together to establish potential applications of quantum computing for secure communication, safe data transfer, and fast operations.

Quantum computing allows exponential military advantage for space applications. Satellites equipped with quantum capabilities are expected to enable carrying out offensive tactics such as surveillance and reconnaissance at high speed, with more accuracy and enhanced security than ever before. China has been steadily gaining the lead in the use of quantum computing in space applications. Moreover, in a collaborative effort among NASA, Google, and Universities Space Research Association (USRA), the Quantum Artificial Intelligence Laboratory (QuAIL) of NASA installed a quantum system offered by D-Wave Systems in 2012 to explore the potential of quantum computers to tackle optimization problems related to computations that are difficult or impossible to mitigate using traditional supercomputers. NASA focuses on machine learning, pattern recognition, mission planning and scheduling, distributed navigation and coordination, and system diagnostics and anomaly detection applications of quantum computing.

### 11.2.1 AEROSPACE & DEFENSE: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- The increased use of quantum computing software in the aerospace & defense industry to perform multiple operations simultaneously is expected to drive the growth of the quantum computing software market.
- Engineers within the aerospace industry and many other industries expect the processing power of quantum computers to be able to greatly improve computer modeling and simulation. This consequently would fuel the adoption of quantum computing software and services in the vertical.
- The use of quantum computing software by militaries is expected to enhance the security of battlefield communication. Moreover, naval forces are expected to benefit from quantum computing software as they could navigate in the ocean undetected. Research centers in Canada are currently working on quantum radar technology, which is expected to help the military detect threats over large distances.

**TABLE 47 AEROSPACE & DEFENSE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)**

| Region        | 2017        | 2018        | 2019         | CAGR (2017–2019) |
|---------------|-------------|-------------|--------------|------------------|
| North America | 2.88        | 3.63        | 4.86         | 29.8%            |
| Europe        | 1.86        | 2.40        | 3.29         | 32.9%            |
| APAC          | 2.27        | 3.07        | 4.42         | 39.6%            |
| RoW           | 0.13        | 0.17        | 0.24         | 33.3%            |
| <b>Total</b>  | <b>7.14</b> | <b>9.26</b> | <b>12.80</b> | <b>33.9%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 48 AEROSPACE & DEFENSE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)**

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| North America | 6.55         | 9.00         | 10.83        | 13.65        | 17.48        | 21.96        | 27.72        | 25.2%            |
| Europe        | 4.54         | 6.36         | 7.82         | 10.08        | 13.23        | 17.06        | 22.14        | 28.3%            |
| APAC          | 6.39         | 9.32         | 11.91        | 15.96        | 21.79        | 29.24        | 39.47        | 33.5%            |
| RoW           | 0.33         | 0.46         | 0.56         | 0.72         | 0.94         | 1.20         | 1.55         | 27.7%            |
| <b>Total</b>  | <b>17.81</b> | <b>25.13</b> | <b>31.12</b> | <b>40.41</b> | <b>53.45</b> | <b>69.47</b> | <b>90.88</b> | <b>29.3%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

### 11.3 BANKING, FINANCIAL SERVICES, AND INSURANCE

In the banking and finance industry, quantum computing is used for risk modeling applications and trading strategies; it is also used to detect market instabilities and optimize trading trajectories, portfolios, and asset pricing and hedging. As the financial sector is difficult to understand, the quantum computing approach is expected to help users understand the complexities of the banking and finance industry. Moreover, it can help traders by suggesting them solutions to overcome financial challenges. Banks and financial institutions, such as hedge fundraisers, are the key adopters of quantum computing systems and services as they help them minimize risks and maximize gains from dynamic portfolios of investments. Potential use cases of quantum computing in the banking and financial industry include portfolio analysis, fraud detection, optimization, and asset valuation, along with cybersecurity system quantum-proofing and high-frequency trading.

The Monte Carlo simulation is a quantum computing application that offers assistance in investment risk analysis and decision-making processes in the banking and finance industry. In 2018, JPMorgan Chase partnered with IBM to explore the capability of a quantum computer used in financial services for trading strategies, portfolio optimization, asset pricing, and risk analysis.

#### 11.3.1 BFSI: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Simulation offers assistance for investment risk analysis and decision-making process in the banking and finance industry, which will drive the adoption of quantum computing software.
- The early adoption of quantum computing software in the BFSI vertical is expected to fuel the growth of the market globally.

**TABLE 49** BFSI: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017         | 2018         | 2019         | CAGR (2017–2019) |
|---------------|--------------|--------------|--------------|------------------|
| North America | 4.74         | 6.08         | 8.32         | 32.6%            |
| Europe        | 2.56         | 3.28         | 4.46         | 32.0%            |
| APAC          | 3.11         | 4.27         | 6.24         | 41.6%            |
| RoW           | 0.03         | 0.04         | 0.06         | 42.5%            |
| <b>Total</b>  | <b>10.44</b> | <b>13.67</b> | <b>19.08</b> | <b>35.2%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 50** BFSI: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p        | 2026-p        | CAGR<br>(2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------------|
| North America | 11.47        | 16.09        | 19.79        | 25.50        | 33.40        | 42.93         | 55.43         | 28.1%               |
| Europe        | 6.12         | 8.51         | 10.39        | 13.29        | 17.30        | 22.13         | 28.48         | 27.3%               |
| APAC          | 9.17         | 13.55        | 17.61        | 24.02        | 33.35        | 45.50         | 62.49         | 35.8%               |
| RoW           | 0.08         | 0.12         | 0.16         | 0.22         | 0.30         | 0.40          | 0.55          | 35.2%               |
| <b>Total</b>  | <b>26.84</b> | <b>38.28</b> | <b>47.95</b> | <b>63.02</b> | <b>84.36</b> | <b>110.97</b> | <b>146.95</b> | <b>30.9%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 11.4 HEALTHCARE & LIFE SCIENCES

Presently, healthcare and pharmaceutical are the flourishing industries in the world owing to the spread of the COVID-19. Governments of various countries have increased their healthcare and pharmaceutical spending to combat the pandemic. The companies operating in this industry focus on the adoption of emerging technologies, such as quantum computing. QUANTUM computing technology enables scientists to develop medical and diagnostics tools that are ultra-precise and ultra-personalized. Moreover, quantum sensors can be used in magnetic resonance imaging (MRI) machines for ultra-precise measurements. These developments can enhance the quality of healthcare services. Various applications of quantum computing in the healthcare industry include optimization of radiotherapy treatments, creation of protein models, generation of targeted cancer drug therapies, and analysis of DNA. These industries deal with a large database containing the personal health records of a number of people, and this data is available on connected systems, which can be hacked. The data can be secured using the quantum entanglement process.

In the pharmaceutical industry, quantum computing is mainly used for the drug discovery process. Simulating a drug is a complex process; hence, a robust and agile computing technology is required to solve problems related to drug discovery in an efficient and timely manner. Quantum computers can enable researchers in the pharmaceutical industry to accelerate the process of drug discovery by understanding complex problems; for instance, they can provide insights for understanding the way cancer drugs interact with tumors and other body cells. This factor is expected to drive the adoption of quantum computing systems and services in the pharmaceutical industry. Accenture PLC, Biogen, and 1QB Information Technologies are working together to carry out advancements in the pharmaceutical industry by developing a quantum-enabled molecular comparison app to speed up the drug discovery for complex neurological conditions, such as multiple sclerosis (MS), Alzheimer's disease, Parkinson's disease, and Lou Gehrig's disease. D-Wave Systems caters to the requirements of companies dealing in the healthcare vertical by offering them quantum computing software and services for the development of personalized medicines and conducting cancer research for optimizing radiotherapy treatments, generating targeted cancer drug therapies, and creating protein models.

The quantum computing technology allows clinicians to integrate a large number of cross-functional datasets with their patient risk factor models. It enables them to analyze environmental databases to evaluate the effect of pollution on patients' health history. Another advantage of quantum computing technology is its ability to process imagery. Analysis of images, such as CT scans, requires more processing power than traditional datasets. With the increased computing power available, clinicians can easily review CT scans over time and quickly identify changes and anomalies.

### 11.4.1 HEALTHCARE & LIFE SCIENCES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Healthcare is one of the industries that holds a lot of potentials to integrate quantum.
- Quantum's ability to compute at scale will allow clinicians to incorporate a large number of cross-functional data sets into their patient risk factor models. This will drive the adoption of quantum computing software in the healthcare and life sciences vertical.

**TABLE 51** HEALTHCARE & LIFE SCIENCES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017-2019 (USD MILLION)

| Region        | 2017       | 2018       | 2019       | CAGR (2017-2019) |
|---------------|------------|------------|------------|------------------|
| North America | 0.9        | 1.3        | 1.8        | 39.0%            |
| Europe        | 0.7        | 0.9        | 1.3        | 36.1%            |
| APAC          | 0.5        | 0.7        | 1.0        | 43.6%            |
| RoW           | NA         | NA         | NA         | NA               |
| <b>Total</b>  | <b>2.1</b> | <b>2.8</b> | <b>4.1</b> | <b>39.1%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 52** HEALTHCARE & LIFE SCIENCES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020-2026 (USD MILLION)

| Region        | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021-2026) |
|---------------|------------|------------|-------------|-------------|-------------|-------------|-------------|------------------|
| North America | 2.6        | 3.8        | 4.9         | 6.6         | 8.9         | 11.8        | 15.8        | 32.7%            |
| Europe        | 1.8        | 2.5        | 3.2         | 4.2         | 5.7         | 7.5         | 10.1        | 31.7%            |
| APAC          | 1.5        | 2.3        | 3.0         | 4.1         | 5.7         | 7.9         | 11.0        | 37.3%            |
| RoW           | NA         | NA         | NA          | 0.2         | 0.3         | 0.4         | 0.5         | NA               |
| <b>Total</b>  | <b>5.9</b> | <b>8.6</b> | <b>11.1</b> | <b>15.1</b> | <b>20.6</b> | <b>27.6</b> | <b>37.4</b> | <b>34.0%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 11.5 ENERGY & UTILITIES

The energy & utilities vertical is one of the prominent users of quantum computing software and services. A highly attractive global outlook of the nuclear and renewable sector is a key factor driving the growth of the quantum computing market for the energy & power vertical. Wind, solar, and hydropower are great for the environment but can be unpredictable due to fluctuations in weather. Thus, hybrid models that optimize across renewable energy sources for reliability, availability, efficiency, and cost are popular. Given a large number of inputs, these models require complex calculations that are difficult for classical computers. Quantum gives renewable energy optimization models significantly more computational muscle, allowing the ability to solve these problems even with thousands of variables and constraints.

The rising global consumption of energy sources is expected to fuel the demand for new and cost-effective energy sources and processes. Developing new energy sources and enhancing the energy delivery process

require complex optimization. Thus, quantum computers are used in such processes as they can provide reliable results in less time. In 2019, ExxonMobil signed a partnership agreement with IBM to identify the potential use of quantum computing for next-generation energy technologies. This development is expected to help optimize operations of power grids, enable highly predictive environmental modeling, and solve quantum chemistry calculations at high speed to discover new materials for highly efficient carbon capture.

### 11.5.1 ENERGY & UTILITIES: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Increasing requirement to develop new energy sources and optimize the energy delivery process is expected to drive the growth of the quantum computing software market.
- Quantum computing software will increase the computational power of utilities, allowing them to solve business problems that were previously too cumbersome to attempt.

**TABLE 53** ENERGY & UTILITIES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 0.54        | 0.70        | 0.95        | 32.5%            |
| Europe        | 0.24        | 0.32        | 0.47        | 39.6%            |
| APAC          | 0.39        | 0.53        | 0.78        | 41.5%            |
| RoW           | 0.05        | 0.06        | 0.09        | 40.5%            |
| <b>Total</b>  | <b>1.22</b> | <b>1.61</b> | <b>2.29</b> | <b>37.1%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 54** ENERGY & UTILITIES: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|------------------|
| North America | 1.31        | 1.84        | 2.26        | 2.91        | 3.81         | 4.89         | 6.31         | 28.0%            |
| Europe        | 0.67        | 0.99        | 1.27        | 1.70        | 2.33         | 3.12         | 4.21         | 33.6%            |
| APAC          | 1.14        | 1.69        | 2.18        | 2.97        | 4.11         | 5.59         | 7.64         | 35.3%            |
| RoW           | 0.13        | 0.19        | 0.25        | 0.34        | 0.47         | 0.64         | 0.85         | 34.7%            |
| <b>Total</b>  | <b>3.26</b> | <b>4.71</b> | <b>5.96</b> | <b>7.92</b> | <b>10.72</b> | <b>14.24</b> | <b>19.02</b> | <b>32.2%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 11.6 CHEMICAL

North America and Europe are lucrative markets for chemicals. The identification and evaluation of the electronic structure of a hydrogen molecule are among the most complex tasks in the chemical industry. The computational costs increase with the increase in the number of atoms. Researchers use quantum computers for calculating the atomic structure-related dimensions of such complex electronic structures. Moreover, these computers can compare the sizes of different molecules, thereby providing a modernized approach for the molecular comparison process. This factor is expected to lead to the increased adoption of quantum computers in the chemical industry globally.

In June 2017, Dow Chemical Company signed an agreement with 1QB Information Technologies to develop quantum computing tools for chemical and material science research and applications. Both these companies are accelerating the deployment of quantum computing solutions across a range of applications relevant to the chemical industry. In 2018, Cambridge Quantum Computing (CQC) collaborated with JSR Corporation (JSR) for a project aimed at calculating the excited states of molecules considering multi-reference characteristics.

### 11.6.1 CHEMICAL: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Quantum chemistry is used across science and engineering fields to accurately predict the properties of molecules and materials. This is expected to drive the adoption of quantum computing software and services in the chemical vertical.
- The current development of new chemical molecules is an inefficient process. Quantum computing could help this process with optimized calculations to understand the development of new chemical molecules. This is expected to drive the growth of the quantum computing software market for the chemical vertical.

**TABLE 55** CHEMICAL: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 1.11        | 1.42        | 1.92        | 31.4%            |
| Europe        | 0.68        | 0.86        | 1.16        | 30.9%            |
| APAC          | 0.56        | 0.75        | 1.07        | 39.0%            |
| RoW           | NA          | NA          | NA          | NA               |
| <b>Total</b>  | <b>2.35</b> | <b>3.02</b> | <b>4.15</b> | <b>33.1%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 56** CHEMICAL: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e      | 2022-p      | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|---------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|---------------------|
| North America | 2.62        | 3.65        | 4.44        | 5.67         | 7.37         | 9.38         | 12.01        | 26.9%               |
| Europe        | 1.58        | 2.18        | 2.63        | 3.34         | 4.30         | 5.45         | 6.93         | 26.1%               |
| APAC          | 1.55        | 2.24        | 2.85        | 3.80         | 5.16         | 6.88         | 9.24         | 32.7%               |
| RoW           | NA          | NA          | NA          | 0.16         | 0.22         | 0.29         | 0.40         | NA                  |
| <b>Total</b>  | <b>5.75</b> | <b>8.06</b> | <b>9.93</b> | <b>12.97</b> | <b>17.05</b> | <b>22.01</b> | <b>28.58</b> | <b>28.8%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 11.7 TRANSPORTATION & LOGISTICS

Several algorithms can be devised to optimize traffic flow, thus reducing the overall traffic at the city level using quantum-inspired approaches. Optimized route guidance minimizes traffic time idling and reduces fuel consumption and carbon emissions. It helps in enabling traffic safety and managing traffic flow by avoiding areas meant for pedestrians and bicycle usage. It also limits traffic in areas of poor visibility. Beyond individual drivers, city planners can also use quantum computing tools to model the impact of major infrastructure projects related to public transportation and road construction on traffic flow. They can also take into account critical scenarios such as evacuation planning while designing the layout of cities.

Ford and Microsoft worked together to significantly reduce both congestion and travel time versus the routing recommendations from Bing Maps using quantum-inspired methods paired with a sophisticated mathematical representation of problems. Preliminary studies with up to 5,000 vehicles show more than a 70% decrease in traffic congestion, as well as a reduction in an average travel time of ~8% using quantum-inspired methods. The quantum-inspired algorithm-optimized routes, represented by heatmaps, show less congestion than the Bing Maps suggestions. While traffic optimization of ~400 vehicles takes 20 seconds on a quantum annealing machine, the quantum-inspired algorithms tackle a comparable problem in only 0.02 seconds on a single CPU core. This speed can be further accelerated using specialized hardware, such as FPGA available in Azure to address the real-world routing complexity. Some applications of quantum computing in the transportation & logistics industry include inventory optimization, supply chain disruptions, and logistics and risk management.

### 11.7.1 TRANSPORTATION & LOGISTICS: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Surged use of quantum-inspired approaches to optimize traffic flow is expected to drive the growth of the quantum computing software market.

**TABLE 57** TRANSPORTATION & LOGISTICS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 1.50        | 1.87        | 2.48        | 28.7%            |
| Europe        | 0.90        | 1.19        | 1.67        | 36.5%            |
| APAC          | 0.70        | 0.97        | 1.41        | 41.4%            |
| RoW           | NA          | NA          | NA          | NA               |
| <b>Total</b>  | <b>3.10</b> | <b>4.02</b> | <b>5.56</b> | <b>33.9%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 58** TRANSPORTATION & LOGISTICS: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| North America | 3.31        | 4.50         | 5.36         | 6.68         | 8.46         | 10.50        | 13.09        | 23.8%            |
| Europe        | 2.37        | 3.41         | 4.29         | 5.67         | 7.62         | 10.07        | 13.37        | 31.5%            |
| APAC          | 2.07        | 3.05         | 3.95         | 5.36         | 7.41         | 10.07        | 13.77        | 35.2%            |
| RoW           | -           | -            | -            | 0.12         | 0.17         | 0.23         | 0.31         | NA               |
| <b>Total</b>  | <b>7.74</b> | <b>10.95</b> | <b>13.60</b> | <b>17.83</b> | <b>23.66</b> | <b>30.87</b> | <b>40.55</b> | <b>29.9%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 11.8 GOVERNMENT

Quantum computing technology is expected to be increasingly deployed worldwide owing to its potential to solve a large number of computational problems in a fraction of a second. Quantum computers, which have been confined to high-tech laboratories of research universities and leading technology firms, are beginning to tackle computational problems arising in the healthcare and government verticals. As the cloud has opened up the quantum program of IBM to the new generation of users, the research partnership program of the company with select academic institutions, government laboratories, and business customers to study the behavior of matter at the molecular level, as well as gain insights related to highly practical problems has been progressing. There will be ample opportunities to use quantum computing software for solving problems around climate change, healthcare, etc. In the private sector, some Europe-based car companies have been testing quantum computing for traffic management scenarios that are expected to benefit governments of large cities in managing traffic flow in the near future.

### 11.8.1 GOVERNMENT: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- An increased number of opportunities to use quantum computing software to solve practical problems of climate change and traffic management is expected to drive the growth of the quantum computing software market.
- The key factor contributing to the growth of the quantum computing software market is the rising investments by governments of different countries to carry out research and development activities related to quantum computing technology.

**TABLE 59** GOVERNMENT: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 1.48        | 1.90        | 2.61        | 33.1%            |
| Europe        | 0.76        | 1.00        | 1.40        | 35.9%            |
| APAC          | 0.93        | 1.29        | 1.91        | 43.1%            |
| RoW           | NA          | NA          | 0.28        | NA               |
| <b>Total</b>  | <b>3.17</b> | <b>4.19</b> | <b>6.20</b> | <b>40.0%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 60** GOVERNMENT: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| North America | 3.61        | 5.09         | 6.29         | 8.13         | 10.69        | 13.78        | 17.86        | 28.5%            |
| Europe        | 1.97        | 2.83         | 3.55         | 4.67         | 6.26         | 8.23         | 10.95        | 31.1%            |
| APAC          | 2.83        | 4.23         | 5.54         | 7.60         | 10.63        | 14.59        | 20.16        | 36.7%            |
| RoW           | 0.41        | 0.59         | 0.76         | 1.01         | 1.38         | 1.85         | 2.50         | 33.4%            |
| <b>Total</b>  | <b>8.83</b> | <b>12.74</b> | <b>16.13</b> | <b>21.41</b> | <b>28.95</b> | <b>38.46</b> | <b>51.46</b> | <b>32.2%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 11.9 EDUCATION

Researchers from academia, the US national laboratories, and the quantum computing field are expected to work together in the coming years with an aim to speed up the fundamental research related to quantum information science. A large number of research activities are being focused on the development of advanced quantum technologies as the goal of quantum information science is to create fault-tolerant quantum computers that can carry out indefinite computational activities without errors.

IBM launched several initiatives to help inspire new students and begin building tomorrow's quantum computing workforce. IBM's Quantum Educators program, in particular, provides professors and students with access to IBM quantum computers as well as the latest learning resources they have developed to help them get started with programming and experimenting on quantum computers.

### 11.9.1 EDUCATION: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- The rise in the number of integrated fundamental quantum information science research activities is projected to fuel market growth.
- Introducing students to quantum computing technology earlier in their academic careers is an opportunity to capture net new interest in the subject.

**TABLE 61** EDUCATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017–2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 0.54        | 0.69        | 0.93        | 31.3%            |
| Europe        | 0.31        | 0.39        | 0.52        | 30.8%            |
| APAC          | 0.44        | 0.58        | 0.80        | 34.9%            |
| RoW           | 0.40        | 0.53        | 0.48        | 9.4%             |
| <b>Total</b>  | <b>1.68</b> | <b>2.18</b> | <b>2.73</b> | <b>27.4%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 62** EDUCATION: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|---------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|------------------|
| North America | 1.27        | 1.76        | 2.14        | 2.74        | 3.55         | 4.52         | 5.77         | 26.8%            |
| Europe        | 0.71        | 0.98        | 1.18        | 1.50        | 1.93         | 2.45         | 2.87         | 24.1%            |
| APAC          | 1.12        | 1.56        | 1.86        | 2.31        | 2.88         | 3.50         | 4.20         | 21.9%            |
| RoW           | 0.68        | 0.99        | 1.27        | 1.22        | 1.67         | 2.25         | 3.05         | 25.1%            |
| <b>Total</b>  | <b>3.77</b> | <b>5.29</b> | <b>6.46</b> | <b>7.76</b> | <b>10.04</b> | <b>12.71</b> | <b>15.90</b> | <b>24.6%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 12 QUANTUM COMPUTING SOFTWARE MARKET, BY REGION

---

### KEY FINDINGS

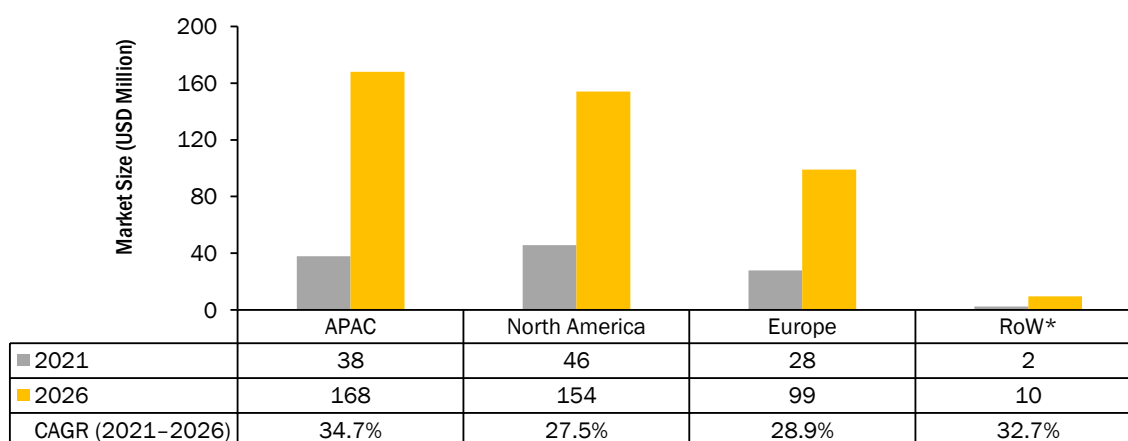
- The quantum computing software market in North America is expected be valued at USD 45.8 million in 2021 and reach USD 154.0 million by 2026, registering a CAGR of 27.5% from 2021 to 2026.
- APAC is expected to hold the largest market share by 2026.
- The significant growth of the quantum computing software market in APAC can be attributed to the increasing demand for quantum computing solutions and services in emerging economies such as China and South Korea for use in different applications in the aerospace & defense, healthcare and life sciences, and energy & utility verticals in the coming years.
- The quantum computing software market in APAC is expected be valued at USD 37.9 million in 2021 and reach USD 168.0 million by 2026, exhibiting a CAGR of 34.7% from 2021 to 2026.
- The growth of the quantum computing software market in North America can be attributed to the fact that the region is home to the leading players of the market; quantum computing solutions and services are also witnessing increased adoption in the aerospace & defense, chemical, and BFSI industries of the region.

## 12.1 INTRODUCTION

The quantum computing software market has been segmented into five regions: North America, Europe, APAC, and RoW. APAC is projected to register the highest CAGR of 34.7% from 2021 to 2026. The growth of the quantum computing software market in APAC can be attributed to the growing adoption of quantum computing solutions and services in various industries such as energy & utilities, aerospace & defense, healthcare & life sciences, chemical, and BFSI in emerging economies of the region. North America accounted for the largest share of 40.2% of the quantum computing software market in 2020. The growth of the market in this region can be attributed to the presence of key quantum computing solution and service providers.

The outbreak and widespread of the COVID-19 pandemic have led to severe disruptions in the market. Governments and private enterprises worldwide have implemented numerous measures to contain the spread of the virus by imposing travel bans and restrictions, quarantines, shelter-in-place orders, and factory and office shutdowns. These measures and other public and private responses to restrict the propagation of the coronavirus have impacted, and may further impact, the operations of companies, vendors, and suppliers. The spread of the COVID-19 is expected to adversely impact the ability of manufacturers of consumer electronics and automobiles to perform critical functions, such as carrying out research and development of new products, manufacturing of product components, and assembling of products, along with distributing and selling them.

**FIGURE 31** APAC TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD



\*RoW includes Latin America, the Middle East, and Africa.

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 63** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2017-2019 (USD MILLION)

| Region        | 2017        | 2018        | 2019        | CAGR (2017-2019) |
|---------------|-------------|-------------|-------------|------------------|
| North America | 13.7        | 17.5        | 23.9        | 31.9%            |
| Europe        | 8.0         | 10.3        | 14.2        | 33.5%            |
| APAC          | 8.9         | 12.1        | 17.6        | 40.8%            |
| *RoW          | 0.6         | 0.8         | 1.1         | 37.5%            |
| <b>Total</b>  | <b>31.2</b> | <b>40.8</b> | <b>56.9</b> | <b>35.0%</b>     |

\*RoW includes Latin America, the Middle East, and Africa.

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 64** QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY REGION, 2020–2026 (USD MILLION)

| Region        | 2020        | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|---------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| North America | 32.8        | 45.8         | 56.0         | 71.8         | 93.7         | 119.8        | 154.0        | 27.5%               |
| Europe        | 19.7        | 27.8         | 34.3         | 44.5         | 58.6         | 76.0         | 99.1         | 28.9%               |
| APAC          | 25.8        | 37.9         | 48.9         | 66.1         | 91.1         | 123.3        | 168.0        | 34.7%               |
| *RoW          | 1.6         | 2.4          | 3.0          | 4.0          | 5.4          | 7.2          | 9.7          | 32.7%               |
| <b>Total</b>  | <b>79.9</b> | <b>113.8</b> | <b>142.2</b> | <b>186.4</b> | <b>248.8</b> | <b>326.3</b> | <b>430.7</b> | <b>30.5%</b>        |

e: estimated; p: projected

\*RoW includes Latin America, the Middle East, and Africa.

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 12.2 NORTH AMERICA

The US and Canada are the leading contributors to the growth of the quantum computing software market in North America. This region is a key market for quantum computing software and services as it is home to a number of key players such as D-Wave Systems, 1QB Information Technologies, IBM, and Amazon. Additionally, the Government of the US is actively supporting advancements in quantum computing applications. In December 2018, the Government of the US passed a bill that was aimed to invest ~USD 1.2 billion in the development of quantum technology. The government of the country is expected to leverage quantum computing for solving significant and complex problems and challenges related to molecular modeling, machine learning, physics, material sciences, chemical simulations, data discovery, etc.

Moreover, Canada-based D-Wave Systems is the only supplier of commercial quantum computers in the world. The company offers its products to the most advanced organizations of the world, including Lockheed Martin Corporation, Google, National Aeronautics and Space Administration (NASA), and Los Alamos National Laboratory. Apart from D-Wave Systems, there are a number of other companies that bring innovations to this market space. For instance, the Dow Chemical Company partnered with 1QB Information Technologies Inc. to develop quantum computing tools for the chemical and material science technology spaces. These partnerships and innovations are expected to gradually drive the growth of the quantum computing software market in North America in the coming years.

### 12.2.1 NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- The establishment of the US as an early adopter of quantum computing software owing to its highly developed economy and keenness to adopt new and advanced technologies will drive the growth of the quantum computing software market in North America.
- Increased investments by the Government of Canada in the development of quantum computing to boost the growth of the Canadian market.

### 12.2.2 NORTH AMERICA: COVID-19 IMPACT

- North America is highly impacted by the COVID-19 pandemic. According to the Bureau of Economic Analysis, which is maintained by the US Department of Commerce, 11.6% of the US economic output comes from the manufacturing vertical. Due to the lockdown, various manufacturing firms have halted their operations and are using few cloud services.
- Various government authorities in North America have implemented numerous measures to contain the spread of the virus, including the imposition of travel bans and restrictions, quarantines, and shelter-in-place orders, along with limited business activities, which have hampered the growth of the regional market.

### 12.2.3 NORTH AMERICA: REGULATORY IMPLICATIONS

This section includes the laws and regulations implemented by government bodies to safeguard personal data and privacy in North America. These regulations help maintain data quality and prevent fraudulent activities by ensuring data transparency among defined entities.

#### 12.2.3.1 Health Insurance Portability and Accountability Act of 1996

The Health Insurance Portability and Accountability Act (HIPAA) was signed and brought into law by President Bill Clinton in 1996. The act was passed onto the US legislation to safeguard and protect medical information by maintaining data privacy. The five sections included in the act are as follows:

- HIPAA Title I protects the health insurance cover of individuals and prohibits group health plans from denying coverage to any individual.
- HIPAA Title II establishes national standards for processing electronic healthcare transactions.
- HIPAA Title III includes provisions related to tax and guidelines for medical care.
- HIPAA Title IV includes provisions for individuals with pre-existing conditions and those seeking continued coverage.
- HIPAA Title V includes provisions on company-owned life insurance and treatment of those who lose their US citizenship for income tax purposes.

#### 12.2.3.2 California Consumer Privacy Act

The California Consumer Privacy Act (CCPA) of 2018 gives consumers more control over the personal information that businesses collect about them. This landmark law securing new privacy rights for consumers in California is as follows:

- The right to know about personal information is helpful for a business to collect consumers' information.
- The right to delete personal information collected from consumers.
- The right to opt-out of the sale of their personal information.
- The right to non-discriminate for exercising their CCPA rights.

#### 12.2.3.3 Gramm–Leach–Bliley Act

The Gramm–Leach–Bliley Act (GLB Act) was implemented in 1999 by the US federal government. It is also known as the Financial Services Modernization Act. The major objective of this act was to maintain the security of private information pertaining to every individual that is handled by financial institutes. It includes provisions to protect the consumer's personal financial information held by banks, securities companies, and insurance companies. Under the jurisdiction of the Federal Trade Commission (FTC), these financial institutions are required to design, implement, and maintain an information security program to protect the privacy and integrity of customer data. Failure to comply can result in dire consequences for

businesses, including imprisonment of directors and officers for up to 5 years, penalties up to USD 100,000 per violation, or both.

#### **12.2.3.4 Health Information Technology for Economic and Clinical Health Act**

The Health Information Technology for Economic and Clinical Health (HITECH) Act was signed and brought into law by US President Barack Obama in 2009 as a part of the American Recovery and Reinvestment Act. The act was passed into the US legislation to promote the use of Electronic Health Records (EHR) and support technology. The Subtitle D of the HITECH Act defines privacy and security concerns related to the electronic transmission of health records.

#### **12.2.3.5 Sarbanes Oxley Act**

The Sarbanes-Oxley (SOX) Act of 2002 is a law the US Congress passed on July 30 of that year to help protect investors from fraudulent financial reporting by corporations. It came in response to financial scandals in the early 2000s involving publicly traded companies, such as Enron Corporation, Tyco International plc, and WorldCom.

- The act created strict new rules for accountants, auditors, and corporate officers and imposed more stringent recordkeeping requirements.
- The act also added new criminal penalties for violating securities laws.

#### **12.2.3.6 United States Securities and Exchange Commission**

The Securities and Exchange Commission (SEC) is an independent agency of the United States federal government, created by Section 4 of the Securities Exchange Act 1934. SEC is responsible for enforcing the federal securities laws, proposing securities rules, and regulating the securities industry. It has three missions: to protect investors, ensure orderly, fair, and efficient markets, and facilitate capital formation. SEC enforces the statutory requirement for public companies and other regulated companies to submit quarterly and annual reports as well as other periodic reports. It makes the reports available to the public through the Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system. This regulation increases the scope of services firms to analyze data proactively and address the growing scrutiny and enforcement. Regulatory analytics is an element of Enterprise Fraud and Misuse Management (EFM), which is capable of detecting fraud in real time. The Division of Economic and Risk Analysis (DERA) of SEC, created in September 2009, integrates financial economics and data analysis to detect market risks and frauds.

#### **12.2.3.7 International Organization for Standardization 27001**

The International Organization for Standardization (ISO) 27001 compliance provides a risk-based framework of policies and procedures for establishing, implementing, operating, monitoring, reviewing, maintaining, and improving the Information Security Management System (ISMS). The procedures and policies include defining a security policy and the scope of the ISMS, conducting risk assessment, managing identified risks, selecting control objectives and controls to be implemented, and preparing a statement of applicability. ISO 27001 guidelines cover various areas, such as access control, audit, incident response, and system and information integrity. The 11 major controls required as a part of ISO 27001 compliance are security policy; organization of information security; asset management; HR security; physical and environmental security; communications and operations management; access control; information systems acquisition, development, and maintenance; information security incident management; business continuity management; and compliance.

#### 12.2.3.8 California Consumer Privacy Act

The State of California passed a digital privacy law granting consumers more control over the insights into the spread of their personal information online, creating one of the most significant regulations overseeing the data collection practices of technology companies in the US. The California Consumer Privacy Act (CCPA) broadly defines the term 'personal information' as information that identifies, relates to, or is capable of being associated with, or could reasonably be linked directly or indirectly, with a particular consumer or household. To make it even more daunting, companies only had time until January 2020 to meet this regulation. The legislation came into effect in January 2020 and empowered consumers to sue companies in the occurrence of a data breach. It provides the state's attorney general with more authority to fine companies that do not adhere to these new regulations.

CCPA requires businesses to delete all personal information about a consumer upon request. Several organizations today are using or planning to use customer analytics to store most of their customer data, and to have a comprehensive view of their customers and businesses and power downstream data science, ML, and business analytics. Businesses that fail to comply with CCPA requirements by January 1, 2020, could be subject to lawsuits and civil penalties.

#### 12.2.3.9 Federal Information Security Management Act

The Federal Information Security Management Act (FISMA) of 2002 enforces stringent standards to ensure the security, confidentiality, and integrity of the US federal property and information. It sets the requirements for each federal agency to create, document, and implement programs that ensure security for the agencies' data and the systems that support the agencies' operations and assets. This extends to state governments that manage federal programs, contractors, and other non-governmental organizations working with and funded by the federal government. Examples of federal programs that states manage are Medicare, Medicaid, and unemployment insurance.

#### 12.2.3.10 Federal Information Processing Standards

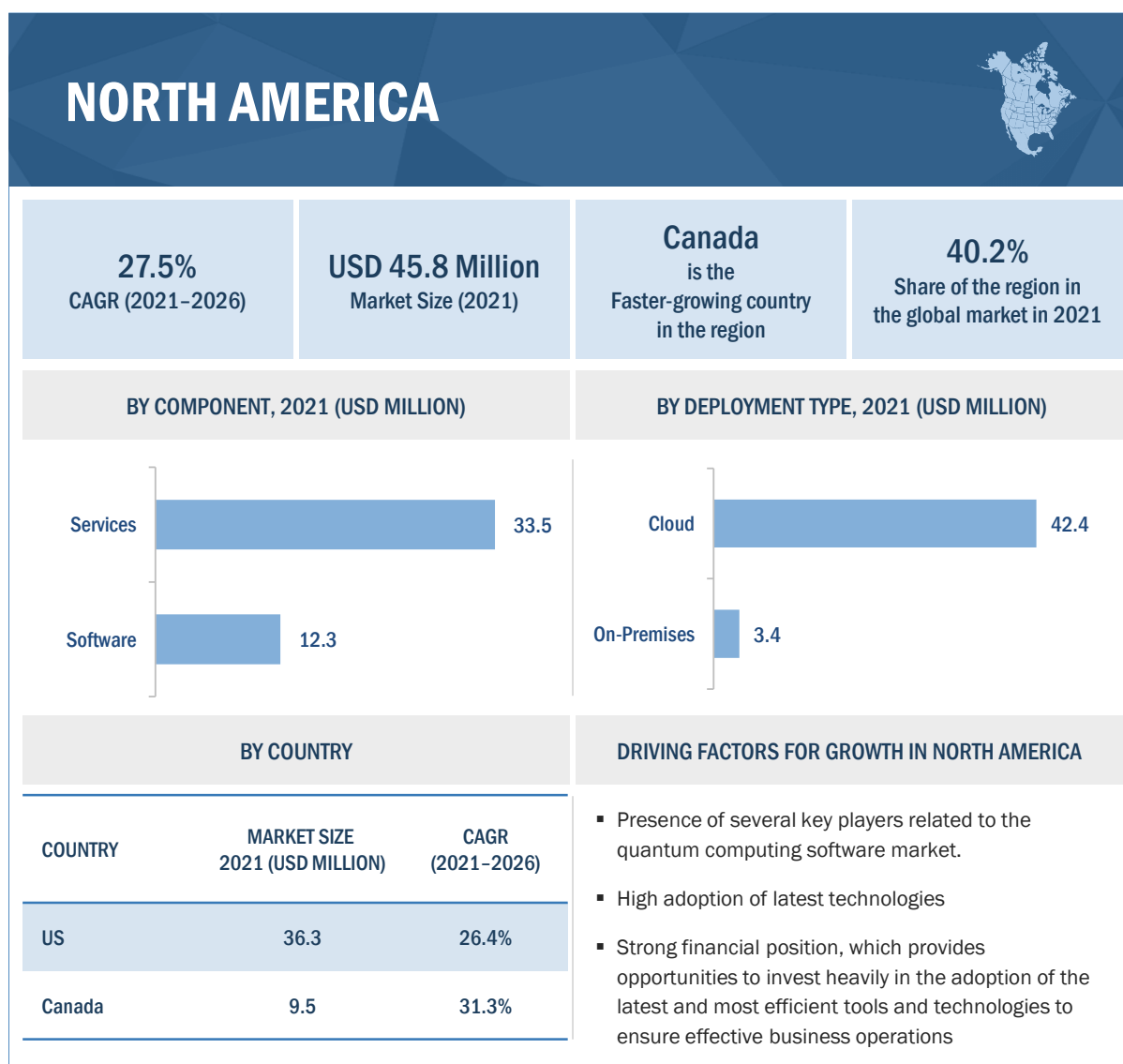
Created by the National Institute of Standards and Technology's (NIST's) Computer Security Division, the Federal Information Processing Standards (FIPS) established a data security and computer system standard that organizations must adhere to as per the Federal Information Security Management Act of 2002 (FISMA).

To become FIPS compliant, a US government agency or contractor's computer systems must meet the requirements outlined in the FIPS publications numbered 140, 180, 186, 197, 198, 199, 200, 201, and 202.

- FIPS 140 covers cryptographic module and testing requirements in both hardware and software.
- FIPS 180 specifies how organizations can be FIPS compliant when using secure hash algorithms for computing a condensed message.
- FIPS 186 is a group of algorithms for generating digital signatures.
- FIPS 197 is a standard that created the Advanced Encryption Standard, a publicly accessible cipher approved by the National Security Agency (NSA) for top secret information.
- FIPS 198 is about a mechanism for message authentication that utilizes cryptographic hash functions.
- FIPS 199 standardizes how federal agencies categorize and secure information.
- FIPS 200 is a standard that helps federal agencies with risk management through levels of information security based on risk levels.
- FIPS 201 specifies the standard for common identification for federal employees and contractors.

- FIPS 202 gives the specifications for the Secure Hash Algorithm-3 (SHA-3). It consists of four cryptographic hash functions and two extendable-output functions.

**FIGURE 32** NORTH AMERICA MARKET SNAPSHOT



Source: Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 65** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|--------------|-------------|-------------|-------------|------------------|
| Software     | 3.8         | 4.9         | 6.6         | 30.6%            |
| Services     | 9.9         | 12.7        | 17.3        | 32.4%            |
| <b>Total</b> | <b>13.7</b> | <b>17.5</b> | <b>23.9</b> | <b>31.9%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 66** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|--------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|---------------------|
| Software     | 8.9         | 12.3        | 14.9        | 18.9        | 24.4        | 30.9         | 39.3         | 26.1%               |
| Services     | 23.9        | 33.5        | 41.1        | 52.9        | 69.2        | 88.9         | 114.7        | 27.9%               |
| <b>Total</b> | <b>32.8</b> | <b>45.8</b> | <b>56.0</b> | <b>71.8</b> | <b>93.7</b> | <b>119.8</b> | <b>154.0</b> | <b>27.5%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 67** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|-----------------------|------------|-------------|-------------|------------------|
| Quantum as a Services | 3.4        | 4.4         | 6.1         | 35.0%            |
| Consulting Services   | 6.5        | 8.3         | 11.2        | 31.1%            |
| <b>Total</b>          | <b>9.9</b> | <b>12.7</b> | <b>17.3</b> | <b>32.4%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 68** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p       | CAGR<br>(2021–2026) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|---------------------|
| Quantum as a Services | 8.6         | 12.3        | 15.3        | 20.1        | 26.7        | 34.8        | 45.7         | 30.1%               |
| Consulting Services   | 15.3        | 21.2        | 25.8        | 32.8        | 42.5        | 54.1        | 69.1         | 26.6%               |
| <b>Total</b>          | <b>23.9</b> | <b>33.5</b> | <b>41.1</b> | <b>52.9</b> | <b>69.2</b> | <b>88.9</b> | <b>114.7</b> | <b>27.9%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 69** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------|-------------|-------------|-------------|------------------|
| On-premises     | 1.1         | 1.4         | 1.8         | 29.2%            |
| Cloud           | 12.6        | 16.2        | 22.1        | 32.2%            |
| <b>Total</b>    | <b>13.7</b> | <b>17.5</b> | <b>23.9</b> | <b>31.9%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 70** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|-----------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|---------------------|
| On-premises     | 2.5         | 3.4         | 4.0         | 5.0         | 6.4         | 8.0          | 10.0         | 24.4%               |
| Cloud           | 30.3        | 42.4        | 52.0        | 66.8        | 87.3        | 111.8        | 144.0        | 27.7%               |
| <b>Total</b>    | <b>32.8</b> | <b>45.8</b> | <b>56.0</b> | <b>71.8</b> | <b>93.7</b> | <b>119.8</b> | <b>154.0</b> | <b>27.5%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 71** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-------------------|-------------|-------------|-------------|------------------|
| SMEs              | 6.2         | 8.0         | 10.9        | 32.9%            |
| Large Enterprises | 7.6         | 9.6         | 13.0        | 31.1%            |
| <b>Total</b>      | <b>13.7</b> | <b>17.5</b> | <b>23.9</b> | <b>31.9%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 72** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|-------------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|---------------------|
| SMEs              | 15.1        | 21.2        | 26.2        | 33.8        | 44.3        | 57.1         | 73.9         | 28.4%               |
| Large Enterprises | 17.7        | 24.5        | 29.9        | 38.1        | 49.3        | 62.7         | 80.1         | 26.7%               |
| <b>Total</b>      | <b>32.8</b> | <b>45.8</b> | <b>56.0</b> | <b>71.8</b> | <b>93.7</b> | <b>119.8</b> | <b>154.0</b> | <b>27.5%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 73** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|------------------|-------------|-------------|-------------|------------------|
| Optimization     | 6.6         | 8.3         | 11.2        | 30.6%            |
| Machine Learning | 1.6         | 2.2         | 3.0         | 35.3%            |
| Simulation       | 3.3         | 4.3         | 6.0         | 33.3%            |
| Others*          | 2.1         | 2.7         | 3.7         | 31.2%            |
| <b>Total</b>     | <b>13.7</b> | <b>17.5</b> | <b>23.9</b> | <b>31.9%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 74** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR (2021–2026) |
|------------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|------------------|
| Optimization     | 15.2        | 21.1        | 25.5        | 32.3        | 41.6        | 52.6         | 66.5         | 25.9%            |
| Machine Learning | 4.2         | 6.0         | 7.6         | 10.0        | 13.4        | 17.5         | 23.1         | 30.7%            |
| Simulation       | 8.2         | 11.6        | 14.4        | 18.7        | 24.6        | 31.8         | 41.3         | 28.8%            |
| Others           | 5.0         | 7.0         | 8.5         | 10.9        | 14.1        | 18.0         | 23.1         | 26.9%            |
| <b>Total</b>     | <b>32.8</b> | <b>45.8</b> | <b>56.0</b> | <b>71.8</b> | <b>93.7</b> | <b>119.8</b> | <b>154.0</b> | <b>27.5%</b>     |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 75** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|----------------------------|-------------|-------------|-------------|------------------|
| Aerospace & Defense        | 2.9         | 3.6         | 4.9         | 29.8%            |
| BFSI                       | 4.7         | 6.1         | 8.3         | 32.6%            |
| Healthcare & Life Sciences | 0.9         | 1.3         | 1.8         | 39.0%            |
| Energy & Utility           | 0.5         | 0.7         | 1.0         | 32.5%            |
| Chemical                   | 1.1         | 1.4         | 1.9         | 31.4%            |
| Transportation & Logistics | 1.5         | 1.9         | 2.5         | 28.7%            |
| Government                 | 1.5         | 1.9         | 2.6         | 33.1%            |
| Education                  | 0.5         | 0.7         | 0.9         | 31.3%            |
| <b>Total</b>               | <b>13.7</b> | <b>17.5</b> | <b>23.9</b> | <b>31.9%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 76** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|----------------------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|---------------------|
| Aerospace & Defense        | 6.6         | 9.0         | 10.8        | 13.6        | 17.5        | 22.0         | 27.7         | 25.2%               |
| BFSI                       | 11.5        | 16.1        | 19.8        | 25.5        | 33.4        | 42.9         | 55.4         | 28.1%               |
| Healthcare & Life Sciences | 2.6         | 3.8         | 4.9         | 6.6         | 8.9         | 11.8         | 15.8         | 32.7%               |
| Energy & Utility           | 1.3         | 1.8         | 2.3         | 2.9         | 3.8         | 4.9          | 6.3          | 28.0%               |
| Chemical                   | 2.6         | 3.6         | 4.4         | 5.7         | 7.4         | 9.4          | 12.0         | 26.9%               |
| Transportation & Logistics | 3.3         | 4.5         | 5.4         | 6.7         | 8.5         | 10.5         | 13.1         | 23.8%               |
| Government                 | 3.6         | 5.1         | 6.3         | 8.1         | 10.7        | 13.8         | 17.9         | 28.5%               |
| Education                  | 1.3         | 1.8         | 2.1         | 2.7         | 3.5         | 4.5          | 5.8          | 26.8%               |
| <b>Total</b>               | <b>32.8</b> | <b>45.8</b> | <b>56.0</b> | <b>71.8</b> | <b>93.7</b> | <b>119.8</b> | <b>154.0</b> | <b>27.5%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

**TABLE 77** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)

| Country      | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|--------------|-------------|-------------|-------------|------------------|
| US           | 11.3        | 14.3        | 19.3        | 30.9%            |
| Canada       | 2.5         | 3.3         | 4.6         | 36.7%            |
| <b>Total</b> | <b>13.7</b> | <b>17.5</b> | <b>23.9</b> | <b>31.9%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 78** NORTH AMERICA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)

| Country      | 2020-e      | 2021-p      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|--------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|---------------------|
| US           | 26.2        | 36.3        | 44.1        | 56.0        | 72.4        | 91.8         | 117.0        | 26.4%               |
| Canada       | 6.6         | 9.5         | 11.9        | 15.8        | 21.2        | 28.0         | 37.0         | 31.3%               |
| <b>Total</b> | <b>32.8</b> | <b>45.8</b> | <b>56.0</b> | <b>71.8</b> | <b>93.7</b> | <b>119.8</b> | <b>154.0</b> | <b>27.5%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

## 12.2.4 US

The US is the most developed economy in North America. It is expected to contribute the maximum share to the growing quantum computing software market. The US is an early adopter of quantum computing software, as it is a highly developed economy and technologically advanced country. The US has been one of the potential markets for quantum computing software as it is home to a number of leading players such as IBM, Microsoft, and Google. IBM has invested at Raleigh, North Carolina (US), as it is one of the five global IBM Q hubs of the company. This hub was established to advance quantum computing and involves a collaboration between IBM and top Fortune 500 companies, national research laboratories, and leading universities. At the hub, students work alongside scientists, researchers, and engineers to carry out advancements in the state of the future computers. The current 20 qubit IBM Q system is expected to be followed by a next-generation 50 qubit prototype.

Moreover, the funds and investments from the Government of the US are also driving the growth of the quantum computing software market in the country. For instance, the US Department of Energy (DoE) has released Funding Opportunity Announcements (FOA) for both government laboratories and non-government organizations that carry out research activities aimed at developing new algorithms and software for quantum computers.

**TABLE 79** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|--------------|-------------|-------------|-------------|------------------|
| Software     | 2.9         | 3.7         | 4.9         | 29.5%            |
| Services     | 8.3         | 10.6        | 14.4        | 31.3%            |
| <b>Total</b> | <b>11.3</b> | <b>14.3</b> | <b>19.3</b> | <b>30.9%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 80** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p       | CAGR (2021–2026) |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|------------------|
| Software     | 6.6         | 9.0         | 10.8        | 13.6        | 17.4        | 21.8        | 27.5         | 24.9%            |
| Services     | 19.6        | 27.3        | 33.2        | 42.4        | 55.0        | 70.0        | 89.5         | 26.8%            |
| <b>Total</b> | <b>26.2</b> | <b>36.3</b> | <b>44.1</b> | <b>56.0</b> | <b>72.4</b> | <b>91.8</b> | <b>117.0</b> | <b>26.4%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 81** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|-----------------------|------------|-------------|-------------|------------------|
| Quantum as a Services | 2.7        | 3.5         | 4.8         | 33.9%            |
| Consulting Services   | 5.7        | 7.1         | 9.6         | 30.1%            |
| <b>Total</b>          | <b>8.3</b> | <b>10.6</b> | <b>14.4</b> | <b>31.3%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 82** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Quantum as a Services | 6.6         | 9.4         | 11.7        | 15.1        | 20.0        | 25.8        | 33.6        | 29.0%            |
| Consulting Services   | 13.0        | 17.9        | 21.6        | 27.2        | 35.0        | 44.2        | 56.0        | 25.7%            |
| <b>Total</b>          | <b>19.6</b> | <b>27.3</b> | <b>33.2</b> | <b>42.4</b> | <b>55.0</b> | <b>70.0</b> | <b>89.5</b> | <b>26.8%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 83** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------|-------------|-------------|-------------|------------------|
| On-premises     | 1.1         | 1.4         | 1.8         | 27.2%            |
| Cloud           | 10.1        | 12.9        | 17.5        | 31.3%            |
| <b>Total</b>    | <b>11.3</b> | <b>14.3</b> | <b>19.3</b> | <b>30.9%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 84** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p       | CAGR (2021–2026) |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|------------------|
| On-premises     | 2.4         | 3.2         | 3.8         | 4.7         | 5.8         | 7.1         | 8.8          | 22.1%            |
| Cloud           | 23.8        | 33.1        | 40.3        | 51.4        | 66.6        | 84.7        | 108.2        | 26.8%            |
| <b>Total</b>    | <b>26.2</b> | <b>36.3</b> | <b>44.1</b> | <b>56.0</b> | <b>72.4</b> | <b>91.8</b> | <b>117.0</b> | <b>26.4%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 85** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-------------------|-------------|-------------|-------------|------------------|
| SMEs              | 4.7         | 6.1         | 8.3         | 32.1%            |
| Large Enterprises | 6.5         | 8.2         | 11.0        | 30.0%            |
| <b>Total</b>      | <b>11.3</b> | <b>14.3</b> | <b>19.3</b> | <b>30.9%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 86** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p       | CAGR (2021–2026) |
|-------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|------------------|
| SMEs              | 11.3        | 15.8        | 19.4        | 24.9        | 32.4        | 41.4        | 53.2         | 27.5%            |
| Large Enterprises | 14.9        | 20.5        | 24.7        | 31.2        | 40.0        | 50.4        | 63.8         | 25.5%            |
| <b>Total</b>      | <b>26.2</b> | <b>36.3</b> | <b>44.1</b> | <b>56.0</b> | <b>72.4</b> | <b>91.8</b> | <b>117.0</b> | <b>26.4%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 87** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|------------------|-------------|-------------|-------------|------------------|
| Optimization     | 5.2         | 6.5         | 8.7         | 29.5%            |
| Machine Learning | 1.6         | 2.0         | 2.8         | 33.7%            |
| Simulation       | 3.0         | 3.8         | 5.2         | 32.1%            |
| Others*          | 1.5         | 1.9         | 2.6         | 30.1%            |
| <b>Total</b>     | <b>11.3</b> | <b>14.3</b> | <b>19.3</b> | <b>30.9%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 88** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p       | CAGR<br>(2021–2026) |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|---------------------|
| Optimization     | 11.7        | 16.0        | 19.2        | 24.1        | 30.7        | 38.5        | 48.2         | 24.7%               |
| Machine Learning | 3.9         | 5.5         | 6.9         | 8.9         | 11.8        | 15.3        | 19.9         | 29.2%               |
| Simulation       | 7.1         | 10.0        | 12.2        | 15.7        | 20.5        | 26.2        | 33.7         | 27.6%               |
| Others*          | 3.5         | 4.8         | 5.8         | 7.4         | 9.5         | 11.9        | 15.2         | 25.7%               |
| <b>Total</b>     | <b>26.2</b> | <b>36.3</b> | <b>44.1</b> | <b>56.0</b> | <b>72.4</b> | <b>91.8</b> | <b>117.0</b> | <b>26.4%</b>        |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 89** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|----------------------------|-------------|-------------|-------------|------------------|
| Aerospace & Defense        | 2.4         | 2.9         | 3.9         | 28.8%            |
| BFSI                       | 3.7         | 4.7         | 6.3         | 31.5%            |
| Healthcare & Life Sciences | 1.0         | 1.3         | 1.9         | 36.3%            |
| Energy & Utility           | 0.4         | 0.6         | 0.8         | 31.4%            |
| Chemical                   | 0.7         | 0.9         | 1.2         | 30.1%            |
| Transportation & Logistics | 1.5         | 1.8         | 2.4         | 28.1%            |
| Government                 | 1.2         | 1.5         | 2.1         | 32.0%            |
| Education                  | 0.4         | 0.6         | 0.7         | 30.2%            |
| <b>Total</b>               | <b>11.3</b> | <b>14.3</b> | <b>19.3</b> | <b>30.9%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 90** US: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p       | CAGR<br>(2020–2026) |
|----------------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|---------------------|
| Aerospace & Defense        | 5.2         | 7.1         | 8.5         | 10.6        | 13.5        | 16.8        | 21.1         | 24.2%               |
| BFSI                       | 8.6         | 12.0        | 14.7        | 18.8        | 24.4        | 31.1        | 39.8         | 27.0%               |
| Healthcare & Life Sciences | 2.6         | 3.8         | 4.7         | 6.2         | 8.3         | 10.9        | 14.3         | 30.6%               |
| Energy & Utility           | 1.0         | 1.5         | 1.8         | 2.3         | 2.9         | 3.8         | 4.8          | 26.9%               |
| Chemical                   | 1.6         | 2.2         | 2.6         | 3.3         | 4.2         | 5.4         | 6.8          | 25.9%               |
| Transportation & Logistics | 3.2         | 4.3         | 5.1         | 6.3         | 8.0         | 9.9         | 12.3         | 23.4%               |
| Government                 | 2.9         | 4.0         | 4.9         | 6.3         | 8.3         | 10.6        | 13.6         | 27.4%               |
| Education                  | 1.0         | 1.4         | 1.7         | 2.1         | 2.7         | 3.4         | 4.3          | 25.4%               |
| <b>Total</b>               | <b>26.2</b> | <b>36.3</b> | <b>44.1</b> | <b>56.0</b> | <b>72.4</b> | <b>91.8</b> | <b>117.0</b> | <b>26.4%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

## 12.2.5 CANADA

Canada plays a vital role in the growth of the quantum computing software market in North America. D-Wave Systems, a market leader in quantum computing, and the Institute for Quantum Computing (IQC), a scientific research institute at the University of Waterloo, are based in Canada. Moreover, the space sector of the country is focusing on the development of new technologies to improve the lives of its citizens. The provincial government of Ontario and the government of the country have both invested ~USD 1.0 billion in the last decade to support such developments in Canada. In March 2019, D-Wave Systems launched the first real-time online quantum computing application called Leap. Leap2 is the latest addition to the quantum cloud services of the company that virtualizes quantum computing with a classical computer and broadband connection.

Canada is actively focusing on adopting and promoting the latest technologies in the country. According to Coldwell Banker Richard Ellis (CBRE), Toronto, followed by Ottawa, is the fastest-growing Canadian technology market, with over 241,000 tech workers, representing an increase of 51.5% over the past 5 years. Hence, Canada is projected to provide various opportunities for the growth of the quantum computing software market during the forecast period.

**TABLE 91** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|--------------|------------|------------|------------|------------------|
| Software     | 0.7        | 0.9        | 1.2        | 35.9%            |
| Services     | 1.8        | 2.4        | 3.4        | 37.0%            |
| <b>Total</b> | <b>2.5</b> | <b>3.3</b> | <b>4.6</b> | <b>36.7%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 92** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|--------------|------------|------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Software     | 1.7        | 2.5        | 3.1         | 4.1         | 5.5         | 7.2         | 9.4         | 30.5%               |
| Services     | 4.8        | 7.0        | 8.8         | 11.7        | 15.7        | 20.8        | 27.5        | 31.6%               |
| <b>Total</b> | <b>6.6</b> | <b>9.5</b> | <b>11.9</b> | <b>15.8</b> | <b>21.2</b> | <b>28.0</b> | <b>37.0</b> | <b>31.3%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 93** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|-----------------------|------------|------------|------------|------------------|
| Quantum as a Services | 0.6        | 0.8        | 1.2        | 38.3%            |
| Consulting Services   | 1.2        | 1.6        | 2.2        | 36.4%            |
| <b>Total</b>          | <b>1.8</b> | <b>2.4</b> | <b>3.4</b> | <b>37.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 94** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020       | 2021-e     | 2022-p     | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|-----------------------|------------|------------|------------|-------------|-------------|-------------|-------------|---------------------|
| Quantum as a Services | 1.7        | 2.5        | 3.1        | 4.2         | 5.7         | 7.6         | 10.1        | 32.7%               |
| Consulting Services   | 3.1        | 4.5        | 5.7        | 7.5         | 10.1        | 13.2        | 17.4        | 31.0%               |
| <b>Total</b>          | <b>4.8</b> | <b>7.0</b> | <b>8.8</b> | <b>11.7</b> | <b>15.7</b> | <b>20.8</b> | <b>27.5</b> | <b>31.6%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 95** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|-----------------|------------|------------|------------|------------------|
| On-premises     | 0.3        | 0.4        | 0.5        | 34.1%            |
| Cloud           | 2.2        | 2.9        | 4.1        | 37.1%            |
| <b>Total</b>    | <b>2.5</b> | <b>3.3</b> | <b>4.6</b> | <b>36.7%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 96** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|-----------------|------------|------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| On-premises     | 0.7        | 1.0        | 1.3         | 1.7         | 2.2         | 2.8         | 3.7         | 28.4%               |
| Cloud           | 5.8        | 8.4        | 10.7        | 14.1        | 19.0        | 25.1        | 33.3        | 31.7%               |
| <b>Total</b>    | <b>6.6</b> | <b>9.5</b> | <b>11.9</b> | <b>15.8</b> | <b>21.2</b> | <b>28.0</b> | <b>37.0</b> | <b>31.3%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 97** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|-------------------|------------|------------|------------|------------------|
| SMEs              | 1.2        | 1.6        | 2.2        | 37.4%            |
| Large Enterprises | 1.3        | 1.7        | 2.4        | 36.1%            |
| <b>Total</b>      | <b>2.5</b> | <b>3.3</b> | <b>4.6</b> | <b>36.7%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 98** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|-------------------|------------|------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| SMEs              | 3.2        | 4.6        | 5.9         | 7.8         | 10.5        | 13.9        | 18.5        | 31.9%               |
| Large Enterprises | 3.4        | 4.8        | 6.1         | 8.0         | 10.7        | 14.0        | 18.5        | 30.8%               |
| <b>Total</b>      | <b>6.6</b> | <b>9.5</b> | <b>11.9</b> | <b>15.8</b> | <b>21.2</b> | <b>28.0</b> | <b>37.0</b> | <b>31.3%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 99** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|------------------|------------|------------|------------|------------------|
| Optimization     | 1.1        | 1.4        | 2.0        | 35.2%            |
| Machine Learning | 0.4        | 0.5        | 0.7        | 39.6%            |
| Simulation       | 0.7        | 0.9        | 1.3        | 38.0%            |
| Others *         | 0.3        | 0.4        | 0.6        | 35.9%            |
| <b>Total</b>     | <b>2.5</b> | <b>3.3</b> | <b>4.6</b> | <b>36.7%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 100** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|------------------|------------|------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Optimization     | 2.8        | 3.9        | 4.9         | 6.4         | 8.5         | 11.0        | 14.3        | 29.5%            |
| Machine Learning | 1.0        | 1.5        | 1.9         | 2.6         | 3.6         | 4.8         | 6.5         | 34.2%            |
| Simulation       | 1.9        | 2.8        | 3.5         | 4.7         | 6.4         | 8.5         | 11.4        | 32.5%            |
| Others *         | 0.9        | 1.3        | 1.6         | 2.1         | 2.8         | 3.6         | 4.8         | 30.7%            |
| <b>Total</b>     | <b>6.6</b> | <b>9.5</b> | <b>11.9</b> | <b>15.8</b> | <b>21.2</b> | <b>28.0</b> | <b>37.0</b> | <b>31.3%</b>     |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 101** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|----------------------------|-------------|-------------|-------------|------------------|
| Aerospace & Defense        | 0.44        | 0.58        | 0.80        | 34.2%            |
| BFSI                       | 0.75        | 1.00        | 1.42        | 37.5%            |
| Healthcare & Life Sciences | 0.22        | 0.30        | 0.44        | 42.4%            |
| Energy & Utility           | 0.10        | 0.13        | 0.18        | 37.3%            |
| Chemical                   | 0.15        | 0.20        | 0.28        | 36.2%            |
| Transportation & Logistics | 0.32        | 0.41        | 0.57        | 33.9%            |
| Government                 | 0.27        | 0.36        | 0.51        | 37.9%            |
| Education                  | 0.22        | 0.29        | 0.41        | 36.3%            |
| <b>Total</b>               | <b>2.47</b> | <b>3.27</b> | <b>4.62</b> | <b>36.7%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 102** CANADA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2020–2026) |
|----------------------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Aerospace & Defense        | 1.11        | 1.58        | 1.95         | 2.53         | 3.33         | 4.29         | 5.54         | 28.6%               |
| BFSI                       | 2.03        | 2.95        | 3.74         | 4.98         | 6.72         | 8.90         | 11.83        | 32.0%               |
| Healthcare & Life Sciences | 0.66        | 0.98        | 1.28         | 1.76         | 2.44         | 3.32         | 4.53         | 35.8%               |
| Energy & Utility           | 0.26        | 0.38        | 0.48         | 0.64         | 0.86         | 1.14         | 1.52         | 31.9%               |
| Chemical                   | 0.40        | 0.57        | 0.72         | 0.94         | 1.26         | 1.66         | 2.18         | 30.8%               |
| Transportation & Logistics | 0.79        | 1.12        | 1.38         | 1.79         | 2.34         | 3.01         | 3.88         | 28.2%               |
| Government                 | 0.72        | 1.05        | 1.34         | 1.79         | 2.42         | 3.22         | 4.29         | 32.4%               |
| Education                  | 0.58        | 0.83        | 1.05         | 1.38         | 1.85         | 2.43         | 3.20         | 30.9%               |
| <b>Total</b>               | <b>6.55</b> | <b>9.46</b> | <b>11.95</b> | <b>15.80</b> | <b>21.23</b> | <b>27.95</b> | <b>36.95</b> | <b>31.3%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

## 12.3 EUROPE

The quantum computing software market in Europe has been studied for Germany, the UK, France, and the Rest of Europe. The scope of Rest of Europe includes Switzerland, Russia, Italy, Austria, and Sweden. The region accounted for the share of 24.4% of the quantum computing software market in 2021. The growth of the market in this region can be attributed to the increasing investments by governments of countries of the region to increase the number of quantum computing projects in Europe, including the UK Quantum Technologies Programme and the NQuTech initiative of the Netherlands. Moreover, in October 2018, the European Commission announced to invest USD 1.1 billion in a quantum computing flagship project to fuel the adoption of quantum technologies across the region. This flagship project focuses on the use of quantum technologies in communication, computing, sensing, and simulation applications. The commission has been supporting the development of quantum technologies for the past 20 years in the region, using a cumulative investment reaching ~USD 608 million. The European Commission also aided the Future and Emerging Technologies (FET)–Work Program for 2016–2017—to contribute to the development and the adoption of quantum computing technology in the region in the coming years. Additionally, in October 2018, the European Union announced the first USD 150 million of its quantum flagship initiative to be split between 20 continent-wide consortia over the next 3 years to develop new types of quantum sensors, communication products, and computers.

European organizations are increasingly focusing on investing in innovations and are among the early adopters of leading technologies. Like North America, enterprises in this region are witnessing progressive growth in the implementation of advanced technologies due to high technical expertise and substantial IT budgets. With the emergence of Industry 4.0, the region is witnessing strong growth in the manufacturing industry vertical. Germany is considered one of the major manufacturing hubs for automotive, machinery and equipment, and electrical equipment.

### 12.3.1 EUROPE: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Developments in industries to bring significant changes in adoption rate of quantum computing technology in the UK are expected to fuel regional market growth.
- Initiatives undertaken by the Government of Germany to spread awareness about the importance of quantum computing will drive the growth of the quantum computing software market in the region.

### 12.3.2 EUROPE: COVID-19 IMPACT

- The specifics of the extensive long-term impact of the coronavirus are currently unknown, but the pandemic has the potential to materially impact supply chain operations across a wide variety of companies in Europe.
- Prominent European countries, such as the UK, Germany, France, Italy, and Spain, have been adversely impacted by this global pandemic. The virus has infected millions, and the total number of fatalities is surging with each passing day. This severe impact on human life and health has resulted in a major economic slowdown; it is expected that the entire European region will witness a significant recession in the post-COVID-19 period.
- Enterprise software expenditure is projected to witness limited growth. IT consulting and professional services would be negatively impacted due to travel restrictions and project delays. Each month of lockdown is projected to limit economic activities by 25–45% as compared to the normal levels.

### 12.3.3 EUROPE: REGULATORY IMPLICATIONS

This section includes the laws and regulations implemented by government bodies to safeguard personal data and privacy in Europe. These regulations help maintain data quality and prevent fraudulent activities by ensuring data transparency among defined entities.

#### 12.3.3.1 General Data Protection Regulation

The General Data Protection Regulation (GDPR) (Regulation [EU] 2016/679) is a regulation by which the European Parliament, the Council of the European Union (EU), and the EC ensure the safety of EU citizens from privacy and data breaches. GDPR extends the scope of the EU data protection law for all foreign organizations that process the data of EU citizens. It delivers the coordination of the data protection regulations throughout the EU, making it easier for non-European companies to comply with the regulations. However, if the regulations are not introduced, companies may have to pay 4% of their worldwide turnover as a penalty. Key privacy and data protection requirements of GDPR include:

- Confidentiality of collected data to ensure data protection
- Safe transfer of data across borders
- Seeking consent of subjects for data processing
- Providing data breach notifications

GDPR would impose consistent data security law for all the EU members and the companies marketing goods and services to EU residents, irrespective of the location.

### 12.3.3.2 European Committee for Standardization

The European Committee for Standardization (CEN) is a government body that works with the ISO and was established to develop IoT standards in Europe. It imposes rules and regulations for integrating the data collected from sensors with the existing RFID-driven barcode systems.

### 12.3.3.3 European Technical Standards Institute

The European Technical Standards Institute (ETSI) is an independent standardization organization in the telecommunications industry in Europe. It imposes globally applicable ICT standards on equipment makers and network operators based on radio, fixed, mobile, broadcast, and internet technologies. The new technical committee has developed standards for Machine-to-Machine (M2M) communications, wireless sensor networks, and RFID. TR 1030290, TR 103 315, and TR 102 857 are a few standards imposed by the ETSI team to help regulate information and technology services and RFID deployment.

**TABLE 103** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|--------------|------------|-------------|-------------|------------------|
| Software     | 2.1        | 2.7         | 3.6         | 32.3%            |
| Services     | 5.9        | 7.7         | 10.6        | 34.0%            |
| <b>Total</b> | <b>8.0</b> | <b>10.3</b> | <b>14.2</b> | <b>33.5%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 104** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Software     | 5.0         | 7.0         | 8.5         | 10.9        | 14.2        | 18.2        | 23.5        | 27.6%            |
| Services     | 14.8        | 20.8        | 25.8        | 33.6        | 44.4        | 57.7        | 75.5        | 29.4%            |
| <b>Total</b> | <b>19.7</b> | <b>27.8</b> | <b>34.3</b> | <b>44.5</b> | <b>58.6</b> | <b>76.0</b> | <b>99.1</b> | <b>28.9%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 105** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017       | 2018       | 2019        | CAGR (2017–2019) |
|-----------------------|------------|------------|-------------|------------------|
| Quantum as a Services | 1.9        | 2.5        | 3.5         | 36.7%            |
| Consulting Services   | 4.0        | 5.2        | 7.1         | 32.7%            |
| <b>Total</b>          | <b>5.9</b> | <b>7.7</b> | <b>10.6</b> | <b>34.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 106** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Quantum as a Services | 5.0         | 7.2         | 9.1         | 12.0        | 16.2        | 21.4        | 28.4        | 31.6%               |
| Consulting Services   | 9.8         | 13.6        | 16.7        | 21.5        | 28.2        | 36.4        | 47.1        | 28.1%               |
| <b>Total</b>          | <b>14.8</b> | <b>20.8</b> | <b>25.8</b> | <b>33.6</b> | <b>44.4</b> | <b>57.7</b> | <b>75.5</b> | <b>29.4%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 107** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|-----------------|------------|-------------|-------------|------------------|
| On-premises     | 0.8        | 1.0         | 1.4         | 30.6%            |
| Cloud           | 7.2        | 9.3         | 12.9        | 33.9%            |
| <b>Total</b>    | <b>8.0</b> | <b>10.3</b> | <b>14.2</b> | <b>33.5%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 108** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| On-premises     | 1.8         | 2.5         | 3.1         | 3.9         | 5.0         | 6.2         | 7.9         | 25.6%               |
| Cloud           | 17.9        | 25.3        | 31.3        | 40.6        | 53.7        | 69.7        | 91.1        | 29.3%               |
| <b>Total</b>    | <b>19.7</b> | <b>27.8</b> | <b>34.3</b> | <b>44.5</b> | <b>58.6</b> | <b>76.0</b> | <b>99.1</b> | <b>28.9%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 109** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|-------------------|------------|-------------|-------------|------------------|
| SMEs              | 3.4        | 4.4         | 6.1         | 34.7%            |
| Large Enterprises | 4.6        | 5.9         | 8.1         | 32.7%            |
| <b>Total</b>      | <b>8.0</b> | <b>10.3</b> | <b>14.2</b> | <b>33.5%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 110** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size        | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|--------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| <b>SMEs</b>              | 8.5         | 12.1        | 15.0        | 19.6        | 26.1        | 34.1        | 44.8        | 30.0%               |
| <b>Large Enterprises</b> | 11.2        | 15.7        | 19.3        | 24.8        | 32.5        | 41.9        | 54.3        | 28.1%               |
| <b>Total</b>             | <b>19.7</b> | <b>27.8</b> | <b>34.3</b> | <b>44.5</b> | <b>58.6</b> | <b>76.0</b> | <b>99.1</b> | <b>28.9%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 111** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application             | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|-------------------------|------------|-------------|-------------|------------------|
| <b>Optimization</b>     | 3.7        | 4.8         | 6.5         | 31.3%            |
| <b>Machine Learning</b> | 0.8        | 1.1         | 1.5         | 38.1%            |
| <b>Simulation</b>       | 2.2        | 2.8         | 3.9         | 35.0%            |
| <b>Others *</b>         | 1.3        | 1.7         | 2.3         | 34.6%            |
| <b>Total</b>            | <b>8.0</b> | <b>10.3</b> | <b>14.2</b> | <b>33.5%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 112** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application             | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|-------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| <b>Optimization</b>     | 8.8         | 12.2        | 14.8        | 18.8        | 24.4        | 31.0        | 39.6        | 26.6%               |
| <b>Machine Learning</b> | 2.2         | 3.2         | 4.1         | 5.4         | 7.4         | 10.0        | 13.4        | 33.4%               |
| <b>Simulation</b>       | 5.5         | 7.8         | 9.8         | 12.8        | 17.1        | 22.4        | 29.5        | 30.4%               |
| <b>Others *</b>         | 3.2         | 4.6         | 5.7         | 7.4         | 9.7         | 12.6        | 16.5        | 29.2%               |
| <b>Total</b>            | <b>19.7</b> | <b>27.8</b> | <b>34.3</b> | <b>44.5</b> | <b>58.6</b> | <b>76.0</b> | <b>99.1</b> | <b>28.9%</b>        |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 113** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018         | 2019         | CAGR (2017–2019) |
|----------------------------|-------------|--------------|--------------|------------------|
| Aerospace & Defense        | 1.86        | 2.40         | 3.29         | 32.9%            |
| BFSI                       | 2.56        | 3.28         | 4.46         | 32.0%            |
| Healthcare & Life Sciences | 0.68        | 0.89         | 1.26         | 36.1%            |
| Energy & Utility           | 0.24        | 0.32         | 0.47         | 39.6%            |
| Chemical                   | 0.68        | 0.86         | 1.16         | 30.9%            |
| Transportation & Logistics | 0.90        | 1.19         | 1.67         | 36.5%            |
| Government                 | 0.76        | 1.00         | 1.40         | 35.9%            |
| Education                  | 0.31        | 0.39         | 0.52         | 30.8%            |
| <b>Total</b>               | <b>7.98</b> | <b>10.32</b> | <b>14.23</b> | <b>33.5%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 114** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2020–2026) |
|----------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| Aerospace & Defense        | 4.54         | 6.36         | 7.82         | 10.08        | 13.23        | 17.06        | 22.14        | 28.3%            |
| BFSI                       | 6.12         | 8.51         | 10.39        | 13.29        | 17.30        | 22.13        | 28.48        | 27.3%            |
| Healthcare & Life Sciences | 1.78         | 2.55         | 3.20         | 4.22         | 5.67         | 7.47         | 10.10        | 31.7%            |
| Energy & Utility           | 0.67         | 0.99         | 1.27         | 1.70         | 2.33         | 3.12         | 4.21         | 33.6%            |
| Chemical                   | 1.58         | 2.18         | 2.63         | 3.34         | 4.30         | 5.45         | 6.93         | 26.1%            |
| Transportation & Logistics | 2.37         | 3.41         | 4.29         | 5.67         | 7.62         | 10.07        | 13.37        | 31.5%            |
| Government                 | 1.97         | 2.83         | 3.55         | 4.67         | 6.26         | 8.23         | 10.95        | 31.1%            |
| Education                  | 0.71         | 0.98         | 1.18         | 1.50         | 1.93         | 2.45         | 2.87         | 24.1%            |
| <b>Total</b>               | <b>19.75</b> | <b>27.80</b> | <b>34.33</b> | <b>44.47</b> | <b>58.65</b> | <b>75.98</b> | <b>99.06</b> | <b>28.9%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

**TABLE 115** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)

| Country         | 2017       | 2018        | 2019        | CAGR<br>(2015–2019) |
|-----------------|------------|-------------|-------------|---------------------|
| UK              | 2.4        | 3.1         | 4.3         | 32.8%               |
| Germany         | 3.0        | 3.8         | 5.1         | 30.0%               |
| France          | 1.5        | 2.0         | 2.8         | 35.9%               |
| *Rest of Europe | 1.0        | 1.4         | 2.0         | 42.2%               |
| <b>Total</b>    | <b>8.0</b> | <b>10.3</b> | <b>14.2</b> | <b>33.5%</b>        |

\*Rest of Europe includes Netherlands, Italy, Belgium, Finland, and Russia.

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 116** EUROPE: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)

| Country         | 2020-e      | 2021-p      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| UK              | 5.9         | 8.3         | 10.2        | 13.1        | 17.2        | 22.2        | 28.7        | 28.2%               |
| Germany         | 6.9         | 9.5         | 11.3        | 14.2        | 18.2        | 22.8        | 28.7        | 24.9%               |
| France          | 3.9         | 5.7         | 7.1         | 9.3         | 12.5        | 16.5        | 21.8        | 31.0%               |
| *Rest of Europe | 3.0         | 4.4         | 5.7         | 7.8         | 10.8        | 14.6        | 19.8        | 35.1%               |
| <b>Total</b>    | <b>19.7</b> | <b>27.8</b> | <b>34.3</b> | <b>44.5</b> | <b>58.6</b> | <b>76.0</b> | <b>99.1</b> | <b>28.9%</b>        |

e: estimated; p: projected

\*Rest of Europe includes Netherlands, Italy, Belgium, Finland, and Russia.

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

### 12.3.4 GERMANY

Germany, being a developed economy, is expected to offer major opportunities to quantum computing software and service providers. The country has the highest Gross Domestic Product (GDP) among European countries and has witnessed digitalization in all areas of business. Germany is a technology hub, hence, presents major opportunities for quantum computing software solution providers. The adoption of new technologies in Germany is increasing owing to its highly developed research and development landscape. Moreover, the Federal Ministry of Education and Research (BMBF) of the country is spreading awareness about the importance of quantum computing technology for the growth of its economy. The Government of Germany entered into an agreement with the Government of Canada to establish quantum computing and machine learning networks in July 2018. The collaboration marks the involvement of centers of Helmholtz Association, DESY, and Forschungszentrum Jülich from Germany, and TRIUMF, TRIUMF Innovations, D-Wave Systems, and 1Qbit from Canada. This international collaboration is expected to encourage developments in large-scale computing, big data, machine learning, and quantum computing.

Germany has a highly developed research and development landscape for emerging technologies such as quantum computing and Industry 4.0. Moreover, in March 2020, Fraunhofer-Gesellschaft and IBM announced the signing of the agreement aimed at advancing quantum computing in Germany. The

collaboration will provide companies and research institutions access to IBM quantum computers in Germany and the US under the umbrella of a nationwide Fraunhofer competence network.

Along with this, Bundesministerium für Bildung und Forschung (BMBF), the Federal Ministry of Education and Research of Germany, selected a few pilot quantum computing projects to showcase the important developments taking place in the field of quantum computing technology. One of the first projects that were selected started in May 2017, and it was named as optclock—an optical single-ion clock. In October 2019, Germany-based Volkswagen collaborated with the customers of Canada-based D-Wave Systems to use the D-Wave computer to understand the traffic pattern of 10,000 taxis in Beijing. These projects and developments in the field of quantum computing are expected to increase the share of Germany in the quantum computing software market in the next 5 years.

**TABLE 117** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|--------------|------------|------------|------------|------------------|
| Software     | 0.8        | 0.9        | 1.2        | 28.1%            |
| Services     | 2.3        | 2.9        | 3.9        | 30.6%            |
| <b>Total</b> | <b>3.0</b> | <b>3.8</b> | <b>5.1</b> | <b>30.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 118** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|--------------|------------|------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Software     | 1.7        | 2.2        | 2.6         | 3.3         | 4.1         | 5.1         | 6.3         | 22.9%            |
| Services     | 5.3        | 7.2        | 8.7         | 11.0        | 14.1        | 17.7        | 22.5        | 25.5%            |
| <b>Total</b> | <b>6.9</b> | <b>9.5</b> | <b>11.3</b> | <b>14.2</b> | <b>18.2</b> | <b>22.8</b> | <b>28.7</b> | <b>24.9%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 119** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------------|-------------|-------------|-------------|------------------|
| Quantum as a Services | 0.68        | 0.87        | 1.18        | 31.3%            |
| Consulting Services   | 1.59        | 2.01        | 2.70        | 30.2%            |
| <b>Total</b>          | <b>2.27</b> | <b>2.88</b> | <b>3.88</b> | <b>30.6%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 120** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020        | 2021-e      | 2022-p      | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|-----------------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|---------------------|
| Quantum as a Services | 1.60        | 2.22        | 2.68        | 3.41         | 4.40         | 5.57         | 7.09         | 26.2%               |
| Consulting Services   | 3.65        | 5.01        | 6.01        | 7.57         | 9.69         | 12.18        | 15.38        | 25.2%               |
| <b>Total</b>          | <b>5.26</b> | <b>7.22</b> | <b>8.70</b> | <b>10.97</b> | <b>14.09</b> | <b>17.74</b> | <b>22.47</b> | <b>25.5%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 121** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------|-------------|-------------|-------------|------------------|
| On-premises     | 0.33        | 0.41        | 0.54        | 26.9%            |
| Cloud           | 2.70        | 3.41        | 4.58        | 30.3%            |
| <b>Total</b>    | <b>3.03</b> | <b>3.82</b> | <b>5.12</b> | <b>30.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 122** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|-----------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| On-premises     | 0.71        | 0.94        | 1.10         | 1.35         | 1.67         | 2.04         | 2.49         | 21.5%               |
| Cloud           | 6.20        | 8.51        | 10.23        | 12.88        | 16.51        | 20.76        | 26.23        | 25.3%               |
| <b>Total</b>    | <b>6.91</b> | <b>9.45</b> | <b>11.33</b> | <b>14.23</b> | <b>18.18</b> | <b>22.79</b> | <b>28.73</b> | <b>24.9%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 123** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-------------------|-------------|-------------|-------------|------------------|
| SMEs              | 1.06        | 1.35        | 1.82        | 31.0%            |
| Large Enterprises | 1.97        | 2.47        | 3.30        | 29.4%            |
| <b>Total</b>      | <b>3.03</b> | <b>3.82</b> | <b>5.12</b> | <b>30.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 124** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|-------------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| SMEs              | 2.48        | 3.41        | 4.12         | 5.22         | 6.72         | 8.48         | 10.77        | 25.8%               |
| Large Enterprises | 4.43        | 6.04        | 7.21         | 9.01         | 11.46        | 14.31        | 17.96        | 24.4%               |
| <b>Total</b>      | <b>6.91</b> | <b>9.45</b> | <b>11.33</b> | <b>14.23</b> | <b>18.18</b> | <b>22.79</b> | <b>28.73</b> | <b>24.9%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 125** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|------------------|-------------|-------------|-------------|------------------|
| Optimization     | 1.42        | 1.76        | 2.33        | 27.8%            |
| Machine Learning | 0.36        | 0.47        | 0.65        | 33.7%            |
| Simulation       | 0.73        | 0.93        | 1.26        | 31.6%            |
| Others *         | 0.52        | 0.65        | 0.88        | 30.9%            |
| <b>Total</b>     | <b>3.03</b> | <b>3.82</b> | <b>5.12</b> | <b>30.0%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 126** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2021–2026) |
|------------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Optimization     | 3.09        | 4.15        | 4.88         | 6.02         | 7.55         | 9.30         | 11.49        | 22.6%               |
| Machine Learning | 0.90        | 1.27        | 1.57         | 2.03         | 2.67         | 3.44         | 4.47         | 28.6%               |
| Simulation       | 1.72        | 2.38        | 2.89         | 3.68         | 4.75         | 6.03         | 7.70         | 26.4%               |
| Others *         | 1.20        | 1.65        | 1.99         | 2.50         | 3.20         | 4.02         | 5.07         | 25.2%               |
| <b>Total</b>     | <b>6.91</b> | <b>9.45</b> | <b>11.33</b> | <b>14.23</b> | <b>18.18</b> | <b>22.79</b> | <b>28.73</b> | <b>24.9%</b>        |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 127** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|----------------------------|-------------|-------------|-------------|------------------|
| Aerospace & Defense        | 0.59        | 0.73        | 0.98        | 29.2%            |
| BFSI                       | 1.03        | 1.29        | 1.71        | 28.5%            |
| Healthcare & Life Sciences | 0.26        | 0.33        | 0.45        | 32.5%            |
| Energy & Utility           | 0.09        | 0.12        | 0.17        | 35.9%            |
| Chemical                   | 0.20        | 0.24        | 0.32        | 26.6%            |
| Transportation & Logistics | 0.34        | 0.44        | 0.60        | 32.8%            |
| Government                 | 0.29        | 0.37        | 0.50        | 32.3%            |
| Education                  | 0.24        | 0.30        | 0.39        | 28.7%            |
| <b>Total</b>               | <b>3.03</b> | <b>3.82</b> | <b>5.12</b> | <b>30.0%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 128** GERMANY: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2020–2026) |
|----------------------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|------------------|
| Aerospace & Defense        | 1.31        | 1.79        | 2.13         | 2.66         | 3.38         | 4.21         | 5.27         | 24.2%            |
| BFSI                       | 2.28        | 3.08        | 3.65         | 4.54         | 5.73         | 7.09         | 8.83         | 23.4%            |
| Healthcare & Life Sciences | 0.62        | 0.87        | 1.06         | 1.35         | 1.76         | 2.24         | 2.93         | 27.6%            |
| Energy & Utility           | 0.24        | 0.34        | 0.42         | 0.55         | 0.72         | 0.94         | 1.22         | 29.4%            |
| Chemical                   | 0.41        | 0.55        | 0.64         | 0.78         | 0.97         | 1.18         | 1.44         | 21.1%            |
| Transportation & Logistics | 0.83        | 1.16        | 1.42         | 1.81         | 2.36         | 3.02         | 3.88         | 27.3%            |
| Government                 | 0.69        | 0.96        | 1.17         | 1.49         | 1.94         | 2.47         | 3.17         | 27.0%            |
| Education                  | 0.52        | 0.71        | 0.84         | 1.05         | 1.33         | 1.65         | 1.98         | 22.8%            |
| <b>Total</b>               | <b>6.91</b> | <b>9.45</b> | <b>11.33</b> | <b>14.23</b> | <b>18.18</b> | <b>22.79</b> | <b>28.73</b> | <b>24.9%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

## 12.3.5 UK

The UK is considered highly competitive in Europe in terms of emerging technologies such as quantum computing. The rising GDP of the country indicates the growth of its economy that has a positive impact on various industries such as BFSI, energy and utilities, and chemical. The development of industries in the UK is expected to bring significant changes in the adoption rate of new technologies such as quantum computing software in the country. This, in turn, is anticipated to fuel the growth of the quantum computing market in the UK in the coming years. Increasing investments by the UK Government in these technologies are also expected to fuel the growth of the market. For instance, in the Autumn Statement 2013, the UK Government launched a 5-year program and invested USD 298 million to implement quantum computing technology in real-world applications. Further, four hubs—Manchester and Leeds from the North, Brighton from the South, and Bristol from the West—invested USD 133 million in the projects aimed at exploring the

properties of quantum mechanics. These hubs are managed by the Universities of Birmingham, Glasgow, York, and Oxford. Such initiatives and developments in the UK are expected to contribute to the growth of the quantum computing software market in the country during the forecast period.

The UK Government is aware of the beneficial impacts of quantum computing technology in applications ranging from computing to timekeeping, navigation to civil engineering, and transport to healthcare. Moreover, the Networked Quantum Information Technologies Hub (NQIT) is the largest of the four hubs of the UK National Quantum Technology Program. This hub is developed using an investment of USD 305 million by the government of the country to fuel the growth of the quantum technology market in the UK. It is working toward building a quantum computer demonstrator, the Q20:20 engine, which exhibits a networked, hybrid light-matter approach for quantum information processing. The UK also has various research centers that are engaged in developing quantum computers. For instance, Oxford Quantum, a center at Oxford University, is one of the largest and most diverse centers in the country for quantum research, with 38 separate research teams and ~200 researchers. Apart from that, the use of QCaaS offered by various quantum service providers is expected to contribute to the growth of the quantum computing software market in the UK during the forecast period.

**TABLE 129** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|--------------|------------|------------|------------|------------------|
| Software     | 0.7        | 0.9        | 1.2        | 31.6%            |
| Services     | 1.8        | 2.3        | 3.1        | 33.2%            |
| <b>Total</b> | <b>2.4</b> | <b>3.1</b> | <b>4.3</b> | <b>32.8%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 130** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|--------------|------------|------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Software     | 1.6        | 2.2        | 2.7         | 3.5         | 4.5         | 5.8         | 7.4         | 27.0%            |
| Services     | 4.3        | 6.1        | 7.5         | 9.6         | 12.7        | 16.4        | 21.3        | 28.6%            |
| <b>Total</b> | <b>5.9</b> | <b>8.3</b> | <b>10.2</b> | <b>13.1</b> | <b>17.2</b> | <b>22.2</b> | <b>28.7</b> | <b>28.2%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 131** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------------|-------------|-------------|-------------|------------------|
| Quantum as a Services | 0.56        | 0.73        | 1.02        | 34.6%            |
| Consulting Services   | 1.19        | 1.53        | 2.09        | 32.6%            |
| <b>Total</b>          | <b>1.75</b> | <b>2.26</b> | <b>3.11</b> | <b>33.2%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 132** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|-----------------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|------------------|
| Quantum as a Services | 1.42        | 2.02        | 2.51        | 3.27        | 4.35         | 5.67         | 7.44         | 29.8%            |
| Consulting Services   | 2.89        | 4.04        | 4.95        | 6.37        | 8.34         | 10.72        | 13.88        | 28.0%            |
| <b>Total</b>          | <b>4.31</b> | <b>6.05</b> | <b>7.46</b> | <b>9.64</b> | <b>12.68</b> | <b>16.39</b> | <b>21.32</b> | <b>28.6%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 133** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------|-------------|-------------|-------------|------------------|
| On-premises     | 0.32        | 0.40        | 0.53        | 29.8%            |
| Cloud           | 2.12        | 2.73        | 3.76        | 33.2%            |
| <b>Total</b>    | <b>2.43</b> | <b>3.13</b> | <b>4.29</b> | <b>32.8%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 134** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|-----------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|------------------|
| On-premises     | 0.72        | 0.98        | 1.18         | 1.48         | 1.89         | 2.37         | 2.98         | 24.9%            |
| Cloud           | 5.21        | 7.31        | 9.01         | 11.64        | 15.32        | 19.80        | 25.75        | 28.6%            |
| <b>Total</b>    | <b>5.92</b> | <b>8.29</b> | <b>10.18</b> | <b>13.12</b> | <b>17.20</b> | <b>22.16</b> | <b>28.73</b> | <b>28.2%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 135** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|-------------------|------------|------------|------------|------------------|
| SMEs              | 1.0        | 1.3        | 1.7        | 33.4%            |
| Large Enterprises | 1.5        | 1.9        | 2.6        | 32.4%            |
| <b>Total</b>      | <b>2.4</b> | <b>3.1</b> | <b>4.3</b> | <b>32.8%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 136** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|-------------------|------------|------------|-------------|-------------|-------------|-------------|-------------|------------------|
| SMEs              | 2.4        | 3.4        | 4.2         | 5.4         | 7.1         | 9.2         | 11.9        | 28.7%            |
| Large Enterprises | 3.5        | 4.9        | 6.0         | 7.7         | 10.1        | 13.0        | 16.8        | 27.8%            |
| <b>Total</b>      | <b>5.9</b> | <b>8.3</b> | <b>10.2</b> | <b>13.1</b> | <b>17.2</b> | <b>22.2</b> | <b>28.7</b> | <b>28.2%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 137** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|------------------|------------|------------|------------|------------------|
| Optimization     | 1.1        | 1.4        | 1.9        | 30.5%            |
| Machine Learning | 0.2        | 0.3        | 0.5        | 37.4%            |
| Simulation       | 0.6        | 0.8        | 1.1        | 34.4%            |
| Others *         | 0.5        | 0.6        | 0.9        | 33.6%            |
| <b>Total</b>     | <b>2.4</b> | <b>3.1</b> | <b>4.3</b> | <b>32.8%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 138** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020       | 2021-e     | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|------------------|------------|------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Optimization     | 2.5        | 3.5        | 4.2         | 5.3         | 6.8         | 8.6         | 10.9        | 25.7%               |
| Machine Learning | 0.7        | 0.9        | 1.2         | 1.6         | 2.2         | 2.9         | 3.9         | 32.6%               |
| Simulation       | 1.5        | 2.2        | 2.7         | 3.5         | 4.7         | 6.1         | 8.0         | 29.7%               |
| Others *         | 1.2        | 1.7        | 2.1         | 2.7         | 3.5         | 4.6         | 5.9         | 28.4%               |
| <b>Total</b>     | <b>5.9</b> | <b>8.3</b> | <b>10.2</b> | <b>13.1</b> | <b>17.2</b> | <b>22.2</b> | <b>28.7</b> | <b>28.2%</b>        |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 139** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|----------------------------|-------------|-------------|-------------|------------------|
| Aerospace & Defense        | 0.47        | 0.60        | 0.82        | 32.0%            |
| BFSI                       | 0.83        | 1.06        | 1.43        | 31.3%            |
| Healthcare & Life Sciences | 0.21        | 0.27        | 0.38        | 35.4%            |
| Energy & Utility           | 0.07        | 0.10        | 0.14        | 38.8%            |
| Chemical                   | 0.16        | 0.20        | 0.26        | 29.3%            |
| Transportation & Logistics | 0.27        | 0.36        | 0.50        | 35.7%            |
| Government                 | 0.23        | 0.30        | 0.42        | 35.2%            |
| Education                  | 0.19        | 0.24        | 0.33        | 31.5%            |
| <b>Total</b>               | <b>2.43</b> | <b>3.13</b> | <b>4.29</b> | <b>32.8%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 140** UK: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020        | 2021-e      | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2020–2026) |
|----------------------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Aerospace & Defense        | 1.13        | 1.57        | 1.91         | 2.45         | 3.19         | 4.09         | 5.27         | 27.5%               |
| BFSI                       | 1.95        | 2.71        | 3.28         | 4.18         | 5.42         | 6.90         | 8.83         | 26.7%               |
| Healthcare & Life Sciences | 0.53        | 0.76        | 0.95         | 1.25         | 1.66         | 2.18         | 2.93         | 31.0%               |
| Energy & Utility           | 0.20        | 0.29        | 0.38         | 0.50         | 0.68         | 0.91         | 1.22         | 32.9%               |
| Chemical                   | 0.36        | 0.48        | 0.58         | 0.72         | 0.92         | 1.14         | 1.44         | 24.3%               |
| Transportation & Logistics | 0.71        | 1.02        | 1.27         | 1.67         | 2.24         | 2.94         | 3.88         | 30.7%               |
| Government                 | 0.59        | 0.84        | 1.05         | 1.38         | 1.83         | 2.40         | 3.17         | 30.4%               |
| Education                  | 0.45        | 0.62        | 0.76         | 0.97         | 1.25         | 1.60         | 1.98         | 26.0%               |
| <b>Total</b>               | <b>5.92</b> | <b>8.29</b> | <b>10.18</b> | <b>13.12</b> | <b>17.20</b> | <b>22.16</b> | <b>28.73</b> | <b>28.2%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

### 12.3.6 FRANCE

France has a large and diversified industrial and agricultural base. It is also one of the leading countries in the global automotive industry. It is the fifth-largest economy in the world. The country has emerged as an enterprise hub in the European region. The surging demand for advanced technologies for secure communication in various industries is fueling the growth of the quantum computing software market in France. Moreover, the presence of Airbus SAS that uses quantum computing technology to speed up research activities related to aircraft is also fueling the growth of the quantum computing software market in France. There is also a research center for quantum computing established in the country.

The Government of France and the Government of Australia signed a Memorandum of Understanding (MoU) in May 2018 for the countries to collectively work toward the development of a silicon quantum computer. Under this MoU, Silicon Quantum Computing (SQC)—the first quantum computing company of Australia—and the Commissariat à l'Energie Atomique et aux Energies Alternatives (CEA)—a research and development organization in France—formed a joint venture related to silicon CMOS quantum computing technology to focus on the development and the commercialization of quantum computing technology.

### 12.3.7 REST OF EUROPE

The scope of the Rest of Europe market includes Netherlands, Italy, Belgium, Finland, and Russia. In June 2019, representatives of seven countries of the European Union, including Belgium, Germany, Italy, Luxembourg, Malta, the Netherlands, and Spain, signed a declaration at the Digital Assembly Bucharest held in Romania. This declaration included an agreement of these countries to carry out the development and the deployment of quantum communication infrastructure (QCI) across the European Union. This infrastructure is expected to enable the transmission and ultra-secure storage of information and data and link communication assets across the union.

In December 2019, Russia invested ~USD 790 million to carry out advancements in the field of quantum computing. It marks the official entry of the country into the quantum marketplace. This investment is expected to provide leading Russia-based researchers with funding over the next 5 years to develop practical quantum technologies and realize the supremacy of quantum computing. This investment is a part of a large research and development investment initiative undertaken by Russia to improve its technological and economic standing worldwide. Russia has joined the US, European Union (EU), China, and Germany in the field of technology to gain and retain its quantum advantages.

## 12.4 ASIA PACIFIC

APAC is expected to register the highest CAGR in the quantum computing software market during the forecast period due to the growing adoption of quantum computing software. This region is expected to contribute 33.3% share to the quantum computing software market in 2021. China, Japan, and South Korea are the key countries contributing to the growth of the quantum computing software market in APAC. The growing per capita income of the countries of the region, along with the large-scale industrialization and urbanization, are the factors driving the growth of the quantum computing software market in APAC. Similarly, the rising demand for energy and the increasing requirement of quantum cryptography to secure mobile transactions are expected to fuel the adoption of quantum computing software and services in the energy & utility and BFSI industries of APAC.

### 12.4.1 ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- Substantial investments by China in the development of quantum computers are expected to drive the growth of the quantum computing software market.

### 12.4.2 ASIA PACIFIC: COVID-19 IMPACT

- While China was the epicenter of the outbreak, the economic impact was felt across APAC. The COVID-19 pandemic has slowed down a majority of the economies, such as China, India, Singapore, Japan, and Australia; however, these countries are expected to see increased adoption of AI and analytics.
- Several countries, such as China, India, and Japan, are leveraging the benefits of information-intensive AI and ML technologies for different verticals. For instance, Shenzhen-based WeBank is applying analytics and AI techniques to data from satellite imagery for developing indexes and other investment tools related to environmental risks for Chinese companies.
- While the expenditure on technology solutions in APAC has increased, a setback is witnessed due to the recent COVID-19 pandemic. With COVID-19 lockdowns, enterprises in the region are focusing on investing in technology-enabled platforms to continue working from home, which has influenced the adoption of digital solutions in this region.

### 12.4.3 ASIA PACIFIC: REGULATORY IMPLICATIONS

This section includes the laws and regulations implemented by government bodies to safeguard personal data and privacy in APAC. These regulations help maintain data quality and prevent fraudulent activities by ensuring data transparency among defined entities.

#### 12.4.3.1 Privacy Commissioner for Personal Data

The office of the Privacy Commissioner for Personal Data (PCPD) is the authority enforcing the Personal Data (Privacy) Ordinance (PDPO) in Hong Kong. In September 2017, the PCPD hosted the 39th International Conference of Data Protection and Privacy Commissioners, the annual gathering of data protection authorities worldwide.

The PCPD pursues an active agenda of public education and the publication of compliance guidance, including a strong focus on technology-related issues. The PCPD is also closely monitoring the implementation of the GDPR, stating that Hong Kong must consider the need to establish a comparable framework and mechanism interoperable with international data protection authorities without compromising economic and technological development.

#### 12.4.3.2 Act on the Protection of Personal Information

Japan's Act on the Protection of Personal Information (APPI) dates to 2003 and stands as one of Asia's oldest laws in this area. In the wake of a series of high-profile data security breaches and revelations of unlawful sales of personal data in Japan, the Japanese government passed extensive reforms to the APPI in September 2015. The reforms implemented in May 2017 are as follows:

- The appointment of an independent, dedicated data protection regulatory authority
- The expansion of the definition of “personal data” to include biometric data
- The introduction of a concept of special care required personal information (i.e., the concept of sensitive personal data) that will be subject to enhanced protections
- The introduction of restrictions on cross-border transfers of personal data, which will now require: (i) data subject consent; (ii) export to a jurisdiction having the benefit of an adequacy finding; or (iii) satisfaction of other criteria to be specified by the new regulatory authority, including certification under the APEC CBPR program
- The introduction of data anonymization regulations that will require organizations making use of anonymized personal data to publicly announce the items of data being anonymized and establish internal and external rules for managing “reidentification risk,” i.e., the risk that the data is processed in such a way to enable the identification of anonymous data subjects.

#### 12.4.3.3 Critical Information Infrastructure

Singapore's Cybersecurity Bill was passed into law on February 5, 2018, providing a framework for the regulation of providers of Critical Information Infrastructure (CII). The Cyber Security Agency of Singapore will define CII on a sectoral basis. Owners of CII are required to take certain protective measures and comply with reporting requirements, and are ultimately responsible for the security of their CII. This will not extend to multinationals with Singapore offices that are supported by infrastructure located overseas.

#### 12.4.3.4 International Organization for Standardization 27001

The International Organization for Standardization (ISO) 27001 compliance provides a risk-based framework of policies and procedures for establishing, implementing, operating, monitoring, reviewing, maintaining, and improving the Information Security Management System (ISMS). The procedures and policies include defining a security policy and the scope of ISMS, conducting risk assessments, managing identified risks, selecting control objectives and controls to be implemented, and preparing a statement of applicability. ISO 27001 guidelines cover various areas, such as access control, audit, incident response, and system and information integrity. The key 114 controls required as a part of ISO 27001 compliance are security policy, organization of information security, asset management, HR security, physical and environmental security, communications and operations management, access control, information security incident management, business continuity management, compliance, and information systems acquisition, development, and maintenance.

#### 12.4.3.5 Personal Data Protection Act

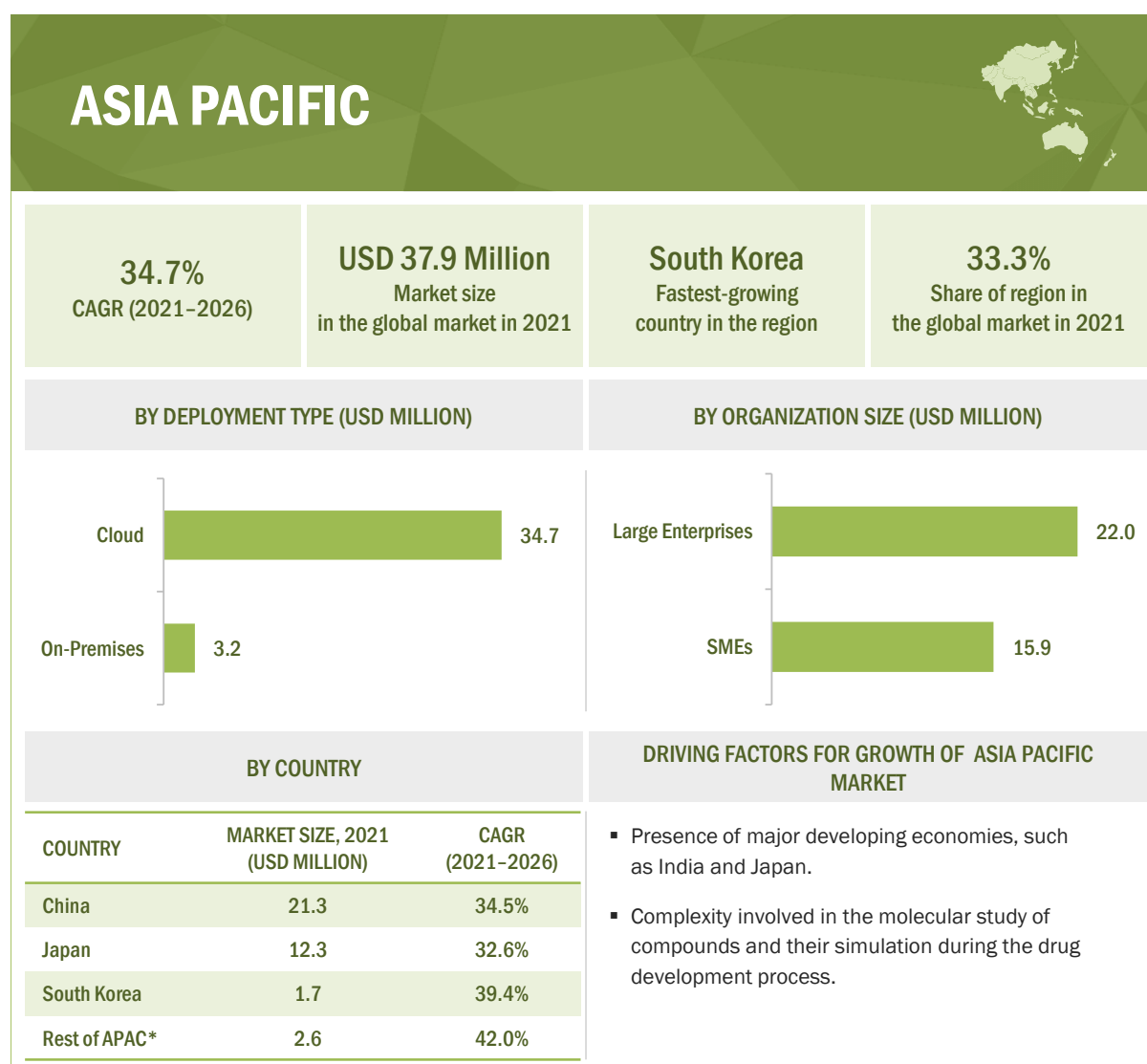
The Personal Data Protection Act (PDPA) establishes a data protection law that comprises various rules governing the collection, use, disclosure, and care of personal data in Singapore. It recognizes both the rights of individuals to protect their personal data, including rights of access and correction, and the need for organizations to collect, use, or disclose personal data for legitimate and reasonable purposes. PDPA provides a national Do Not Call (DNC) registry for establishments. The DNC registry enables individuals to register their Singapore telephone numbers to opt-out of receiving marketing phone calls; mobile text messages, such as Short Message Service (SMS) or Multimedia Messaging Service (MMS); and faxes from organizations.

PDPA covers personal data stored in electronic and non-electronic forms. The data protection provisions in PDPA (parts III to VI) generally do not apply to the following:

- Any individual acting on a personal or domestic basis
- Any employee acting in the course of his/her employment with an organization
- Any public agency or an organization in the course of acting on behalf of a public agency in relation to the collection, use, or disclosure of personal data. Companies can refer to the Personal Data Protection (Statutory Bodies) Notification 2013 for the list of specified public agencies.
- Business contact information: This refers to an individual's name, position name or title, business telephone number, business address, business electronic mail address or business fax number, and any other similar information about the individual, not provided by the individual solely for his/her purposes.

These rules act as a baseline law operating as part of the law of Singapore. They do not supersede existing statutes, such as the Banking Act and Insurance Act, but work in conjunction with them and the common law.

**FIGURE 33 ASIA PACIFIC: MARKET SNAPSHOT**



\*Rest of APAC includes India, Australia and New Zealand (ANZ), Singapore, Malaysia, and Taiwan

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 141** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|--------------|------------|-------------|-------------|------------------|
| Software     | 2.2        | 3.0         | 4.3         | 38.9%            |
| Services     | 6.7        | 9.1         | 13.4        | 41.5%            |
| <b>Total</b> | <b>8.9</b> | <b>12.1</b> | <b>17.6</b> | <b>40.8%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 142** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR (2021–2026) |
|--------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|------------------|
| Software     | 6.2         | 8.9         | 11.4        | 15.1        | 20.5        | 27.3         | 36.6         | 32.6%            |
| Services     | 19.6        | 28.9        | 37.5        | 51.0        | 70.6        | 96.0         | 131.4        | 35.3%            |
| <b>Total</b> | <b>25.8</b> | <b>37.9</b> | <b>48.9</b> | <b>66.1</b> | <b>91.1</b> | <b>123.3</b> | <b>168.0</b> | <b>34.7%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 143** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017       | 2018       | 2019        | CAGR (2017–2019) |
|-----------------------|------------|------------|-------------|------------------|
| Quantum as a Services | 2.0        | 2.8        | 4.1         | 42.5%            |
| Consulting Services   | 4.7        | 6.4        | 9.3         | 41.1%            |
| <b>Total</b>          | <b>6.7</b> | <b>9.1</b> | <b>13.4</b> | <b>41.5%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 144** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p       | CAGR (2021–2026) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|------------------|
| Quantum as a Services | 6.0         | 8.9         | 11.6        | 15.9        | 22.2        | 30.4        | 41.9         | 36.2%            |
| Consulting Services   | 13.6        | 20.0        | 25.9        | 35.0        | 48.4        | 65.6        | 89.5         | 34.9%            |
| <b>Total</b>          | <b>19.6</b> | <b>28.9</b> | <b>37.5</b> | <b>51.0</b> | <b>70.6</b> | <b>96.0</b> | <b>131.4</b> | <b>35.3%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 145** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|-----------------|------------|-------------|-------------|------------------|
| On-premises     | 0.8        | 1.1         | 1.5         | 38.9%            |
| Cloud           | 8.1        | 11.1        | 16.1        | 41.0%            |
| <b>Total</b>    | <b>8.9</b> | <b>12.1</b> | <b>17.6</b> | <b>40.8%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 146** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR (2021–2026) |
|-----------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|------------------|
| On-premises     | 2.2         | 3.2         | 4.1         | 5.5         | 7.4         | 9.9          | 13.2         | 32.7%            |
| Cloud           | 23.5        | 34.7        | 44.8        | 60.6        | 83.7        | 113.4        | 154.7        | 34.9%            |
| <b>Total</b>    | <b>25.8</b> | <b>37.9</b> | <b>48.9</b> | <b>66.1</b> | <b>91.1</b> | <b>123.3</b> | <b>168.0</b> | <b>34.7%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 147** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017       | 2018        | 2019        | CAGR (2017–2019) |
|-------------------|------------|-------------|-------------|------------------|
| SMEs              | 3.6        | 4.9         | 7.2         | 42.6%            |
| Large Enterprises | 5.3        | 7.2         | 10.4        | 39.6%            |
| <b>Total</b>      | <b>8.9</b> | <b>12.1</b> | <b>17.6</b> | <b>40.8%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 148** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p       | 2026-p       | CAGR (2021–2026) |
|-------------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|------------------|
| SMEs              | 10.7        | 15.9        | 20.8        | 28.5        | 39.6        | 54.3         | 74.7         | 36.2%            |
| Large Enterprises | 15.1        | 22.0        | 28.1        | 37.7        | 51.4        | 69.0         | 93.2         | 33.5%            |
| <b>Total</b>      | <b>25.8</b> | <b>37.9</b> | <b>48.9</b> | <b>66.1</b> | <b>91.1</b> | <b>123.3</b> | <b>168.0</b> | <b>34.7%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 149** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017        | 2018         | 2019         | CAGR (2017–2019) |
|------------------|-------------|--------------|--------------|------------------|
| Optimization     | 4.62        | 6.16         | 8.72         | 37.4%            |
| Machine Learning | 1.07        | 1.50         | 2.24         | 44.9%            |
| Simulation       | 1.96        | 2.72         | 4.04         | 43.6%            |
| Others *         | 1.24        | 1.76         | 2.64         | 45.6%            |
| <b>Total</b>     | <b>8.89</b> | <b>12.14</b> | <b>17.64</b> | <b>40.8%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 150** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p        | 2026-p        | CAGR (2021–2026) |
|------------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|------------------|
| Optimization     | 12.43        | 17.82        | 22.41        | 29.58        | 39.74        | 52.46         | 69.71         | 31.4%            |
| Machine Learning | 3.37         | 5.09         | 6.76         | 9.41         | 13.34        | 18.57         | 26.04         | 38.6%            |
| Simulation       | 6.01         | 9.01         | 11.85        | 16.36        | 22.98        | 31.72         | 44.08         | 37.4%            |
| Others *         | 3.96         | 5.96         | 7.83         | 10.76        | 15.02        | 20.53         | 28.15         | 36.4%            |
| <b>Total</b>     | <b>25.77</b> | <b>37.88</b> | <b>48.86</b> | <b>66.11</b> | <b>91.08</b> | <b>123.29</b> | <b>167.97</b> | <b>34.7%</b>     |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 151** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018         | 2019         | CAGR (2017–2019) |
|----------------------------|-------------|--------------|--------------|------------------|
| Aerospace & Defense        | 2.27        | 3.07         | 4.42         | 39.6%            |
| BFSI                       | 3.11        | 4.27         | 6.24         | 41.6%            |
| Healthcare & Life Sciences | 0.49        | 0.68         | 1.01         | 43.6%            |
| Energy & Utility           | 0.39        | 0.53         | 0.78         | 41.5%            |
| Chemical                   | 0.56        | 0.75         | 1.07         | 39.0%            |
| Transportation & Logistics | 0.70        | 0.97         | 1.41         | 41.4%            |
| Government                 | 0.93        | 1.29         | 1.91         | 43.1%            |
| Education                  | 0.44        | 0.58         | 0.80         | 34.9%            |
| <b>Total</b>               | <b>8.89</b> | <b>12.14</b> | <b>17.64</b> | <b>40.8%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 152** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p        | 2026-p        | CAGR<br>(2020–2026) |
|----------------------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------------|
| Aerospace & Defense        | 6.39         | 9.32         | 11.91        | 15.96        | 21.79        | 29.24         | 39.47         | 33.5%               |
| BFSI                       | 9.17         | 13.55        | 17.61        | 24.02        | 33.35        | 45.50         | 62.49         | 35.8%               |
| Healthcare & Life Sciences | 1.50         | 2.25         | 2.96         | 4.08         | 5.74         | 7.92          | 11.00         | 37.3%               |
| Energy & Utility           | 1.14         | 1.69         | 2.18         | 2.97         | 4.11         | 5.59          | 7.64          | 35.3%               |
| Chemical                   | 1.55         | 2.24         | 2.85         | 3.80         | 5.16         | 6.88          | 9.24          | 32.7%               |
| Transportation & Logistics | 2.07         | 3.05         | 3.95         | 5.36         | 7.41         | 10.07         | 13.77         | 35.2%               |
| Government                 | 2.83         | 4.23         | 5.54         | 7.60         | 10.63        | 14.59         | 20.16         | 36.7%               |
| Education                  | 1.12         | 1.56         | 1.86         | 2.31         | 2.88         | 3.50          | 4.20          | 21.9%               |
| <b>Total</b>               | <b>25.77</b> | <b>37.88</b> | <b>48.86</b> | <b>66.11</b> | <b>91.08</b> | <b>123.29</b> | <b>167.97</b> | <b>34.7%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

**TABLE 153** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)

| Country       | 2017        | 2018         | 2019         | CAGR (2017–2019) |
|---------------|-------------|--------------|--------------|------------------|
| China         | 4.98        | 6.80         | 9.90         | 41.0%            |
| Japan         | 3.07        | 4.13         | 5.91         | 38.8%            |
| South Korea   | 0.36        | 0.50         | 0.76         | 45.8%            |
| Rest of APAC* | 0.49        | 0.70         | 1.08         | 48.5%            |
| <b>Total</b>  | <b>8.89</b> | <b>12.14</b> | <b>17.64</b> | <b>40.8%</b>     |

Note: \*Rest of APAC includes India, Australia and New Zealand (ANZ), Singapore, Malaysia, and Taiwan

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 154** ASIA PACIFIC: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)

| Country       | 2020-e       | 2021-p       | 2022-p       | 2023-p       | 2024-p       | 2025-p        | 2026-p        | CAGR<br>(2021–2026) |
|---------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------------|
| China         | 14.46        | 21.26        | 27.40        | 37.04        | 50.96        | 68.84         | 93.56         | 34.5%               |
| Japan         | 8.50         | 12.31        | 15.63        | 20.82        | 28.23        | 37.60         | 50.39         | 32.6%               |
| South Korea   | 1.14         | 1.74         | 2.32         | 3.25         | 4.63         | 6.49          | 9.15          | 39.4%               |
| Rest of APAC* | 1.66         | 2.57         | 3.50         | 4.99         | 7.25         | 10.35         | 14.87         | 42.0%               |
| <b>Total</b>  | <b>25.77</b> | <b>37.88</b> | <b>48.86</b> | <b>66.11</b> | <b>91.08</b> | <b>123.29</b> | <b>167.97</b> | <b>34.7%</b>        |

e: estimated; p: projected

Note: \*Rest of APAC includes India, Australia and New Zealand (ANZ), Singapore, Malaysia, and Taiwan

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

#### 12.4.4 CHINA

China is one of the fastest-growing economies in the world, with a large population base and continuously increasing per capita income. China has witnessed an increase in technological investments in APAC. According to the World Bank report, China's Global Index ranking is projected to be the highest among developing countries. Since 2000, China's contribution to global GDP growth (in purchasing-power-parity terms) has been higher than that of the US, and the country accounted for over half of the combined contribution of India, Brazil, and Russia, the three next-largest emerging economies. China is a growing economy that has been rapidly adopting the latest technologies as part of its digital transformation process. The financial strength of the country enables it to make significant investments in emerging technologies such as quantum computing. China has already developed the fastest supercomputer in the world. Moreover, the researchers in the country have developed a quantum computer using multiple photons, which increases the speed and scale of computing. With this invention, China is strengthening its technology reach. The country is also making substantial investments in the development of quantum computers. Increased support and initiatives undertaken by the government of the country have led to a surge in the number of patents filed by China-based companies pertaining to quantum computing technology in the last few years. Companies such as Tencent, Alibaba, and Baidu have also entered the field of quantum computing. Moreover, with the help of the quantum computer offered by Alibaba, China is expecting to witness progress in the areas of weapon development, astronomy, etc.

The China-based e-commerce company, Alibaba, has developed the Alibaba Quantum Laboratory (AQL), which was unveiled in 2015, along with the Chinese Academy of Sciences. This laboratory is contributing to the growth of the quantum computing software market in the country. China is also in the process of making some investments in its quantum research efforts. For instance, the country invested USD 10.0 billion for the construction of the National Laboratory for Quantum Information Sciences; the laboratory opened in 2020. Such factors are expected to drive the growth of the quantum computing software market in China during the forecast period.

**TABLE 155 CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)**

| Component    | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|--------------|------------|------------|------------|------------------|
| Software     | 1.1        | 1.5        | 2.1        | 39.3%            |
| Services     | 3.9        | 5.3        | 7.8        | 41.4%            |
| <b>Total</b> | <b>5.0</b> | <b>6.8</b> | <b>9.9</b> | <b>41.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 156 CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)**

| Component    | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Software     | 3.1         | 4.5         | 5.7         | 7.6         | 10.3        | 13.7        | 18.4        | 32.8%            |
| Services     | 11.4        | 16.8        | 21.7        | 29.5        | 40.7        | 55.1        | 75.1        | 34.9%            |
| <b>Total</b> | <b>14.5</b> | <b>21.3</b> | <b>27.4</b> | <b>37.0</b> | <b>51.0</b> | <b>68.8</b> | <b>93.6</b> | <b>34.5%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 157** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------------|-------------|-------------|-------------|------------------|
| Quantum as a Services | 1.24        | 1.69        | 2.44        | 40.2%            |
| Consulting Services   | 2.64        | 3.64        | 5.33        | 42.0%            |
| <b>Total</b>          | <b>3.88</b> | <b>5.32</b> | <b>7.77</b> | <b>41.4%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 158** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR (2021–2026) |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| Quantum as a Services | 3.55         | 5.19         | 6.65         | 8.94         | 12.22        | 16.41        | 22.16        | 33.7%            |
| Consulting Services   | 7.84         | 11.61        | 15.07        | 20.52        | 28.43        | 38.69        | 52.97        | 35.5%            |
| <b>Total</b>          | <b>11.39</b> | <b>16.80</b> | <b>21.72</b> | <b>29.46</b> | <b>40.65</b> | <b>55.10</b> | <b>75.13</b> | <b>34.9%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 159** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|-----------------|------------|------------|------------|------------------|
| On-premises     | 0.5        | 0.7        | 1.0        | 38.1%            |
| Cloud           | 4.4        | 6.1        | 8.9        | 41.3%            |
| <b>Total</b>    | <b>5.0</b> | <b>6.8</b> | <b>9.9</b> | <b>41.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 160** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| On-premises     | 1.5         | 2.2         | 2.7         | 3.6         | 4.8         | 6.4         | 8.4         | 31.4%            |
| Cloud           | 13.0        | 19.1        | 24.7        | 33.5        | 46.1        | 62.5        | 85.1        | 34.8%            |
| <b>Total</b>    | <b>14.5</b> | <b>21.3</b> | <b>27.4</b> | <b>37.0</b> | <b>51.0</b> | <b>68.8</b> | <b>93.6</b> | <b>34.5%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 161** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|-------------------|------------|------------|------------|------------------|
| SMEs              | 2.1        | 2.8        | 4.1        | 40.2%            |
| Large Enterprises | 2.9        | 4.0        | 5.8        | 41.6%            |
| <b>Total</b>      | <b>5.0</b> | <b>6.8</b> | <b>9.9</b> | <b>41.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 162** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|-------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| SMEs              | 6.0         | 8.7         | 11.2        | 15.0        | 20.5        | 27.6        | 37.2        | 33.7%            |
| Large Enterprises | 8.5         | 12.5        | 16.2        | 22.0        | 30.4        | 41.3        | 56.3        | 35.1%            |
| <b>Total</b>      | <b>14.5</b> | <b>21.3</b> | <b>27.4</b> | <b>37.0</b> | <b>51.0</b> | <b>68.8</b> | <b>93.6</b> | <b>34.5%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 163** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017       | 2018       | 2019       | CAGR (2017–2019) |
|------------------|------------|------------|------------|------------------|
| Optimization     | 2.6        | 3.4        | 4.9        | 37.3%            |
| Machine Learning | 0.6        | 0.8        | 1.3        | 45.0%            |
| Simulation       | 1.1        | 1.5        | 2.3        | 43.8%            |
| Others *         | 0.7        | 1.0        | 1.5        | 46.3%            |
| <b>Total</b>     | <b>5.0</b> | <b>6.8</b> | <b>9.9</b> | <b>41.0%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 164** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Optimization     | 6.9         | 10.0        | 12.5        | 16.5        | 22.1        | 29.0        | 38.4        | 31.0%               |
| Machine Learning | 1.9         | 2.9         | 3.8         | 5.3         | 7.5         | 10.4        | 14.5        | 38.4%               |
| Simulation       | 3.4         | 5.1         | 6.6         | 9.2         | 12.9        | 17.7        | 24.6        | 37.2%               |
| Others *         | 2.3         | 3.4         | 4.4         | 6.1         | 8.6         | 11.7        | 16.1        | 36.7%               |
| <b>Total</b>     | <b>14.5</b> | <b>21.3</b> | <b>27.4</b> | <b>37.0</b> | <b>51.0</b> | <b>68.8</b> | <b>93.6</b> | <b>34.5%</b>        |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 165** CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|----------------------------|-------------|-------------|-------------|------------------|
| Aerospace & Defense        | 1.17        | 1.58        | 2.28        | 39.6%            |
| BFSI                       | 1.74        | 2.39        | 3.50        | 41.7%            |
| Healthcare & Life Sciences | 0.27        | 0.38        | 0.57        | 43.7%            |
| Energy & Utility           | 0.22        | 0.30        | 0.44        | 41.4%            |
| Chemical                   | 0.31        | 0.42        | 0.60        | 39.1%            |
| Transportation & Logistics | 0.44        | 0.61        | 0.89        | 41.4%            |
| Government                 | 0.47        | 0.66        | 0.97        | 43.4%            |
| Education                  | 0.35        | 0.46        | 0.65        | 36.9%            |
| <b>Total</b>               | <b>4.98</b> | <b>6.80</b> | <b>9.90</b> | <b>41.0%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 166 CHINA: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)**

| Vertical                   | 2020         | 2021-e       | 2022-p       | 2023-p       | 2024-p       | 2025-p       | 2026-p       | CAGR<br>(2020–2026) |
|----------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Aerospace & Defense        | 3.30         | 4.80         | 6.13         | 8.20         | 11.18        | 14.95        | 20.12        | 3.30                |
| BFSI                       | 5.14         | 7.60         | 9.85         | 13.39        | 18.52        | 25.16        | 34.38        | 5.14                |
| Healthcare & Life Sciences | 0.84         | 1.26         | 1.66         | 2.29         | 3.21         | 4.42         | 6.13         | 0.84                |
| Energy & Utility           | 0.64         | 0.94         | 1.22         | 1.65         | 2.28         | 3.09         | 4.21         | 0.64                |
| Chemical                   | 0.87         | 1.26         | 1.60         | 2.13         | 2.89         | 3.84         | 5.15         | 0.87                |
| Transportation & Logistics | 1.30         | 1.92         | 2.49         | 3.37         | 4.66         | 6.31         | 8.61         | 1.30                |
| Government                 | 1.45         | 2.16         | 2.83         | 3.89         | 5.44         | 7.46         | 10.29        | 1.45                |
| Education                  | 0.92         | 1.30         | 1.62         | 2.11         | 2.79         | 3.61         | 4.68         | 0.92                |
| <b>Total</b>               | <b>14.46</b> | <b>21.26</b> | <b>27.40</b> | <b>37.04</b> | <b>50.96</b> | <b>68.84</b> | <b>93.56</b> | <b>14.46</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

## 12.4.5 JAPAN

Japan focuses on emerging technologies to make the country technologically advanced. Japan has companies such as Quantum Biosystems Inc. and H-bar Consultants (a consultancy specializing in the new field of quantum technology). The presence of such prominent players in the quantum computing industry in the country is expected to drive the growth of the quantum computing software market in Japan in the coming years. Moreover, companies such as NTT, NEC, and Fujitsu are working on the development of quantum computing technology. In November 2017, NTT unveiled the first quantum computer prototype of the country with a potential processing speed of 100 supercomputers using only 1 kilowatt of power. A few Japan-based companies are also actively involved in the manufacturing of quantum computer components. For instance, Coax Co. supplies superconducting cables that carry signals used to control the bits. These are specially designed to conduct very little heat so that the delicate quantum state of qubits is not disrupted inside fridges.

## 12.4.6 SOUTH KOREA

The quantum computing software market in South Korea is expected to be driven by government initiatives and the adoption of quantum computing technology by key manufacturers of consumer electronics, including Samsung Electronics and LG Electronics. In early 2019, the government of the country announced an investment of ~USD 40 million over the next 5 years for the development of next-generation ICT proprietary technologies, including core quantum computing technology. On January 31, 2019, the Ministry of Science and ICT (MSIT) of South Korea announced that its quantum computing technology development project and the next-generation information computing technology development project for 2019 have been finalized and are ready to kick off. Through the development of key quantum computing technologies, the government of the country plans to complete the demonstration of a practical five-qubit (qubit information unit) quantum computer system with more than 90% reliability by 2023.

### 12.4.7 REST OF ASIA PACIFIC

The scope of rest of APAC includes India, Australia and New Zealand (ANZ), Singapore, Malaysia, and Taiwan. These countries are among the most competitive economies in the region and are expected to witness increased adoption of quantum computing software and services. In December 2015, the Government of Australia invested USD 20 million for the development of quantum-computing technology as a part of its National Innovation and Science Agenda (NISA). The first quantum computing company in Australia, namely, Silicon Quantum Computing, was established in May 2017 to accelerate the commercialization of silicon quantum computing technology. The company was founded with over USD 63 million capital funding from the Australian Commonwealth Government, the University of New South Wales (UNSW) Sydney, the Commonwealth Bank of Australia, Telstra Corporation, and the State Government of New South Wales.

## 12.5 REST OF THE WORLD

The quantum computing software market in RoW has been studied for the Middle East and Africa and Latin America. In RoW, the Middle East holds significant growth potential for the quantum computing software market, as there is a rise in applications of quantum computing in different industries. The Gulf countries collectively have the largest oil industry in the world. The use of quantum computing software and services in the energy and power industry is expected to play a vital role in carrying out tasks such as surveillance and monitoring, and data communication with far-stationed/remote oilfields. The quantum computing software market in RoW is projected to grow at a high rate in the near future owing to the increased demand for quantum computing software and services in the energy and power, and transportation and logistics industries.

### 12.5.1 REST OF THE WORLD: QUANTUM COMPUTING SOFTWARE MARKET DRIVERS

- The establishment of Brazil Quantum as the first quantum community in Brazil to solve problems regarding quantum computing is expected to fuel the adoption of quantum computing software and services.

### 12.5.2 REST OF THE WORLD: COVID-19 IMPACT

- The UAE's Ministry of Economy has set up an interactive online tool to help companies address the impact of the COVID-19 pandemic, support business continuity, and promote sound business practices as part of efforts to shore up the private sector.
- Healthcare providers in the region have adopted advanced technologies to diagnose diseases more accurately and administer medical treatment with more efficiency. Big data and AI are being utilized in the healthcare sector for contact tracing and vaccine development to curb the effects of COVID-19.
- Gulf Cooperation Council (GCC) members, such as Saudi Arabia, Bahrain, Oman, Qatar, Kuwait, and the UAE, are increasing the use of AI tools to stop the spread of the virus. They are increasingly deploying sophisticated speed cameras, drones, and robots to ensure movement is limited and social distancing is in place.
- The central government and central bank of the UAE have announced a package of USD 34 billion to revive the economy.
- The Government of KSA has announced USD 20 billion to support economic growth in the country.
- A project has been developed by the Social Science Academy of Nigeria to leverage data science and AI tools for developing a better understanding of the impact of COVID-19 on gender-based violence and support women who are at the greatest risk.

**TABLE 167** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2017–2019 (USD MILLION)

| Component    | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|--------------|-------------|-------------|-------------|------------------|
| Software     | 0.17        | 0.22        | 0.31        | 36.3%            |
| Services     | 0.43        | 0.58        | 0.83        | 38.0%            |
| <b>Total</b> | <b>0.60</b> | <b>0.80</b> | <b>1.14</b> | <b>37.5%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 168** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COMPONENT, 2020–2026 (USD MILLION)

| Component    | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Software     | 0.44        | 0.64        | 0.80        | 1.06        | 1.42        | 1.88        | 2.49        | 31.4%            |
| Services     | 1.18        | 1.72        | 2.20        | 2.94        | 4.00        | 5.35        | 7.20        | 33.1%            |
| <b>Total</b> | <b>1.63</b> | <b>2.36</b> | <b>3.00</b> | <b>3.99</b> | <b>5.42</b> | <b>7.23</b> | <b>9.69</b> | <b>32.7%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 169** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2017–2019 (USD MILLION)

| Service               | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------------|-------------|-------------|-------------|------------------|
| Quantum as a Services | 0.13        | 0.17        | 0.24        | 37.2%            |
| Consulting Services   | 0.30        | 0.41        | 0.58        | 38.3%            |
| <b>Total</b>          | <b>0.43</b> | <b>0.58</b> | <b>0.83</b> | <b>38.0%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 170** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY SERVICE, 2020–2026 (USD MILLION)

| Service               | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Quantum as a Services | 0.35        | 0.50        | 0.64        | 0.85        | 1.15        | 1.53        | 2.05        | 32.4%            |
| Consulting Services   | 0.83        | 1.22        | 1.55        | 2.09        | 2.85        | 3.82        | 5.15        | 33.5%            |
| <b>Total</b>          | <b>1.18</b> | <b>1.72</b> | <b>2.20</b> | <b>2.94</b> | <b>4.00</b> | <b>5.35</b> | <b>7.20</b> | <b>33.1%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 171** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2017–2019 (USD MILLION)

| Deployment Mode | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-----------------|-------------|-------------|-------------|------------------|
| On-premises     | 0.07        | 0.09        | 0.13        | 35.6%            |
| Cloud           | 0.53        | 0.71        | 1.01        | 37.8%            |
| <b>Total</b>    | <b>0.60</b> | <b>0.80</b> | <b>1.14</b> | <b>37.5%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 172** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY DEPLOYMENT MODE, 2020–2026 (USD MILLION)

| Deployment Mode | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| On-premises     | 0.19        | 0.27        | 0.33        | 0.44        | 0.59        | 0.77        | 1.02        | 30.7%            |
| Cloud           | 1.44        | 2.09        | 2.66        | 3.55        | 4.83        | 6.45        | 8.67        | 32.9%            |
| <b>Total</b>    | <b>1.63</b> | <b>2.36</b> | <b>3.00</b> | <b>3.99</b> | <b>5.42</b> | <b>7.23</b> | <b>9.69</b> | <b>32.7%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 173** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2017–2019 (USD MILLION)

| Organization Size | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|-------------------|-------------|-------------|-------------|------------------|
| SMEs              | 0.29        | 0.39        | 0.55        | 38.1%            |
| Large Enterprises | 0.31        | 0.42        | 0.59        | 37.0%            |
| <b>Total</b>      | <b>0.60</b> | <b>0.80</b> | <b>1.14</b> | <b>37.5%</b>     |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 174** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY ORGANIZATION SIZE, 2020–2026 (USD MILLION)

| Organization Size | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|-------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| SMEs              | 0.79        | 1.15        | 1.47        | 1.97        | 2.68        | 3.58        | 4.83        | 33.2%            |
| Large Enterprises | 0.84        | 1.21        | 1.53        | 2.03        | 2.74        | 3.64        | 4.86        | 32.2%            |
| <b>Total</b>      | <b>1.63</b> | <b>2.36</b> | <b>3.00</b> | <b>3.99</b> | <b>5.42</b> | <b>7.23</b> | <b>9.69</b> | <b>32.7%</b>     |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 175** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2017–2019 (USD MILLION)

| Application      | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|------------------|-------------|-------------|-------------|------------------|
| Optimization     | 0.33        | 0.43        | 0.61        | 35.6%            |
| Machine Learning | 0.08        | 0.12        | 0.17        | 41.0%            |
| Simulation       | 0.12        | 0.16        | 0.23        | 39.3%            |
| Others *         | 0.07        | 0.09        | 0.13        | 39.1%            |
| <b>Total</b>     | <b>0.60</b> | <b>0.80</b> | <b>1.14</b> | <b>37.5%</b>     |

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 176** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY APPLICATION, 2020–2026 (USD MILLION)

| Application      | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR (2021–2026) |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| Optimization     | 0.86        | 1.23        | 1.54        | 2.02        | 2.71        | 3.56        | 4.71        | 30.9%            |
| Machine Learning | 0.25        | 0.36        | 0.47        | 0.65        | 0.90        | 1.23        | 1.70        | 36.0%            |
| Simulation       | 0.34        | 0.50        | 0.64        | 0.86        | 1.19        | 1.60        | 2.18        | 34.4%            |
| Others *         | 0.18        | 0.27        | 0.34        | 0.46        | 0.62        | 0.83        | 1.10        | 32.7%            |
| <b>Total</b>     | <b>1.63</b> | <b>2.36</b> | <b>3.00</b> | <b>3.99</b> | <b>5.42</b> | <b>7.23</b> | <b>9.69</b> | <b>32.7%</b>     |

e: estimated; p: projected

Note: \*Others includes Quantum Chemistry and Quantum Finance

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 177** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2017–2019 (USD MILLION)

| Vertical                   | 2017        | 2018        | 2019        | CAGR (2017–2019) |
|----------------------------|-------------|-------------|-------------|------------------|
| Aerospace & Defense        | 0.13        | 0.17        | 0.24        | 33.3%            |
| BFSI                       | 0.03        | 0.04        | 0.06        | 42.5%            |
| Healthcare & Life Sciences | 0.00        | 0.00        | 0.00        | NA               |
| Energy & Utility           | 0.05        | 0.06        | 0.09        | 40.5%            |
| Chemical                   | 0.00        | 0.00        | 0.00        | NA               |
| Transportation & Logistics | 0.00        | 0.00        | 0.00        | NA               |
| Government                 | 0.00        | 0.00        | 0.28        | NA               |
| Education                  | 0.40        | 0.53        | 0.48        | 9.4%             |
| <b>Total</b>               | <b>0.60</b> | <b>0.80</b> | <b>1.14</b> | <b>37.5%</b>     |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 178** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY VERTICAL, 2020–2026 (USD MILLION)

| Vertical                   | 2020        | 2021-e      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2020–2026) |
|----------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Aerospace & Defense        | 0.33        | 0.46        | 0.56        | 0.72        | 0.94        | 1.20        | 1.55        | 27.7%               |
| BFSI                       | 0.08        | 0.12        | 0.16        | 0.22        | 0.30        | 0.40        | 0.55        | 35.2%               |
| Healthcare & Life Sciences | 0.00        | 0.00        | 0.00        | 0.20        | 0.27        | 0.35        | 0.47        | NA                  |
| Energy & Utility           | 0.13        | 0.19        | 0.25        | 0.34        | 0.47        | 0.64        | 0.85        | 34.7%               |
| Chemical                   | 0.00        | 0.00        | 0.00        | 0.16        | 0.22        | 0.29        | 0.40        | NA                  |
| Transportation & Logistics | 0.00        | 0.00        | 0.00        | 0.12        | 0.17        | 0.23        | 0.31        | NA                  |
| Government                 | 0.41        | 0.59        | 0.76        | 1.01        | 1.38        | 1.85        | 2.50        | NA                  |
| Education                  | 0.68        | 0.99        | 1.27        | 1.22        | 1.67        | 2.25        | 3.05        | 25.1%               |
| <b>Total</b>               | <b>1.63</b> | <b>2.36</b> | <b>3.00</b> | <b>3.99</b> | <b>5.42</b> | <b>7.22</b> | <b>9.68</b> | <b>32.6%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

**TABLE 179** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2017–2019 (USD MILLION)

| Country       | 2017        | 2018        | 2019        | CAGR<br>(2015–2019) |
|---------------|-------------|-------------|-------------|---------------------|
| Latin America | 0.18        | 0.24        | 0.33        | 35.4%               |
| MEA           | 0.42        | 0.56        | 0.81        | 38.4%               |
| <b>Total</b>  | <b>0.60</b> | <b>0.80</b> | <b>1.14</b> | <b>37.5%</b>        |

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

**TABLE 180** ROW: QUANTUM COMPUTING SOFTWARE MARKET SIZE, BY COUNTRY, 2020–2026 (USD MILLION)

| Country       | 2020-e      | 2021-p      | 2022-p      | 2023-p      | 2024-p      | 2025-p      | 2026-p      | CAGR<br>(2021–2026) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Latin America | 0.47        | 0.66        | 0.83        | 1.09        | 1.45        | 1.91        | 2.52        | 30.6%               |
| MEA           | 1.16        | 1.69        | 2.17        | 2.91        | 3.97        | 5.32        | 7.17        | 33.5%               |
| <b>Total</b>  | <b>1.63</b> | <b>2.36</b> | <b>3.00</b> | <b>3.99</b> | <b>5.42</b> | <b>7.23</b> | <b>9.69</b> | <b>32.7%</b>        |

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, which may result in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

### 12.5.3 LATIN AMERICA

Latin America includes Brazil, Mexico, and the Rest of Latin America. Brazil and Mexico offer growth opportunities for quantum computing software vendors. The adoption of technological solutions is low in Latin America, but it is projected to increase in the coming years, primarily due to high investments in technologies by various organizations.

Brazil Quantum is the first community of Brazil related to quantum computing founded in July 2020. It aims to solve some crucial problems regarding quantum computing in the country. Brazil Quantum focuses on promoting and creating quantum computing content in the Portuguese language. Beyond writing blog posts and tutorials, this community also runs an interview series (BQ Talks) with quantum computing experts. Brazil Quantum is continuously coming up with new and relevant partnerships to articulate its future projects in the best way possible that include hackathons, career events, and online courses.

Colombia came up with a startup, Spin Quantum Tech, which was founded in 2018. It is part of a large Bernier Group SAS, a company that offers solutions to business problems through the use of the latest cutting-edge technologies. Spin Quantum Tech promises to cater to the technological requirements for cybersecurity, data compression, artificial intelligence, and chemical molecule simulation applications using algorithms based on quantum computing.

### 12.5.4 MIDDLE EAST AND AFRICA

The investments in quantum computing research activities should be a feature of all national strategic agendas, and this is particularly the case in the Middle East, where research budgets are relatively modest, yet the rewards of early investments are expected to be considerable. Quantum computing software and services also hold a promise for other economic priorities in the Middle East as countries in the region are making efforts to reduce their dependence on oil. In June 2018, the Dubai Electricity and Water Authority (DEWA) announced plans to work with Microsoft to develop quantum-based products to address the energy optimization requirements, as classical computers have serious limitations in this regard. This made DEWA the first organization outside the US to participate in the Microsoft Quantum program. As a part of the deal, Microsoft is working closely with DEWA to identify the areas that can expect the significant impact of quantum computing. Moreover, Dubai is planning to emerge as a technology hub for the region owing to increased startup activities to encourage local innovations. According to the Ministry of State for Artificial Intelligence of the UAE, science and technology are expected to have a significant impact on the region in the future. This is expected to contribute to the strong position of the Middle East to capitalize on other emerging technologies that promise to transform the region through the development of smart cities.

The National Institute for Theoretical Physics (NITheP) had over 30 summer schools in South Africa, which exposed students to theoretical science. It partnered with the Centre for High Performance Computing (CHPC) in 2019 and included quantum computing and computational science streams in their curriculum. These joint schools sponsored students from Africa with engineering, computer science, biology, informatics, and statistics backgrounds. The second joint school in 2020 proved to be successful with a diverse audience and significant focus on quantum computing with speakers from leading international organizations.

The Wits University was the first African partner for the quantum computing network of IBM. This partnership paved the way for 15 other African universities to have access to quantum computers of IBM. It boosted quantum computing research and experiments in the African content. The partnership also gave researchers from Africa the opportunity to be at the forefront of technological advancements in quantum computing and encouraged the adoption of quantum computing software and services in the region.

## 13 COMPETITIVE LANDSCAPE

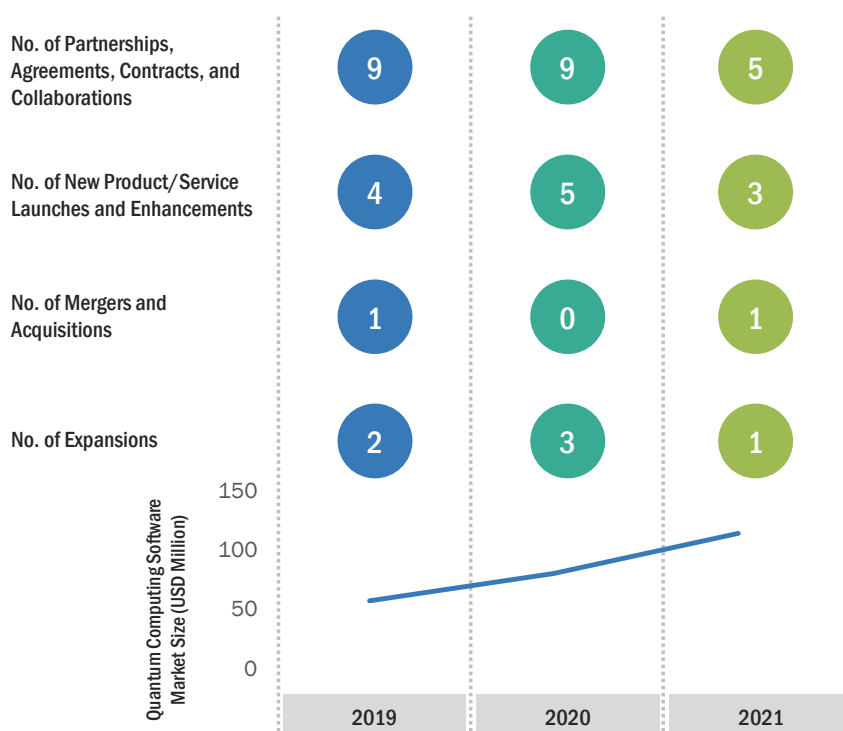
### 13.1 OVERVIEW

This section of the report provides a broad level of understanding pertaining to the competitive leadership mapping of the key vendors and start-ups in the quantum computing software market. It also covers the detailed company profiles to gauge the key strategies adopted by market players to strengthen their market position. It comprises the business overviews of vendors and company snapshots that include basic financial information, as well as solutions offered by vendors and their recent developments to provide innovative software and services to grow in the market. These players have adopted organic as well as inorganic growth strategies to expand their global presence and increase their share in the quantum computing software market. Major market players generate a significant portion of their revenues from North America, followed by Europe and APAC.

### 13.2 MARKET EVALUATION FRAMEWORK

The competitive landscape chapter analyzes and provides an overview of the prevailing competitive scenario in the quantum computing software market. The quantum computing software market, being at the nascent stage, is growing at a rapid rate across all regions. For gaining more market share and ensuring a better market position, vendors and players are adopting several growth strategies, which are highlighted in the figure below. The strategies adopted by major players primarily revolve around new product launches and enhancements, partnerships and collaborations, and mergers & acquisitions with other market players. The competitive landscape also studies the key growth strategies adopted by the market players between 2019 and 2021 to expand their presence and increase their market share in the global quantum computing software market. MarketsandMarkets' market evaluation framework captures the key data from 2019 to 2021 for the quantum computing software market.

**FIGURE 34** QUANTUM COMPUTING SOFTWARE MARKET EVALUATION FRAMEWORK



Note: The numbers in bubbles indicate the total number of market developments by market players who hold nearly 70% of the total market share.

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

### 13.3 MARKET RANKING

Players in the quantum computing software market were identified through secondary research. The share of each market player, in terms of value, was inferred from annual reports, presentations, and secondary research sources. Primary research interviews were conducted with key opinion leaders, such as CEOs, directors, industry experts, and other executives to validate revenue and market share.

The figure below shows the leading vendors in the global quantum computing software market:

**FIGURE 35** MARKET RANKING IN 2020



Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

### 13.4 MARKET SHARE ANALYSIS

Players in the global quantum computing software market were identified through secondary research, and their market share, in terms of value, was inferred from their annual reports, presentations, and other secondary sources. Primary research interviews were conducted with key opinion leaders in the industry, such as CEOs, directors, industry experts, and other executives, to validate revenue and market share. The following table illustrates the market share of the top 5 players in the global quantum computing software market.

**TABLE 181** QUANTUM COMPUTING SOFTWARE MARKET: DEGREE OF COMPETITION

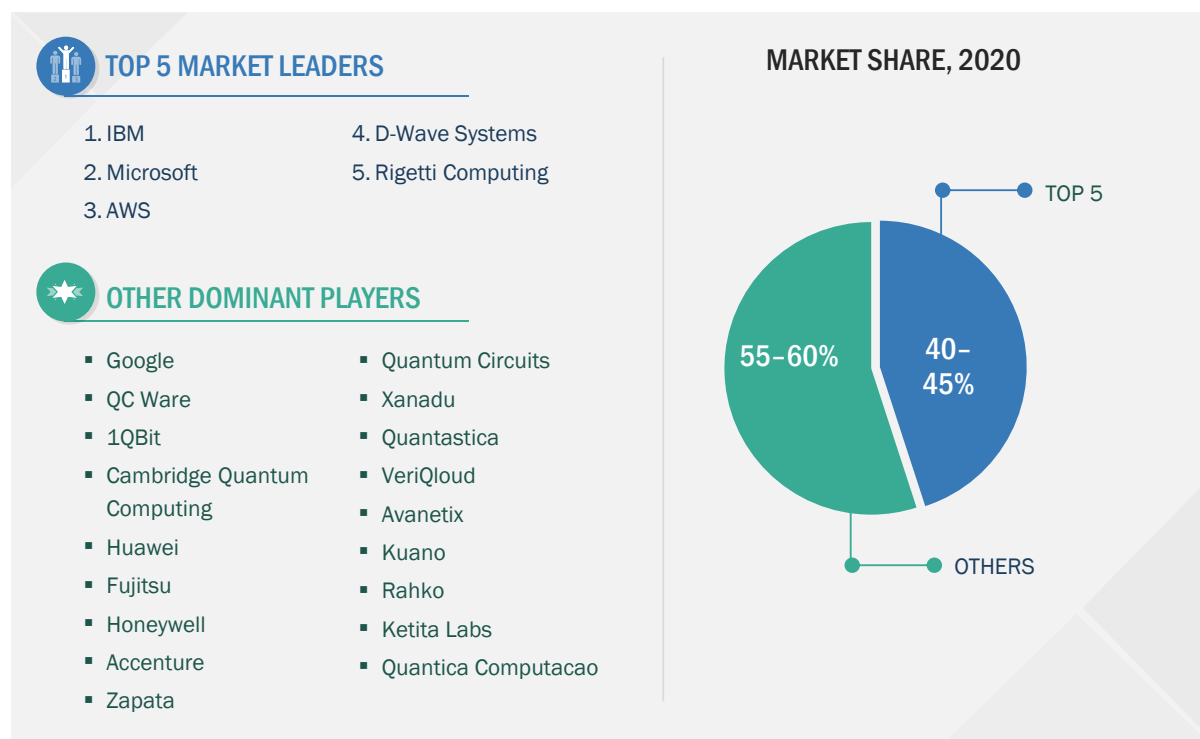
| DEGREE OF COMPETITION               | COMPETITIVE |
|-------------------------------------|-------------|
| Total Market Share of Top-5 players | 40–45%      |
| IBM                                 | 15–20%      |
| D-Wave Systems                      | 10–15%      |
| Microsoft                           | 10–12%      |
| AWS                                 | 8–10%       |
| Rigetti Computing                   | 6–7%        |

Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

The market being at the theoretical phase is highly fragmented, majorly by the emergence of a large number of start-ups in the market. The top 5 players constitute only 40–45% of the global quantum computing software market, with other dominant players comprising 55–60% of the global market. In addition, the market is witnessing the entry of new players and comprises a large number of small- and

medium-sized quantum computing software players, leading to noticeable market fragmentation. The following figure shows the market share of the global quantum computing software market in 2020.

**FIGURE 36** MARKET SHARE ANALYSIS OF COMPANIES IN QUANTUM COMPUTING SOFTWARE MARKET



Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets Analysis

IBM is one of the prominent players in the Quantum Computing Software market. The company was founded in 1911 and headquartered in New York, US. The company is a multinational technology and consulting corporation that offers infrastructure, hosting, and consulting services. The company in 2016 made a quantum computer available to the public by connecting it to the cloud. In September 2019, it opened a Quantum Computation Center. The Quantum Computation Center offers about 100 IBM clients, academic institutions, and more than 200,000 registered users access to this cutting-edge technology through a collaborative effort called the IBM Q Network and Qiskit, IBM's open-source development platform for quantum computing. Through these efforts, IBM is exploring the ways quantum computing can address the most complicated problems faced while training the workforce to use this technology.

D-Wave Systems was founded in 1999 and is headquartered in British Columbia, Canada. D-Wave is one of the leaders in the development and delivery of quantum computing systems, software, and services and is the commercial supplier of quantum computers. D-Wave Systems renders quantum computing systems and services for logistics, artificial intelligence, material sciences, drug discovery, cybersecurity, fault detection, and financial modeling applications. D-Wave Systems offers customer support, site preparation, site installation, training package, and on-site engineering support services to its customers. The products of the company are used primarily in optimization, machine learning, simulation, pattern recognition, anomaly detection, and financial analysis applications.

## 13.5 KEY MARKET DEVELOPMENTS

**FIGURE 37** KEY DEVELOPMENTS IN QUANTUM COMPUTING SOFTWARE MARKET FOR 2019–2021

| COMPANY NAME      | ORGANIC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | INORGANIC                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | NEW PRODUCT LAUNCHES/ PRODUCT ENHANCEMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ACQUISITIONS                                                                                                                                                                                                                                                                                                                                         | AGREEMENTS, COLLABORATIONS, AND PARTNERSHIPS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| IBM               | In June 2020, IBM Quantum launched the IBM Quantum Researchers Program to enable academic researchers to perform advanced scientific work in quantum.                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                      | In January 2020, IBM partnered with Daimler AG, the parent company of Mercedes-Benz, to enhance the capacity and increase the charging speed of batteries of electric vehicles. These companies used a quantum computer to model the dipole moment of three lithium-containing molecules that paves the way for the development of the next-generation lithium-sulfur (Li-S) batteries that will be more powerful, long-lasting, and cost-effective than lithium-ion batteries.  |
| D-WAVE SYSTEMS    | In February 2021, D-Wave Systems has announced the introduction of Leap quantum cloud service in Singapore. Developers, researchers, and businesses in Singapore can now access D-Wave's Advantage quantum computer, hybrid quantum/classical solvers, and the QAE in real time via Leap to increase the development of business-critical, in-production hybrid applications.                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                      | In May 2021, CINECA, the Italian inter-university consortium, and D-Wave have announced a formal collaboration to offer Italian universities, researchers, and developers extended access to practical quantum computing technology and resources through D-Wave's Leap quantum cloud service.                                                                                                                                                                                                                                                                      |
| MICROSOFT         | In May 2020, Azure Quantum of Microsoft is moving to a limited preview, which will allow selected customers and partners to develop their programs and skillsets to tackle business challenges that quantum computing is expected to solve. Developers interested in becoming quantum-ready can access new learning modules available free on Microsoft Learn.                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                      | In April 2021, Ally Financial joined  Azure Quantum program to transform customer experiences in the financial industry. Ally is working with Microsoft's Quantum team to solve optimization problems and explore quantum computing.                                                                                                                                                                                                                                            |
| AWS               | In August 2020, Amazon Web Services announced the release of Amazon Braket, which can be accessed through both classically powered circuit simulators and quantum computers of D-Wave Systems, IonQ, and Rigetti Computing..                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| RIGETTI COMPUTING | In December 2020, Rigetti released Quil-T, making gate definitions and pulse parameters completely editable across the entire Rigetti Aspen-8 processor. This product development gives QCS users of Rigetti a fine-grained control over the parameters used to enact every gate operation, thereby unlocking the ability to experiment with optimal controls, as well as control precise pulse timing. It also facilitates the development of novel gate schemes that can significantly reduce overall gate depth. | In July 2019, Rigetti acquired QxBranh. This acquisition combines the leadership of QxBranh in customer-facing application development with the leading hardware and software stack of Rigetti. By joining their forces, these companies have tightened the feedback loop in both directions and have accelerated progress toward quantum advantage. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Note –  indicates developments that have a significant impact on the market

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

### 13.5.1 PRODUCT LAUNCHES AND ENHANCEMENTS

The major companies that have incorporated new product launches and enhancements strategy include D-Wave Systems, Rigetti Computing, Honeywell, AWS, IBM, Microsoft, Cambridge Quantum Computing, and QC Ware.

**TABLE 182** QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS, 2019–2021

| MONTH YEAR     | COMPANY NAME      | PRODUCT NAME                    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|-------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| February 2021  | D-Wave Systems    | Leap Quantum Cloud Service      | D-Wave Systems has announced the introduction of Leap quantum cloud service in Singapore. Developers, researchers, and businesses in Singapore can now access D-Wave's Advantage quantum computer, hybrid quantum/classical solvers, and the QAE in real time via Leap to increase the development of business-critical, in-production hybrid applications.                                                                                                                                                      |
| December 2020  | Rigetti Computing | Quil-T                          | Rigetti released Quil-T, making gate definitions and pulse parameters completely editable across the entire Rigetti Aspen-8 processor. This product development gives QCS users of Rigetti a fine-grained control over the parameters used to enact every gate operation, thereby unlocking the ability to experiment with optimal controls, as well as control precise pulse timing. It also facilitates the development of novel gate schemes that can significantly reduce overall gate depth.                |
| October 2020   | Honeywell         | System Model H1                 | Honeywell announced a breakthrough in quantum computing technology with the introduction of its next-generation quantum computer, the System Model H1. This next-generation computer features differentiated quantum charge-coupled device (QCCD) trapped-ion technology of Honeywell. It is strategically designed to be rapidly upgraded throughout its lifetime.                                                                                                                                              |
| September 2020 | D-Wave Systems    | Quantum Computing Platform      | D-Wave Systems announced the general availability of its next-generation quantum computing platform, incorporated with new hardware, software, and tools to enable and accelerate the delivery of in-production quantum computing applications. Available in the Leap quantum cloud service, the platform includes the Advantage quantum system, with more than 5000 qubits and 15-way qubit connectivity, in addition to an expanded hybrid solver service that can run problems with up to 1 million variables |
| August 2020    | AWS               | Amazon Braket                   | Amazon Web Services announced the release of Amazon Braket, which can be accessed through both classically powered circuit simulators and quantum computers of D-Wave Systems, IonQ, and Rigetti Computing.                                                                                                                                                                                                                                                                                                      |
| June 2020      | IBM               | IBM Quantum Researchers Program | IBM Quantum launched the IBM Quantum Researchers Program to enable academic researchers to perform advanced scientific work in quantum.                                                                                                                                                                                                                                                                                                                                                                          |

|                |                             |                    |                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|-----------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2020      | D-Wave Systems              | Leap2              | D-Wave Systems announced the immediate availability of Leap2, the first quantum cloud service designed for developers and organizations to easily develop and deploy real-world hybrid quantum applications with practical impact.                                                                                                                                         |
| May 2020       | Microsoft                   | Azure Quantum      | Azure Quantum of Microsoft is moving to a limited preview, which will allow selected customers and partners to develop their programs and skillsets to tackle business challenges that quantum computing is expected to solve. Developers interested in becoming quantum-ready can access new learning modules available free on Microsoft Learn.                          |
| April 2020     | Cambridge Quantum Computing | t ket) 0.5 Version | Cambridge Quantum Computing announced the release of a new version of its class-leading quantum software platform. The release of the t ket) 0.5 version brings with it important enhancements to its existing capabilities, as well as a variety of new features.                                                                                                         |
| September 2019 | QC Ware                     | Forge              | QC Ware launched the public beta of its Forge cloud service. Forge was developed by the company to enable large enterprises and public-sector organizations to start developing quantum skills and prepare for the potential disruptions that quantum computing will bring to the market.                                                                                  |
| January 2019   | IBM                         | IBM Q System One   | IBM unveiled IBM Q System One, the integrated universal approximate quantum computing system designed for scientific and commercial use. IBM also announced plans to open its first IBM Q Quantum Computation Center for commercial clients in Poughkeepsie, New York, in 2019.                                                                                            |
| May 2019       | IBM                         | IBM Q Experience   | IBM introduced a number of significant enhancements to its IBM Q Experience quantum cloud services and software platform. The public beta launched by the company enables existing and new users to make circuits and application-building tools. These services further accelerate the movement of research and development from IBM Q systems to practical applications. |

Source: Press Releases

## 13.5.2 DEALS

The major companies that have incorporated acquisition, partnership, and agreement strategy include Honeywell, Cambridge Quantum Computing, D-Wave Systems, IBM, Fujitsu, QC Ware, Microsoft, Rigetti Computing, and IBM.

**TABLE 183** QUANTUM COMPUTING SOFTWARE MARKET: DEALS, 2019–2021

| MONTH-YEAR | DEAL TYPE | COMPANY NAME 1              | COMPANY NAME 2              | DESCRIPTION                                                                                                                                                                                                                                                                                                                                               |
|------------|-----------|-----------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2021  | Merger    | Honeywell Quantum Solutions | Cambridge Quantum Computing | Honeywell announced that Honeywell Quantum Solutions (HQS) and Cambridge Quantum Computing (CQC) will combine to form what the companies said will be the largest, most advanced standalone quantum computing company in the world, setting the pace for what is projected to become a USD 1 trillion quantum computing industry over the next 3 decades. |

|               |               |                             |                             |                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|---------------|-----------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2021      | Collaboration | D-Wave Systems              | CINECA                      | CINECA, the Italian inter-university consortium, and D-Wave have announced a formal collaboration to offer Italian universities, researchers, and developers extended access to practical quantum computing technology and resources through D-Wave's Leap quantum cloud service.                                                                                                        |
| April 2021    | Collaboration | QC Ware                     | Goldman Sachs               | The collaboration of Goldman Sachs and QC Ware has brought a new way to price risky assets within reach of quantum computers. Goldman Sachs and QC Ware researchers have designed new, robust quantum algorithms that outperform state-of-the-art classical algorithms for Monte Carlo simulations and can be used on near-term quantum hardware expected to be available in 5–10 years. |
| April 2021    | Partnership   | D-Wave Systems              | NEC Australia               | NEC Australia and D-Wave have partnered with the Australian Department of Defense to provide quantum computing capabilities to optimize last-mile resupply.                                                                                                                                                                                                                              |
| April 2021    | Partnership   | Microsoft                   | Ally Financial              | Ally Financial joined Azure Quantum program to transform customer experiences in the financial industry. Ally is working with Microsoft's Quantum team to solve optimization problems and explore quantum computing.                                                                                                                                                                     |
| December 2020 | Collaboration | QC Ware                     | IonQ                        | QC Ware and IonQ collaborated to perform an experiment that demonstrated that machine learning on near-term quantum computers can achieve the same or improved level of accuracy as classical computers and can be faster than them.                                                                                                                                                     |
| November 2020 | Collaboration | Fujitsu                     | University of Toronto       | Fujitsu Laboratories Limited, in collaboration with the University of Toronto, has successfully developed a new parallel search technology to help achieve megabit-class performance for large-scale problems, representing an important technical milestone for its Digital Annealer.                                                                                                   |
| November 2020 | Partnership   | Cambridge Quantum Computing | National Physics Laboratory | Cambridge Quantum Computing confirmed that the National Physics Laboratory (NPL) of the UK has announced that it will be supporting the company in the development of its novel photonics-based product for generating certifiable quantum entropy (commonly referred to as certifiable QRNG).                                                                                           |
| June 2020     | Partnership   | D-Wave Systems              | NEC Corporation             | NEC Corporation and D-Wave Systems announced that they had begun joint activities to combine the computing power of NEC's systems with the quantum computing power of D-Wave's systems, software, and cloud service to bring those combined capabilities to customers in Japan. NEC has made a USD 10 million investment in D-Wave in connection with this initiative.                   |
| January 2020  | Partnership   | IBM                         | Daimler AG                  | IBM partnered with Daimler AG, the parent company of Mercedes-Benz, to enhance the capacity and increase the charging speed of batteries of electric vehicles. These companies used a quantum computer to model the dipole moment of three lithium-containing molecules that paves the way for the development of                                                                        |

|               |               |                   |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|---------------|-------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |               |                   |              | the next-generation lithium-sulfur (Li-S) batteries that will be more powerful, long-lasting, and cost-effective than lithium-ion batteries.                                                                                                                                                                                                                                                                                                        |
| November 2019 | Partnership   | IBM               | Unitary Fund | IBM partnered with the Unitary Fund to provide grants and priority access to certain IBM Q systems. Similar to the quantum computing mission of IBM, the Unitary Fund aims to create a quantum technology industry that benefits most people in terms of the adoption of quantum technology.                                                                                                                                                        |
| November 2019 | Collaboration | IBM               | Wells Fargo  | IBM collaborated with Wells Fargo to focus on research and learning that are intended to enhance the artificial intelligence and quantum computing capabilities of the company. As a result of this collaboration, Wells Fargo plans to accelerate its learning to carry out innovations that reimagine the future of financial services by making them simple, fast, safe, and convenient for users.                                               |
| July 2019     | Acquisition   | Rigetti Computing | QxBranch     | Rigetti acquired QxBranch. This acquisition combines the leadership of QxBranch in customer-facing application development with the leading hardware and software stack of Rigetti. By joining their forces, these companies have tightened the feedback loop in both directions and have accelerated progress toward quantum advantage.                                                                                                            |
| March 2019    | Collaboration | IBM               | CERN         | IBM collaborated with CERN to make innovations in computing solutions.                                                                                                                                                                                                                                                                                                                                                                              |
| January 2019  | PARTNERSHIP   | IBM               | EXXONMOBIL   | IBM partnered with EXXONMOBIL to carry out advancements in quantum computing, as well as explore its scientific and business applications. These advancements aim at the development of next-generation energy and manufacturing technologies. Advancements in quantum computing are expected to allow EXXONMOBIL to address challenging computational problems in a range of applications, including the optimization of power grids of countries. |

Source: Press Releases

## 13.6 COMPANY EVALUATION MATRIX

The company evaluation matrix provides information about the major players in the quantum computing software market. It outlines the findings and analysis of how well each market vendor performs within the predefined competitive leadership mapping criteria. The vendor evaluations are based on two broad categories: strength of product portfolio and business strategy excellence. Each category carries various criteria based on which the vendors have been evaluated. The evaluation criteria considered under the strength of product portfolio include the breadth and depth of product offerings; product features and functionalities; delivery; scalability; and sales, support, and customer engagement. The evaluation criteria considered under business strategy excellence include geographic presence, viability, growth strategy, partner ecosystem, and roadmap across the globe.

**TABLE 184** COMPANY PRODUCT FOOTPRINT

| COMPANY NAME                | COMPONENT SCORE | APPLICATION FOOTPRINT | REGION FOOTPRINT | TOTAL SCORE |
|-----------------------------|-----------------|-----------------------|------------------|-------------|
| IBM                         | 4.00            | 4.00                  | 4.00             | 4.00        |
| Microsoft                   | 4.00            | 3.00                  | 4.00             | 3.67        |
| AWS                         | 4.00            | 3.00                  | 4.00             | 3.67        |
| Google                      | 4.00            | 4.00                  | 4.00             | 4.00        |
| D-Wave                      | 4.00            | 4.00                  | 2.40             | 3.47        |
| Rigetti                     | 4.00            | 3.00                  | 2.40             | 3.13        |
| QC Ware                     | 4.00            | 3.00                  | 1.60             | 2.87        |
| 1QBIT                       | 4.00            | 3.00                  | 0.80             | 2.60        |
| Cambridge Quantum Computing | 4.00            | 3.00                  | 2.40             | 3.13        |
| Huawei                      | 4.00            | 4.00                  | 4.00             | 4.00        |
| Fujitsu                     | 4.00            | 1.00                  | 4.00             | 3.00        |
| Honeywell                   | 4.00            | 4.00                  | 4.00             | 4.00        |
| Accenture                   | 4.00            | 3.00                  | 4.00             | 3.67        |
| Zapata                      | 2.00            | 2.00                  | 0.80             | 1.60        |
| Quantum Circuits            | 4.00            | 3.00                  | 0.80             | 2.60        |
| Xanadu                      | 4.00            | 2.00                  | 0.80             | 2.27        |

Source: MarketsandMarkets Analysis

**TABLE 185** COMPANY APPLICATION FOOTPRINT

| COMPANY                     | OPTIMIZATION | SIMULATION | MACHINE LEARNING | OTHERS | OVERALL APPLICATION FOOTPRINT |
|-----------------------------|--------------|------------|------------------|--------|-------------------------------|
| IBM                         | Y            | Y          | Y                | Y      | 4.00                          |
| Microsoft                   | Y            | Y          | Y                | N      | 3.00                          |
| AWS                         | Y            | Y          | Y                | N      | 3.00                          |
| Google                      | Y            | Y          | Y                | Y      | 4.00                          |
| D-Wave                      | Y            | Y          | Y                | Y      | 4.00                          |
| Rigetti                     | Y            | Y          | N                | Y      | 3.00                          |
| QC Ware                     | Y            | Y          | Y                | N      | 3.00                          |
| 1QBIT                       | Y            | Y          | Y                | N      | 3.00                          |
| Cambridge Quantum Computing | Y            | Y          | Y                | N      | 3.00                          |
| Huawei                      | Y            | Y          | Y                | Y      | 4.00                          |
| Fujitsu                     | Y            | N          | N                | N      | 1.00                          |
| Honeywell                   | Y            | Y          | Y                | Y      | 4.00                          |
| Accenture                   | Y            | Y          | Y                | N      | 3.00                          |
| Zapata                      | Y            | Y          | N                | N      | 2.00                          |
| Quantum Circuits            | Y            | Y          | Y                | N      | 3.00                          |
| Xanadu                      | N            | N          | Y                | Y      | 2.00                          |

Source: MarketsandMarkets Analysis

**TABLE 186** COMPANY REGION FOOTPRINT

| COMPANY                     | NORTH AMERICA | EUROPE | ASIA PACIFIC | MIDDLE EAST & AFRICA | LATIN AMERICA | OVERALL REGION FOOTPRINT |
|-----------------------------|---------------|--------|--------------|----------------------|---------------|--------------------------|
| IBM                         | Y             | Y      | Y            | Y                    | Y             | 4.00                     |
| Microsoft                   | Y             | Y      | Y            | Y                    | Y             | 4.00                     |
| AWS                         | Y             | Y      | Y            | Y                    | Y             | 4.00                     |
| Google                      | Y             | Y      | Y            | Y                    | Y             | 4.00                     |
| D-Wave                      | Y             | Y      | Y            | N                    | N             | 2.40                     |
| Rigetti                     | Y             | Y      | Y            | N                    | N             | 2.40                     |
| QC Ware                     | Y             | Y      | N            | N                    | N             | 1.60                     |
| 1QBIT                       | Y             | N      | N            | N                    | N             | 0.80                     |
| Cambridge Quantum Computing | Y             | Y      | Y            | N                    | N             | 2.40                     |

|                  |   |   |   |   |   |      |
|------------------|---|---|---|---|---|------|
| Huawei           | Y | Y | Y | Y | Y | 4.00 |
| Fujitsu          | Y | Y | Y | Y | Y | 4.00 |
| Honeywell        | Y | Y | Y | Y | Y | 4.00 |
| Accenture        | Y | Y | Y | Y | Y | 4.00 |
| Zapata           | Y | N | N | N | N | 0.80 |
| Quantum Circuits | Y | N | N | N | N | 0.80 |
| Xanadu           | Y | N | N | N | N | 0.80 |

Source: MarketsandMarkets Analysis

**TABLE 187** COMPANY COMPONENT FOOTPRINT

| COMPANY NAME                | SOFTWARE | SERVICES | TOTAL SCORE |
|-----------------------------|----------|----------|-------------|
| IBM                         | Y        | Y        | 4.00        |
| Microsoft                   | Y        | Y        | 4.00        |
| AWS                         | Y        | Y        | 4.00        |
| Google                      | Y        | Y        | 4.00        |
| D-Wave                      | Y        | Y        | 4.00        |
| Rigetti                     | Y        | Y        | 4.00        |
| QC Ware                     | Y        | Y        | 4.00        |
| 1QBIT                       | Y        | Y        | 4.00        |
| Cambridge Quantum Computing | Y        | Y        | 4.00        |
| Huawei                      | Y        | Y        | 4.00        |
| Fujitsu                     | Y        | Y        | 4.00        |
| Honeywell                   | Y        | Y        | 4.00        |
| Accenture                   | Y        | Y        | 4.00        |
| Zapata                      | Y        | N        | 2.00        |
| Quantum Circuits            | Y        | Y        | 4.00        |
| Xanadu                      | Y        | Y        | 4.00        |

Note: Component (Software and Services) Score has been decided on the basis of feature, tools provided and other factors

Source: MarketsandMarkets Analysis

### 13.6.1 STAR

IBM, Microsoft, D-Wave Systems, Rigetti Computing, AWS, and Google are the leaders in the quantum computing software market and are recognized as star companies. They offer strong solution portfolios in the quantum computing software market. These vendors have been marking their presence in the market by offering customized software and services as per the requirements of end users and adopting various growth strategies to achieve advanced infrastructure in the quantum computing software market.

### 13.6.2 EMERGING LEADER

Honeywell, Fujitsu, QC Ware, and Huawei are recognized as emerging leaders in the quantum computing software market. They possess innovative services to cater to future quantum computing software demands. The companies under this category are concerned about their product portfolio and have a high potential to build strong business strategies to expand their business and stay at par with the stars. These vendors have been offering quantum computing software and solutions to their customers with an aim to fulfill their demands. Emerging leaders have been at the forefront in terms of the deployment of their services.

### 13.6.3 PERVASIVE

Accenture is recognized as a pervasive company in the quantum computing software market. Pervasive companies offer innovative services and have an extensive network of channel partners and resellers to increase the deployment of their services across a multitude of verticals. Pervasive companies have been consistently witnessing revenue growth in the quantum computing software market, and their market position is boosted through organic and inorganic ventures.

### 13.6.4 PARTICIPANT

Participant companies are vendors with niche product offerings that have started to gain a foothold in the market. They do not have strong business strategies as compared to other established vendors. They are new entrants in the market and, hence, require some time to gain significant traction in the market. IQBit, Quantum Circuits, Xanadu, Zapata, and Cambridge Quantum Computing are some of the key players considered in this category.

**FIGURE 38** GLOBAL QUANTUM COMPUTING SOFTWARE MARKET, COMPANY EVALUATION MATRIX



Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

## 13.7 START-UP/SME EVALUATION MATRIX, 2021

The start-ups/SMEs selected for the matrix were founded after 2015. Vendor evaluations are based on two broad categories—strength of product portfolio and business strategy excellence. Each category carries various criteria based on which the vendors have been evaluated. The evaluation criteria considered under the strength of product portfolio include the breadth and depth of service offerings, features and functionality (based on application areas served), delivery and deployment of services (based on organization size and deployment type), and service innovation (based on strategies for new product development plans). The evaluation criteria considered under business strategy excellence include reach (based on geographic presence), channel strategy (based on channel partners that the vendors cater to and inorganic growth comprising of partnerships, collaborations, and acquisitions), funding (collected based on rounds in USD million), and vision alignment.

### 13.7.1 PROGRESSIVE COMPANY

Quantastica, VeriQloud, Avonetix, and Rahko are progressive companies in the quantum computing software market. They offer a strong services portfolio in this market and have been marking their presence by offering customized services as per user requirements.

### 13.7.2 RESPONSIVE COMPANY

Aliro Quantum, Multiverse Computing, and Ketita Labs are recognized as responsive companies in the quantum computing software market. They possess innovative services to cater to future quantum computing software demands.

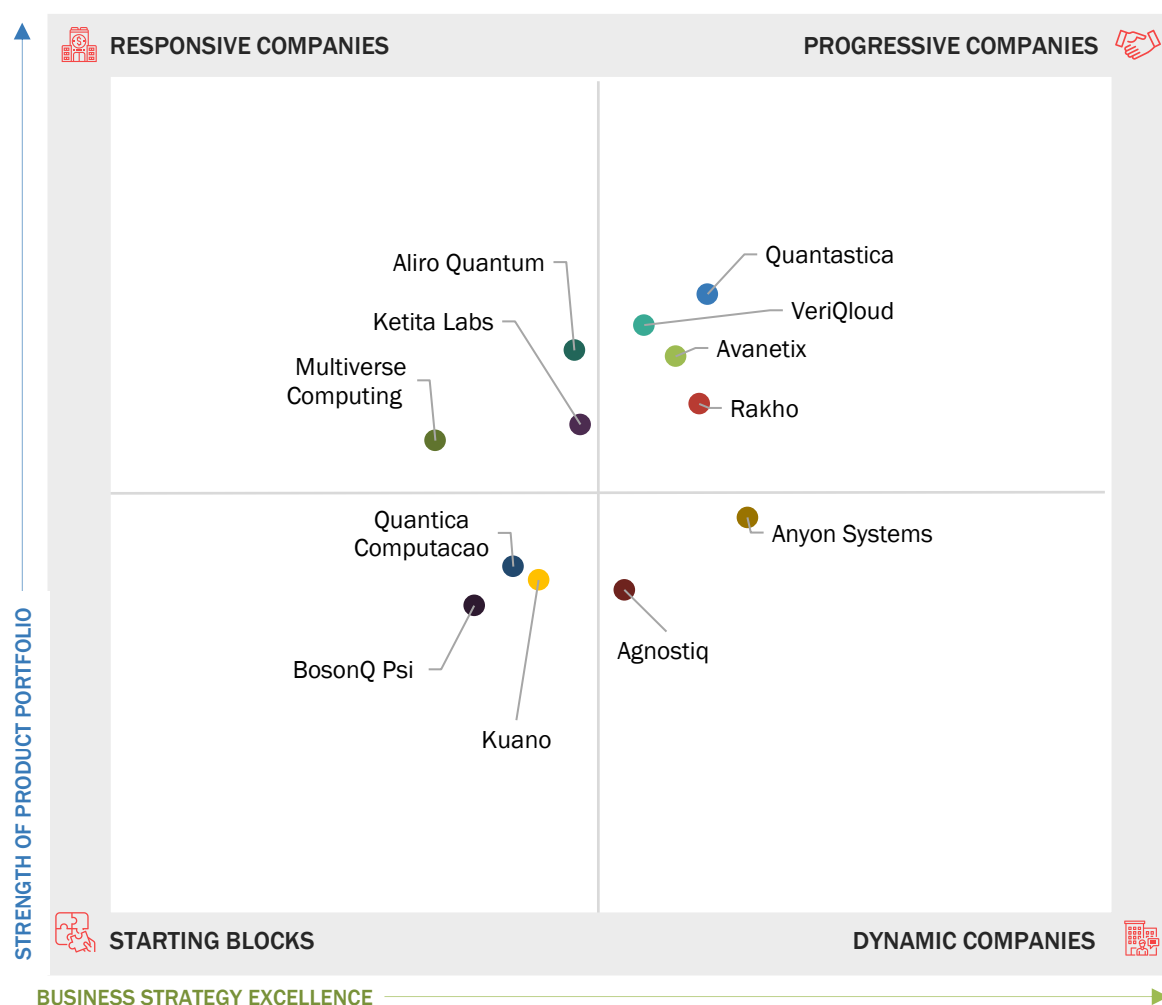
### 13.7.3 DYNAMIC COMPANY

Anyon Systems and Agnostiq are recognized as dynamic companies in the quantum computing software market. They offer an innovative services portfolio and have an extensive network of channel partners and resellers to increase the deployment of their services.

### 13.7.4 STARTING BLOCK

Quantica Computacao, Kuano, and BosonQ are fall into this quadrant. Most of the starting blocks have been undertaking multiple acquisitions for boosting their sales capabilities across regions to offer integrated services to a wide range of clients.

**FIGURE 39** GLOBAL QUANTUM COMPUTING SOFTWARE: STARTUP/SME EVALUATION MATRIX, 2021



Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

## 14 COMPANY PROFILES

### 14.1 INTRODUCTION

In this section, we have provided a brief overview of the major players operating in the quantum computing software market. It includes information about solutions and services, key strengths, weaknesses, competitive threats, and recent developments during the last 3 years of the top five market players. We have also analyzed the organic and inorganic growth strategies adopted by various players in the market to expand their global presence and increase the overall market share. The major players in the quantum computing software market generate a significant part of their revenue from North America, followed by Europe and APAC

### 14.2 MAJOR PLAYERS

#### 14.2.1 IBM

##### 14.2.1.1 Business overview

IBM was founded in 1911 and is headquartered in New York, US. It is a multinational technology and consulting corporation that offers infrastructure, hosting, and consulting services. The company operates through five major business segments: Cloud and Cognitive Software, Global Business Services, Global Technology Services, Systems, and Global Financing. IBM Cloud has emerged as a platform of choice for all business applications, as it is AI compatible. It is a unifying platform that integrates IBM's capabilities with a single architecture and spans over public and private cloud platforms. With this powerful cloud platform, the company can cater to the requirements of different businesses across the globe. IBM caters to various verticals, including aerospace & defense, education, healthcare, oil & gas, automotive, electronics, insurance, retail and consumer products, banking and finance, energy and utilities, life sciences, telecommunications, media and entertainment, chemical, government, manufacturing, travel & transportation, construction, and metals & mining. The company has a strong presence in the Americas, Europe, MEA, and APAC and clients in more than 175 countries.

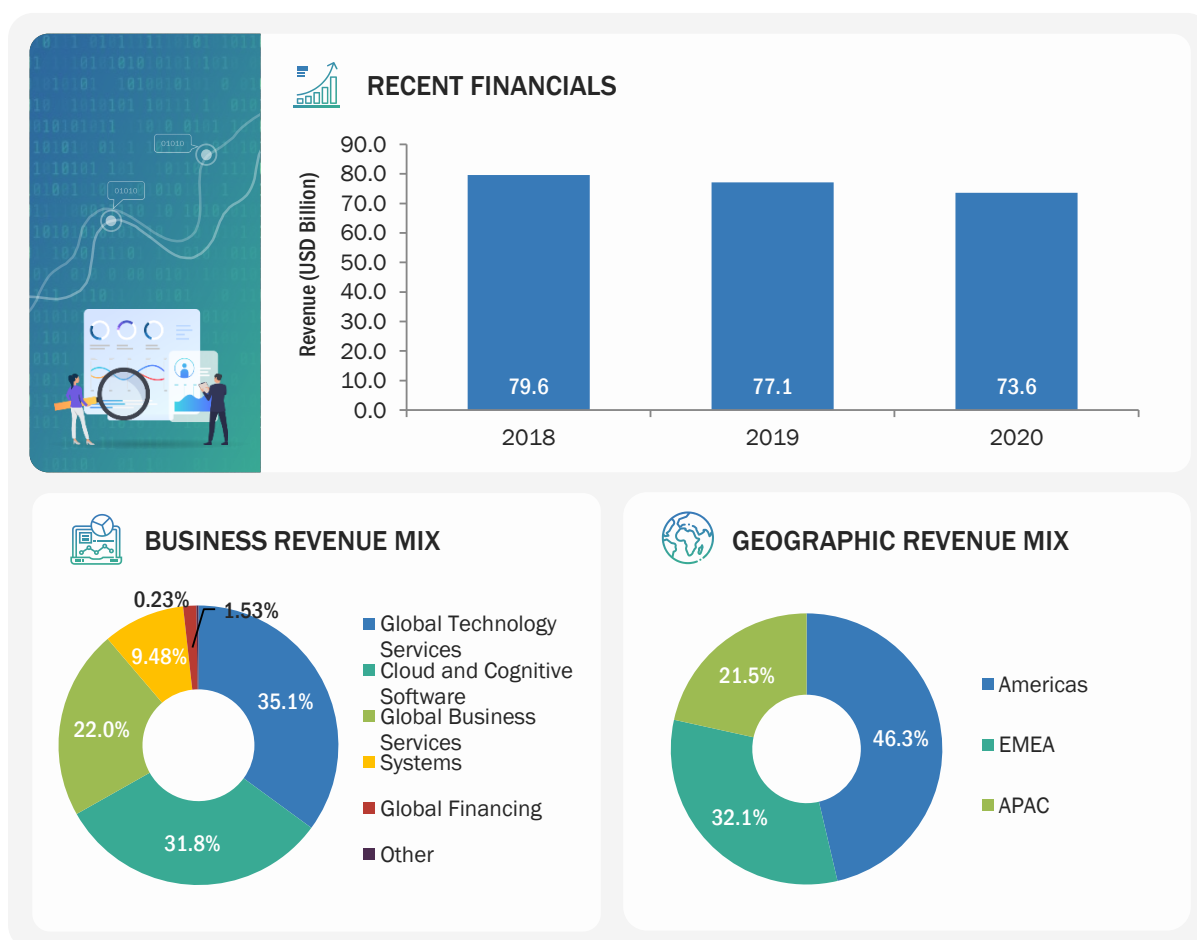
IBM is one of the major players in the quantum computing ecosystem. The company in 2016 made a quantum computer available to the public by connecting it to the cloud. In September 2019, it opened a Quantum Computation Center. The Quantum Computation Center offers about 100 IBM clients, academic institutions, and more than 200,000 registered users access to this cutting-edge technology through a collaborative effort called the IBM Q Network and Qiskit, IBM's open-source development platform for quantum computing. Through these efforts, IBM is exploring the ways quantum computing can address the most complicated problems faced while training the workforce to use this technology.

**TABLE 188 IBM: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE—2020 |                    |
|--------------------------|--------------------|
| Founded:                 | 1911               |
| Headquarters:            | New York, US       |
| No. of Employees:        | 345,000            |
| Overall Annual Revenue   | USD 73.6 Billion   |
| R&D Expenditure:         | USD 6.0 Billion    |
| Ownership:               | Public (NYSE: IBM) |

Source: Company Website, Annual Reports, and SEC Filings

**FIGURE 40** IBM: COMPANY SNAPSHOT



Note: The company's fiscal year ends on December 31. The snapshot includes information for the overall company

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 189** IBM: PRODUCTS OFFERED

| PRODUCT                                   | DESCRIPTION                                                                                                                                                                               | PRODUCT TYPE   | APPLICATION                                                                                                                  |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------|
| IBM Q System One                          | IBM Q System One enables universal superconducting quantum computers to operate beyond the confines of research laboratories.                                                             | Quantum System | <ul style="list-style-type: none"> <li>Machine Learning</li> <li>Optimization</li> <li>Materials</li> <li>Finance</li> </ul> |
| IBM Quantum Services                      | IBM Quantum Services gives secure, portable, containerized runtime and programming tools to access the latest world-leading cloud-based quantum systems and simulators via the IBM Cloud. | Service        | <ul style="list-style-type: none"> <li>Machine Learning</li> <li>Optimization</li> <li>Materials</li> <li>Finance</li> </ul> |
| Quantum Information Software Kit (Qiskit) | IBM Q simulators and quantum devices can be easily accessed over the cloud through Qiskit and an open-source quantum programming framework.                                               | Software       | <ul style="list-style-type: none"> <li>Machine learning</li> <li>Optimization</li> <li>Materials</li> <li>Finance</li> </ul> |

Source: Company Website

### 14.2.1.2 Recent developments

**TABLE 190** IBM: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME                    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                |
|----------------|--------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2020      | IBM          | IBM Quantum Researchers Program | IBM Quantum launched the IBM Quantum Researchers Program to enable academic researchers to perform advanced scientific work in quantum.                                                                                                                                                                                                                                    |
| January 2019   | IBM          | IBM Q System One                | IBM unveiled IBM Q System One, the integrated universal approximate quantum computing system designed for scientific and commercial use. IBM also announced plans to open its first IBM Q Quantum Computation Center for commercial clients in Poughkeepsie, New York, in 2019.                                                                                            |
| May 2019       | IBM          | IBM Q Experience                | IBM introduced a number of significant enhancements to its IBM Q Experience quantum cloud services and software platform. The public beta launched by the company enables existing and new users to make circuits and application-building tools. These services further accelerate the movement of research and development from IBM Q systems to practical applications. |

Source: Press Releases

**TABLE 191** IBM: QUANTUM COMPUTING SOFTWARE MARKET: DEALS

| MONTH YEAR    | DEAL TYPE     | COMPANY 1 | COMPANY 2    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|---------------|-----------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| January 2020  | Partnership   | IBM       | Daimler AG   | IBM partnered with Daimler AG, the parent company of Mercedes-Benz, to enhance the capacity and increase the charging speed of batteries of electric vehicles. These companies used a quantum computer to model the dipole moment of three lithium-containing molecules that paves the way for the development of the next-generation lithium-sulfur (Li-S) batteries that will be more powerful, long-lasting, and cost-effective than lithium-ion batteries. |
| November 2019 | Partnership   | IBM       | Unitary Fund | IBM partnered with the Unitary Fund to provide grants and priority access to certain IBM Q systems. Similar to the quantum computing mission of IBM, the Unitary Fund aims to create a quantum technology industry that benefits most people in terms of the adoption of quantum technology.                                                                                                                                                                   |
| November 2019 | Collaboration | IBM       | Wells Fargo  | IBM collaborated with Wells Fargo to focus on research and learning that are intended to enhance the artificial intelligence and quantum computing capabilities of the company. As a result of this collaboration, Wells Fargo plans to                                                                                                                                                                                                                        |

|              |               |     |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|---------------|-----|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |               |     |            | accelerate its learning to carry out innovations that reimagine the future of financial services by making them simple, fast, safe, and convenient for users.                                                                                                                                                                                                                                                                                       |
| March 2019   | Collaboration | IBM | CERN       | IBM collaborated with CERN to make innovations in computing solutions.                                                                                                                                                                                                                                                                                                                                                                              |
| January 2019 | PARTNERSHIP   | IBM | EXXONMOBIL | IBM partnered with EXXONMOBIL to carry out advancements in quantum computing, as well as explore its scientific and business applications. These advancements aim at the development of next-generation energy and manufacturing technologies. Advancements in quantum computing are expected to allow EXXONMOBIL to address challenging computational problems in a range of applications, including the optimization of power grids of countries. |

Source: Press Releases

### 14.2.1.3 IBM: COVID-19-related developments

IBM is constantly monitoring developments related to the COVID-19 outbreak. In March 2020, the company developed a new, interactive global dashboard to show the spread of COVID-19 across the world. The COVID-19 data reflected in the dashboard is obtained from state and local governments and the WHO. The dashboard is designed to offer the public, researchers, and government officials detailed, localized, and current information on COVID-19. IBM has also launched Watson Works to address the challenges of returning to workplaces. Watson Works is a curated set of products embedded with the Watson AI model and applications to help companies navigate many aspects of the return-to-workplace challenge due to lockdowns put in place to slow the spread of COVID-19.

### 14.2.1.4 MnM view

#### 14.2.1.4.1 Key strengths/Right to win

IBM is one of the major quantum computing firms. Research, theory, and computer science are three of the company's core areas of expertise. These areas of expertise aid the company in gaining realistic quantum skills and the benefits of being an early quantum adopter. For the 28th straight year, IBM led the industry in US patents, the majority of them in AI, cloud and hybrid cloud, quantum computing, and security. IBM leads in the quantum computing market, with nearly 300,000 registered users and more than 140 organizations in its network.

Research is a critical component of IBM's commitment to delighting its clients with the solutions and industry expertise they need to compete and grow. IBM is optimally positioned for growth with its hybrid cloud, AI, and service offerings. Emerging technologies, e.g., quantum, along with IBM's ongoing innovations in security and energy efficiency, will boost the next phase of its clients' digital transformations as they derive even greater business value from their data.

#### 14.2.1.4.2 Strategic choices made

IBM's research and development (R&D) operations differentiate it from its competitors. In 2020, IBM invested approximately 9% of the total revenue in R&D, focusing on high-growth, high-value opportunities. Investment in future technologies is an important part of IBM's strategy. Hybrid cloud and AI are the technologies of today, but IBM's investment in quantum computing positions it to win in the next era of computing. The company published plans to develop a quantum processor with more than 1,000 qubits by 2023, which will give it a first-mover advantage. IBM Research continues to develop new technologies and improve its existing ones, ensuring that it stays in the frontline of technological innovation and helps enterprises achieve their digital transformations.

#### 14.2.1.4.3 Weaknesses and competitive threats

The revenue generation of IBM is highly dependent on the American market as the Americas contributes to 46.3% of its total revenues. As IBM is continuously investing in new technologies and key strategic areas to encourage its revenue growth, the risk of failure of innovative initiatives is also high, which could impact its competitiveness. As the company has a global presence, its business and operations could be impacted by local legal, economic, political, and health conditions. Though IBM is a well-advanced technology-driven company, changes in the regulatory landscape could hamper its overall growth. Government regulations and restrictions across regions may restrict the reach of its solution and service offerings. One of the major threats to IBM's future is competition in the cloud ecosystem, which has some major players.

## 14.2.2 MICROSOFT

### 14.2.2.1 Business overview

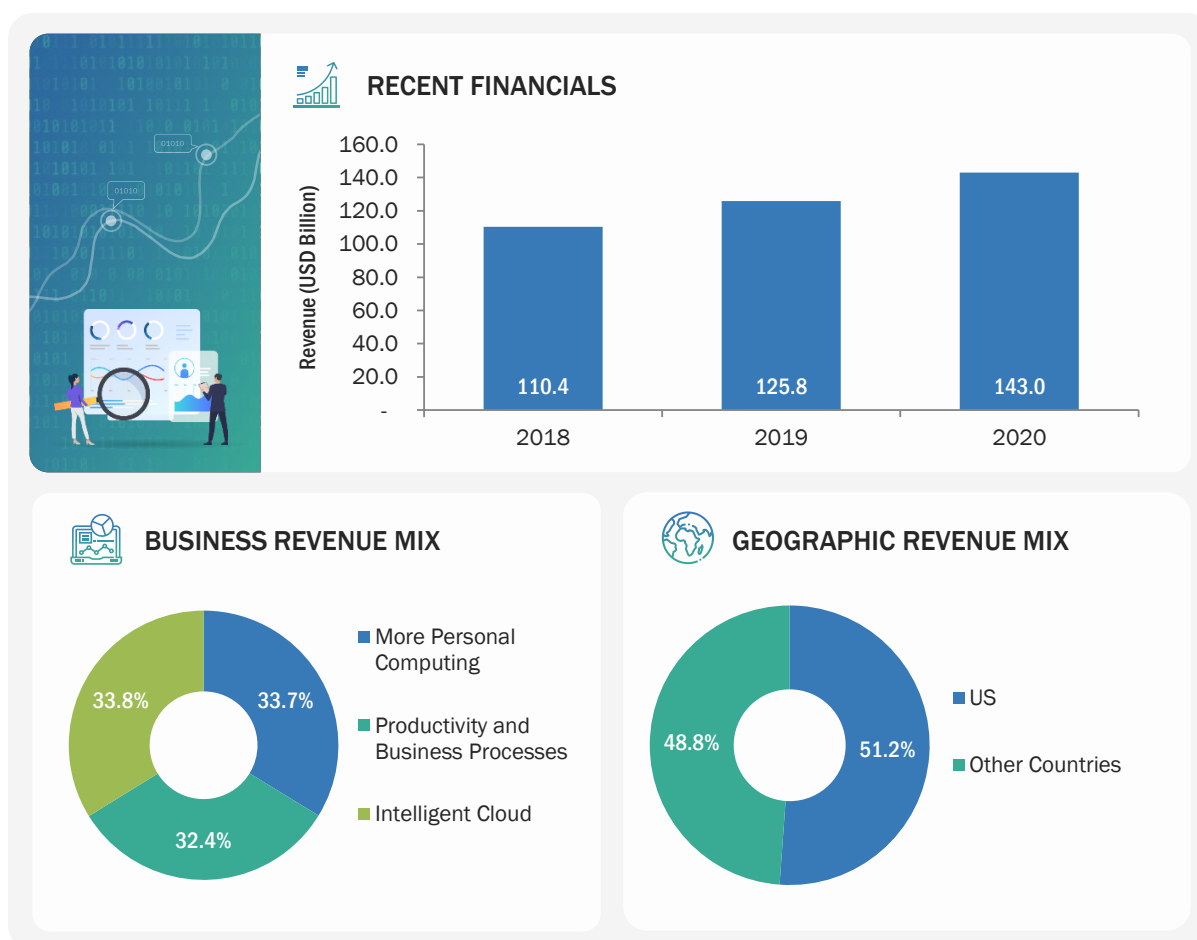
Microsoft was founded in 1975 and is headquartered in Washington, US. The company develops software, devices, and solutions and offers related services to compete in the era of the intelligent cloud and intelligent edge. With the increase in investments in mix-reality cloud, Microsoft enables its customers to digitalize their business processes. The company's offerings include cloud-based solutions that provide customers with software, platforms, and content. Its product portfolio includes operating systems (OS), cross-device productivity applications, server applications, business solution applications, desktop and server management tools, software development tools, and video games. Microsoft's platforms and tools help drive the productivity of small businesses, the competitiveness of large businesses, and the efficiency of the public sector. The company works on three pillars: privacy, cybersecurity, and responsible AI to provide tools and frameworks for its customers to encourage policy change. It focuses on investing in data centers and other hybrid and edge infrastructure to enhance its service offerings. Microsoft operates through three segments: Productivity and Business Processes, Intelligent Cloud, and More Personal Computing. The company caters to a wide range of verticals, including finance and insurance, manufacturing and retail, energy and utilities, media and entertainment, public, healthcare, and IT and telecommunications. It marks a presence in more than 190 countries across North America, APAC, Latin America, MEA, and Europe.

**TABLE 192** MICROSOFT: BUSINESS OVERVIEW

| COMPANY AT A GLANCE—2020 |                       |
|--------------------------|-----------------------|
| Year Founded             | 1975                  |
| Headquarters             | Washington, US        |
| No. of Employees         | 163,000               |
| Overall Annual Revenue   | USD 143.0 Billion     |
| R&D Expenditure          | USD 19.3 Billion      |
| Ownership                | Public (NASDAQ: MSFT) |

Source: Company Website

**FIGURE 41** MICROSOFT: COMPANY SNAPSHOT



Note: The company's fiscal year ends on June 30. The snapshot includes the company's overall information.

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 193** MICROSOFT: PRODUCTS OFFERED

| PRODUCT                 | DESCRIPTION                                                                                                                                                                                                                                                                    | PRODUCT TYPE      | APPLICATION                                                                                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Azure Quantum           | Azure Quantum is a Microsoft Azure service that is used to run quantum computing programs or solve optimization problems in the cloud. Using the Azure Quantum tools and SDKs, one can create quantum programs and run them against different quantum simulators and machines. | Service           | <ul style="list-style-type: none"> <li>Energy and Utilities</li> <li>Financial Services Optimization Solutions</li> <li>Transportation and Logistics</li> </ul> |
| Quantum Development Kit | The quantum development kit is a set of enterprise-grade tools to build, debug, and optimize quantum apps and algorithms.                                                                                                                                                      | Development Tools | <ul style="list-style-type: none"> <li>Energy and Utilities</li> <li>Financial Services Optimization Solutions</li> <li>Transportation and Logistics</li> </ul> |

Source: Company Website

### 14.2.2.2 Recent developments

**TABLE 194** MICROSOFT: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME            | DESCRIPTION                                                                                                                                                                                                                                                                                                                                       |
|----------------|--------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2020       | Microsoft    | Azure Quantum           | Azure Quantum of Microsoft is moving to a limited preview, which will allow selected customers and partners to develop their programs and skillsets to tackle business challenges that quantum computing is expected to solve. Developers interested in becoming quantum-ready can access new learning modules available free on Microsoft Learn. |
| February 2018  | Microsoft    | Quantum Development Kit | Microsoft launched updates to its Quantum Development Kit, including support for macOS and Linux, additional open-source libraries, and interoperability with Python. These updates brought the power of quantum computing to even more developers on more platforms.                                                                             |

Source: Press Releases

**TABLE 195** MICROSOFT: QUANTUM COMPUTING SOFTWARE MARKET: DEALS

| MONTH YEAR | DEAL TYPE   | COMPANY 1 | COMPANY 2      | DESCRIPTION                                                                                                                                                                                                          |
|------------|-------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| April 2021 | Partnership | Microsoft | Ally Financial | Ally Financial joined Azure Quantum program to transform customer experiences in the financial industry. Ally is working with Microsoft's Quantum team to solve optimization problems and explore quantum computing. |

Source: Press Releases

### 14.2.2.3 Microsoft: COVID-19-related developments

Microsoft is constantly monitoring developments related to the COVID-19 outbreak. The company is focusing on various initiatives for COVID-19. For instance, in June 2020, Microsoft announced a new global skills initiative required for the COVID-19 scenario. The objective of the initiative is to provide extensive access to digital skills for improving economic recovery, particularly for people who lost their jobs. The initiative would provide immediate steps to help those looking to reskill and pursue an in-demand job, combining existing and new resources from LinkedIn, GitHub, and Microsoft. The company is also focusing on various partnership strategies for tackling the COVID-19 situation. For instance, in July 2020, Citrix and Microsoft partnered to reimagine new flexible workplaces during the COVID-19 pandemic situation. Under the partnership, Microsoft has selected Citrix Workspace as a preferred digital workspace solution, and Citrix has selected Microsoft Azure as a preferred cloud platform for shifting its existing on-premises customers to Microsoft Azure to enable them to work anywhere across devices. In June 2020, Microsoft partnered with Adaptive Biotechnologies to launch the ImmuneCODE database to share an immune response status. The partnership focuses on providing advanced solutions for diagnosing, treating, and preventing the virus and improving the existing research efforts that mainly focus on the biology of the virus. This data would be made available for free to any researcher, public health official, or organization worldwide via an open data access portal.

#### 14.2.2.4 MnM view

##### 14.2.2.4.1 Key strengths/Right to win

Microsoft is one of the leading providers of quantum computing solutions. It has completely transformed quantum computing by integrating it with software applications, cloud services, and business network providers. The company has a direct presence in more than 170 countries across the world. It invests a major portion of its annual revenue in research and development. Microsoft has been working on scalable quantum computing for nearly two decades, creating its first quantum computing group known as Station Q in 2006. The company is investing in scalable quantum computing for over a decade. Over decades of research and development, Microsoft has achieved advancements across every layer of the quantum stack, including software, applications, devices, and controls, and is delivering high-end solutions through quantum-inspired classical computing.

##### 14.2.2.4.2 Strategic choices made

Microsoft offers integrated quantum computing services, software, and solutions. It plans to enhance its business solutions by leveraging resources to expand its product portfolio and develop high-value systems and services that meet the specific requirements of its customers. The company is focusing on increasing the adoption of QCaaS for data center applications by partnering with companies offering quantum systems. Microsoft carries out continuous innovations and develops solutions that address the technical requirements of its clients. The company leverages the advantages of IT platforms to increase its management efficiency and achieve business growth. It focuses on organic strategies to enhance its position in the quantum computing market. In May 2020, the company launched Azure Quantum with limited preview, which allows selected customers and partners to develop their programs and skillsets to tackle business challenges that quantum computing can solve. Developers interested in becoming quantum-ready can access new learning modules available free on Microsoft Learn.

##### 14.2.2.4.3 Weaknesses and competitive threats

Increasing competition in the quantum computing market from new players and established companies poses a threat to Microsoft. The company has a high dependency on channel partners and third-party vendors for access to quantum systems. The development and implementation of high-cost complex solutions by Microsoft may affect their adoption rate in organizations with limited resources.

### 14.2.3 AWS

#### 14.2.3.1 Business overview

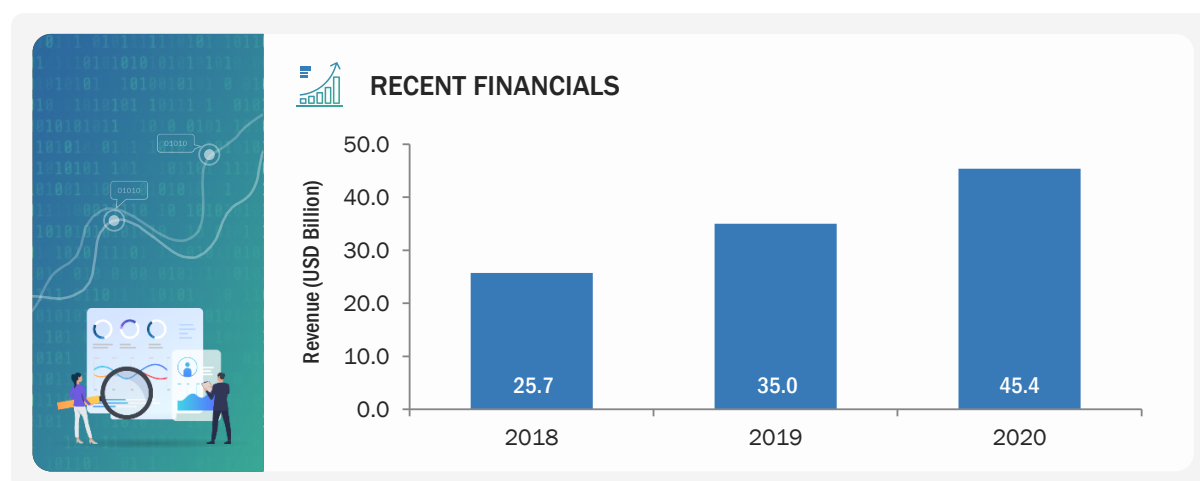
Amazon Web Services (AWS) was founded in 2006 and is headquartered in Washington, US. The company is a subsidiary of Amazon and primarily offers cloud computing services in the form of web services. It offers a wide range of products and services to customers in 190 countries. AWS offers over 175 fully featured services for compute, storage, databases, networking, analytics, robotics, ML, AI, IoT, mobile, security, hybrid, virtual and augmented reality, media, application development, deployment, and management from 76 availability zones within 24 geographies. It also offers website and web apps; mobile services; back-up, storage and archive, and financial services; and digital media. The company provides IoT solutions for clients from the industrial, consumer, and commercial verticals. These solutions analyze data from IoT devices. The company caters to various other verticals, including media and entertainment, automotive, education, BFSI, game tech, government, healthcare and life sciences, manufacturing, retail, telecommunications, oil & gas, and power utilities. It has a presence in North America, Europe, APAC, MEA, and Latin America.

**TABLE 196** AWS: BUSINESS OVERVIEW

| COMPANY AT A GLANCE—2020 |                     |
|--------------------------|---------------------|
| Year Founded             | 2006                |
| Headquarters             | Washington, US      |
| Overall Annual Revenue   | USD 45.4 Billion    |
| Ownership                | Subsidiary (Amazon) |

Source: Company Website

**FIGURE 42** AWS: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on December 31. The snapshot includes information for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 197** AWS: PRODUCTS OFFERED

| PRODUCT       | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                         | PRODUCT TYPE              | APPLICATION                                                                                                                          |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Amazon Braket | Amazon Braket is a fully managed quantum computing service that helps researchers and developers get started with the technology to accelerate research and discovery of quantum computing. Amazon Braket provides a development environment for users to explore and develop quantum algorithms, test them on quantum circuit simulators, and run them on different quantum hardware technologies. | Quantum Computing Service | <ul style="list-style-type: none"> <li>▪ Molecular Simulation</li> <li>▪ Optimization</li> <li>▪ Quantum Machine Learning</li> </ul> |

Source: Company Website

### 14.2.3.2 Recent developments

**TABLE 198** AWS: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME  | DESCRIPTION                                                                                                                                                                                                 |
|----------------|--------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| August 2020    | AWS          | Amazon Braket | Amazon Web Services announced the release of Amazon Braket, which can be accessed through both classically powered circuit simulators and quantum computers of D-Wave Systems, IonQ, and Rigetti Computing. |

Source: Press Releases

### 14.2.3.3 AWS: COVID-19-related developments

In response to the COVID-19 pandemic, AWS is playing a crucial role. Hospital networks, pharmaceutical companies, and research labs are using AWS to provide care for patients, explore treatments, and mitigate the impact of COVID-19 in many other ways. Academic institutions worldwide are transitioning from in-person to virtual classrooms and are opting for AWS to help ensure continuity of learning. Governments are leveraging AWS as a secure platform to build new capabilities to end this pandemic. In May 2020, AWS launched AWS Diagnostic Development Initiative, a program to support customers working to bring more accurate diagnostic solutions in the market for COVID-19. AWS would fund diagnostic research projects that have the potential to blunt future infectious disease outbreaks.

### 14.2.3.4 MnM view

#### 14.2.3.4.1 Key strengths/Right to win

The strategically balanced product offerings of Amazon reduce its business risks and enable it to tap the opportunities in new and existing markets. The company focuses on its growth and, as such, is expanding its business in emerging economies. This will not only increase the revenue of the company but is expected to expand its customer base as well, outside the domestic market. Amazon has a wide geographic presence across the world. A major portion (~69%) of its revenue is generated from the US.

#### 14.2.3.4.2 Strategic choices made

Amazon has partnered and collaborated with similar high-tech companies and start-ups to expand the reach of its platforms to adjacent market segments. For instance, in August 2020, Amazon Web Services announced the release of Amazon Braket, which can be accessed through both classically powered circuit simulators and quantum computers of D-Wave Systems, IonQ, and Rigetti.

#### 14.2.3.4.3 Weaknesses and competitive threats

Amazon is highly dependent on its partners for developing quantum computing applications. It faces intense competition from players such as IBM and Microsoft in the quantum computing market. The company faces a challenge to cope with the rapidly changing product demand in a dynamic technological landscape.

## 14.2.4 D-WAVE SYSTEMS

### 14.2.4.1 Business overview

D-Wave Systems was founded in 1999 and is headquartered in British Columbia, Canada. D-Wave is one of the leaders in the development and delivery of quantum computing systems, software, and services and is the commercial supplier of quantum computers. The company has its presence in several locations, including the US, Europe, and Asia.

The company offers a full stack of systems, software, developer tools, and services to enable enterprises, governments, laboratories, and academic institutions to access the power of quantum computing. Moreover, it has various premier customers, including Lockheed Martin, Google, NASA Ames, Volkswagen, DENSO, USRA, USC, Los Alamos National Laboratory, and Oak Ridge National Laboratory. D-Wave Systems renders quantum computing systems and services for logistics, artificial intelligence, material sciences, drug discovery, cybersecurity, fault detection, and financial modeling applications. D-Wave Systems offers customer support, site preparation, site installation, training package, and on-site engineering support services to its customers. The products of the company are used primarily in optimization, machine learning, simulation, pattern recognition, anomaly detection, and financial analysis applications.

**TABLE 199 D-WAVE SYSTEMS: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE—2021 |                          |
|--------------------------|--------------------------|
| Year Founded             | 1999                     |
| Headquarters             | British Columbia, Canada |
| Ownership                | Private                  |

Source: Company Website

Note: This is a privately held company. Hence, the financial information is not available in the public domain.

**TABLE 200 D-WAVE SYSTEMS: PRODUCTS OFFERED**

| PRODUCT             | DESCRIPTION                                                                                                                                                                                            | PRODUCT TYPE    | APPLICATION                                                                                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Advantage           | It has 5000+ qubits and 15-way qubit connectivity, which powers large-scale, in-production hybrid quantum applications.                                                                                | Quantum Systems | <ul style="list-style-type: none"> <li>Logistics</li> <li>Artificial Intelligence</li> <li>Materials Sciences</li> <li>Drug Discovery</li> <li>Cybersecurity</li> <li>Fault Detection</li> <li>Financial Modeling</li> </ul> |
| D-Wave 2000Q System | This quantum computer leverages quantum dynamics to accelerate and enable the development of new methods for solving discrete optimization, sampling, material science, and machine learning problems. | Quantum Systems | <ul style="list-style-type: none"> <li>Optimization</li> <li>Machine Learning</li> <li>Pattern Recognition and Anomaly Detection</li> <li>Cybersecurity</li> </ul>                                                           |

|      |                                                                                                                                                                         |             |                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Leap | Leap is a quantum cloud service and a quantum application environment (QAE). It is the only cloud-based service to provide real-time access to a live quantum computer. | Quantum QAE | <ul style="list-style-type: none"> <li>Image Analysis</li> <li>Financial Analysis</li> <li>Software/Hardware Verification and Validation</li> </ul> |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

Source: Company Website

#### 14.2.4.2 Recent developments

**TABLE 201** D-WAVE SYSTEMS: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME   | PRODUCT NAME               | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|----------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| February 2021  | D-Wave Systems | Leap Quantum Cloud Service | D-Wave Systems has announced the introduction of Leap quantum cloud service in Singapore. Developers, researchers, and businesses in Singapore can now access D-Wave's Advantage quantum computer, hybrid quantum/classical solvers, and the QAE in real time via Leap to increase the development of business-critical, in-production hybrid applications.                                                                                                                                                      |
| September 2020 | D-Wave Systems | Quantum Computing Platform | D-Wave Systems announced the general availability of its next-generation quantum computing platform, incorporated with new hardware, software, and tools to enable and accelerate the delivery of in-production quantum computing applications. Available in the Leap quantum cloud service, the platform includes the Advantage quantum system, with more than 5000 qubits and 15-way qubit connectivity, in addition to an expanded hybrid solver service that can run problems with up to 1 million variables |
| June 2020      | D-Wave Systems | Leap2                      | D-Wave Systems announced the immediate availability of Leap2, the first quantum cloud service designed for developers and organizations to easily develop and deploy real-world hybrid quantum applications with practical impact.                                                                                                                                                                                                                                                                               |
| September 2019 | D-Wave Systems | Advantage                  | D-Wave Systems announced the launch of its fifth-generation, 5,000-qubit system during the annual user conference, Qubits, of the company held in Newport, Rhode Island. The new quantum computer, Advantage, has been made available for on-premise deployments using the Leap cloud service of D-Wave Systems.                                                                                                                                                                                                 |
| June 2019      | D-Wave Systems | D-Wave Hybrid              | D-Wave Systems announced the general availability of D-Wave Hybrid, a simple open-source hybrid workflow platform for developing and running quantum-classical hybrid applications.                                                                                                                                                                                                                                                                                                                              |

Source: Press Releases

**TABLE 202** D-WAVE SYSTEM: QUANTUM COMPUTING SOFTWARE MARKET: DEALS

| MONTH YEAR    | DEAL TYPE     | COMPANY 1      | COMPANY 2          | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                            |
|---------------|---------------|----------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2021      | Collaboration | D-Wave Systems | CINECA             | CINECA, the Italian inter-university consortium, and D-Wave have announced a formal collaboration to offer Italian universities, researchers, and developers extended access to practical quantum computing technology and resources through D-Wave's Leap quantum cloud service.                                                                                      |
| April 2021    | Partnership   | D-Wave Systems | NEC Australia      | NEC Australia and D-Wave have partnered with the Australian Department of Defense to provide quantum computing capabilities to optimize last-mile resupply.                                                                                                                                                                                                            |
| June 2020     | Partnership   | D-Wave Systems | NEC Corporation    | NEC Corporation and D-Wave Systems announced that they had begun joint activities to combine the computing power of NEC's systems with the quantum computing power of D-Wave's systems, software, and cloud service to bring those combined capabilities to customers in Japan. NEC has made a USD 10 million investment in D-Wave in connection with this initiative. |
| November 2019 | Agreement     | D-Wave Systems | NEC Corporation    | D-Wave Systems announced that it had signed agreements with NEC Corporation to drive the development of hybrid services that combine the best features of quantum and classical computers and make it easy to incorporate quantum capabilities with integrated workflows.                                                                                              |
| December 2019 | Collaboration | D-Wave Systems | Amazon Web Service | D-Wave Systems announced the expansion of the Leap API into Amazon Braket, a new, fully managed Amazon Web Services (AWS) solution that allows scientists, researchers, and developers to begin experimenting with computers from quantum hardware providers in a single place.                                                                                        |
| July 2019     | Contract      | D-Wave Systems | Sigma-I            | D-Wave Systems and Sigma-I (Japan) announced the signing of the largest-ever global quantum cloud contract. As a result of this contract, Sigma-I is offering quantum consulting services and access to the D-Wave 2000Q family of systems to companies, universities, and research laboratories throughout Japan using quantum cloud service, Leap of D-Wave Systems. |
| May 2018      | Agreement     | D-Wave Systems | MDR Corporation    | D-Wave Systems and MDR Corporation signed an agreement to use the quantum computers of D-Wave to develop quantum computing applications for machine learning and optimization problems. Combining the expertise of MDR in developing tools and applications with that of D-Wave in quantum computing has benefitted Japan-based organizations to add                   |

value to their businesses through this advanced technology.

Source: Press Releases

**TABLE 203 D-WAVE SYSTEMS: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS**

| MONTH YEAR   | DEAL TYPE            | COMPANY NAME   | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|----------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 2020    | Geographic Expansion | D-Wave Systems | D-Wave Systems announced its latest geographic expansion of the Leap quantum cloud service to India and Australia. Developers, researchers, and businesses in these countries can now access D-Wave 2000Q quantum computers, hybrid solvers, and QAE in real time through Leap to drive the development of business-critical, in-production hybrid applications.                                             |
| October 2019 | Geographic Expansion | D-Wave Systems | D-Wave Systems announced that it will house its first Leap quantum cloud-based system outside of North America at the Forschungszentrum Jülich Supercomputing Center in Jülich, Germany. The system marks the cornerstone of the new Jülich UNified Infrastructure for Quantum computing (JUNIQ) lab designed to offer practically usable quantum computers to researchers in Germany and throughout Europe. |
| March 2019   | Geographic Expansion | D-Wave System  | D-Wave Systems announced the geographic expansion of its Leap quantum cloud service, thereby enabling users of some countries to access the D-Wave 2000Q system and its QAE. The 33 new countries that can access it include all 28 member states of the European Union, Japan, Iceland, Liechtenstein, Norway, and Switzerland.                                                                             |
| May 2018     | Geographic Expansion | D-Wave System  | D-Wave Systems established its Quadrant business unit focused on providing machine learning services that make state-of-the-art deep learning accessible to companies across a wide range of industries and applications. Quadrant offers the services of its in-house experts to help customers get the benefit of leading-edge machine learning solutions.                                                 |

Source: Press Releases

#### 14.2.4.3 D-Wave Systems: COVID-19-related developments

In March 2020, D-Wave Systems announced the immediate availability of free access to its quantum systems through the Leap quantum cloud service for anyone responding to the COVID-19 crisis.

#### 14.2.4.4 MnM view

##### 14.2.4.4.1 Key strengths/Right to win

D-wave Systems has a strong foothold in the quantum computing software market. Its robust network of channel partners and distributors has strengthened its reach across the world. The company focuses on enhancing its quantum computing product portfolio through innovations. D-wave Systems has introduced various products and upgrades in the market to strengthen its portfolio. The company has been granted over 160 US patents, and it has published over 100 peer-reviewed papers in leading scientific journals to date. D-wave Systems offers various solutions to support supply chain management functions. It offers quantum systems and quantum application environment (QAE), either on the cloud or on premises. The company caters to the requirements of its existing on-premise client base. In the past few years, D-wave

Systems strengthened its cloud-based portfolio and launched several products that can be accessed through the cloud for catering to the requirements of QCaaS-based businesses. As organizations are moving toward cloud-based solutions, the cloud-based Leap and Leap2 offerings of the company are expected to gain traction among enterprises worldwide.

#### **14.2.4.4.2 Strategic choices made**

D-wave Systems keeps on upgrading its capabilities in the quantum computing software market by introducing new solutions and services. In February 2020, the company announced the immediate availability of Leap2, which is the first quantum cloud service designed for developers and organizations to enable them to easily develop and deploy real-world hybrid quantum applications with practical impact on real-time problems. D-wave Systems focuses on the market in APAC, and as such, has made alliances in technologically growing countries, such as India and Australia. For instance, in July 2020, D-Wave Systems announced its latest geographic expansion of the Leap quantum cloud service to India and Australia. Developers, researchers, and businesses in these countries can now access D-Wave 2000Q quantum computers, hybrid solvers, and QAE in real time through Leap for driving the development of their business-critical, in-production hybrid applications.

#### **14.2.4.4.3 Weaknesses and competitive threats**

D-Wave Systems is facing intense competition from established and niche players in the quantum computing software market, including IBM and Google. The company may also face challenges in complying with different trade regulations and requirements of countries of the world.

## 14.2.5 RIGETTI COMPUTING

### 14.2.5.1 Business overview

Rigetti Computing was founded in 2013 and is headquartered in California, US. Rigetti Computing designs and manufactures superconducting quantum-integrated circuits. It develops quantum computers, as well as superconducting quantum processors that power them. The machines of the company can be integrated with any public, private, or hybrid cloud through the quantum cloud services (QCS) platform. It is a full-stack quantum computing company that provides an integrated computing environment.

Rigetti Computing develops algorithms for quantum computing that focus on application areas such as machine learning, logistics, healthcare and pharmaceuticals, and chemicals. The company also delivers a set of tools, such as Quil, pyQuil, and Quilc, which help solve optimization problems.

**TABLE 204 RIGETTI COMPUTING: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE—2020 |                |
|--------------------------|----------------|
| Year Founded             | 2013           |
| Headquarters             | California, US |
| Ownership                | Private        |

Source: Company Website

**TABLE 205 RIGETTI COMPUTING: PRODUCTS OFFERED**

| PRODUCT | DESCRIPTION                                                                                                                                                                                                                                                                                                            | PRODUCT TYPE  | APPLICATION                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUIL    | Rigetti-developed quantum instruction language, Quil, provides a quantum programming abstraction for QPU. Quil, which most closely resembles an assembly language for quantum programming, combines gate- and pulse-level descriptions of quantum circuits while integrating classical instructions and shared memory. | QCS Platform  | <ul style="list-style-type: none"> <li>Finance</li> <li>Energy and Power</li> <li>Chemicals</li> <li>Pharmaceuticals</li> <li>Artificial Intelligence</li> </ul> |
| Quil-T  | Quil-T is an extension of the Quil language. It offers support for controlling the continuous-time dynamics of the microwave signals sent to the quantum integrated circuits.                                                                                                                                          | QCS Platform  | <ul style="list-style-type: none"> <li>Finance</li> <li>Energy and Power</li> <li>Chemicals</li> <li>Pharmaceuticals</li> <li>Artificial Intelligence</li> </ul> |
| pyQuil  | pyQuil is a Python library for writing and running quantum programs using Quil.                                                                                                                                                                                                                                        | Software Tool | <ul style="list-style-type: none"> <li>Finance</li> <li>Energy and Power</li> <li>Chemicals</li> <li>Pharmaceuticals</li> <li>Artificial Intelligence</li> </ul> |
| Quilc   | Quilc is an optimizing compiler for gate-based quantum programs.                                                                                                                                                                                                                                                       | Software Tool | <ul style="list-style-type: none"> <li>Finance</li> <li>Energy and Power</li> <li>Chemicals</li> </ul>                                                           |

|                                         |                                                                                                                                    |                          |                                                                                                                                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |                                                                                                                                    |                          | <ul style="list-style-type: none"> <li>Pharmaceuticals</li> <li>Artificial Intelligence</li> </ul>                                                               |
| QVM                                     | The quantum virtual machine (QVM) is an open-source implementation of a quantum computation simulator.                             | Software Tool            | <ul style="list-style-type: none"> <li>Finance</li> <li>Energy and Power</li> <li>Chemicals</li> <li>Pharmaceuticals</li> <li>Artificial Intelligence</li> </ul> |
| Rigetti Forest Software Development Kit | Rigetti Forest Software Development Kit includes pyQuil, the Rigetti Quil Compiler (Quilc), and the Quantum Virtual Machine (qvm). | Software Development Kit | <ul style="list-style-type: none"> <li>Finance</li> <li>Energy and Power</li> <li>Chemicals</li> <li>Pharmaceuticals</li> <li>Artificial Intelligence</li> </ul> |

Source: Company Website

#### 14.2.5.2 Recent developments

**TABLE 206** RIGETTI COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME      | PRODUCT NAME      | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|-------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| December 2020  | Rigetti Computing | Quil-T            | Rigetti released Quil-T, making gate definitions and pulse parameters completely editable across the entire Rigetti Aspen-8 processor. This product development gives QCS users of Rigetti a fine-grained control over the parameters used to enact every gate operation, thereby unlocking the ability to experiment with optimal controls, as well as control precise pulse timing. It also facilitates the development of novel gate schemes that can significantly reduce overall gate depth. |
| September 2018 | Rigetti Computing | Quantum Computers | Rigetti announced that it is developing 128-qubit quantum computers with low error rates. These systems are based on scalable 16-, 32-, and 128-qubit Aspen quantum processors.                                                                                                                                                                                                                                                                                                                   |
| February 2018  | Rigetti Computing | Forest            | Rigetti Computing released a new version of Forest, their quantum software platform. Forest 1.3 offers upgraded developer tools, improved stability, and faster execution.                                                                                                                                                                                                                                                                                                                        |

Source: Press Releases

**TABLE 207 RIGETTI COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: DEALS**

| MONTH YEAR | DEAL TYPE   | COMPANY 1         | COMPANY 2 | DESCRIPTION                                                                                                                                                                                                                                                                                                                              |
|------------|-------------|-------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 2019  | Acquisition | Rigetti Computing | QxBranch  | Rigetti acquired QxBranch. This acquisition combines the leadership of QxBranch in customer-facing application development with the leading hardware and software stack of Rigetti. By joining their forces, these companies have tightened the feedback loop in both directions and have accelerated progress toward quantum advantage. |

Source: Press Releases

**TABLE 208 RIGETTI COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS**

| MONTH YEAR     | DEAL TYPE            | COMPANY NAME      | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|----------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2020 | Geographic Expansion | Rigetti Computing | Rigetti announced that it will lead a USD 14 million consortium to accelerate the commercialization of quantum computing in the UK. The 3-year program will develop and operate the first quantum computer in the UK and make it available to its partners and customers over the cloud. This computer is expected to be useful for practical applications in the fields of machine learning, material simulation, and finance |
| August 2020    | Funding              | Rigetti Computing | Rigetti raised USD 79 Million Series C funding for its quantum computing platform.                                                                                                                                                                                                                                                                                                                                             |

Source: Press Releases

### 14.2.5.3 MnM view

#### 14.2.5.3.1 Key strengths/Right to win

Rigetti Computing focuses on research and development and continuously launches innovative products in the quantum computing market. In September 2020, the company established a research facility in the UK and launched a 3-year program to develop and operate the first quantum computer in the UK and make it available to its partners and customers over the cloud. This computer is expected to be useful for practical applications in the fields of machine learning, material simulation, and finance. Rigetti Computing mainly focuses on keeping its customers updated with the latest market trends, such as progressions in quantum computers, chips, QCS platforms, and software tools. In December 2020, it released Quil-T, making gate definitions and pulse parameters completely editable across the entire Rigetti Aspen-8 processor.

#### 14.2.5.3.2 Strategic choices made

Rigetti Computing has adopted various organic and inorganic growth strategies to stay ahead in the competition. Through its research and development (R&D) investments and strategic acquisitions, the company is better equipped than ever to solve quantum challenges. This, in turn, leads to new and attractive opportunities for Rigetti Computing to register profitable growth in the coming years. The acquisition of businesses, assets, products, or technologies and new product launches are key business strategies of the company that allow it to complement its existing product offerings, expand its market coverage, and enhance its technological capabilities.

#### 14.2.5.3.3 Weaknesses and competitive threats

Rigetti Computing faces intense technological and pricing competition from the manufacturers of quantum computing systems such as IBM, IonQ, and Honeywell. These companies compete with Rigetti Computing in terms of technological innovations, brand strength, product portfolio diversity, product performance, technical support, delivery capabilities, customer service quality, and reliability. Thus, intense technological and pricing competition could affect the market share and operating results of Rigetti Computing.

## 14.2.6 GOOGLE

### 14.2.6.1 Business overview

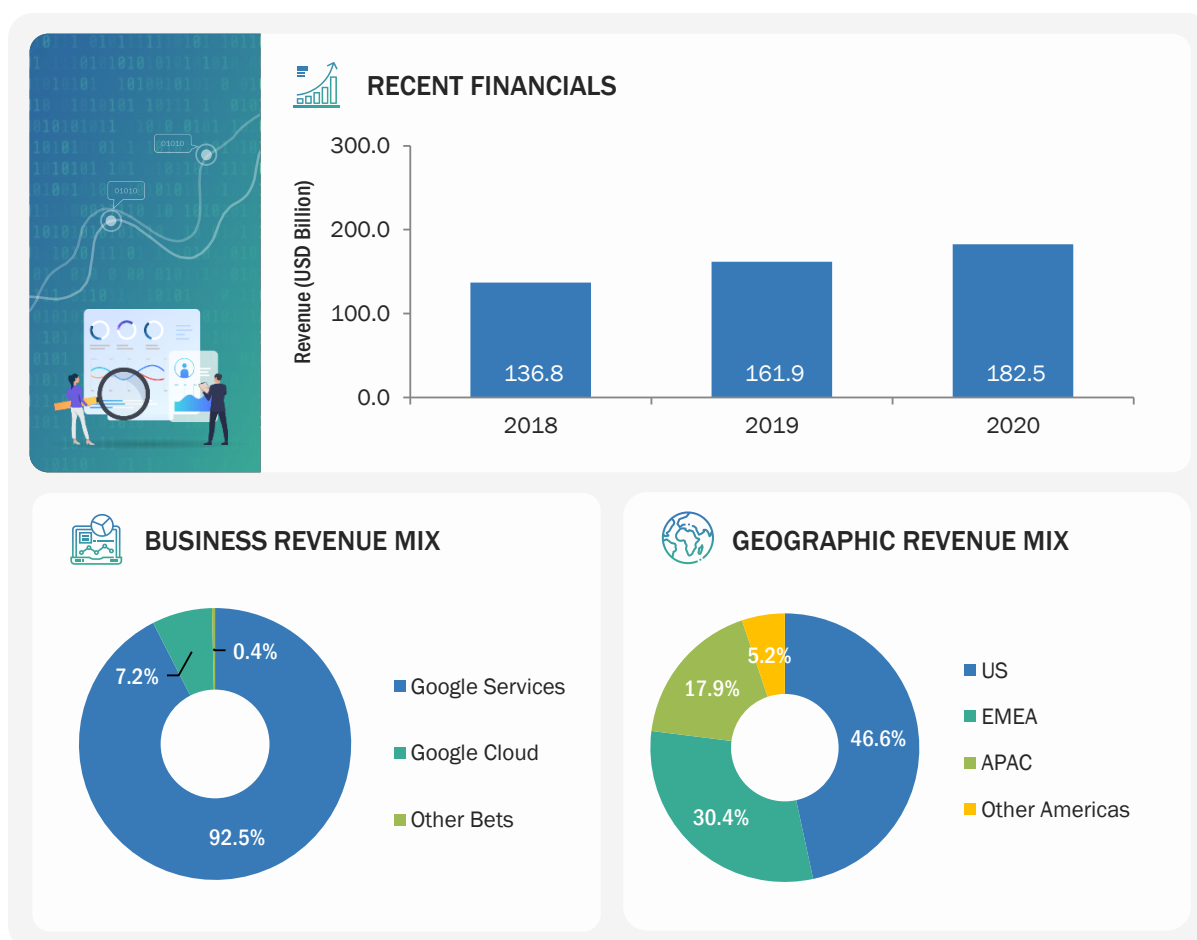
Google, a publicly held company, was founded in 1998 and is headquartered in California, US. The company's parent organization, Alphabet, was founded in 2015. Google's product innovations help increase the use of its services, and the company is one of the most recognized brands in the world. Its core products and platforms include Android, Chrome, Gmail, Google Drive, Google Maps, Google Play, Search, and YouTube, and each of them has over 1 billion active users every month. Google has built a platform in the cloud and invests in infrastructure, security, data management, analytics, and AI. The company's global clientele is spread across verticals such as automotive, BFSI, retail and eCommerce, education, energy, engineering, entertainment, environment, food, beverage, government, healthcare, manufacturing, media, telecommunications & IT, transportation, and travel & hospitality. As more digital experiences are being built in the cloud, the company's cloud products are helping enterprises of all sizes to benefit from the latest technologies and efficiently run their businesses. Google invests in platforms, such as Android mobile OS, Chrome browser, Chrome OS, and Daydream Virtual Reality (VR). It operates through 70 offices in more than 50 countries worldwide. In February 2020, the company completed the acquisition of Looker, a unified platform provider for business intelligence, data applications, and embedded analytics.

**TABLE 209**    **GOOGLE: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE—2020 |                        |
|--------------------------|------------------------|
| Year Founded             | 1998                   |
| Headquarters             | California, US         |
| No. of Employees         | 135,301                |
| Overall Annual Revenue   | USD 182.5 Billion      |
| R&D Expenditure          | USD 27.6 Billion       |
| Ownership                | Public (NASDAQ: GOOGL) |

Source: Company Website

**FIGURE 43** GOOGLE: COMPANY SNAPSHOT



Note: The company's fiscal year ends on December 31. The snapshot includes information for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 210** GOOGLE: PRODUCTS OFFERED

| PRODUCT                   | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                 | PRODUCT TYPE              | APPLICATION |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------|
| Cirq                      | Cirq is a Python software library for writing, manipulating, and optimizing quantum circuits and then running them on quantum computers and quantum simulators. The two main types of simulations that Cirq supports are pure state and mixed state. Cirq provides useful abstractions for dealing with today's noisy intermediate-scale quantum computers, where details of the hardware are vital for achieving state-of-the-art results. | Framework                 | NA          |
| Quantum Computing Service | Quantum Computing Service gives customers access to Google's quantum computing hardware. Quantum Computing Service uses Cirq, an open-source Python framework for creating quantum programs and running them against simulators or against real quantum computers.                                                                                                                                                                          | Quantum Computing Service | NA          |

Source: Company Website

### 14.2.6.2 Recent developments

**TABLE 211** GOOGLE: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME | DESCRIPTION                                                                                                                                                                                                                     |
|----------------|--------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| March 2018     | Google       | Bristlecone  | Research at Google resulted in the development of Bristlecone, the newest 72-qubit quantum processor. It is equipped with superconducting qubits and has a chip-based scalable architecture targeting 2-qubit gate error <0.5%. |

Source: Press Releases

**TABLE 212** GOOGLE: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME      | DESCRIPTION                                                                                                                                                                                                                |
|----------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2021       | Google       | Quantum AI Campus | Google has unveiled its new Quantum AI campus in Santa Barbara, California. This campus includes the first quantum data center, quantum hardware research laboratories, and quantum processor chip fabrication facilities. |

## 14.2.7 HONEYWELL

### 14.2.7.1 Business overview

Honeywell was founded in 1906 and is headquartered in North Carolina, US. It is one of the key players, which commercialize technologies that address the requirements of energy, safety, security, air travel, productivity, and global urbanization applications. The company is uniquely positioned to blend physical products with software to serve the pharmaceutical, chemical, finance, aerospace & defense, oil & gas, data centers, manufacturing, and telecommunication industries globally.

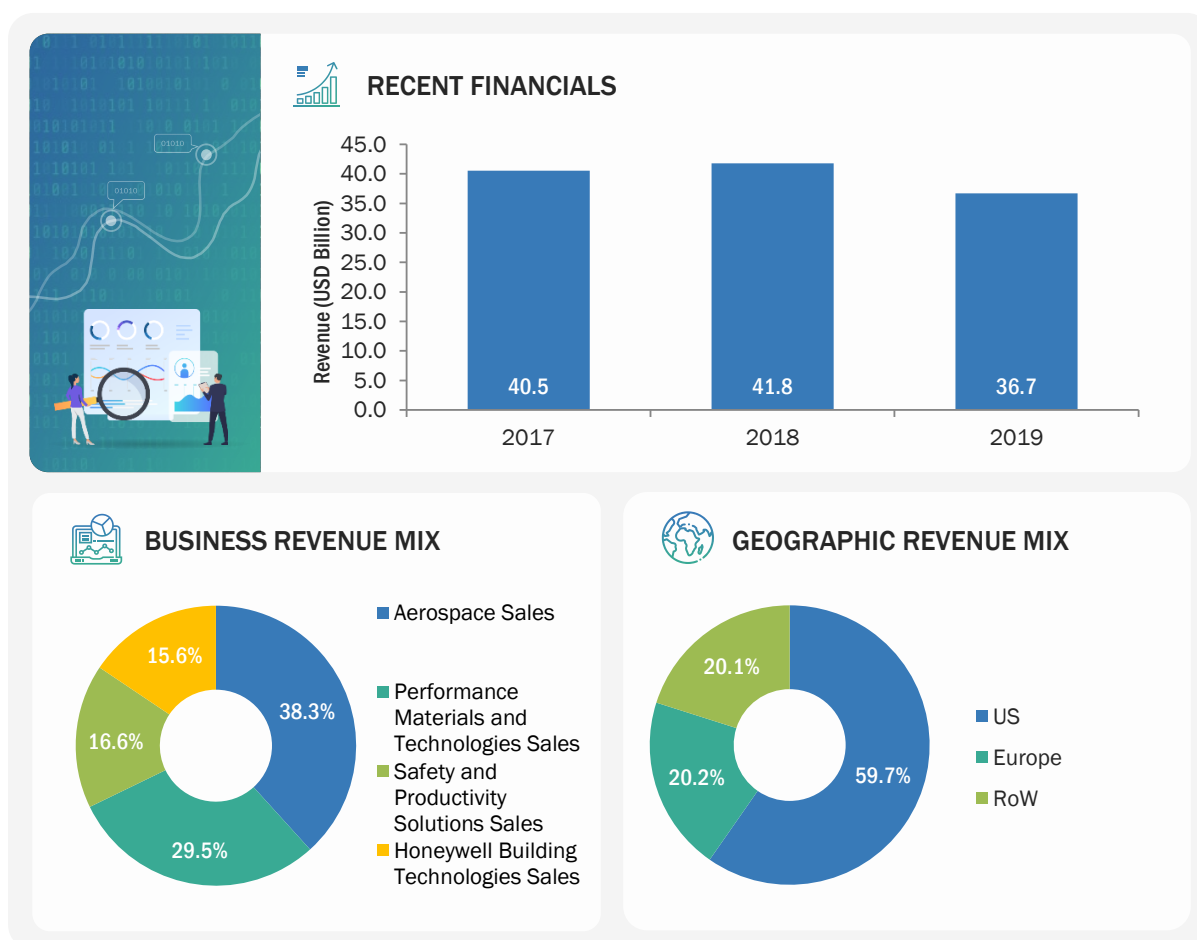
The company operates through four segments, namely, Aerospace, Honeywell Building Technologies, Performance Materials and Technologies, and Safety and Productivity Solutions. The safety products offered by the company include personal protection equipment, apparel, and footwear designed for work, play, and outdoor activities. It also offers gas detection technology and cloud-based notification and emergency messaging services. The productivity solutions offered by Honeywell include mobile devices; software for collecting and computing data, along with thermal printing; supply chain and warehouse automation equipment, software, and solutions; custom-engineered sensors, switches, and controls for sensing and productivity solutions; and software-based data and asset management productivity solutions. Honeywell Forge solutions digitally automate quantum processes to improve the efficiency of systems while reducing downtime and safety costs.

**TABLE 213 HONEYWELL: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE—2019 |                    |
|--------------------------|--------------------|
| Year Founded             | 1906               |
| Headquarters             | North Carolina, US |
| No. of Employees         | 113,000            |
| Overall Annual Revenue   | USD 36.7 Billion   |
| Ownership                | Public (NYSE: HON) |

Source: Company Website

**FIGURE 44 HONEYWELL: COMPANY SNAPSHOT**



Note: The company's fiscal year ends on December 31. The snapshot includes information for the overall company

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 214 HONEYWELL: PRODUCTS OFFERED**

| PRODUCT         | DESCRIPTION                                                                       | PRODUCT TYPE      | APPLICATION                                                                                                        |
|-----------------|-----------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------|
| System Model HØ | This model has 6 qubits, and its coherence time ranges from >106 secs to 2 secs.  | Quantum Computers | <ul style="list-style-type: none"> <li>Pharmaceuticals</li> <li>Chemicals</li> <li>Finance</li> </ul>              |
| System Model H1 | This model has 10 qubits, and its coherence time ranges from >106 secs to 3 secs. | Quantum Computers | <ul style="list-style-type: none"> <li>Oil and Gas</li> <li>Data Centers</li> <li>Aerospace and Defense</li> </ul> |

Source: Company Website

## 14.2.7.2 Recent developments

**TABLE 215 HONEYWELL: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS**

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME    | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                         |
|----------------|--------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| October 2020   | Honeywell    | System Model H1 | Honeywell announced a breakthrough in quantum computing technology with the introduction of its next-generation quantum computer, the System Model H1. This next-generation computer features differentiated quantum charge-coupled device (QCCD) trapped-ion technology of Honeywell. It is strategically designed to be rapidly upgraded throughout its lifetime. |

Source: Press Releases

**TABLE 216 HONEYWELL: QUANTUM COMPUTING SOFTWARE MARKET: DEALS**

| MONTH YEAR | DEAL TYPE | COMPANY 1                   | COMPANY 2                   | DESCRIPTION                                                                                                                                                                                                                                                                                                                                               |
|------------|-----------|-----------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2021  | Merger    | Honeywell Quantum Solutions | Cambridge Quantum Computing | Honeywell announced that Honeywell Quantum Solutions (HQS) and Cambridge Quantum Computing (CQC) will combine to form what the companies said will be the largest, most advanced standalone quantum computing company in the world, setting the pace for what is projected to become a USD 1 trillion quantum computing industry over the next 3 decades. |

Source: Press Releases

## 14.2.8 QC WARE

### 14.2.8.1 Business overview

QC Ware was founded in 1989 and is headquartered in California, US. QC Ware is an application software company that was established by a group of aerospace and finance professionals, physicists, computer scientists, and quantum algorithm specialists. It develops hardware-agnostic enterprise software solutions for quantum computers. Moreover, the company offers a quantum computing platform-as-a-service (PaaS), which allows users to integrate quantum computing with any application using codes. The software solutions of QC Ware are being adopted in finance, aerospace, and cybersecurity applications.

**TABLE 217 QC WARE: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE—2020 |                |
|--------------------------|----------------|
| Year Founded             | 2014           |
| Headquarters             | California, US |
| Ownership                | Private        |

Source: Company Website

**TABLE 218 QC WARE: PRODUCTS OFFERED**

| PRODUCT | DESCRIPTION                                                                                                                     | PRODUCT TYPE           | APPLICATION                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forge   | Forge enables algorithm implementations, editing, and execution on quantum hardware and simulators in an integrated environment | Cloud Service Platform | <ul style="list-style-type: none"> <li>Binary Optimization</li> <li>Machine Learning and Artificial Intelligence</li> <li>Chemistry Simulation</li> </ul> |

Source: Company Website

### 14.2.8.2 Recent developments

**TABLE 219 QC WARE: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS**

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME | DESCRIPTION                                                                                                                                                                                                                                                                               |
|----------------|--------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2019 | QC Ware      | Forge        | QC Ware launched the public beta of its Forge cloud service. Forge was developed by the company to enable large enterprises and public-sector organizations to start developing quantum skills and prepare for the potential disruptions that quantum computing will bring to the market. |

Source: Press Releases

**TABLE 220**    **QC WARE: QUANTUM COMPUTING SOFTWARE MARKET: DEALS**

| MONTH YEAR    | DEAL TYPE     | COMPANY 1 | COMPANY 2                     | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|---------------|-----------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| April 2021    | Collaboration | QC Ware   | Goldman Sachs                 | The collaboration of Goldman Sachs and QC Ware has brought a new way to price risky assets within reach of quantum computers. Goldman Sachs and QC Ware researchers have designed new, robust quantum algorithms that outperform state-of-the-art classical algorithms for Monte Carlo simulations and can be used on near-term quantum hardware expected to be available in 5–10 years.                               |
| April 2021    | Collaboration | QC Ware   | Air Force Research Laboratory | Air Force Research Laboratory has partnered with QC Ware to apply quantum machine learning in identifying flight patterns of unmanned aircraft                                                                                                                                                                                                                                                                         |
| December 2020 | Collaboration | QC Ware   | IonQ                          | QC Ware and IonQ collaborated to perform an experiment that demonstrated that machine learning on near-term quantum computers can achieve the same or improved level of accuracy as classical computers and can be faster than them.                                                                                                                                                                                   |
| November 2020 | Collaboration | QC Ware   | Total                         | QC Ware announced a joint research collaboration with Total, one of the world's largest energy companies, to explore the potential of quantum optimization algorithms. The quantum engineers of QC Ware and researchers of Total worked together on a project focused on the potential of quantum computing to model continuous variables to run optimization use cases that are critical for the operations of Total. |
| July 2018     | Partnership   | QC Ware   | Cirq                          | QC Ware announced that its quantum computing cloud service had been fully integrated with the Cirq interface and simulator. This integration allowed applications and development tools of QC Ware to run seamlessly on the quantum computing simulator and future hardware of Google.                                                                                                                                 |
| April 2018    | Partnership   | QC Ware   | IBM                           | QC Ware partnered with IBM to access its quantum computers. This partnership enabled quantum computers to reach their full potential in a short time.                                                                                                                                                                                                                                                                  |

Source: Press Releases

## 14.2.9 1QBIT

### 14.2.9.1 Business overview

1QBit was founded in 2012 and is headquartered in California, US. 1QBit is one of the leading global providers of quantum software solutions. It was founded by a group of intellectually curious entrepreneurs and researchers during the primary stages of quantum computing. The company offers consulting services to the finance, energy, and life sciences industries. It provides an application programming interface (API) developed on a hardware-agnostic platform to help users solve complexities while interfacing with multiple types of hardware.

1QBit works with Fortune 500 clients and various leading hardware providers to solve industrial problems pertaining to optimization, simulation, and machine learning. 1QBit's team of researchers, industry specialists, and software developers understand the unique strengths of each type of quantum computing architecture and identify problems within an industry that is well suited to benefit from these new approaches to processing.

**TABLE 221 1QBIT: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE—2021 |                |
|--------------------------|----------------|
| Year Founded             | 2012           |
| Headquarters             | California, US |
| Ownership                | Private        |

Source: Company Website

**TABLE 222 1QBIT: PRODUCTS OFFERED**

| PRODUCT                        | DESCRIPTION                                                                                                                                                                                                                                                 | PRODUCT TYPE          | APPLICATION                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1QLOUD                         | The 1Qloud platform is a computational bridge that connects intractable industry problems with novel quantum-inspired optimization solutions that use the most advanced hardware.                                                                           | Optimization Platform | <ul style="list-style-type: none"> <li>Life Sciences</li> <li>Advanced Materials</li> <li>Finance</li> <li>Logistics</li> </ul> |
| QEMIST                         | Quantum-enabled molecular ab initio simulation toolkit (QEMIST) is an innovative solution to fundamental and intractable problems in chemistry, that is, ab initio simulation of molecules.                                                                 | Simulation Toolkit    | <ul style="list-style-type: none"> <li>Material Sciences</li> <li>Drug Discovery</li> </ul>                                     |
| XrAI                           | XrAI is a quality-control and clinical-decision support tool that uses deep learning to expedite the way frontline care workers identify lung-related abnormalities. It highlights regions of abnormality in the X-ray images of the chest.                 | Radiology AI Tool     | <ul style="list-style-type: none"> <li>Healthcare</li> </ul>                                                                    |
| Quantum Opportunity Assessment | Quantum opportunity assessment helps in identifying problems that can be resolved through various quantum computing architectures. It also provides a recommendation for the technologies that will have an impact on the business partners of the company. | Service               | NA                                                                                                                              |

Source: Company Website

### 14.2.9.2 Recent developments

**TABLE 223** 1QBIT: QUANTUM COMPUTING SOFTWARE MARKET: DEALS

| MONTH YEAR    | DEAL TYPE     | COMPANY 1 | COMPANY 2 | DESCRIPTION                                                                                                                                                                                                         |
|---------------|---------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2019      | Collaboration | 1QBit     | Microsoft | 1QBit and Microsoft collaborated to disrupt material innovations with the 1QBit OpenQEMIST platform and the Microsoft quantum development kit.                                                                      |
| December 2018 | Partnership   | 1QBit     | Microsoft | 1QBit partnered with Microsoft to continue enhancing its offerings to explore solutions for intractable industry problems as its hardware-agnostic platform leverages the best hardware available for each problem. |

Source: Press Releases

**TABLE 224** 1QBIT: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS

| MONTH YEAR | DEAL TYPE | COMPANY NAME | DESCRIPTION                                                                                                                                                                                                                                                                                                                                          |
|------------|-----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2020   | Funding   | 1QBit        | 1QBit, a global leader in advanced computing and software development, and its partners representing health authorities from East to West received funding from the Digital Technology Supercluster to accelerate the clinical deployment of XrAI, the first radiology-based AI tool to be certified as a Class III medical device by Health Canada. |

Source: Press Releases

## 14.2.10 CAMBRIDGE QUANTUM COMPUTING

### 14.2.10.1 Business overview

Cambridge Quantum Computing was founded in 2014 and is headquartered in London, England. Cambridge Quantum Computing develops tools for the commercialization of quantum technologies. It has proficiency in various areas such as quantum information processing, artificial intelligence, optimization, and pattern recognition. The company provides operating systems for quantum computers. In addition, Cambridge Quantum Computing focuses on the development of algorithms and source codes that are related to cryptography, financial services, and medicine and biotechnology. It has a presence in the US, Europe, and Japan. The company has long-term and committed clients that are predominantly Fortune 500 companies and governments. The company offers its clients tools to tackle cutting-edge research problems related to the discovery of new materials. It combines its expertise in quantum software, specifically quantum development platforms, quantum chemistry enterprise applications (EUMEN), quantum machine learning (QML), and quantum augmented cybersecurity (IronBridge).

**TABLE 225 CAMBRIDGE QUANTUM COMPUTING: BUSINESS OVERVIEW**

| COMPANY AT A GLANCE-2021 |                 |
|--------------------------|-----------------|
| Year Founded             | 2014            |
| Headquarters             | London, England |
| Ownership                | Private         |

Source: Company Website

**TABLE 226 CAMBRIDGE QUANTUM COMPUTING: PRODUCTS OFFERED**

| PRODUCT    | DESCRIPTION                                                                                                                                                                                                                                                     | PRODUCT TYPE                          | APPLICATION                                                                                                                                          |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| t ket>     | t ket> is an architecture-agnostic quantum software stack and the best-in-class compiler. It translates machine-independent algorithms into executable circuits, thereby optimizing the physical qubit layout while reducing the number of required operations. | Quantum Software Development Platform | <ul style="list-style-type: none"> <li>Chemistry</li> <li>Material Sciences</li> <li>Finance</li> <li>Optimization</li> </ul>                        |
| EUMEN      | It combines the proprietary technology and unique interdisciplinary expertise of the company for quantum information and computational chemistry.                                                                                                               | Quantum Chemistry                     | <ul style="list-style-type: none"> <li>Pharmaceuticals</li> <li>Specialty Chemicals</li> <li>Performance Materials</li> <li>Agrochemicals</li> </ul> |
| IronBridge | IronBridge is quantum encryption used to develop methods to provide cybersecurity to current and post-quantum data by solving the most fundamental vulnerabilities in cryptographic protocols and procedures.                                                   | Quantum Encryption                    | <ul style="list-style-type: none"> <li>IoT Devices</li> <li>Certificate Key Generation</li> <li>Quantum Watermarking</li> </ul>                      |

Source: Company Website

### 14.2.10.2 Recent developments

**TABLE 227** CAMBRIDGE QUANTUM COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME                | PRODUCT NAME       | DESCRIPTION                                                                                                                                                                                                                                                        |
|----------------|-----------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| April 2020     | Cambridge Quantum Computing | t ket> 0.5 Version | Cambridge Quantum Computing announced the release of a new version of its class-leading quantum software platform. The release of the t ket> 0.5 version brings with it important enhancements to its existing capabilities, as well as a variety of new features. |

Source: Press Releases

**TABLE 228** CAMBRIDGE QUANTUM COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: DEALS

| MONTH YEAR    | DEAL TYPE     | COMPANY 1                   | COMPANY 2                                  | DESCRIPTION                                                                                                                                                                                                                                                                                                                |
|---------------|---------------|-----------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| November 2020 | Partnership   | Cambridge Quantum Computing | National Physics Laboratory                | Cambridge Quantum Computing confirmed that the National Physics Laboratory (NPL) of the UK has announced that it will be supporting the company in the development of its novel photonics-based product for generating certifiable quantum entropy (commonly referred to as certifiable QRNG).                             |
| July 2020     | Partnership   | Cambridge Quantum Computing | Pasqal                                     | Cambridge Quantum Computing partnered with France-based quantum computing start-up, Pasqal. As a result of this partnership, quantum software development platform, t ket> of Cambridge Quantum Computing is expected to be used by the cutting-edge quantum information processors of Pasqal.                             |
| April 2020    | Collaboration | Cambridge Quantum Computing | Total                                      | Cambridge Quantum Computing and Total collaborated on a multi-year project to tackle the most critical problems faced by the earth. This project is being led by the Cambridge-based quantum chemistry team and is expected to benefit from the class-leading quantum software development platform t ket> of the company. |
| February 2020 | Collaboration | Cambridge Quantum Computing | CERN                                       | Cambridge Quantum Computing and CERN launched the Quaternion project to advance the application of quantum technologies to particle physics. As a result of this collaboration, Cambridge Quantum Computing is expected to work closely with CERN scientists on a carefully selected series of projects.                   |
| January 2019  | Partnership   | Cambridge Quantum Computing | Networked Quantum Information Technologies | Cambridge Quantum Computing partnered with Networked Quantum Information Technologies (NQIT) to launch the UK Quantum Readiness Programme (QRP). The National QRP has been designed to provide corporations and organizations based in the UK with the                                                                     |

knowledge required to understand the opportunities presented by quantum computing and associated quantum technologies.

Source: Press Releases

**TABLE 229** CAMBRIDGE QUANTUM COMPUTING: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS

| MONTH AND YEAR | COMPANY NAME                | PRODUCT NAME          | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|-----------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| October 2020   | Cambridge Quantum Computing | Ph.D. Student Program | Cambridge Quantum Computing announced an exciting Ph.D. studentship with a multinational pharmaceutical company, GSK, and the University of Strathclyde to design and implement quantum algorithms for drug design and discovery. The engagement is expected to see all three parties combine their expertise in quantum chemistry and quantum computing to accelerate the development and adoption of quantum computing-based tools in the early stages of drug discovery by presenting the opportunity for a highly unique, industry-based Ph.D. studentship. |

Source: Press Releases

## 14.2.11 FUJITSU

### 14.2.11.1 Business overview

Fujitsu was founded in 1935 and is headquartered in Tokyo, Japan. It is a leading Japanese information and communication technology (ICT) company offering a full range of technology products, solutions, and services. The company operates through three business segments, namely, Technology Solutions, Ubiquitous Solutions, and Device Solutions. The company provides cloud-based services, business services, cybersecurity, computing devices, integrated systems, storage devices, servers, networking hardware, mainframe solutions, and peripheral devices. FUJITSU Cloud Service K5 is Fujitsu's cloud computing platform. It is highly flexible as it is based on open source technologies. It provides both private and public cloud implementation. The company caters to various verticals, such as retail, financial services, manufacturing, and automotive. It caters to customers in over 180 countries. The company has a global presence in Japan, EMEA, the Americas, Asia, and Oceania.

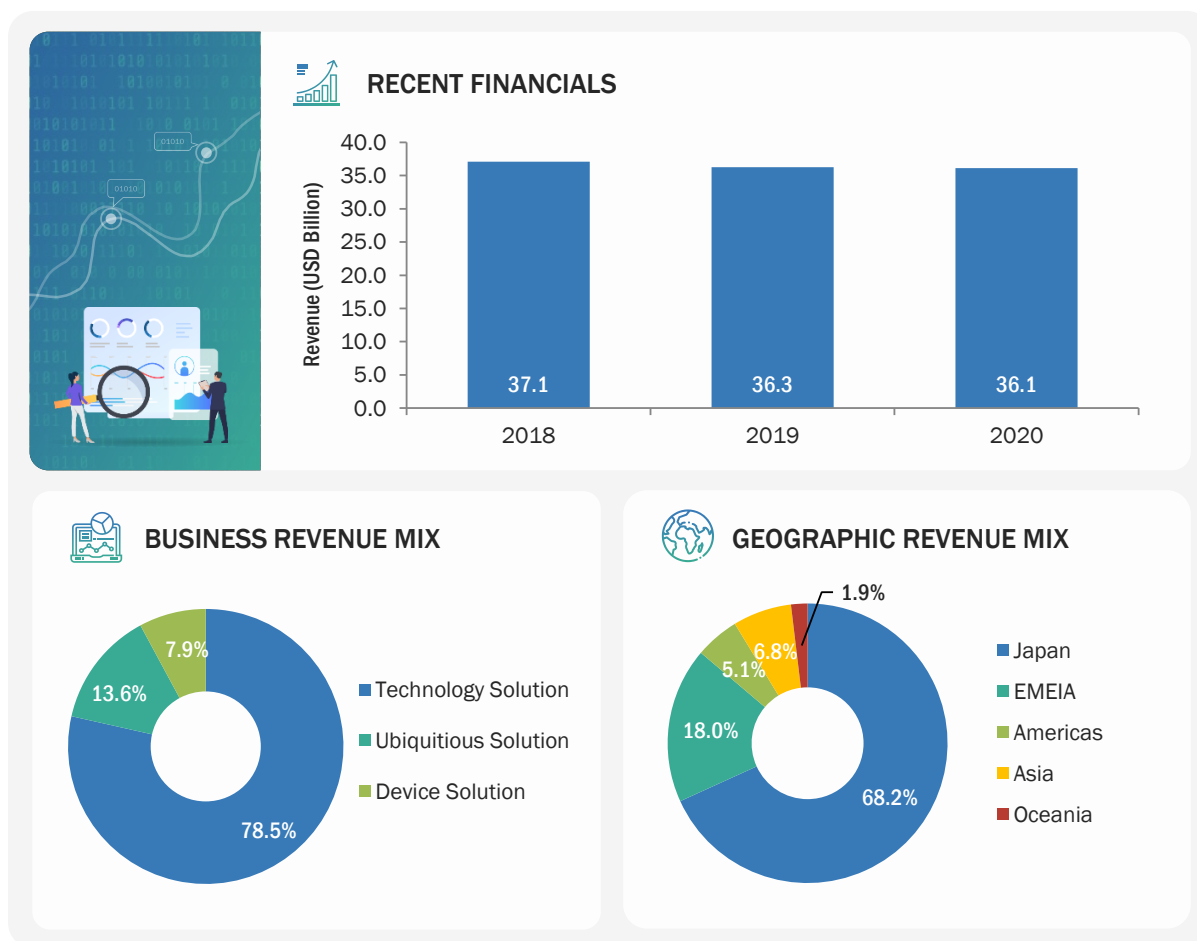
Fujitsu's Quantum-Inspired Digital Annealer solves large-scale complex combinatorial optimization problems in real time. Delivered as an end-to-end service, it can be deployed anywhere from cloud to on-premises. Digital Annealer is fully compatible with existing architectures and fits into existing business workflows and processes.

**TABLE 230** FUJITSU: BUSINESS OVERVIEW

| COMPANY AT A GLANCE—2020 |                    |
|--------------------------|--------------------|
| Year Founded             | 1935               |
| Headquarters             | Tokyo, Japan       |
| No. of Employees         | 129,071            |
| Ownership                | Public (TYO: 6702) |

Source: Company Website

**FIGURE 45** FUJITSU: COMPANY SNAPSHOT



Note: The company's fiscal year ends on March 31. The snapshot includes information for the overall company

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 231** FUJITSU: PRODUCTS OFFERED

| PRODUCT          | DESCRIPTION                                                                                                                                                                                                                                                                  | PRODUCT TYPE | APPLICATION  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| Digital Annealer | It is a ground-breaking solution inspired by the key characteristics of quantum computing—superposition, quantum tunneling, and entanglement—to enable the simultaneous evaluation of an increasing number of potential options and the delivery of lightning-fast insights. | Solution     | Optimization |

Source: Company Website

### 14.2.11.2 Recent developments

**TABLE 232** FUJITSU: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME                   | DESCRIPTION                                                                                                                                                                                                                                                                                                              |
|----------------|--------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| December 2018  | Fujitsu      | Digital Annealer Cloud Service | Fujitsu announced the launch of the second generation of its Fujitsu Quantum-Inspired Computing Digital Annealer Cloud Service. The service utilizes Fujitsu's unique Digital Annealer architecture to quickly solve combinatorial optimization problems, drawing on an innovative design inspired by quantum phenomena. |

Source: Press Releases

**TABLE 233** FUJITSU: QUANTUM COMPUTING SOFTWARE MARKET: DEALS

| MONTH YEAR    | DEAL TYPE     | COMPANY 1 | COMPANY 2             | DESCRIPTION                                                                                                                                                                                                                                                                            |
|---------------|---------------|-----------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| November 2020 | Collaboration | Fujitsu   | University of Toronto | Fujitsu Laboratories Limited, in collaboration with the University of Toronto, has successfully developed a new parallel search technology to help achieve megabit-class performance for large-scale problems, representing an important technical milestone for its Digital Annealer. |

Source: Press Releases

## 14.2.12 HUAWEI

### 14.2.12.1 Business overview

Huawei was founded in 1987 and is headquartered in Shenzhen, China. It is one of the largest telecommunication equipment manufacturers and providers of services and solutions related to the Information and Communications Technology (ICT) industry. The company offers its solutions in more than 170 countries and has an employee base of 121,269. It operates through four core business segments, namely, Consumer Business, Carrier Business, Enterprise Business, and Other. The company has a competitive portfolio for the ICT industry that includes end-to-end enterprise solutions. Moreover, it offers a wide range of products and solutions, including cloud computing, enterprise Long Term Evolution (eLTE) Broadband Trunking, eLTE Broadband Access, Enterprise Cloud Communication, and routers. Huawei's product portfolio caters to various verticals across North America, Europe, APAC, MEA, and Latin America.

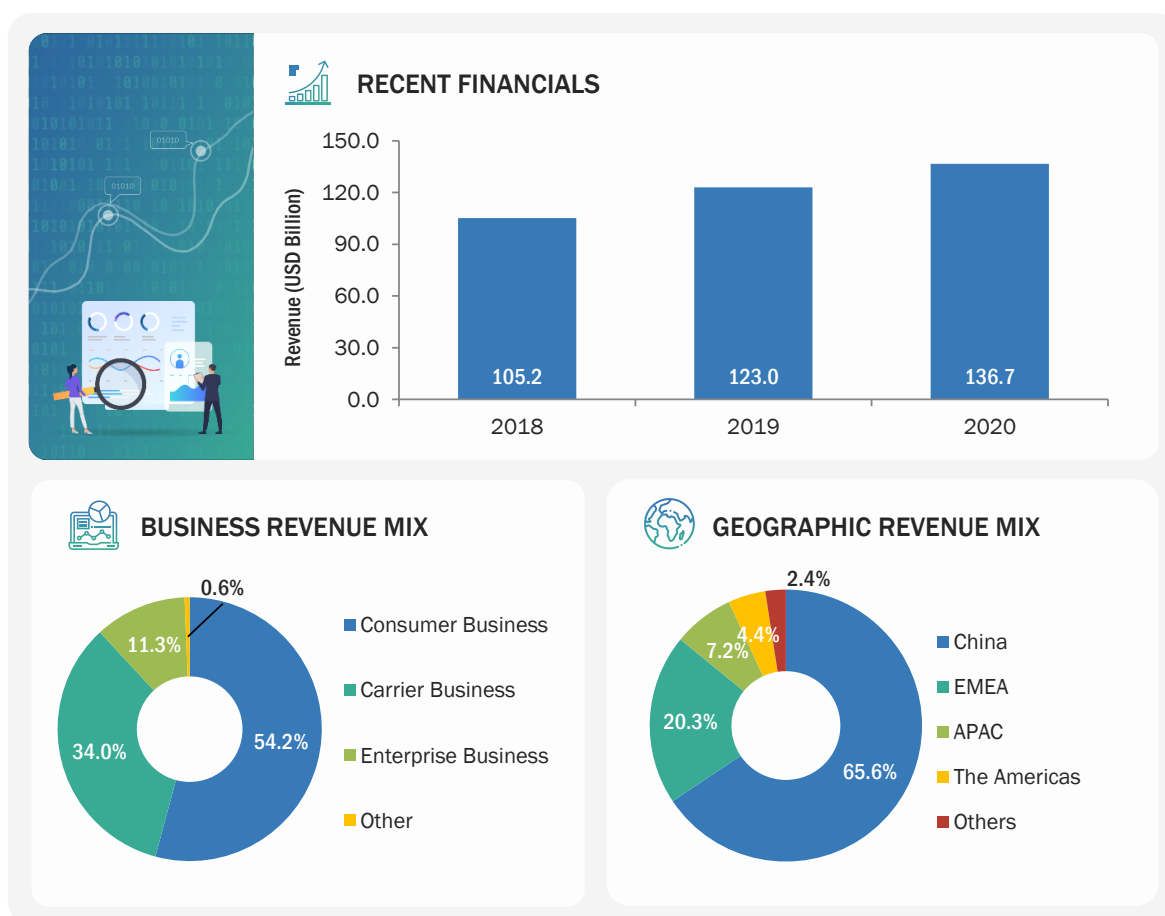
In the quantum computing software market, the company offers the Quantum Computing Simulation HiQ Cloud Service platform. Based on Huawei cloud's powerful computing capability, HiQ offers cloud services with both full- and single-amplitude simulations. The HiQ platform can simulate quantum circuits with at least 42-qubits for full-amplitude simulations and at least 81 qubits for single amplitudes. In addition, for low-depth circuits, the qubit number can reach 169 for single-amplitude simulations.

**TABLE 234** HUAWEI: BUSINESS OVERVIEW

| COMPANY AT A GLANCE—2020 |                    |
|--------------------------|--------------------|
| Year Founded             | 1987               |
| Headquarters             | Shenzhen, China    |
| No. of Employees         | 121,269            |
| Overall Annual Revenue   | USD 136.7 Billion  |
| Ownership                | Public (002502.SZ) |

Source: Company Website

**FIGURE 46** HUAWEI: COMPANY SNAPSHOT



Note: The company's fiscal year ends on December 31. The snapshot includes information for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 235** HUAWEI: PRODUCTS OFFERED

| PRODUCT    | DESCRIPTION                                                                                                                                                                                                                                              | PRODUCT TYPE | APPLICATION                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------|
| Huawei HiQ | Huawei HiQ is an open-source software framework for quantum computing. It aims at providing tools that facilitate inventing, implementing, testing, debugging, and running quantum algorithms using either classical hardware or actual quantum devices. | Platform     | Quantum Computing Simulation |

Source: Company Website

#### 14.2.12.2 Recent developments

**TABLE 236** HUAWEI: QUANTUM COMPUTING SOFTWARE MARKET: PRODUCT LAUNCHES AND ENHANCEMENTS

| MONTH AND YEAR | COMPANY NAME | PRODUCT NAME | DESCRIPTION                                                                                                                                                                        |
|----------------|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| October 2018   | Huawei       | Huawei HiQ   | Huawei released a cloud service platform for quantum computing simulation, HiQ, which includes a quantum computing simulator and a quantum programming framework on the simulator. |

Source: Press Releases

### 14.2.13 ACCENTURE

#### 14.2.13.1 Business overview

Accenture was founded in 1989 and is headquartered in Dublin, Ireland. It is a leading global professional services company with high expertise in digital, cloud, and security technologies. Accenture combines unmatched experience and specialized capabilities across more than 40 industries and offers strategy and consulting, interactive, and technological services. Its employee base of ~506,000 serves clients in more than 120 countries to help them build their digital core, transform their operations, and accelerate revenue growth by creating tangible value across their enterprises at speed and scale. The company caters to various verticals, such as aerospace & defense, chemical, healthcare, life sciences, automotive, communication and media, high tech, natural resources, travel, banking, consumer goods and service, industrial equipment, public service, utilities, energy, insurance, and retail. The company has a global presence in North America, Europe, Africa, Asia Pacific, the Middle East, and South America.

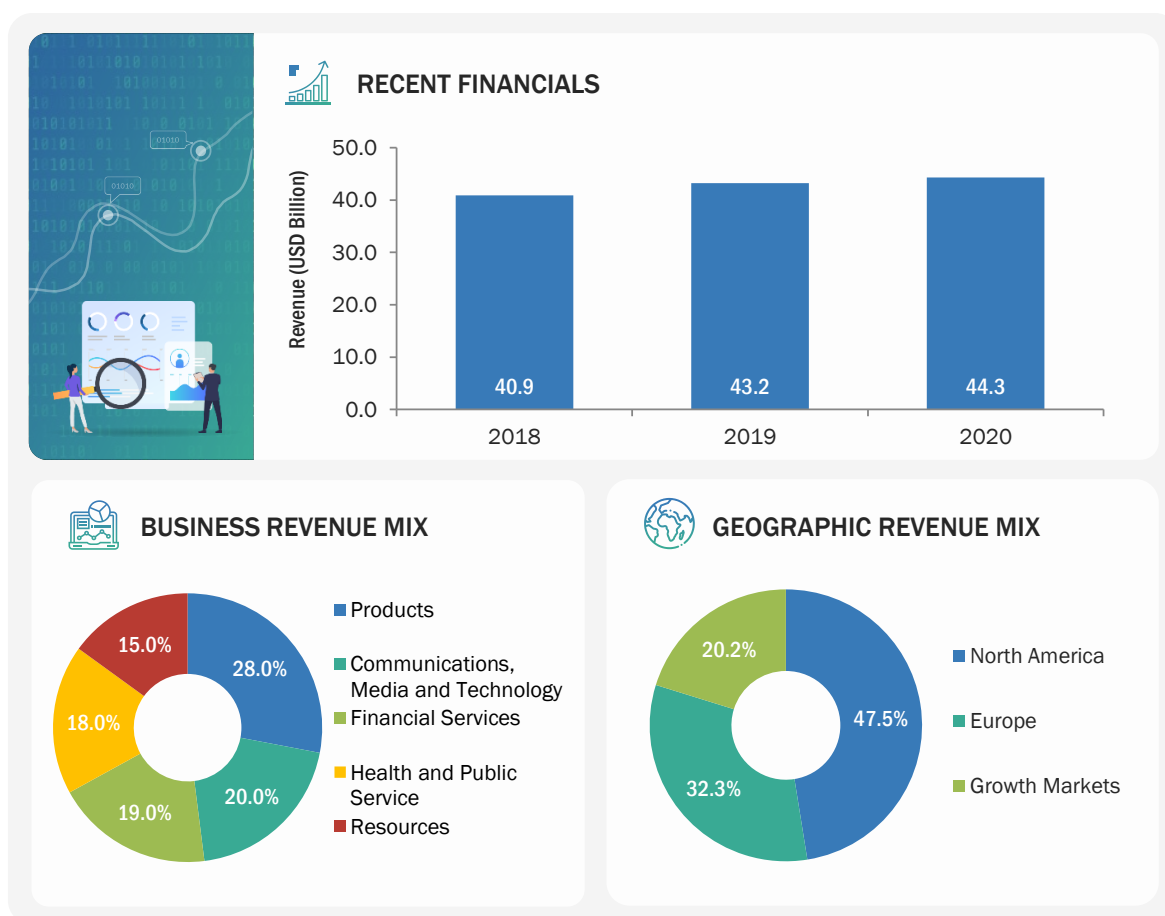
Accenture's quantum partner ecosystem includes software providers, research labs, and academic institutions. The company works with 1QBit, Rigetti, D-Wave, Fujitsu, Google, Microsoft, and IBM Research. Accenture is a member of the IBM Q Network, a group of research labs, start-ups, academic institutions, and Fortune 500 companies pursuing quantum computing. It has also been conducting daylong workshops on quantum computing and other emerging technologies at innovation centers worldwide for hundreds of clients.

**TABLE 237** ACCENTURE: BUSINESS OVERVIEW

| COMPANY AT A GLANCE—2020 |                    |
|--------------------------|--------------------|
| Year Founded             | 1989               |
| Headquarters             | Dublin, Ireland    |
| No. of Employees         | 506,000            |
| Overall Annual Revenue   | USD 44.3 Billion   |
| Ownership                | Public (NYSE: ACN) |

Source: Company Website

**FIGURE 47** ACCENTURE: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on August 31. The snapshot includes information for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

**TABLE 238** ACCENTURE: PRODUCTS OFFERED

| PRODUCT        | DESCRIPTION                                                                                                                                                                                                                 | PRODUCT TYPE | APPLICATION                                                                                                |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------|
| Accenture Labs | Accenture Labs examines the science behind quantum computing, potential use cases in an industry, and recommended steps for business leaders who want to be best positioned when this emerging technology reaches maturity. | Service      | <ul style="list-style-type: none"> <li>Optimization</li> <li>Sampling</li> <li>Machine Learning</li> </ul> |

Source: Company Website

### 14.2.13.2 Recent developments

**TABLE 239** ACCENTURE: QUANTUM COMPUTING SOFTWARE MARKET: OTHER DEVELOPMENTS

| MONTH YEAR | DEAL TYPE | COMPANY NAME | DESCRIPTION                                                                                                                                                                                                                                                           |
|------------|-----------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 2019  | Patent    | Accenture    | Accenture has been granted a US patent for a "quantum computing machine learning module" that trains artificial intelligence (AI) models to determine when and how computational tasks would be best handled by quantum computing versus classical computing methods. |

Source: Press Releases

## 14.3 OTHER PLAYERS

### 14.3.1 RIVERLANE

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded                   | 2017                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Headquarters              | UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Ownership                 | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Business Overview         | Riverlane is a quantum computing software company, which spun out of the University of Cambridge in 2017. The company offers consulting services to its clients, as well as develops new quantum algorithms for solving high-value problems in machine learning and quantum chemistry. In collaboration with Dividiti, Riverlane developed a quantum collective knowledge (QCK), an SDK that facilitates fair and reproducible benchmarking and co-designing, as well as optimization of quantum software and hardware. The company develops simulation engines for microscopic systems that accurately account for quantum effects. Riverlane has collaborated with leading quantum computing players such as the University of Bristol, Google, and IBM for quantum computing. |
| Products/Services Offered | Operating Systems for Quantum Computers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Geographic Presence       | UK (Laboratory Location)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Source: Company Website

### 14.3.2 ZAPATA COMPUTING

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded                   | 2017                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Headquarters              | Massachusetts, US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Ownership                 | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Business Overview         | Zapata Computing was spun out of Harvard in 2017 to commercialize the quantum algorithms research. Zapata Computing develops quantum algorithms for an exponential increase in the computational power of enterprise applications in quantum computers.<br><br>With Fortune 100 and Global 1000 companies, Zapata Computing is developing solutions for a wide range of industries, including chemistry, logistics, finance, oil & gas, aviation, pharmaceutical, and materials. Zapata's quantum platform Orquestra(R) combines a powerful software platform and quantum algorithm libraries to deliver real-world advances in computational power for applications – particularly in chemistry, machine learning, and optimization. Orquestra enables users to compose quantum workflows and orchestrate their execution across classical and quantum technologies. |
| Products/Services Offered | Orquestra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Geographic Presence       | US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Source: Company Website

### 14.3.3 QUANTUM CIRCUITS

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded                   | 2015                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Headquarters              | US                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Ownership                 | Private                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Business Overview         | Quantum Circuits has a long-term goal to develop, manufacture, and sell practical quantum computers based on superconducting devices. It is led by experts who carry out quantum information processing using solid-state devices. These experts have decades-long records of innovating and developing new hardware, techniques, and fundamental concepts that are the foundation of quantum computing with superconducting circuits. |
| Products/Services Offered | <ul style="list-style-type: none"> <li>Quantum as a Service</li> <li>Quantum Circuits</li> </ul>                                                                                                                                                                                                                                                                                                                                       |
| Geographic Presence       | US                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Source: Company Website

### 14.3.4 QUANTICA COMPUTACAO

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded                   | 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Headquarters              | Tamil Nadu, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Ownership                 | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Business Overview         | Quantic Computacao's primary objective is to develop a software platform to utilize future quantum computer technology. Quantic Computacao develops essential software tools, algorithms, and components that can be used with quantum computers. It caters to the requirements of the banking and finance, healthcare, and media industries. The company implements algorithms for research applications, such as optimizing supply chain distribution and planning, advertising schedules, and carrying out revenue optimization and forecasting. |
| Products/Services Offered | <ul style="list-style-type: none"> <li>Alchemy (Quantum Virtual Simulators)</li> <li>Quantum Algorithms</li> <li>Machine Learning and Artificial Intelligence</li> </ul>                                                                                                                                                                                                                                                                                                                                                                            |
| Geographic Presence       | India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Source: Company Website

### 14.3.5 XANADU

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded                   | 2016                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Headquarters              | Ontario, Canada                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Ownership                 | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Business Overview         | <p>XANADU is a photonic quantum computing and advanced artificial intelligence company. It designs quantum silicon photonic chips and integrates them with the existing hardware for full-stack quantum computing. The photonic chips developed by XANADU are used to generate, control, and measure photons that enable extremely fast computation. The company uses laser light traveling through nano-scale channels to power quantum processor chips.</p> <p>Today, enterprises and researchers can begin using Xanadu's photonic quantum computers through the Xanadu Quantum Cloud (XQC) service and Strawberry Fields application library. The company is also advancing the field of quantum machine learning (QML) through the development of PennyLane, an open-source project that has become a leading software library among quantum researchers and developers.</p> |
| Products/Services Offered | <ul style="list-style-type: none"> <li>▪ Xanadu Quantum Cloud</li> <li>▪ Strawberry Fields</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Geographic Presence       | Canada                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Source: Company Website

### 14.3.6 QUANTASTICA

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded             | 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Headquarters        | Finland, Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Ownership           | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Business Overview   | <p>Quantastica provides software tools and solutions to help people utilize hybrid quantum-classical computing techniques. The company's main products are Quantum Programming Studio—a web-based graphical user interface for designing quantum algorithms—and Qubit Toaster—a high-performance quantum circuit simulator. The second but not less important offering is quantum-classical software solutions for customers preparing for the quantum revolution.</p> <p>Quantastica also released a software called Forest Backend for Qiskit that allows users to run Qiskit programs on Rigetti simulators and quantum computers by changing only two lines of code.</p> |
| Products Offered    | <ul style="list-style-type: none"> <li>▪ Quantum Programming Studio</li> <li>▪ Qubit Toaster</li> <li>▪ Quantum Algorithm Development</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Geographic Presence | Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Source: Company Website

### 14.3.7 VERIQLLOUD

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded             | 2017                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Headquarters        | Montrouge, France                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Ownership           | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Business Overview   | <p>VeriQloud is a company developing applications and software for quantum networks. VeriQloud develops innovative solutions based on optimized network architecture for specific use cases. It has developed Qline, a quantum communication architecture with very light hardware. Qline offers the same capabilities as standard quantum key distribution networks at a lower cost for small-scale networks.</p> <p>VeriQloud is involved in several international quantum initiatives. It is working on the applications within the Quantum Internet Alliance, a project led by TU Delft. In this context, VeriQloud hosts the Quantum Protocol Zoo, a free encyclopedia of quantum communication protocols, available at <a href="http://wiki.veriqcloud.com">wiki.veriqcloud.com</a>. VeriQloud is also involved in the feasibility study of the future European Quantum Communication Infrastructure.</p> |
| Products Offered    | Qline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Geographic Presence | Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Source: Company Website

### 14.3.8 AVANETIX

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded             | 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Headquarters        | Berlin, Germany                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Ownership           | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Business Overview   | <p>AVANETIX provides a vertically integrated and industry-specific application software stack that uses classical optimization methods, machine learning, and quantum computing to solve supply chain challenges. Its software utilizes a hybrid approach of quantum and classical technology as powerful technologies in a unique, vertically integrated, and industry-specific application software stack to simulate and optimize complex scenarios exponentially faster. It is supporting enterprises in the automotive, manufacturing, and logistics industries to automate and optimize their supply chains to increase productivity and efficiency.</p> |
| Products Offered    | AVANETIX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Geographic Presence | Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Source: Company Website

### 14.3.9 KUANO

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded             | 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Headquarters        | London, England                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Ownership           | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Business Overview   | Kuano is an AI company headquartered in London, England, using quantum simulations to create molecular design solutions for challenging enzyme targets. Enzymes represent the largest clinically and commercially validated class of drug targets. It offers a discovery platform that combines target quantum mapping and AI-led chemistry to enable the faster development of better drug candidates. The company's platform helps healthcare providers to get better and safer drug candidates to cure their patients faster. |
| Products Offered    | Kuano Drug Design Platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Geographic Presence | UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Source: Company Website

### 14.3.10 RAHKO

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded             | 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Headquarters        | London, UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Ownership           | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Business Overview   | <p>Rahko is a technology company specializing in quantum machine learning, quantum chemistry, quantum software, and quantum hardware engineering. The company develops a platform that allows to test and prototype algorithms and access quantum computers. Rahko has a strong focus on solving real-world, commercially valuable problems in partnership with several of the world's largest quantum hardware manufacturers and with some of the world's largest companies whose future capacity to compete depends on their capabilities in chemical simulation. Rahko is making drug discovery faster and cheaper today with quantum-inspired machine learning, and highly powerful and accurate tomorrow with quantum computing.</p> <p>Rahko brings together three key technologies in its quantum discovery platform Hyrax: computational chemistry, machine learning, and quantum computing.</p> |
| Products Offered    | Hyrax Platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Geographic Presence | UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Source: Company Website

### 14.3.11 KETITA LABS

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded             | 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Headquarters        | Estonia, Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Ownership           | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Business Overview   | Ketita Labs is an Estonian university spin-off working on hybrid quantum-classical algorithms for near-term quantum computing. Ketita Labs designs and develops quantum software products for consumers and writes quantum software tools. AD-Sim is Ketita's in-house tool that is used in the research on Analog-Digital quantum control. It will start making it available for cloud use in the next couple of months. Ketita Labs offers consulting services for decision-makers, researchers, and educators from the private and public sectors. |
| Products Offered    | AD-Sim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Geographic Presence | Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Source: Company Website

### 14.3.12 ALIRO QUANTUM

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded             | 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Headquarters        | Massachusetts, US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Ownership           | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Business Overview   | Aliro Quantum is a quantum networking platform company that spun out of NarangLab at the Harvard University. Aliro is leading the charge on quantum network market creation by offering the foundational technologies needed for organizations worldwide to build powerful quantum systems. Aliro, an Air Force Research grant recipient, is designing quantum network simulation and emulation tools while partnering with national labs and hardware vendors, including Air Force Research Labs, IBM Q Network, Rigetti, Honeywell Quantum Solutions, and Hyperion Research, to make scalable quantum computing accessible. |
| Products Offered    | Aliro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Geographic Presence | North America                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Source: Company Website

## 15 APPENDIX

### 15.1 DISCUSSION GUIDE

Currently, we are conducting a global study on the quantum computing software market. It features a detailed analysis of opportunities, market drivers, restraints, competitive landscape, and key industry players' profiles.

Name:

Designation:

Company Name:

Phone No:

Email ID:

Fax No:

**Q. 1. What are the top three drivers, restraints, opportunities, and challenges for your company in the quantum computing software market?**

| SR. NO. | DRIVERS | RESTRAINTS | OPPORTUNITIES | CHALLENGES |
|---------|---------|------------|---------------|------------|
| 1       |         |            |               |            |
| 2       |         |            |               |            |
| 3       |         |            |               |            |

Comments: \_\_\_\_\_

**Q. 2. Which region, according to you, would offer maximum opportunities for quantum computing software vendors? Kindly provide a market share out of 100%.**

| SR. NO. | QUANTUM COMPUTING SOFTWARE MARKET - REGION | ATTRACTIVENESS OF OPPORTUNITY (MARKET SHARE OUT OF 100%) - 2021 | ATTRACTIVENESS OF OPPORTUNITY (MARKET SHARE OUT OF 100%) - 2026 |
|---------|--------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| 1       | North America                              |                                                                 |                                                                 |
| 2       | Europe                                     |                                                                 |                                                                 |
| 3       | APAC                                       |                                                                 |                                                                 |
| 4       | MEA                                        |                                                                 |                                                                 |
| 5       | Latin America                              |                                                                 |                                                                 |

Comments: \_\_\_\_\_

**Q. 3. Which component, according to you, would offer maximum opportunities in the quantum computing software market? Kindly provide a market share out of 100%.**

| SR. NO. | COMPONENT | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) -<br>2021 | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) -<br>2026 |
|---------|-----------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1       | Software  |                                                                       |                                                                       |
| 2       | Services  |                                                                       |                                                                       |

Comments: \_\_\_\_\_

**Q. 4. Which services, according to you, would offer maximum opportunities in the quantum computing software market? Kindly provide a market share out of 100%.**

| SR. NO. | SERVICE                        | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2021 | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2026 |
|---------|--------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| 1       | Quantum as a Service<br>(QaaS) |                                                                    |                                                                    |
| 2       | Consulting Services            |                                                                    |                                                                    |

Comments: \_\_\_\_\_

**Q. 5. Which deployment mode, according to you, would offer maximum opportunities in the quantum computing software market? Kindly provide a market share out of 100%.**

| SR. NO. | DEPLOYMENT MODE | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2021 | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2026 |
|---------|-----------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| 1       | Cloud           |                                                                    |                                                                    |
| 2       | On-Premises     |                                                                    |                                                                    |

Comments: \_\_\_\_\_

**Q. 6. Which organizations, according to you, would offer maximum opportunities in the quantum computing software market? Kindly provide a market share out of 100%.**

| SR. NO. | ORGANIZATION SIZE | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) -<br>2021 | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) -<br>2026 |
|---------|-------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1       | Large Enterprises |                                                                       |                                                                       |
| 2       | SMEs              |                                                                       |                                                                       |

Comments: \_\_\_\_\_

**Q. 7. Which application, according to you, would offer maximum opportunities in the quantum computing software market? Kindly provide a market share out of 100%.**

| SR. NO. | APPLICATION                                    | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2021 | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2026 |
|---------|------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| 1       | Optimization                                   |                                                                    |                                                                    |
| 2       | Simulation                                     |                                                                    |                                                                    |
| 3       | ML                                             |                                                                    |                                                                    |
| 4       | Others (Quantum Chemistry and Quantum Finance) |                                                                    |                                                                    |

Comments: \_\_\_\_\_

**Q. 8. Which technology, according to you, would offer maximum opportunities in the quantum computing software market? Kindly provide a market share out of 100%.**

| SR. NO. | TECHNOLOGY                                  | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2021 | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2026 |
|---------|---------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| 1       | Super Conducting Qubits                     |                                                                    |                                                                    |
| 2       | Trapped Ions                                |                                                                    |                                                                    |
| 3       | Quantum Annealing                           |                                                                    |                                                                    |
| 4       | Other Technology (Topological and Photonic) |                                                                    |                                                                    |

Comments: \_\_\_\_\_

**Q. 9. Which vertical, according to you, would offer maximum opportunities in the quantum computing software market? Kindly provide a market share out of 100%.**

| SR. NO. | VERTICAL                   | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2021 | ATTRACTIVENESS OF OPPORTUNITY<br>(MARKET SHARE OUT OF 100%) - 2026 |
|---------|----------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| 1       | BFSI                       |                                                                    |                                                                    |
| 2       | Government                 |                                                                    |                                                                    |
| 3       | Healthcare & Life Sciences |                                                                    |                                                                    |
| 4       | Energy & Utility           |                                                                    |                                                                    |
| 5       | Transportation & Logistics |                                                                    |                                                                    |
| 6       | Education                  |                                                                    |                                                                    |
| 7       | Aerospace & Defense        |                                                                    |                                                                    |
| 8       | Chemical                   |                                                                    |                                                                    |

Comments: \_\_\_\_\_

**Q. 10.** Could you please name your competitors in the quantum computing software market?

Comments: \_\_\_\_\_

**Q. 11.** What is the expected growth rate of the quantum computing software market during the forecast period?

Comments: \_\_\_\_\_

## 15.2 KNOWLEDGE STORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

Knowledge Store contains MarketsandMarkets' unique market insights. As a subscriber, you have access to all of the published titles and any new ones added over the course of the subscription period. You can subscribe to specific number of reports or domains of your interest.

### KNOWLEDGE STORE ENABLES YOU TO:

- View & download all subscribed reports from a single, online platform
- Contact our analysts to answer any questions related to our reports or request a custom research
- Share comments on specific reports with the other users of your organization
- Suggest titles/topics to our research teams that you would like us to cover in our future reports
- Identify reports published on the high growth markets within your industry

Get started now by requesting a demo and learning more about Knowledge store <https://www.mnmks.com>

### MARKETSANDMARKETS KNOWLEDGE STORE SNAPSHOT



UK +44 20396 89007 US +1 206 480 1717 [mnmksupport@marketsandmarkets.com](mailto:mnmksupport@marketsandmarkets.com)

**KNOWLEDGESTORE**  
World's no.1 platform for High Growth Niche Markets

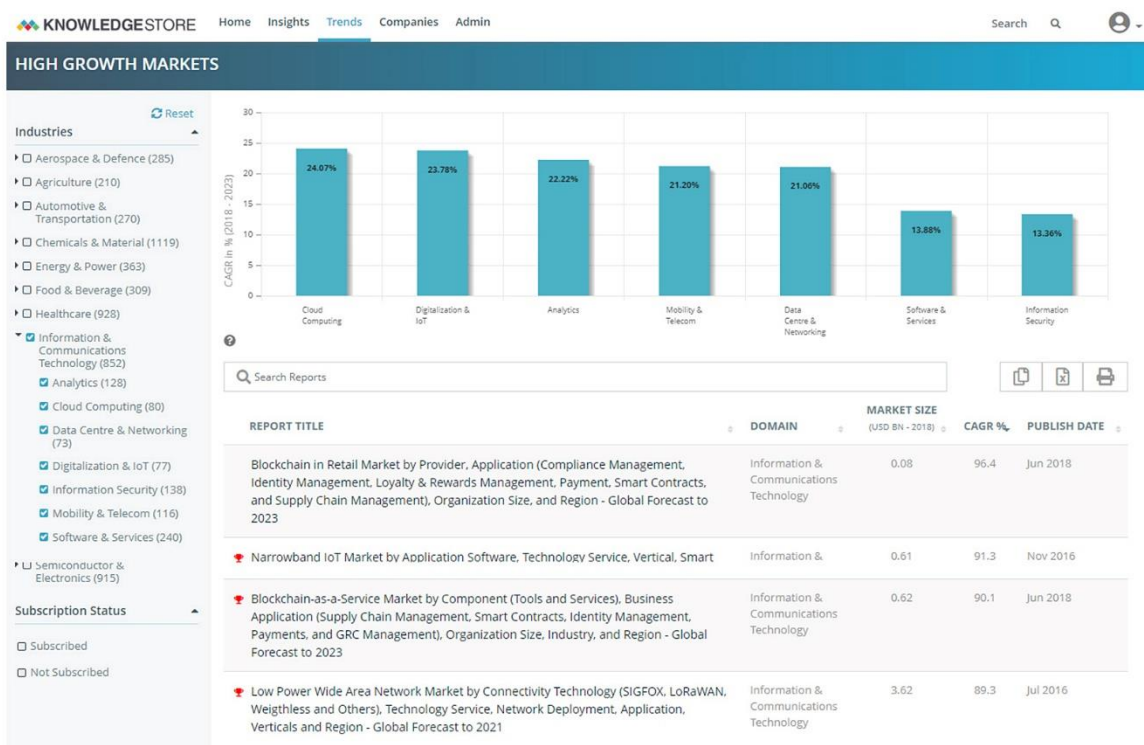
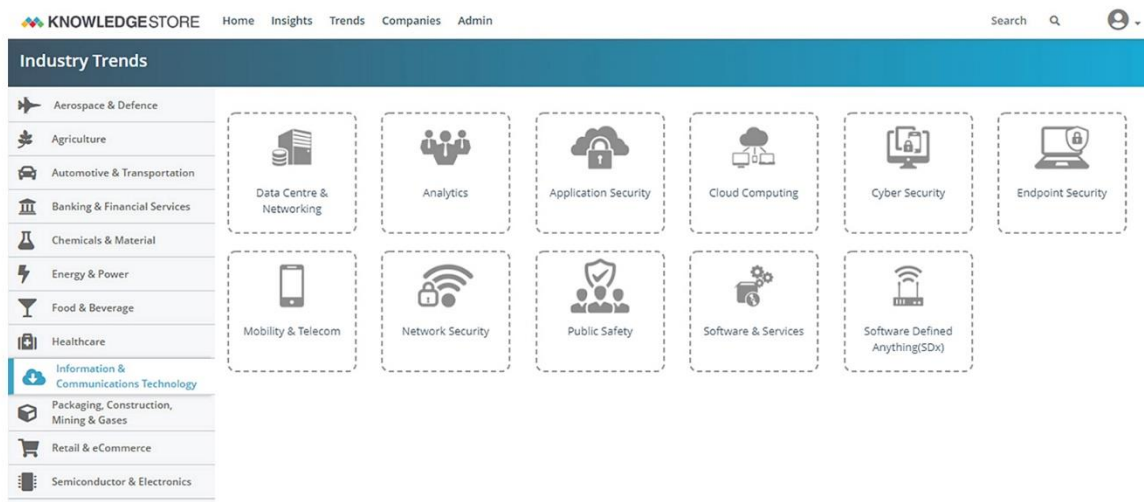
Username Password [Login](#) [Forgot Password?](#)

**IMPACT YOUR REVENUES**  
WITH EXCLUSIVE, IN-DEPTH MARKET INTELLIGENCE

80% of the Global 2,000 use MarketsandMarkets to identify new revenue opportunities in high growth niche markets. Shouldn't you?

[REQUEST TRIAL](#) [WATCH VIDEO](#)

## MARKETSANDMARKETS KNOWLEDGE STORE: INFORMATION & COMMUNICATION TECHNOLOGY SNAPSHOT



### 15.3 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations as per the company's specific needs. The following customization options are available for the report:

#### PRODUCT ANALYSIS

- Product Matrix which gives a detailed comparison of the product portfolio of each company

#### COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to 5)

### 15.4 RELATED REPORTS

| SR. NO | REPORT TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | PUBLISHED DATE |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1      | <b>QUANTUM CRYPTOGRAPHY MARKET - GLOBAL FORECAST TO 2025</b><br>By Component (Solutions and Services), Services (Consulting and Advisory, Deployment and Integration, and Support and Maintenance), Security Type (Network and Application Security), Vertical & Region<br><a href="https://www.marketsandmarkets.com/Market-Reports/quantum-cryptography-market-45857130.html">https://www.marketsandmarkets.com/Market-Reports/quantum-cryptography-market-45857130.html</a> | JUL 2020       |
| 2      | <b>DEEP LEARNING MARKET - GLOBAL FORECAST TO 2023</b><br>By Offering (Hardware, Software, and Services), Application (Image Recognition, Signal Recognition, Data Mining), End-User Industry (Security, Marketing, Healthcare, Fintech, Automotive, Law), and Geography<br><a href="https://www.marketsandmarkets.com/Market-Reports/deep-learning-market-107369271.html">https://www.marketsandmarkets.com/Market-Reports/deep-learning-market-107369271.html</a>             | MAR 2018       |

## 15.5 AUTHOR DETAILS

### Balaji Sreedhar

*Associate Vice President,  
Information Communication  
& Technology*

With an experience of over 14 years, he has been associated with ICT research for more than 10 years now. Prior to joining Markets and Markets, he has worked with firms like Frost and Sullivan, AMI-Partners, TNS Market Research and global IT major, IBM, where he worked for more than 5 years, and his last role saw him as Market Segment Manager. He holds rich experience in the technology sector working on several customs and syndicated research assignments with large Multinational Organizations.

### Sameer Bhatnagar

*Associate Director,  
Information Communication  
& Technology*

The role encompasses handling syndicated and custom research projects in the ICT domain. Having more than 14 years of experience with 10+ years in research, Sameer joined MarketsandMarkets from Forrester Research, where he was working as a Consultant, serving the Business Technology (BT) Professionals role. The role saw him working on varied consulting/custom-research projects and communicating directly with clients, internal resources (i.e., sales, research), and third-party vendors to address queries and manage expectations to create quality deliverables. Sameer has singlehandedly driven projects that include deliverable's in the shape of data modeling, trend analysis, end-user requirements, competitive profiling, etc.

### Contributors

Abhinav Dhruwa  
Research Manager  
Sushmit Chakraborty  
Team Lead  
Anchal Kushwaha  
Research Analyst  
Kiran Neoge  
Senior Research Associate

Disclaimer: MarketsandMarkets™ provides strategic analysis services to a select group of customers in response to orders. Our customers acknowledge when ordering that these strategic analysis services are solely for internal use and not for general publication or disclosure to any third party. MarketsandMarkets™ does not endorse any vendor, product, or service profiled in its publications. MarketsandMarkets™ strategic analysis constitutes estimations and projections based on secondary and primary research and are therefore subject to variations. MarketsandMarkets™ disclaims all warranties, expressed or implied, with respect to this research, including any warranties of merchantability or fitness, for any particular purpose. MarketsandMarkets™ takes no responsibility for incorrect information supplied to it by manufacturers or users.

Trademarks, copyrights, and other forms of intellectual property belong to MarketsandMarkets™ or their respective owners and are protected by law. Under no circumstance may any of these be reproduced, copied, or circulated in any form, without the prior written approval of MarketsandMarkets™ or its owner—as the case may be. No part of this strategic analysis service may be given, lent, resold, or disclosed to any third party, without express permission from MarketsandMarkets™.

Reproduction and/or transmission in any form and by any means, including photocopying, mechanical, electronic, recording, or otherwise, without the permission of MarketsandMarkets™, is prohibited.

For information regarding permission, contact:  
Tel: +1-888-600-6441